

Study exploring the context, challenges, opportunities, and trends in algorithmic management

VT-2022-035

under the multiple framework contracts EMPL/2020/OP/0016

Final Report

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Contact: Andrea GLORIOSO

E-mail: empl-b1-unit@ec.europa.eu

European Commission
B-1049 Brussels

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Authors:

Agnė Paliokaitė, Aleksandr Christenko, Antonio Aloisi, Phoebe Moore, Nastazja Potocka-Sionek, Robert S. Donoghue, Pijus Krūminas, Diana Čop, Jan Bednorz, Ildar Daminov, Aistė Ragaliauskaitė.

Contributors:

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Introduction

The present study was carried out to provide the Commission with a comprehensive, evidence-based analysis of the context, challenges, opportunities, and trends in algorithmic management (AM) in the workplace, as well as potential gaps regarding the relevance and effectiveness of the existing legal and policy responses to it. The temporal scope of the project is until the end of September 2023. The specific objectives of the study are to:

- Provide an overview of the concepts and typologies involved in algorithmic management. These are discussed in Chapter 1. Algorithmic management and the world of work, section Concepts and typologies of algorithmic management of the study.
- Carry out an extensive quantitative overview and analysis of the extent to which AM tools are used in the EU. This is discussed in Chapter 1. Algorithmic management and the world of work, section 1.2. 1.2. Prevalence of AM in the EU.
- Prepare an extensive analysis of the challenges and opportunities presented by AM tools for workers and employers. These are discussed in Chapter 2. Opportunities and challenges.
- Provide an overview and analysis of the relevant EU *acquis*, national legislation and social partner agreements. These are covered in Chapter 3. Legal and policy framework.
- Identify likely future scenarios for the evolution of algorithmic management, including how such evolution might affect the responses to AM of workers, employers, and public policies. These are described in Chapter 4. Future Trends.
- Prepare conclusions, underpinned by results of the above analysis. These are provided in the report's Conclusions.

The evidence base and overall findings of this report rely on the following sources and methods (for more details see *Annex 1: Methodological annex*):

- A detailed and comprehensive literature review of documents directly and indirectly relating to AM.
- An extensive review of existing legislation relevant to AM at EU and national levels, including five case studies on selected case law.
- Analysis of a large body of quantitative data directly and indirectly covering algorithmic management, including the European Survey of Enterprises on New and Emerging Risks 2019 (ESENER-3), the European Company Survey 2019 (ECS-2019), the European Working Conditions Telephone Survey 2021 (EWCTS-2021), and the Digital Economy and Society Index (DESI).
- A survey of employees, which yielded 1,279 complete responses.
- A survey of employers, which yielded 158 complete responses.
- A Delphi survey of academics, employer and employee representatives, AM tool developers, employers, and regulators. This survey covered current usage of AM, drivers of and barriers to change, and future trends. In total, 130 respondents provided complete responses.
- Interviews with national stakeholders and the creators of AM tools. In total, 56 such interviews were carried out.
- Interviews with academics and experts in the field. A total of 15 interviews were carried out.

- Six country-level case studies, exploring AM usage and the relevant regulatory contexts in Lithuania, Germany, Netherlands, Poland, Spain, and Sweden.
- Three case studies on AM tools, focusing on recruitment, employee monitoring and surveillance, and employee management.
- Four workshops, including three thematic workshops focusing on (i) data protection and the exercise of collective rights; (ii) the occupational safety and health implications of using AM tools in the workplace; and (iii) HR and people analytics. In addition, a final workshop was conducted with the client and relevant stakeholders from the European Commission.

Significant efforts were dedicated to building a broad and solid evidence base for the study. However, several challenges and constraints were encountered. The main two issues encountered were as follows.

First, given the novelty and complexity of the topic, there is a limited amount of data available, and some of the available data are inconsistent between data sources. For example, according to ECS-2019, around 27 % of organisations monitor worker performance; meanwhile, according to ESENER-3, only 8.2 % of companies do so. To a certain extent, this disparity may be explained by variations in the sampling methods of the two surveys, but this does not fully explain the discrepancies. Different sources of data, including qualitative and quantitative ones, were used to mitigate this issue, which was corroborated as far as possible.

Second, the surveys conducted received a low rate of responses, particularly from employers. This is probably due to survey fatigue, lack of motivation, and a tense relationship between employees and employers in some countries during the implementation of the study. In an attempt to boost the response rate, the study team tried to increase the potential number of respondents and carried out extensive follow-up activity. With regard to the number of respondents, invitations were sent out to more than 1,900 and 1,700 workers' and employers' representatives, respectively, at the Member State level. In addition, 27,000 HR managers were contacted from companies operating within the EU, and the 10 main EU-level employer and employee representative organisations were contacted. In terms of follow-up, several rounds of follow-up emails were sent out, and an extensive calling campaign was conducted. Nevertheless, the response rate remained rather low. Due to these limitations, outputs from the surveys are predominantly used throughout the report to support findings from other sources. For more information on the challenges encountered, mitigations strategies employed, and the methodology used, see *Annex 1: Methodological annex*.

Table 0-1 below describes how the various chapters and sections of the report reflect the implementation of study tasks, as outlined in the Tender Specifications.

Table 0-1 – Study Map

Structure of the report	Respective study tasks covered
Introduction	
1. Algorithmic management and the world of work	
1.1. Concepts and typologies of AM	**WP1 Task 1:** Detailed and comprehensive mapping of the literature concerning AM **WP1 Task 3:** Quantified overview of the extent of the use of AM tools in the EU **WP1 Task 5:** Case studies **WP3:** Consultations
1.2. Prevalence of AM in the EU	

Structure of the report	Respective study tasks covered
2. Opportunities and challenges	
2.1. Methodology	**WP1 Task 1:** Detailed and comprehensive mapping of the literature concerning AM **WP1 Task 2:** Identification of challenges and opportunities **WP1 Task 5:** Case studies **WP3:** Consultations
2.2. Opportunities posed by AM for workers and employers	
2.3. Challenges posed by AM for workers and employers	
2.4. Effects of AM on the public sector and social security systems	
2.5. General effects of AM on broader society	
2.6. General effects of AM on competitiveness and consumers	
3. Legal and policy framework	
3.1. Methodology	**WP2 Task 1:** Mapping of policy and legislation addressing AM **WP2 Task 2:** Analysis of EU legislation concerning AM **WP3:** Consultations
3.2. Working conditions	
3.3. Data protection and privacy	
3.4. Fairness and discrimination	
3.5. Workers' collective rights	
3.6. Effects on broader society	
4. Future trends	
4.1 Assessing the possible future evolution of AM	**WP1 Task 4:** Future trends in AM **WP3:** Consultations
4.2. Likely effects of further growth in AM usage	
Conclusions and recommendations	
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Annex 1: Methodological annex	**WP1, WP2, WP3**
Annex 2: Literature mapping	**WP1 Task 1:** Detailed and comprehensive mapping of the literature concerning AM
Annex 3: Synopsis report covering all stakeholder consultations	**WP3:** Consultations
Annex 4: Summaries of research projects	**WP1 Task 1:** Detailed and comprehensive mapping of the literature concerning AM
Annex 5: Country case study reports	**WP1 Task 3:** Quantified overview of the extent of the use of AM tools in the EU
Annex 6: Quantified data on AM adoption in the six countries selected for the case studies	**WP1 Task 3:** Quantified overview of the extent of the use of AM tools in the EU
Annex 7: Case study reports on AM tools	**WP1 Task 5:** Case studies
Annex 8: Legislation and policy mapping	**WP2 Task 1:** Mapping of policy and legislation addressing AM

Structure of the report	Respective study tasks covered
Annex 9: Workers' survey questionnaire	**WP3:** Consultations
Annex 10: Employers' survey questionnaire	**WP3:** Consultations
Annex 11: Delphi survey questionnaire	**WP1 Task 4:** Future trends in AM
Annex 12: Interview questionnaires	**WP3:** Consultations
Annex 13: Factual summary of the workers' survey	**WP3:** Consultations
Annex 14: Factual summary of the employers' survey	**WP3:** Consultations
Annex 15: Factual summary of the Delphi survey	**WP1 Task 4:** Future trends in AM
Annex 16: Factual summaries of the workshops	**WP3:** Consultations
Annex 17: Trends in the use of AI, robotics and digital technology	**WP1 Task 4:** Future trends in AM

1. Algorithmic management and the world of work

This chapter of the report explores the main concepts, definitions and typologies of algorithmic management, as well as discusses the prevalence of such technologies across the EU and beyond. It is structured as follows:

- Concepts and typologies of algorithmic management – describes the conceptualisation of algorithmic management and details how it can be disaggregated into managerial[1] functions.
- 1.2. 1.2. Prevalence of AM in the EU – provides an overview of the usage of AM, highlighting the prevalence of such tools in different countries, organisations, and for different types of workers.

1.1. Concepts and typologies of algorithmic management

Key findings:

- There is no singular, widely agreed definition of what algorithmic management (AM) is, but a large number of definitions agree that AM is related to the automation of managerial tasks.
- It may be more advantageous to look at AM not as a single concept but through the range of managerial functions automated using AM tools. These may include: (i) recruitment; (ii) work/task scheduling; (iii) nudging/directing (e.g. algorithms that suggest to workers how to complete tasks); (iv) worker monitoring/surveillance; (v) worker evaluation; (vi) talent management/training; (vii) rewarding workers; and (viii) worker termination/firing/dismissal. This approach simplifies the analysis and enables data to be collected from workers and employers who might not be familiar with the concept of AM, but know when algorithms are used for various managerial functions. It also streamlines policy analysis.

1.1.1 Definition of key terms

According to Mateescu and Nguyen (2019), **algorithmic management (AM) refers to 'a diverse set of technological tools and techniques to remotely manage workforces, relying on data collection and surveillance of workers to enable automated or semi-automated decision-making'**.[2] Based on this definition, algorithmic management refers to a shift whereby tasks that have traditionally been performed by (human) managers are carried out through, and often by, algorithms. These algorithms can rely on a list of simple programmed procedures (e.g. if a person completes a task, the algorithm can give him/her a new one from a list of predefined tasks). Alternatively, they may be complex, including machine learning or artificial intelligence (AI)-based approaches (e.g. some algorithms use AI to assess the emotional well-being of a worker based on their communication patterns).[3]

AM tools can interact with workers through hardware (e.g. wearables that track employees) and software (e.g. computer programs that allocate tasks to workers). Such tools may also operate completely automatically; namely, once the algorithms have been programmed, they make decisions and interact with workers without any intervention or supervision from

[1] 'Managerial' as used in the current report is a term that refers to *practices of management*, broadly conceived. A literature within Management Studies uses the term 'managerial' to refer to management practices within 'new public management' (see e.g. Moore P.V., Joyce S. (2019) Black box or hidden abode? The expansion and exposure of platform work managerialism. *Review of International Political Economy*). The current report does not use the term in this way.

[2] Mateescu, A., & Nguyen, A. (2019). Explainer: Algorithmic management in the workplace. Data & Society Research Institute.

[3] Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek, E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work. Report prepared for EU-OSHA.

managers.[4] Algorithms can also operate in a semi-automated fashion, whereby they provide data, information and/or recommendations relating to certain managerial activities to managers, who make final decisions.[5] Although Mateescu and Nguyen (2019) highlight that AM is connected to decision-making, it can also be used in other ways, such as to collect data about workers. Hence, expanding on the definition by Mateescu and Nguyen (2019), some AM tools are not simply '…relying on data collection and surveillance of workers…', but their whole purpose may be to collect such data. For more information on the various managerial functions for which AM can be used, see the following subsection 1.1.2. Functions and typologies of AM.

The definition by Mateescu and Nguyen (2019), with the small modifications outlined above, is used, where relevant, as the operational definition for this study. However, it is only one of many, as **there is no widely adopted definition of what algorithmic management is**. For example, a similar definition comes from Wood (2021), who conceptualised AM as 'software to automate organisational functions traditionally carried out by human managers, identified in both platform work and conventional employment settings'.[6] While this definition shares many similarities with that of Mateescu and Nguyen (2019), Wood (2021) explicitly mentions platform work (working arrangements under which individuals provide their services through online platforms), from which many algorithmic management tools originate[7]. However, this definition is also limited, as it focuses only on software – even though hardware, such as wearables, cameras and sensors, which can be used to monitor workers, are also widely utilised in AM.

Alternatively, according to Baiocco et al. (2022), 'algorithmic management is the use of computer-programmed procedures to coordinate labour'.[8] Similarly, Hossein and Sutherland (2019) describe AM as 'a method of organising and coordinating extremely large groups of workers and clients in an automated way'.[9] Both authors provide simple and short definitions focusing on coordination. However, by focusing on coordination and, in the case of Sutherland (2019), also on organisation, important aspects of worker management may be omitted, such as analysing workers' performance, engagement, satisfaction, etc. In addition, although the definition applied by Sutherland (2019) mentions that AM is used on 'extremely large groups', such technologies are sometimes employed in smaller companies, as will be discussed in Section 1.2. 1.2. Prevalence of AM in the EU. In addition, algorithmic management has also recently been mentioned and discussed in the Explanatory Memorandum of the Proposal for a Directive on improving working conditions in platform work.[10] In this document, AM is defined as 'automated systems to match supply and demand for work, […] assign tasks, monitor, evaluate and take decisions for the people working through [digital labour platforms]'.[11] However, this definition is limited to platform work; currently, in the EU legal sphere, there is no definition of AM that is applicable to regular work.

[4] Prassl, J. (2019). What if your boss was an algorithm? Economic Incentives, Legal Challenges, and the Rise of Artificial Intelligence at Work. *Comparative Labor Law and Policy Journal*, *41*(1).

[5] Lenaerts, K., Waeyaert, W., Smits, I., & Hauben, H. (2021). Digital platform work and occupational safety and health: A policy brief. Digital platform work: occupational safety and health policy and practice for risk prevention and management.

[6] Wood, A.J. (2021). *Algorithmic management consequences for work organisation and working conditions* (No. 2021/07). JRC Working Papers Series; Baiocco, S., Fernández-Macías, E., Rani, U., & Pesole, A. (2022). *The algorithmic management of work and its implications in different contexts*. No. 2022/02. JRC Working Papers Series on Labour, Education and Technology.

[7] Aloisi, A., & De Stefano, V. (2022). Your boss is an algorithm: artificial intelligence, platform work and labour. Bloomsbury Publishing.

[8] Baiocco, S., & Fernandez Macias, E. (2022). *The algorithmic management of work: A basic compass, European Commission*, JRC129819.

[9] Jarrahi, M.H., & Sutherland, W. (2019). 'Algorithmic management and algorithmic competencies: Understanding and appropriating algorithms in gig work.' *International Conference on Information*. Springer, Cham.

[10] See: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM%3A2021%3A762%3AFIN

[11] For more information, see the chapter 'Context of the Proposal' in the proposed directive, a link to which can be found above.

Despite the lack of a legal definition that is applicable to all types of workers, by relying on several pieces of EU legislation, it is possible to draw a comprehensive picture of AM 'by proxy'. For example, the **General Data Protection Regulation (GDPR)** provides for an (indirect) definition of automated decision-making, which is strongly connected to AM, borrowed from its predecessor, the Data Protection Directive 95/46/EC. Article 22(1) of the GDPR mentions that 'The data subject shall have the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her, or similarly significantly affects him or her'.[12] This implies that workers are protected from AM tools that significantly affect them. In addition, Article 22(2) also highlights that Article 22(1) shall not apply if the decision 'is necessary for entering into, or performance of, a contract between the data subject and a data controller',[13] which might be understood as a contract between an employer and worker. However, this is only one of several possible interpretations of the aforementioned articles.

In addition, the **draft EU regulation on AI**[14] discusses certain AI-driven practices, including some uses or practices identified as high-risk AI systems – namely, 'AI systems that pose significant risks to the health and safety or fundamental rights of persons'.[15] Among these high-risk systems, Annex III of the proposal includes 'AI systems intended to be used for recruitment or selection of natural persons', and 'AI intended to be used for making decisions on promotion and termination of work-related contractual relationships, for task allocation and monitoring and evaluating performance and behaviour of persons in such relationships'.[16] This implies that AM can be considered a 'high-risk' application of algorithms. MEPs adopted a negotiation position, and discussions began with the Member States.[17] Hence, this regulation may come into force in the upcoming years. For more on how AM is covered in current EU and national legislation, see Chapter 3. Legal and policy framework.

To summarise, **at the level of the workplace, AM tools inform, shape and design human resources and managerial practices in many ways, and their effects are not always intentional or explicit.**[18] Therefore, for research and regulatory purposes, focusing on strict definitions could be problematic, as it creates a risk of being over- or under-inclusive, and is prone to rapid obsolescence, given the highly innovative and fast-paced nature of the phenomenon. For these reasons, **when discussing AM, it is better to focus on those managerial functions that can be automated using algorithms, rather than the abstract concept of algorithmic management**. This is the approach employed in this research. However, where relevant, overarching conclusions concerning AM will be derived using the operational definition outlined previously.

1.1.2. Functions and typologies of AM

The notion of looking at AM through the prism of the managerial functions it automates helps us to define the target of the study more clearly, as well as helping businesses, workers, and social partners to identify those actions that may fall within the remit of AM. Moreover, from a regulatory perspective, such a functionalist approach may facilitate the identification and assessment of instruments of EU law that already regulate one or more of these functions. For example, when AM tools are

[12] See: https://gdpr-info.eu/art-22-gdpr/.

[13] Ibid.

[14] See: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A52021PC0206.

[15] See: https://eur-lex.europa.eu/resource.html?uri=cellar:e0649735-a372-11eb-9585-01aa75ed71a1.0001.02/DOC_1&format=PDF.

[16] See: https://eur-lex.europa.eu/resource.html?uri=cellar:e0649735-a372-11eb-9585-01aa75ed71a1.0001.02/DOC_2&format=PDF.

[17] See: https://www.europarl.europa.eu/news/en/press-room/20230609IPR96212/meps-ready-to-negotiate-first-ever-rules-for-safe-and-transparent-ai.

[18] Jarrahi, M.H., Newlands, G., Lee, M.K., Wolf, C.T., Kinder, E., & Sutherland, W. (2021). 'Algorithmic management in a work context.' *Big Data & Society* 8, no. 2.

(primarily) used in hiring, anti-discrimination and equality law provisions enshrined in a large set of EU Directives could be applied[19] (the same provisions can be enforced when AM is deployed to assign competitive entitlements among workers). When AM tools are predominantly used to organise tasks at work by setting methods and pace, legal provisions covering working conditions, working time, and access to work (e.g. as laid down in the Working Time Directive,[20] the Directive on Transparent and Predictable Working Conditions,[21] the Work-Life Balance Directive[22] and many more) must be used to ensure labour rights. In addition, provisions covering OSH (e.g. the Occupational Health and Safety Framework Directive 89/391/EEC[23]) should be applied, thereby ensuring workers' health and safety.

Based on the literature[24] and interviews, the following is a possible list of managerial functions for which AM can be used:

- **Recruitment** – automating some, or all, activities in the recruitment process. This might include, but is not limited to: automated CV screening; analysing answers provided by potential candidates; as well as analysing how they perform during a recruitment interview; and identifying and contacting potential new candidates for a position through social platforms, such as LinkedIn, etc.
- **Work/task scheduling/distribution** – automatic allocation of tasks or scheduling of shifts for workers. Algorithms can allocate tasks and schedule work in many ways, including allocating newly created tasks to those who are currently not very busy, scheduling shifts based on the availability or preferences of workers, assigning a task to a worker whose skill set is best suited to tackling it, etc.
- **Nudging/directing** – suggesting or prompting/nudging workers to behave in a particular manner. For example, algorithms suggesting how a person could complete a task (e.g. on which tasks to focus first, how to improve productivity), guiding workers to the location of a particular item they need, such as a package in a warehouse or a file in an office setting, etc.
- **Worker monitoring/surveillance** – technologies that allow employers to gather information on the activities and location of workers. This can be done purely through software solutions, such as keyloggers which log every keystroke a worker types, or email scanners that scan the contents of emails, as well as through tools that combine hardware and software, such as wearables and sensors that track workers.
- **Worker evaluation** – using automated tools to evaluate workers' performance, based on selected metrics. This might include evaluating how quickly workers complete tasks, how many breaks they take, assessing evaluations from clients regarding workers' outputs, etc.

[19] For example, Directive 2000/78/EC on equal treatment in employment and occupation; Council Directive 1999/70/EC of 28 June 1999 concerning the framework agreement on fixed-term work concluded by the ETUC, UNICE and CEEP; Council Directive 97/81/EC of 15 December 1997 concerning the Framework Agreement on part-time work concluded by UNICE, CEEP and the ETUC - Annex: Framework agreement on part-time work; Directive 2008/104/EC of the European Parliament and of the Council of 19 November 2008 on temporary agency work.

[20] See: https://ec.europa.eu/social/main.jsp?catId=706&langId=en&intPageId= 205.

[21] See: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32019L1152 .

[22] See: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32019L1158 .

[23] See: https://osha.europa.eu/en/legislation/directives/the-osh-framework-directive/1 .

[24] This discussion is based on a variety of sources, including Kellogg, K.C., Valentine, M.A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. Academy of Management Annals, 14(1), pp.366-410; Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek, E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work. Report prepared for EU-OSHA.; Parent-Rocheleau, X., & Parker, S.K. (2022). Algorithms as work designers: How algorithmic management influences the design of jobs. *Human Resource Management Review*, 32(3), 100838.

- **Talent management/training** – technologies that assess the competencies of workers, identify gaps in knowledge, and/or recommend how these gaps could be filled. Falling under this category are tools that can assess what kinds of skills and knowledge an individual possesses, and if it is sufficient for their current position.
- **Rewarding workers** – systems that dynamically reward workers on the basis of their performance. These might include algorithms that calculate and offer bonuses or promotions to high-performing workers.
- **Worker termination/firing/dismissal** – tools that can identify underperforming workers, and provide recommendations to HR managers regarding terminations.

The managerial functions automated by AM can be further grouped into different typologies, described in Table 1-1 below. Using an aggregated typology rather than only focusing on separate managerial functions provides flexibility – it enables new functions to be introduced into the typology, which may merge due to the rapidly developing algorithmic management phenomenon. However, it is also important to highlight that **although the typologies outlined in the table below provide some examples of how AM functions can be grouped, in many cases these typologies are also limiting.** For example, it is unclear where recruiting would be allocated in the typology proposed by Parent-Rocheleau et al. (2022). Similarly, the typology used by Hoffmann (2022) does not clearly define worker monitoring and surveillance. This is an issue with more concrete typologies that might be exacerbated in the future, given that the rapid development of AM might lead to the introduction of new managerial functions that could be performed by such tools. Similarly, typologies which apply to algorithms in general, such as the one discussed by Unruh et al. (2022), can be too vague to accurately classify various AM functions or to structure a discussion.

Table 1-1 – Typologies of AM

Source	Typology
Baiocco et al. (2022)[25]	Splits algorithmic management into: (i) **Planning and strategy**; (ii) **Allocation of work and direction**; (iii) **Control and monitoring**
Parent-Rocheleau et al. (2022)[26]	Splits AM into six HR and managerial functions: (i) **Monitoring**; (ii) **Goal setting;** (iii) **Performance management**; (iv) **Scheduling**; (v) **Compensation**; (vi) **Job termination**
Hoffmann (2022)[27]	Divides the typology of AM into: (i) **Goals specification;** (ii) **Task specification**; (iii) **Planning specification**; (iv) **Incentivising behaviour**; (v) **Staffing**.
Ministerio de Trabajo y Economía Social (2022)[28]	Splits AI, algorithmic and automated decision-making into: (i) **Selection and recruitment;** (ii) **Monitoring and surveillance;** (iii) **Work direction and management**.
Kellogg et al. (2020)[29]	Splits worker control through algorithms into: (i) **Evaluation**; (ii) **Direction**; (iii) **Discipline.**

[25] Baiocco, S., Fernández-Macías, E., Rani, U., & Pesole, A. (2022). The algorithmic management of work and its implications in different contexts (No. 2022/02). JRC Working Papers Series on Labour, Education and Technology.

[26] Parent-Rocheleau, X., & Parker, S.K. (2022). Algorithms as work designers: How algorithmic management influences the design of jobs. *Human Resource Management Review*, 32(3), 100838.

[27] Nurski, L., & Hoffmann, M. (2022). The impact of artificial intelligence on the nature and quality of jobs, Bruegel.

[28] Ministerio de Trabajo y Economía Social (2022). Información algorítmica en el ámbito laboral. Guía práctica y herramienta sobre la obligación empresarial de información sobre el uso de algoritmos en el ámbito laboral

[29] Kellogg, K.C., Valentine, M.A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. *Academy of Management Annals*, 14(1), 366-410.

Source	Typology
Unruh et al. (2022)[30]	Discusses algorithmic management and splits algorithms into: (i) **Descriptive**, (ii) **Predictive**; (iii) **Prescriptive.**

Source: Authors' elaboration, based on the literature review.

With this in mind, this study employs a modified version of the Kellogg et al. (2020) typology, which is based on the work of Edward (1979),[31] as the latter is directly connected to the world of work, and also allows some flexibility.

[30] Unruh, C.F., Haid, C., Johannes, F., & Büthe, T. (2022, July). Human autonomy in algorithmic management. In: *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 753-762).

[31] Edwards, R. (1979). C*ontested terrain: The transformation of the workplace in the twentieth century*. New York: Basic Books.

Table 1-2 below describes the typology used in this study, which separates managerial functions into three groups – evaluation, direction, and discipline – as well as highlighting which managerial functions fall into which group.

The first group, **evaluation**, predominantly deals with the collection of data and transforming this into information (such as, for example, ranking workers based on criteria of productivity, efficiency, or effectiveness). This, in turn, helps managers to make more informed decisions or can be also used in other types of managerial functions.

The second group, **direction**, covers managerial functions that are directly connected to improving the implementation of work-related activities, as well as being used to fill in the gaps in worker's knowledge. Hence, algorithms that directly interact with workers and try to improve effectiveness, efficiency, etc. fall into this group.

The third group, **discipline**, relates to tools that influence workers' behaviour indirectly, by punishing or rewarding them. Hence, algorithms that are not directly connected with improving individuals' day-to-day work, but which interact indirectly with them, fall under this category.

In Kellogg's typology, recruitment falls under this category, which might introduce some confusion. To alleviate this, recruitment was decoupled from the existing three dimensions and will be treated separately. However, this decoupling did not affect the analysis or data collection activities carried out, as data were collected at the level of function. Hence, this modification to the Kellogg typology only affects how the results are presented.

Table 1-2 – Modified Kellogg et al. (2020) typology of managerial functions that can be automated using AM

Dimensions	Short description	Managerial functions
Evaluation	Collecting data on workers and reviewing their activities. Hence, **'evaluation' describes when an algorithm gathers and synthesises information on workers**.	**Worker monitoring/surveillance**
		Worker evaluation (includes transforming the data collected on workers into a more usable form, such as rankings of workers)
Direction	Providing recommendations, instructions and 'nudges' to guide workers on what tasks they need to perform, and how to perform them efficiently and effectively. Hence, **'direction' describes when algorithms interact directly with how workers do their jobs.**	**Work/task scheduling/distribution**
		Nudging/directing tools
		Talent management/training (includes identifying gaps in knowledge and providing recommendations to workers on how to fill them)
Discipline	Punishing and rewarding workers to promote desired conduct and deter undesired behaviour. Hence, **'discipline' describes when data that are collected about workers are used by managers to make decisions that are not directly connected with the work performed, but indirectly try to improve it.**	**Rewarding workers**
		Worker termination/firing/dismissal
Recruitment	Automating some, or all, activities in the recruitment process	**Recruitment**

Source: Authors' own elaboration, based on Kellogg et al. (2020).

These dimensions are strongly interconnected, and they are not inherently positive or negative. For example, a combination of the dimensions of evaluation and discipline (which include worker monitoring, evaluating worker performance, and disciplining workers), as Kellogg et al. (2020) define them, could be used to identify if workers are correctly wearing their safety equipment and disciplining them if they fail to do so. This can be considered positive if workers give their consent to being monitored on the job in this regard. At the same time, if personal data are collected to discipline workers without first informing them or gaining their consent, this can lead to a plethora of negative consequences, including shifting the power balance between the employer and worker. It can also lead to negative psychological consequences for workers if they find out about such monitoring. To better reflect the ambiguous nature of AM, when discussing these dimensions, the study authors tried to keep the language as neutral as possible and, where relevant, explicitly highlight the opportunities and challenges created by the technologies associated with these dimensions.

This typology, including the dimensions and the relevant managerial functions, is used in this report when discussing the prevalence of AM within the EU (Section 1.2. 1.2. Prevalence of AM in the EU), and future trends in algorithmic management (Chapter 4. Future Trends). When discussing the opportunities and challenges of AM, Chapter 2. Opportunities and challenges also references relevant managerial functions and dimensions. However, given that many different managerial functions can have the same effects on workers (thus implying a high level of overlap), **Chapter 2. Opportunities and challenges introduces a different typology, for the opportunities and challenges of AM**. This additional typology, based on the one presented here, avoids the duplication of information, and is also later used in the analysis of legislation covering AM (see Chapter 3. Legal and policy framework).

1.2. Prevalence of AM in the EU

This section of the report explores the extent of AM usage in the EU-27 (i.e. following Brexit). The analysis highlights its usage across different managerial functions. Where data are available, in addition to highlighting the use of AM in general, the discussion covers its usage across different types of workers and employers, including differences in: (i) **countries / Member States**, (ii) **age groups**, (iii) **genders**, (iv) **levels of education**, (iv) **occupations**, (v) **public/private sectors**, (vi) **economic sectors**, and (vii) **size of the companies**.

Given the lack of a robust, unified dataset on the usage of AM, several sources were employed in our analysis. The first comprises a **literature review of peer-reviewed and high-level documents** and an **analysis of large-scale EU surveys and databases**, which are considered robust sources. The literature review predominantly focused on high-level peer-reviewed publications and reports, while insights from grey literature were only covered if they used a relatively robust methodology (e.g. using large sample size, clearly described methodology, etc.). Concerning the analysis of quantitative data, the study focuses on sources containing direct information on AM usage, such as how algorithms and AI technologies are used for different managerial functions. The table below summarises the quantitative data sources used that include direct insights on AM usage, as well as highlighting the target group, timeline, geographic coverage, and sample size of each source.

Table 1-3 – Overview of data sources analysed that provide direct insights into the prevalence of AM

Data	Description	Examples of relevant variables / questions
European Survey of Enterprises on New and Emerging Risks 2019 (ESENER-3)	A survey was carried out with European businesses to identify prevailing and emerging risks in the workplace, especially those associated with occupational safety and health (OSH). • **Target group** – companies and organisations from all sectors. The dataset only contains information on companies with five or more employees. • **Timeline** – only data on 2019 were used in the analysis, as older ESENER (e.g. ESENER-2014 or 2009) do not provide information about AM. • **Geographical coverage** – EU-28 + Iceland + Macedonia + Norway + Switzerland + Serbia. However, only data on Member States that are part of EU after Brexit will be used in the present analysis. **Sample size*** - 37,460 organisations in the EU-27	Does your establishment use: • Machines, systems, or computers to determine the content or pace of work? • Machines, systems or computers that monitor workers' performance? • Wearable devices, such as smart watches, data glasses or other (embedded) sensors?
European Company Survey 2019 (ECS-2019)	ECS-2019 comprised two comprehensive surveys, one focusing on managers and the other on workers' representatives. These cover various characteristics of work, including types of work organisation, participation, representation, etc. In the present study, only the managers' dataset was analysed. This is because the survey of workers' representatives does not contain relevant information on AM. In addition: • **Target group** – companies and organisations from all sectors apart from the following: A. Agriculture, forestry, and fishing; O. Public administration and defence; compulsory social security; P. Education; Q. Human health and social work activities. The survey only contained data on organisations with 10 or more employees. • **Timeline** – only data on 2019 were used for the present analysis, as older ECS data (e.g. ECS-2013 or 2009) do not	• For how many employees at this establishment is the pace of work determined by machines or computers? • Does this establishment use data analytics to monitor employee performance?

Data	Description	Examples of relevant variables / questions
	provide information on AM. • **Geographical coverage** – EU-28. However, for the present analysis, only data on countries that remained EU Member States after Brexit have been used. • **Sample size*** - 21,172 organisations in the EU-27 (from the survey of managers that was used in the study)	
European Working Conditions Telephone Survey 2021 (EWCTS-2021)	EWCTS-2021 was a modification of the European Working Conditions Survey that was conducted during the COVID-19 pandemic, and covered various aspects of work, including working conditions, well-being, company practices, etc. In addition: • **Target group** – employees and the self-employed. Covers individuals working in enterprises of all sizes. • **Timeline** – only data on 2021 have been used for the present analysis, as older EWCS data (e.g. EWCS-2015 or 2010) do not provide information on AM. • **Geographical coverage** – EU-28 + candidate countries (apart from Ukraine) + Norway + Kosovo + Switzerland. However, for the present analysis, only data on countries that remained EU Member States after Brexit has been used. • **Sample size*** - 58 403 individuals working in the EU-27 (50 684 employees, but 7 434 self-employed, and 285 individuals did not specify their current working arrangement). An argument can be made that the variable selected does not directly measure AM usage. Hence, the study provides a discussion on what motivated the selection of this variable under the heading *Usage by worker type*, which is discussed in subsection *1.2.2. Usage of AM: Evidence from direct sources* of this report.	• To what extent does a computerised system influence what you do in your work?
The Digital Economy and Society Index (DESI)	DESI is a composite index measuring the digital performance and progress of EU countries in terms of connectivity, human capital, the use of internet services, ignorance of digital technologies, and digital public services. It contains information that can be directly connected to AM (see the first point under 'examples of relevant variables/questions'), as well as general information on digitalisation and AI usage in the workplace. It covers: • **Target group** – MS. • **Timeline** – spans the period 2014-2022, but information on the two aforementioned variables is only available for 2021. • **Geographical coverage** – EU-28 + candidate countries (apart from Ukraine + Norway). However, for the present analysis, only EU-27 data were used. • **Sample size** – unknown, as no micro-level data are available.	• Percentage of enterprises that use AI technologies for human resource management or recruitment

(*) Sample sizes highlighted in the table specify the total size of the datasets. However, the sample sizes vary with respect to different variables, given that not all individuals and companies represented in these data provided answers/information with regard to all questions.
Source: Authors' elaboration, based on aforementioned data sources.

As can be observed in Table 1-3 above, no data source covers all managerial functions outlined in subsection 1.1.2. Functions and typologies of AM of this report. In addition, data sources often have different target groups or geographical coverage, and several of them cover different time frames. This, in turn, complicates the analysis, preventing the simple

aggregation of results. Hence, to increase robustness, the aforementioned data sources were complemented with several surveys carried out by the study, as well as proxy data.

First, Visionary Analytics (VA) carried out a **survey targeting employers, and another targeting workers** (henceforth referred to as the 'VA survey of employers' and 'VA survey of workers'). Given the previously noted lack of robustness of these VA surveys, insights from them were used throughout the report predominantly to supplement other sources. Second, a **Delphi survey** was carried out among experts in the field, AM tool creators, employers' and workers' representatives, and regulators. This survey predominantly focused on the future of AM, but it also sought insights into its current usage. Lastly, **proxy data, such as insights on the usage and evolution of advanced technologies**, were also employed to gather additional insights, especially regarding the future usage of AM. Each of these data sources on its own is not robust enough to provide definitive insights on AM. However, by combining information from them with other sources, including those discussed in Table 1-3, a relatively accurate picture regarding the prevalence of AM can be drawn. The table below provides further detail on each of the additional sources used.

Table 1-4 – Overview of data sources analysed that provide additional insights on AM

Data	Description	Examples of relevant variables / questions
VA worker survey	A survey was carried out for the present study, targeting workers throughout the EU-27 who might have experience with AM. The survey explicitly asked questions about AM usage, as well as the opportunities and challenges it creates for workers. In addition: • **Target group** – workers working in organisations of all sizes and from all sectors. • **Timeline** – 2023. • **Geographic coverage** – EU-27, with gaps in some countries (see *Annex 16: Factual summaries of each consultation* for details). • **Sample size*** – 1,270 workers who answered all questions in the survey, and 238 who provided partial answers that could be analysed, as they covered at least the first two blocks of survey questions. **The worker and employer surveys do not provide statistically representative results for the whole EU-27, due to the underrepresentation of certain countries and sectors.** Hence, results from the survey have been used to support other data sources by adding robustness or, in some cases, highlighting possible contradictions.	For a full list of questions used in the workers' survey, see *Annex 9: Workers' survey questionnaire*.
VA employer survey	A survey was carried out for the present study, targeting employers throughout the EU-27 who might have experience with AM. The survey explicitly asked questions about AM usage, as well as the opportunities and challenges it creates for employers. In addition: • **Target group** – representatives (e.g. HR managers, high-ranking managers) of organisations of all sizes and in all sectors, to gain the fullest picture of AM usage. • **Timeline** – 2023. • **Geographic coverage** – EU-27, with gaps in certain countries (see *Annex 16: Factual summaries of each consultation* for details). • **Sample size*** – 155 employers, with around 81 partial answers that could also be analysed.	For a full list of questions used in the employers' survey, see *Annex 10: Employers' survey questionnaire*.

Data	Description	Examples of relevant variables / questions
Delphi survey	A survey targeting experts in the field and relevant stakeholders, covering the current and future usage of AM: • **Target group** – academics, employers' representatives, employers and creators of AM tools, workers' representatives, regulators, policymakers, agency representatives. • **Timeline** – 2023. • **Geographic coverage** – EU-27. • **Sample size*** – 130 full answers.	For a full list of questions used in the Delphi survey, see *Annex 11: Delphi survey questionnaire*.
(Proxy data) the Digital Economy and Society Index (DESI)	*Same as described in Table 1-3 above.*	Questions on the usage of advanced technologies, such as: • AI in the workplace • Usage of cloud computing • Buying computing power to run the enterprise's own software

Given that different data sources differ in terms of robustness, and to ensure a more streamlined discussion, the remainder of this subsection explores insights from the source for each group before concluding with a general summary of the results. More specifically, the remainder of the section is structured as follows:

- 1.2.1. Evidence from direct sources – provides insights from the most robust and direct sources of information covering AM, discussed in Table 1-3.
- Hence, based purely on those sources analysed that look at the usage of AM tools among individuals with different education levels, workers with a higher level of education are more likely to be subject to algorithmic management than those with a lower level of education. One explanation for such a trend is the fact that individuals with higher education more often work in jobs using computers and other smart devices that can be used for algorithmic management. For example, according to EWCTS-2021, 74.1 % of individuals with a Bachelor's, Master's, or doctoral degree use computers 'Always', compared with only 21.1 % of individuals with upper- or lower-secondary, as well as individuals with early childhood education. This greater use of computers by those with high levels of education, in turn, creates greater opportunities for employers to use AM. An alternative explanation is that higher education tends to imply higher job complexity, which, in turn, means workers are more likely to interact with various digital tools that might involve an element of algorithmic management (e.g., workers might use email systems that are monitored by employers). Higher complexity also implies a greater likelihood that decision-making is data-driven, which may also be supported through algorithmic management.
- 1.2.2. Evidence from the Visionary Analytics surveys – discusses the results of the workers' and employers' surveys.
- 1.2.3. Evidence from the Visionary Analytics Delphi survey – discusses the results of the Delphi survey.
- 1.2.4. Evidence from proxies – discusses insights regarding the usage, and the evolution of said usage, of technologies that can be connected to AM, or which can serve as proxies to it.

- 1.2.5. General summary/conclusions – a qualitative summary of the results, based on the quantitative insights discussed in the previous subsections.

The box below summarises the main findings of this section. For an explanation of these findings, see the final subsection 1.2.5. General summary/conclusions.

Key findings:

- There is much conflicting information regarding the usage of AM, which prevents the study from providing an accurate assessment of the prevalence of such tools. However, by aggregating results, giving higher values to more robust data, and taking limitations into account, it can be argued that over one-quarter of companies in the EU-27 use AM. To a large extent these, and other, outputs are also corroborated by the insights from the workshops, especially the final validation workshop.
- Enterprises in the EU use AM most frequently for evaluating worker performance and monitoring workers.
- Although there is a lack of robust data, AM is probably least often used for dismissing workers, which might be motivated by the sensitive nature of this activity.
- Large privately owned companies use AM more frequently than other types of companies.
- There is little robust evidence regarding what types of workers are most often subject to AM.

1.2.1. Evidence from direct sources

This subsection explores the prevalence of AM by reviewing the data that directly indicate how often employers and workers use, or are subject to, algorithmic management. This analysis is carried out by highlighting how frequently various managerial functions are automated through the use of algorithms. The remainder of the subsection covers AM usage in various contexts, including:

- **General AM usage** – usage of AM in general in the EU-27, focusing on the current state of affairs. A discussion of future trends is provided in Chapter 4. Future Trends.
- **Usage by AM type and managerial functions** – explores how extensively different AM tools are used to automate different managerial functions, discussed previously in subsection 1.1.2. Functions and typologies of AM. Information is provided at the level of the EU-27 and of individual Member States.
- **Usage by employer type** – explores the usage of AM tools in different types of companies, including public/private, from different economic sectors, and of different sizes. In this subsection, information on both the distribution of usage and the number of companies employing various tools is discussed. In cases where data are available, insights into how many employees this affects are also provided. Further insights on usage by worker types are provided in the subsection that follows.
- **Usage by worker type** – explores the usage of AM tools with respect to different kinds of workers, including by age, gender, education level and occupation. Where data are available, the number of workers susceptible to a particular technology directly connected to AM is discussed.

General usage of AM

There is a great deal of conflicting information on the usage of AM, which prevents us from providing an accurate assessment of the general usage of AM. Nevertheless, the results can be aggregated to provide a possible extent of its usage, an overview of which is presented in the table below and discussed in detail in the remainder of the section. The table below only presents insights on general usage, while a discussion on the usage by managerial functions is provided in the next subsection.

Table 1-5 – Main findings on the general usage of AM*

Evidence based on different sources
AMPWork (2023)[32]**:** around one in five workers in Germany and one in three workers in Spain are subject to at least one form of algorithmic management.
DESI (2021)[33]**:** around 0.7 % of enterprises with 10 or more workers (excluding the financial sector) use AI for human resource management or recruiting.
OECD 2023 report[34]**:** around 21 % of companies in the finance and manufacturing sectors in Austria, Canada, France, Germany, Ireland, the United Kingdom, and the United States use AI for HR management.

(*) All results provided are for the EU-27 unless specified.

Based on a recent study conducted by the JRC, which carried out an Algorithmic Management and Platform Work survey (AMPWork), 20 % of workers in Germany and 35 % in Spain are subject to at least one form of algorithmic management, with an automated allocation of work being considered the most widespread form of AM use.[35] However, this estimate ignores some forms of AM, such as recruitment, firing or algorithmic training, as well as worker monitoring, which was analysed separately from other algorithmic management activities. In addition to AMPWork, a recent study by the OECD that covered the impact of AI on the workplace, found that 21 % of the companies surveyed operating in the finance and manufacturing sectors in Austria, Canada, France, Germany, Ireland, the United Kingdom and the United States use AI for HR management.[36] However, based on DESI 2021 data, only around 0.7 % of enterprises with 10 or more workers in the EU-27 use AI for human resource management or recruiting (see the figure below for more details).[37] According to DESI, such usage is more prevalent in Scandinavia (e.g., Denmark, Sweden, and Finland), Western Europe (e.g., the Netherlands, Ireland, Luxembourg, and Belgium), and some Southern European countries (e.g. Portugal and Malta). This very low estimate may, to some extent, be explained by DESI's focus on AI, which is only one part of algorithmic management. However, this does not fully explain the disparity, and hence it is likely that DESI underrepresents the usage of AM, as other sources uniformly provide much higher estimates.

[32] For details, see: https://joint-research-centre.ec.europa.eu/jrc-news-and-updates/platformisation-work-accelerated-pandemic-phenomenon-likely-grow-2023-04-17_en

[33] For details see: https://ec.europa.eu/eurostat/databrowser/view/isoc_eb_ain2__custom_8115625/default/table?lang=en

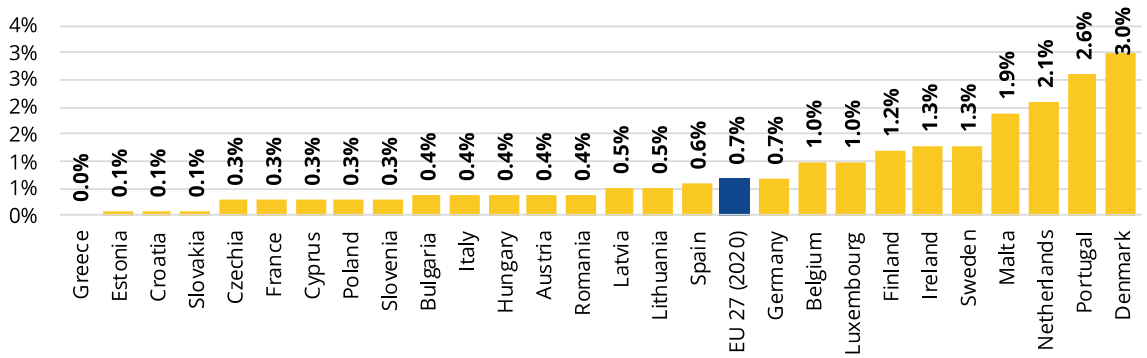
[34] OECD (2023). The impact of AI on the workplace: Main findings from the OECD AI surveys of employers and workers. OECD Social, Employment and Migration Working Papers.

[35] For details, see: https://joint-research-centre.ec.europa.eu/jrc-news-and-updates/platformisation-work-accelerated-pandemic-phenomenon-likely-grow-2023-04-17_en

[36] OECD (2023). The impact of AI on the workplace: Main findings from the OECD AI surveys of employers and workers. OECD Social, Employment and Migration Working Papers.

[37] No information is available on the usage of AI technologies for human resource management or recruiting for 2022.

Figure 1-1: Percentage of enterprises with 10 or more workers using AI for human resource management or recruiting in 2021



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Source: Authors' elaboration based on DESI 2021

In addition, the grey literature provides further insights into AM usage, although the robustness of this information is questionable. First, based on a study by PwC in 2017, which interviewed 1,379 CEOs from 79 countries, it was found that around 40 % of international companies employed AI applications for HR functions.[38] Second, according to Eightfold AI (2021), which surveyed 224 managers from multinational companies, around 50 % of them stated that they use AI-related technologies for talent management.[39] Lastly, a Mercer report in 2019 reported that 88 % of companies globally implement some aspect of AI in HR.[40] As can be seen, the grey literature provides relatively high estimates of AI usage in human resource management. These findings may be to some extent motivated by such literature using less robust methodologies which, for example, do not ensure that respondents can differentiate between AI and simple algorithms, or which may involve some selection bias, focusing more on individuals and companies that are more likely to use AM.

In summary, a great disparity exists between different sources, including robust ones. One reason for this is that different sources employ different sampling strategies. For example, the OECD focuses on two economic sectors and several OECD countries, while DESI covers a wider array of companies[41] throughout the EU. Meanwhile, the JRC report focuses on two countries, Germany, and Spain. These differences are important, as usage differs relatively strongly between countries (as shown in Figure 1-1 above and discussed in detail later), as well as between types of companies. In addition, these disparities can also be explained by the different focuses of these sources, many of which only cover AI, while AM is a more multifaceted concept that includes AI, but also the use of simpler algorithms. Lastly, as previously discussed, AM is a relatively new concept and its use is often not apparent, as good AM tools can perform managerial tasks to the same or higher level than human managers. Hence, some workers may not be aware if such technologies are being used. Similarly, some employers might also be exaggerating when they say they use algorithmic management, as they may be unable to differentiate between simple software and that which is powered by algorithms. Nevertheless, taking the collected data together with insights from other sources, some general and specific conclusions can be made. These are provided in subsection 1.2.5. General summary/conclusions.

[38] PwC (2017). Artificial Intelligence in HR: a No-brainer. Available from: https://www.pwc.at/de/publikationen/verschiedenes/artificial-intelligence-in-hr-a-no-brainer.pdf

[39] Eightfold AI (2021). HR's Future State Report 2021: The Impact of Artificial Intelligence on Talent Processes.

[40] Mercer (2019). Global Talent Trends 2019.

[41] DESI only covers companies in the manufacturing, construction, wholesale and retail Trade, repair of motor vehicles and motorcycles, transportation and storage, accommodation and food service activities, information and communication, and administrative and support service activities sectors.

Usage by AM type and managerial functions

Under this heading, the study explores the usage of AM, based on the typology outlined previously. The table below summarises our main findings, while the paragraphs that follow discuss them in detail, following the modified Kellogg (2020) typology outlined in 1.1.2. Functions and typologies of AM.

Table 1-6 – Main findings on the usage of AM by type and managerial function*

	Evidence based on different sources
Evaluation	**ESENER-3 (2019):** 8.2 % of organisations with five employees or more use machines, systems or computers to monitor workers' performance.
	ECS-2019 (2019): 27.0 % of organisations with 10 workers or more monitor employee performance.
	AMPWork (2023):[42] based on this survey: (i) In Spain, 46 % of workers stated that digital technologies are used to monitor their working time, 39 % stated that digital technologies are used to track their arrival/exit/movements, 20 % stated that their employers use CCTV or webcams, while fewer than 15 % highlighted that their employers use other types of monitoring. In Germany, the corresponding percentages are 50 %, 14 % and 9 %, while fewer than 10 % of workers are subject to other types of monitoring; (ii) in Spain, 11 % of workers have their performance publicly ranked, while in Germany this percentage is 2 %; (iii) 7 % of workers stated that they are subject to performance or customer ratings, while in Germany the figure is 2 %; (iv) 10 % of workers stated that they had their workload reduced or job cancelled on the basic of such rankings, while in Germany this percentage is 2 %; (iv)
	UNI Europa and Central European Labour Studies Institute study by Holubová (2022):[43] 47 % of workers surveyed believe their employers use software to track their physical and digital activities.
	UNI Europa and Central European Labour Studies Institute study by Holubová (2022): 52 % of workers surveyed believe their company uses consumer-sourced rating systems to evaluate performance
	European Employer Survey Report (2021):[44] 29 % of companies surveyed from selected Western European countries reported using data analytics for HR strategy and employee management, including evaluating employee performance.
Direction	**ESENER-3 (2019; these findings are relevant for all managerial functions falling under direction):** 11.8 % of enterprises employing five workers or more state that they use technologies that dictate the pace of work for worker
	ECS-2019 (2019; these findings are relevant for all managerial functions falling under direction): 43.8 % of organisations with 10 workers or more state that the pace of work in their organisation is determined by machines or computers.
	AMPWork (2023): around 11 % of workers in Spain and 2 % in Germany are subject to work direction through digital technologies.
	UNI Europa and Central European Labour Studies Institute study by Holubová (2022): 38 % of workers surveyed believed their employers use automatic projections to schedule shifts, and 35 % believed that their employer uses algorithms to assign tasks.

[42] See: https://publications.jrc.ec.europa.eu/repository/handle/JRC133016

[43] Holubová, B. (2022). Algorithmic Management: awareness, risks, and responses of the social partners. Final Report.

[44] Littler (2022). European Employer Survey Report 2022. Available at: https://www.littler.com/files/2022_littler_european_employer_survey_report.pdf

Evidence based on different sources	
Discipline	**AMPWork (2023):**[45] based on this survey, 8 % of workers in Spain receive bonuses or prizes for performance, while in Germany this percentage is 3 % through AM tools.
Recruitment	**UNI Europa and Central European Labour Studies Institute study by Holubová (2022):** 35 % of workers surveyed highlighted that their company uses CV/resumé screening algorithms to filter applications; 30 % stated that automatic background checks are used in recruitment; and 10 % reported that their company employs automated job interviews without human intervention.
	European Employer Survey Report (2021): 28 % of companies surveyed reported using technology solutions and/or AI tools to support their recruitment and hiring efforts.

(*) All results provided are for the EU-27 unless specified

Evaluation

The evaluation comprises activities relating to the monitoring and evaluation of workers. **Worker monitoring** refers to tracking workers via, for example, time-tracking solutions that record when employees log in and out of work, software that scans workers' emails, GPS tracking tools, and similar. Based on data from ESENER-3 2019, monitoring of workers remains relatively rare, as only around 8.2 % of organisations in the EU-27 with five or more employees use machines, systems, or computers to monitor workers' performance. Extrapolating this percentage to the whole EU-27 population using the methodology described in the ESENER-3 manual[46] (which is also described in Section *1.3. WP1 Task 3: Quantitative overview of the extent of the use of AM tools in the EU* in *Annex 1: Methodological annex*), suggests that around 420,000 organisations do so. Meanwhile, according to ECS-2019, data analytics is used to monitor worker performance in around 27.0 % of organisations with 10 workers or more in the EU-27. Extrapolated using the methodology outlined in the ECS-2019 Sampling and weighted report[47] (which is also described in *Annex 1: Methodological annex*), suggests that around 530,000 organisations monitor employee performance.

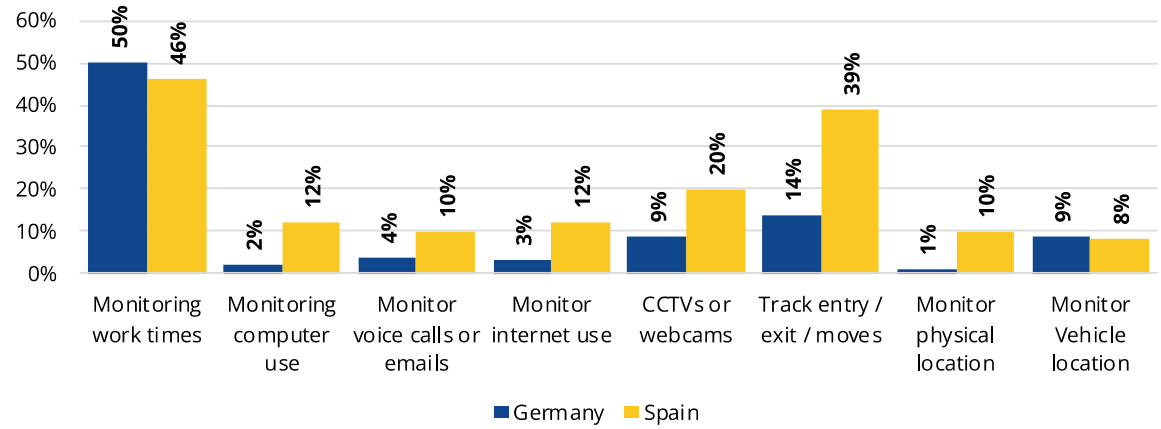
More recent data from the JRC AMPWork report provide higher estimates with more granular insights. Based on its results (summarised in the figure above), the aspect most frequently subject to monitoring is workers' time. In Germany, 50 % of respondents reported that their working time is monitored, compared with 46 % in Spain. The second most widely monitored aspect is workers' arrival/exit/movements, reported by 39 % of workers in Spain and 14 % in Germany. Around 20 % of workers in Spain and 9 % in Germany are also subject to CCTV and webcam monitoring, while other types of monitoring are used less frequently.

[45] See: https://publications.jrc.ec.europa.eu/repository/handle/JRC133016

[46] See: https://oshwiki.osha.europa.eu/en/themes/esener-2019-methodology.

[47] See: https://www.eurofound.europa.eu/sites/default/files/wpef20014.pdf

Figure 1-2: OECD, 2015, EU = EU and its Member States

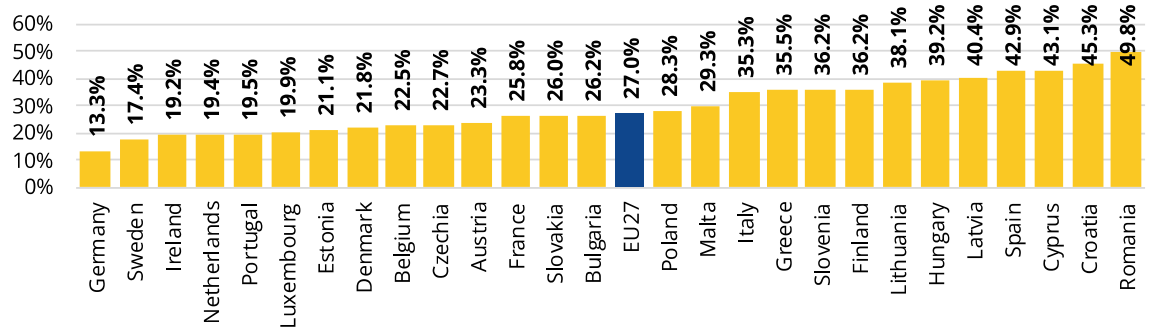


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Source: adapted from Figure 4, p. 22 of the JRC AMPWork report.[48]

With regard to the distribution of monitoring tool usage across the EU-27, some insights can be derived from ECS-2019. These are presented Figure 1-3 below. According to the results of ECS-2019, monitoring is used most frequently in Eastern and Central European countries such as Romania and Croatia, as well as in Southern European countries such as Cyprus and Spain. Such tools are used least in Western European countries such as Germany, Ireland, and the Netherlands. Similarly, according to the ESENER-3 data, worker monitoring is used the least often in Austria, Italy, Germany, and the Netherlands, with 3.8 %, 4.8 %, 5.6 % and 6.5 % of enterprises using such tools, respectively. Meanwhile, employers in Finland, Bulgaria, Lithuania, and Ireland use such technologies the most, with 23 %, 22.0 %, 20.4 %, and 18.1 % of enterprises respectively using them to monitor workers.[49] Hence, no clear trend can be observed, and the only insight that can be derived is that there is wide variation between sources.

Figure 1-3: Percentage of companies using data analytics to monitor employee performance in 2019



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Note: N = 21,172. Data were weighted using the s5_wgt_final weight. Based on ECS 2019 weighted data, around 530,000 organisations in the EU-27 monitor employee performance. For more on weighting, see *Annex 1: Methodological annex.*

Source: Authors' elaboration, based on ECS-2019 data weighted data.

As can be seen, different sources provide relatively different results. To some extent, these discrepancies can be explained by the different times of measurement, with the COVID-19

[48] See: https://publications.jrc.ec.europa.eu/repository/handle/JRC133016

[49] Estimates are based on a sample of 37,460 observations that were weighted in order to extrapolate the results to the whole EU-27.

pandemic accelerating the adoption of monitoring tools in 2020-2021.[50] In addition, different data sources also often employ different sampling strategies, which may have introduced discrepancies. For example, ECS-2019 focused on companies with 10+ employees, while ESENER-3 covers companies with five or more. This is an important difference, given that large companies use AM more frequently, as will be discussed later. Nevertheless, this does not completely explain the disparities, especially those between ESENER-3 and ECS-2019.

Beyond monitoring, **worker evaluation** using algorithms has also started to gain prominence in recent years. More specifically, performance evaluation technologies originate in platform work environments, such as the ride-sharing companies Uber and Lyft. On such platforms, the driver performance scores that are seen by the company– which are not necessarily what users of the platforms see – are based on several metrics, including their ride acceptance rate (around 80 %), cancellation rate (around 5 %), and customer ratings (around 4.6 %).[51] However, **recently, especially since the COVID-19 pandemic, performance evaluation tools have begun to be employed in many classic working arrangements.**[52]

Based on AMPWork data, around 11 % of workers and Spain and 2 % in Germany state that they are subject to digital technologies that rank their performance publicly, while 7 % in Spain and 2 % in Germany are also subject to performance or customer ratings.[53] Similarly, 8 % of workers in Spain state that they have received badges or rewards for performance, while 10 % state that such ratings have impacted how much work they received. In Germany, these percentages are 3 % and 2 %, respectively. In addition, based on Holubová (2022), around 52 % of workers surveyed stated that their company uses consumer-sourced rating systems to evaluate performance. Lastly, according to Littler (2021),[54] who surveyed 758 professionals from the UK, France, Italy, Germany, Spain, Belgium, Netherlands, Poland, Austria, and Iceland, 29 % of companies using AI and data analytics for HR strategy and employee management, including evaluating employee performance.

Direction

Direction refers to those algorithms used for **work/task scheduling, nudging, and talent management**. Based on this, around 11 % of workers in Spain and 2 % in Germany are subject to worker direction via digital technologies. In addition, based on the UNI Europa and Central European Labour Studies Institute study by Holubová (2022), around 38 % and 35 % of workers surveyed believe that their employers use algorithms to schedule shifts and assign tasks, respectively. In addition, given that direction refers to algorithms that directly interact with how workers do their jobs, some insights can be derived from the ECS 2019 and ESENER-3 surveys, which include data on how many employers use technologies that dictate the pace of work. However, this variable captures more than direction or AM in general, as machines used in manufacturing can also dictate the pace of work. Nevertheless, because such technology can also be often used for task allocation and nudging, this variable is considered to be an indicator of the extent to which direction is used.

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[50] Aloisi, A., & De Stefano, V. (2022). Essential jobs, remote work, and digital surveillance: Addressing the COVID-19 pandemic panopticon. *International Labour Review*, 161(2), 289-314.

[51] Lee, M.K., Kusbit, D., Metsky, E., & Dabbish, L. (2015, April). Working with machines: The impact of algorithmic and data-driven management on human workers. In: Proceedings of the 33rd annual ACM conference on human factors in computing systems (pp. 1603-1612); Rosenblat, A., & Stark, L. (2016). Algorithmic labour and information asymmetries: A case study of Uber's drivers. *International Journal of Communication*, 10, 27.

[52] Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek, E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work.

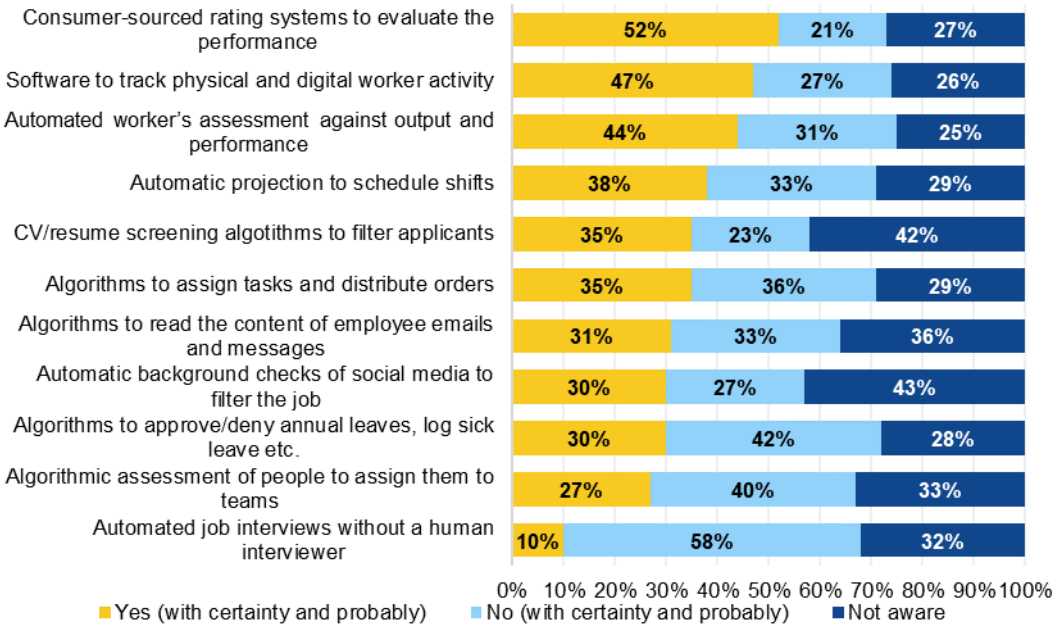
[53] See: https://publications.jrc.ec.europa.eu/repository/handle/JRC133016

[54] Littler (2021). European Employer Survey Report, available from: https://www.littler.com/files/2021_littler_european_employer_survey_report.pdf

According to ESENER-3 (2019), around 11.8 % of enterprises employing five workers or more in the EU-27 use technologies that dictate the pace of work for workers. This equates to around 600,000 such enterprises (i.e. with five or more workers) in the EU-27. However, according to the European Company Survey (ECS 2019), in which respondents were asked to specify if the pace of work in their establishment was determined by machines or computers, such technologies were used in 43.8 % of enterprises with 10 or more workers in the EU-27. This equates to around 875,000 enterprises.

To some extent, the above disparity might be attributed to the different sampling strategies used in the two datasets. However, after harmonising them by removing companies with between five and nine employees from the ESENER-3 dataset, as well as the agriculture, education and human health and social work activities sectors (which were not present in ECS 2019), the percentage of enterprises in which technologies dictate the pace of work increased to 15.2 %. However, this is still far from the 43.8 % seen in the ECS 2019 data. Hence, it does not fully explain the observed differences. In addition, these differences can, to some extent, be explained by the fact that the two surveys used slightly different questionnaires. In the ESENER-3 survey, respondents were asked if such tools are used at all, while in the ECS 2019 respondents were asked to highlight for how many workers the pace of work is determined by machines, ranging from 'None at all', 'Less than 20 %', '20-39 %', '40-59 %', '60-79 %', '80-99 %', and 'All'.[55] However, this does not completely explain the disparity either.

Figure 1-4: Awareness of workers in 2022 of AM usage, by type of tool



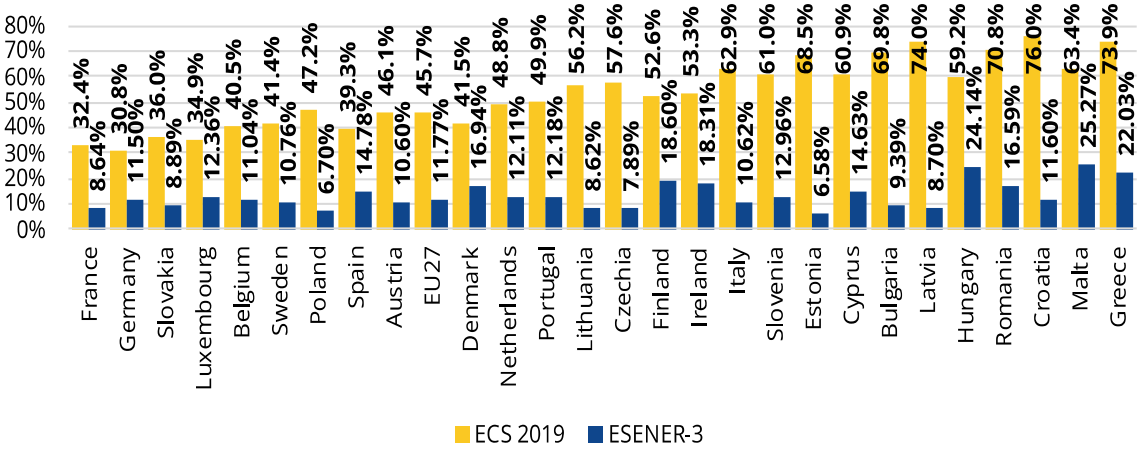
© European Union

Note: N = 1,158.

Source: adapted from Holubová (2022)

[55] More specifically, ESENER-3 asked: 'Does your establishment use any of the following digital technologies for work? Machines, systems, or computer determining the content or pace of work'; ECS-2019 asked: 'For how many employees at this establishment is the pace of work determined by machines or computers? Your best estimate is good enough.'

Figure 1-5: Percentage of companies in the EU-27 in 2019 in which the pace of work for workers is dictated by technologies



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Note 1: for ECS 2019: N = 21,172. Data were weighted using the s5_wgt_final weight. Based on ECS 2019 weighted data, around 875,000 organisations use technologies that dictate the pace of work. For more on weighting see *Annex 1: Methodological annex.*

Note 2: for ESENER-3: N = 37,460. Data was weighted using the estex weight. Based on ESENER-3 data, around 600,000 organisations use technologies that dictate the pace of work for workers. For more on weighting, see *Annex 1: Methodological annex.*

Source: Authors' elaboration based on ESENER-3 and ECS 2019 weighted data.

The available data only allow some insights to be derived from the ESENER-3 or ECS 2019 surveys regarding the distribution of AM management usage for direction across different Member States. Given that the two data sources provide relatively different results, the figure below displays **both** across the EU-27. After averaging the information from the two sources in companies from Western Europe, the usage of such technologies was less prevalent (with some exceptions, such as Poland, which performed similarly to Western European countries). Technologies dictated the pace of work most often in Southern Europe, such as in Greece and Malta, as well as in the Central and Eastern European Member States, including Croatia, Romania, and Hungary. These results imply, to some extent, that the use of AM for worker evaluation is less prevalent in Western and Northern countries, while it is most prevalent in Central and Eastern Europe. However, it may simply indicate a wide variation in results similar to that highlighted when discussing the algorithms used for worker monitoring and evaluation.

Discipline

Discipline, in the context of AM, refers to the use of algorithms to promote desired behaviour on the part of workers through rewards or punishments. One element of this is **rewarding workers**. This is closely connected to worker evaluation but differs from it because worker evaluation refers to assigning a score or ranking to workers, while rewarding refers to acting in response to this score or some other data. Algorithmic rewarding also includes using algorithms to identify the best rewards for workers, such as deciding how much their wages should increase, or what bonus they should receive. In general, there is a lack of data regarding the use of algorithms to reward workers, probably because many workers connect such activity instead with worker evaluation. Although some insights can be derived from AMPWork, according to which 8 % of workers in Spain receive badges or rewards awarded by algorithms based on their performance, compared with only 2 % in Germany.[56]

[56] See: https://publications.jrc.ec.europa.eu/repository/handle/JRC133016

In addition, another dimension of worker discipline is the **use of algorithms for dismissing workers**. However, even less data is available on this, probably due to the sensitive nature of such activities. Hence, insights on discipline through algorithms predominantly come from the surveys conducted for the present study. These are discussed in subsections Hence, based purely on those sources analysed that look at the usage of AM tools among individuals with different education levels, workers with a higher level of education are more likely to be subject to algorithmic management than those with a lower level of education. One explanation for such a trend is the fact that individuals with higher education more often work in jobs using computers and other smart devices that can be used for algorithmic management. For example, according to EWCTS-2021, 74.1 % of individuals with a Bachelor's, Master's, or doctoral degree use computers 'Always', compared with only 21.1 % of individuals with upper- or lower-secondary, as well as individuals with early childhood education. This greater use of computers by those with high levels of education, in turn, creates greater opportunities for employers to use AM. An alternative explanation is that higher education tends to imply higher job complexity, which, in turn, means workers are more likely to interact with various digital tools that might involve an element of algorithmic management (e.g., workers might use email systems that are monitored by employers). Higher complexity also implies a greater likelihood that decision-making is data-driven, which may also be supported through algorithmic management.

1.2.2. Evidence from the Visionary Analytics surveys and 1.2.3. Evidence from the Visionary Analytics Delphi survey.

Recruitment

According to the survey of European enterprises conducted by Littler (2021), 28 % of companies surveyed reported using technology solutions and/or AI tools to support their recruiting and hiring efforts.[57] Meanwhile, according to Holubová's (2022) UNI report, 35 % of workers surveyed highlighted that their company uses CV/resumé screening algorithms to filter applications, 30 % stated that automatic background checks are used in recruitment, while 10 % said that their company employs automated job interviews without human intervention.[58]

Usage by employer type

Under this heading, the usage of AM by different employers is explored – including, where possible, highlighting the usage of different algorithms in managerial functions. Prevalence at the employer level is explored in the **public/private sectors, by economic sector, and among organisations of different sizes.** In addition to providing insights on the share of enterprises using AM, the number of enterprises in the EU-27 that use such tools is also estimated. Over the next few paragraphs, insights into the number of workers who are subject to such tools are periodically provided, though most insights on workers are described below, under 'Usage by worker type'. The table below provides the main quantitative findings with regard to the different types of companies, while the remainder of the subsection goes into further depth.

Table 1-7 – Main findings on the usage of AM, by type of enterprise*

Evidence based on different sources

[57] Littler (2022). European Employer Survey Report 2022. Available at: https://www.littler.com/files/2022_littler_european_employer_survey_report.pdf

[58] Holubová, B. (2022). Algorithmic Management: awareness, risks, and responses of the social partners. Final Report.

	Evidence based on different sources
Private / public	**ESENER-3 (2019):** in 12.5 % of private companies and 7.9 % of public organisations, machines, systems, or computers determine the content or pace of work; 8.6 % of private companies use worker monitoring, compared with 6.4 % of public entities; 4.96 % of private companies use wearable devices such as smartwatches, compared with 4.2 % of public entities. Around 9.9 % of employees in the public sector and 14.7 % of those in the private ones are subject to worker monitoring. Machines, systems, and technologies determine the content and pace of work for around 11.8 % of employees in the public sector and 20.7 % of those in the private sector.
Economic sector	**ESENER-3 (2023):** companies operating in finance and insurance, transportation and storage, water supply, sewerage, waste management and remediation activities use work monitoring most frequently (17.52 %, 17.24 %, and 12.6 % respectively); machines, systems, or computers determine the content or pace of work most frequently in companies from the manufacturing (22.97 %), agriculture, forestry, and fishing (19.61 %), and transportation and storage (15.76 %) sectors. Worker monitoring is used most frequently on employees from the manufacturing sector (in which 22.2 % of workers are subject to this technology), as well as in the transportation and storage (21.7 %), and finance and insurance (19.5 %) sectors. Machines and systems determine the content and pace of work most frequently for employees in manufacturing (39.1 % of workers), mining and quarrying (26.7 %), and the electricity and gas (25.1 %) sectors.
	ECS (2019): enterprises from the following sectors use worker monitoring the most: transportation and storage (35.1 % of such companies are subject to such technologies), information and communication (31.8 %), wholesale and retail (30.6 %); in enterprises from the following sectors, machines are used to dictate the pace of work: manufacturing (71.1 %), mining and quarrying (66.7 %), water supply, sewerage, etc. (54.5 %).
	International Labour Organization (2021)[59]**:** 40 % of workers carrying out gig work highlighted that AM is used in their place of work.
Size	**ESENER-3:** enterprises with 250+ employees use machines, systems, or computers to monitor workers' performance the most (18.74 % of such companies); enterprises with 50-249 employees come second (12.28 %); enterprises with 10-49 employees are third (9.09 %); while enterprises with five to nine employees use such technologies the least (6.46 %). Similar trends are observed with wearable devices such as smartwatches and data glasses (large enterprises, 11.55 %; medium-sized enterprises, 6.37 %; small enterprises,5.4 %; and micro-enterprises, 3.84 %) and with machines, systems and computers determining the content and pace of work (large enterprises, 26.07 %; medium-sized enterprises, 17.12 %; small enterprises, 12.76 %; and micro-enterprises, 9.52 %). Around 19 % of employees in large, 10.3 % in small, and 7.2 % in micro-enterprises are subject to worker monitoring.
	ECS (2019): 39.79 % of large enterprises, 33.59 % of medium-sized, and 25.38 % of small enterprises use data analytics to monitor workers.
	DESI (2021): 3.2 % of large enterprises use AI technologies for human resource management or recruitment, compared with only 1.2 % of medium-sized, and 0.5 % of small companies.

(*) All results provided are for the EU-27 unless specified.

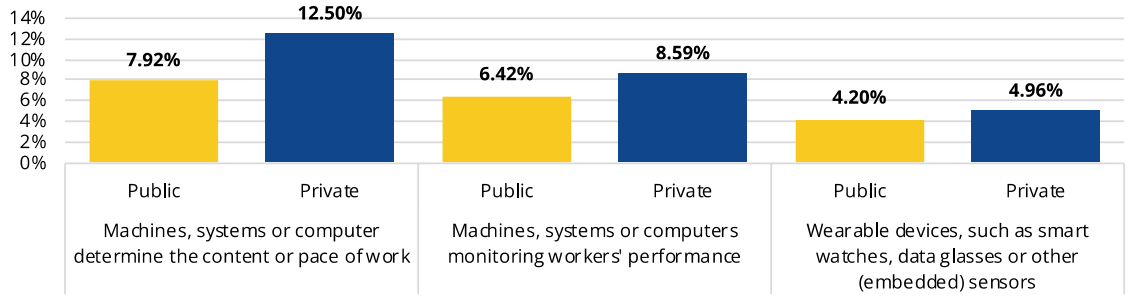
Public/private sector

According to experts interviewed and country case studies carried out for the present study, **the adoption of AM has so far been lower in the public sector than in the private sector**. This finding is supported by quantitative data. For example, according to ESENER-3, which covers machines, systems or computers determining the content or pace of work – which is likely to involve technologies that allocate tasks and schedule work (the results of

[59] Baiocco, S., Fernández-Macías, E., Rani, U., & Pesole, A. (2022). The algorithmic management of work and its implications in different contexts (No. 2022/02). JRC Working Papers Series on Labour, Education and Technology.

which are provided in Figure 1-6 below), the usage of technologies associated with AM is more prevalent among private organisations. This trend is consistent across different technologies, albeit the extent of the difference between public and private sectors varies. For example, in 12.5 % of companies from the private sector with five or more employees, machines, systems or computers are used to determine the content or pace of work, compared with 7.9 % in the public sector (public sector in ESENER-3 included organisations from all sectors). This contrasts with worker monitoring, in which 8.6 % of private companies use such technologies, compared with 6.4 % of public entities. These results are based on the analysis of around 10,000 public and 25,000 private enterprises covered in ESENER-3. Extrapolating these results to the overall population of EU-27 employers on the basis of the methodology outlined in the ESENER-3 manual,[60] somewhere between 34,000 and 63,000 public entities, and somewhere between 208,000 and 525,000 private companies, use algorithms for one of the purposes discussed above.

Figure 1-6: Share of organisations in the EU-27 with five or more workers that use various technologies associated with AM (2019)



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Note: N was around 37,310 (there were minor variations, concerning around 50 observations, between variables). Data were weighted using the estex weight. Based on weighted ESENER-3 data, a total of more than 800,000 public entities and over 4,200,000 private companies in the EU-27 are reflected in these findings. For more on weighting, see *Annex 1: Methodological annex.*
Source: Authors' elaboration based on ESENER-3 weighted data.

Taking these results as a basic estimate of how many workers are affected by such tools, around 9.9 % of employees from the public sector (around 475,000 employees) and around 14.7 % of employees from the private one (around 19 million employees) are subject to algorithms that monitor workers' performance, who also work in companies with five or more employees. Meanwhile, around 11.8 % of employees working in the public sector (570,000), and 20.7 % of employees working in the private sector (26.8 million) are subject to machines, systems and technologies that determine the content or pace of their work.

To a large extent, the disparity between the rates of private and public sector organisations using AM can be explained by the different types concerned. For example, public sector organisations predominantly include enterprises in public administration and defence, education and the human health and social work sectors, while private companies predominantly operate in manufacturing, wholesale, retail, and other service-related sectors. While this difference does not vitiate the conclusions above, it may change the reasoning behind differences in the uptake of AM.

Economic sector

Wide variation in the use of AM exists between economic sectors. On the one hand, according to ESENER-3 (see

Figure 1-7 below), when it comes to **computers determining the content or pace of work**, organisations with a **higher proportion of manual, routine tasks tend to rely more on these technologies.** Specifically, such technologies are used most frequently in the agricultural, manufacturing, and mining and quarrying sectors. These figures translate into

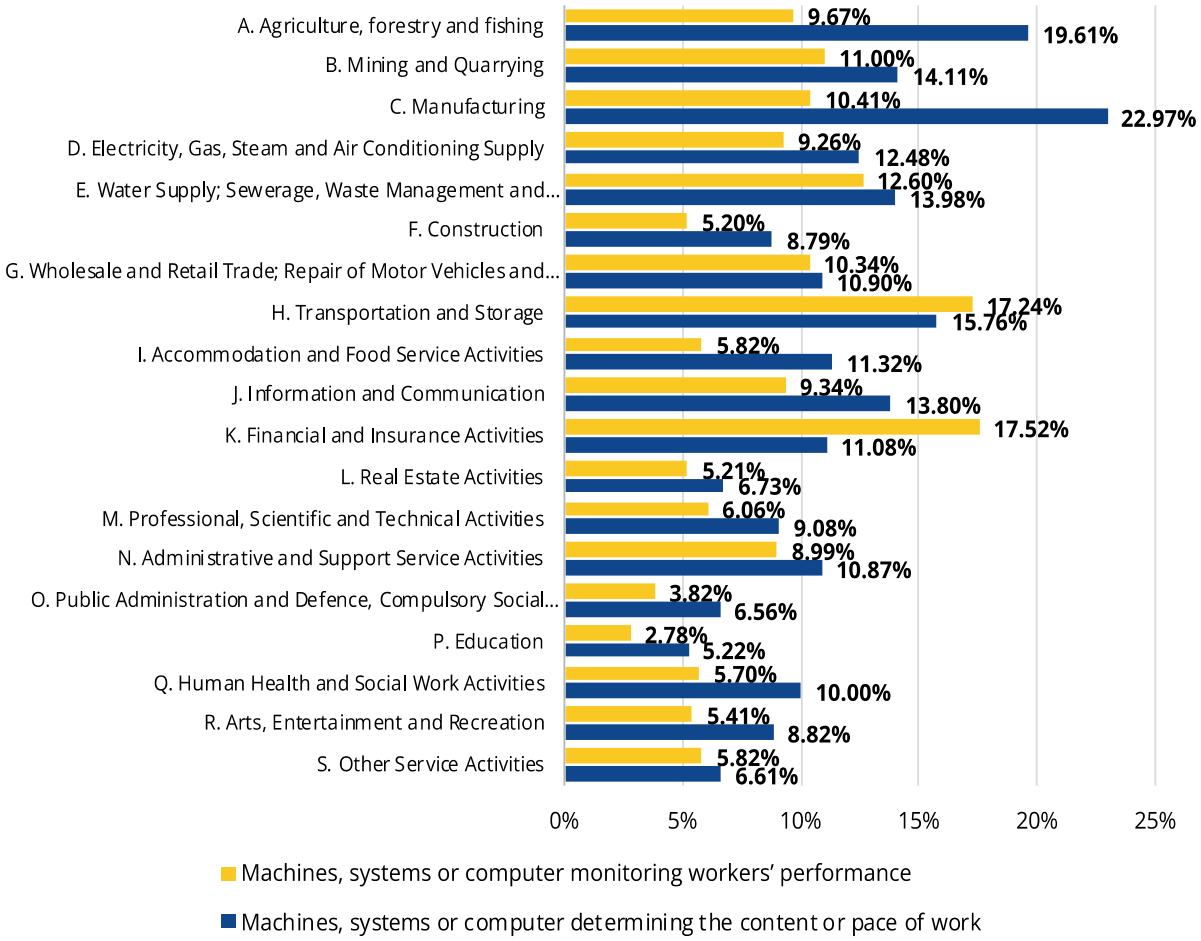
[60] See: https://oshwiki.osha.europa.eu/en/themes/esener-2019-methodology

around 22,000 companies in the agricultural sector across the EU-27 using technologies that determine the pace of work, compared with 142,000 companies in the manufacturing sector. **Monitoring, meanwhile, is used more frequently in the service sector.** For example, worker monitoring is employed most frequently by organisations in finance and insurance (17.52 % of workers) and transportation and storage (17.24 % of workers). This extrapolates to around 24,500 companies in the financial sector across the EU-27, and 41,000 companies in the transportation and storage sectors. Interviewees involved in the present study shared a similar view that such types of companies use AM the most.

Based on ECS-2019 data, the sectors in which companies with 10 or more employees use worker monitoring most frequently are transportation and storage (35.1 % of enterprises, which extrapolates to 45,000 companies across the EU-27); information and communication (31.8 % of enterprises, which extrapolates to 32,500 companies); and wholesale and retail (30.6 % of enterprises, which translates to around 165,000 companies). Also, according to ECS-2019, the pace of work is dictated by machines most frequently in the manufacturing sector (71.1 % of enterprises, which extrapolates to around 290,000 companies across the EU-27); the mining and quarrying sector (66.7 % of enterprises, which extrapolates to 6,000 companies); and the water supply, sewerage, etc. sector (54.5 % of enterprises, which extrapolates to around 12,000 companies).

A similar picture emerges from the perspective of workers, according to data from ESENER-3. Across the EU-27, employees are subject to worker monitoring most frequently in manufacturing (22.2 % of employees, which extrapolates to around 6.5 million workers); transportation and storage (21.7 %, or 2.1 million); and financial and insurance activities (19.5 %, or 1 million). Similarly, the pace and content of work are determined by machines most frequently in manufacturing (39.1 %, or 11.5 million employees); mining and quarrying (26.7 % 130 thousand); and the supply of electricity, gas, steam, etc. (25.1 %, or 270,000).

Figure 1-7: Usage of technologies associated with AM across economic sectors in the EU-27 in 2019



Note: N was 37,249 for the variable covering the pace of work, and 37,288 for the variable covering worker monitoring. Data were weighted using the estex weight. Based on weighted ESENER-3 data, the visual represents the following number of companies for each sector (rounded): A – 113,000; B – 10,000; C – 617,000; D – 16,500; E – 38,000; F – 456,000; G – 1,082,000; H – 240,000; I – 410,000; J – 134,000; K – 139,500; L – 78,500; M – 350,000; N – 226,000; O – 169,500; P – 360,000; Q – 408,000; R – 83,000; S – 159,000. For more on weighting, see *Annex 1: Methodological annex.*
Source: Authors' elaboration based on ESENER-3 weighted data.

To some extent, the differences between sectors can be explained by the fact that various sectors tend to be populated by different sizes of companies. For example, based on ESENER-3 and weighted data, the education sector, which is the sector in which computers are least often used to determine the content or pace of work, and workers are monitored the least, is predominantly composed of small enterprises (10-49 employees); indeed, 57.6 % of companies in this sector are small. In contrast, the agricultural sector, in which computers are most frequently used to determine the pace of work, is predominantly composed of micro-enterprises (organisations of between five and nine employees constitute 55.2 % of all companies in this sector). Similarly, the finance and insurance sector, which often uses worker monitoring, is also mostly composed of micro-enterprises (57.0 %). However, the size of organisations does not completely explain the heavier usage of AM in certain sectors. This is because companies in the manufacturing and mining sectors, where machines, systems or computers determine the content and pace of work, and the transportation sector, where monitoring is used almost as frequently as in the financial sector, are predominantly composed of small companies employing between 10 and 49 employees. However, these results are based on data from prior to the COVID-19 pandemic, which changed the usage of AM tools quite substantially. For example, according to AMPWork,[61] which collected data on algorithmic management in 2023, the education sector uses AM relatively heavily – in particular, automated direction. Similarly,

[61] See: https://publications.jrc.ec.europa.eu/repository/handle/JRC133016

according to AMPWork, knowledge-intensive and hospitality and tourism sectors also rely heavily on automated evaluation tools.

Furthermore, **AM and related tools are used particularly intensely in the gig economy and platform work**, though the extent of this usage is difficult to measure, as platform companies rarely disclose such information. However, some data such practices do exist. Based on AMPWork, almost all platform work companies use some sort of digital technologies to manage workers. In addition, the AMPWork survey found that among those platform workers who said that such work was their main job, between 20 % and 55 % in Spain, and between 5 % and 25 % in Germany reported being monitored. Similarly, according to the International Labour Organisation (2021),[62] in a global survey of workers carried out in 2019-2020, 40 % of workers working through platforms reported being tracked regularly. In addition, AMPWork found that among those platform workers who said that such work was their main job, around 50 % in Spain and 20 % in Germany said they were subject to algorithmic direction, while around 40 % in both countries said they were subject to algorithmic evaluations such as rating systems. This, combined with the fact that 70 million workers globally are registered with platform apps, and annual user growth on online platforms of 26 %,[63] gig and platform workers are likely to be significantly impacted by developments in AM.

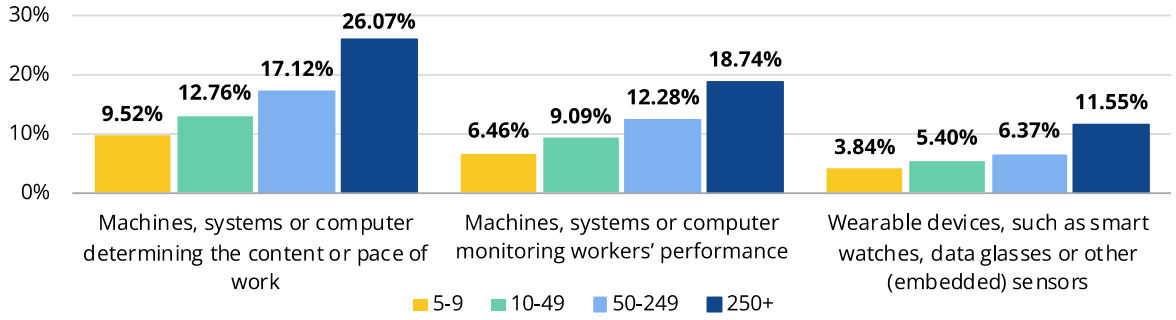
Size of company

According to quantitative data and interviews carried out for the present study, **larger companies use AM more frequently** than smaller ones. For example, according to ESENER-3, around 6.46 % of micro-enterprises (employing five to nine workers) use machines, systems, or computers to monitor workers' performance, compared with almost three times this percentage – 18.74 % – of large organisations (250+ employees). This trend of micro-enterprises using AM three times less often than large ones is relatively consistent across different technologies, as can be seen in Figure 1-8 below. Similar trends are also observed when looking at the number of employees who are subject to AM. Around 19 % of employees in large enterprises (9.5 million employees) are subject to worker monitoring compared with 10.3 % and 7.2 % of employees in small companies and micro-enterprises, respectively (which extrapolates to 6 million and 1.4 million employees, respectively). Other sources also corroborate the conclusion that large enterprises use AM more frequently than smaller ones, albeit with inconsistencies in the extent of how much more.

[62] Baiocco, S., Fernández-Macías, E., Rani, U., & Pesole, A. (2022). The algorithmic management of work and its implications in different contexts (No. 2022/02). JRC Working Papers Series on Labour, Education and Technology.

[63] Wood, A.J., Graham, M., Lehdonvirta, V., & Hjorth, I. (2019). Good gig, bad gig: Autonomy and algorithmic control in the global gig economy. *Work, Employment & Society*, 33(1), 56–75. https://doi.org/10.1177/0950017018785616

Figure 1-8: Usage of technologies associated with AM across organisations with different numbers of employees in the EU-27 in 2019



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Note: N = 37,460. Data were weighted using the estex weight. Based on weighted ESENER-3 data, the visual represents the following number of organisations (rounded): micro (5-9) – 2,400,000; small (10-49) – 2 200 000; medium (50-249) – 430,000; and large (250+) – 65,000. For more on weighting, see *Annex 1: Methodological annex.*

Source: Authors' elaboration based on ESENER-3 weighted data.

According to DESI, 3.2 % of large enterprises use AI technologies for human resource management or recruitment, compared with only 1.2 % of medium-sized and 0.5 % of small companies. Similarly, according to ECS-2019 data, 39.79 % of large enterprises, 33.59 % of medium-sized, and 25.38 % of small enterprises in the EU-27 use data analytics to monitor workers. This translates to around 22,000 large, 95,000 medium-sized, and 410,000 small enterprises in the EU-27 using such technologies. Similar results can also be seen in other datasets.

Usage by worker type

Under this heading, the usage of different technologies associated with AM across different worker types is explored, according to: (i) **age group**, (ii) **gender**, (iii) **occupation**, and (iv) **level of education**. The results of this analysis are summarised in the table below, while further detail is provided in the paragraphs that follow.

Table 1-8 – Main findings on the AM usage by type of employee*

	Evidence based on different sources
Age	**EWCTS (2021):** employees between the ages of 25 and 59 are subject to the influence of computer systems most often.
	Trade Union Congress (TUC) survey (2018)[64]**:** 68 % of employees aged 18-24 and 60 % of those aged 25-34 said that they believed their employers were monitoring them, which constitutes the largest share of workers by age.
Gender	**EWCTS (2021):** 20.6 % of women stated that computer systems influence what they do at their work to a large extent, while 9 % said they did so to some extent. Among men, these percentages were 17.4 % and 10 %, respectively.
	Parkes (2023)[65]: 38 % of low-skilled women workers are subject to worker monitoring, compared with 33 % of men. 68 % of non-unionised men are subject to worker monitoring, compared with 60 % of women.

[64] Trades Union Congress (2018). I'll be watching you: a report on workplace monitoring. London, UK: Trades Union Congress.

[65] Parkes, H. (2023). Watching me, watching you: Worker surveillance in the UK after the pandemic, IPPR, available at: https://www.ippr.org/research/publications/worker-surveillance-after-the-pandemic

Evidence based on different sources	
Occupation	**AMPWork (2023)**[66]: activity monitoring is most prevalent among clerks and professionals, while physical monitoring is most prevalent among different kinds of operators.
	EWCTS (2021): white-collar workers are more often subject to computers influencing what they do at their job, with 25.5 % of managers, 24.7 % of professionals, 23.6 % of technicians and associated professionals, and 31.5 % of clerical support workers stating that computers influence them to large extent.
Education level	**EWCTS (2021):** between 23.4 % and 25.4 % of individuals with a doctorate or equivalent, Master's degree, Bachelor's degree or short-cycle tertiary education say that computer systems influence what they do at work 'to a large extent' This compares with 14.9 % of individuals with upper secondary education. Meanwhile, the corresponding percentages among those with lower secondary, primary, and early childhood education are 10 %, 6.4 %, and 5.8 %, respectively.

(*) All results provided are for the EU-27 unless specified.

Compared with data from the employer's perspective, there is a scarcity of information regarding how extensively individuals report being subjected to AM. For this reason, in the present analyses, initial insights are derived from the EWTCS-2021, and are corroborated where possible with other sources. The EWCTS survey is used as a base because it is one of the few existing datasets that are representative of workers in the EU27 and, it can be argued, covers AM through the following question 'To what extent does a computerised system influence what you do in your work?'.

The usage of computer systems that influence what workers do in their jobs can be connected to algorithmic management directly through algorithms that allocate tasks or nudge workers, which fall under such systems. Algorithmic management may also indirectly affect what workers do in their jobs through, for example, monitoring, worker evaluation, and similar. In addition, this question captures the degree to which digital technologies affect workers and not simply how often they use certain digital technologies. However, an argument can also be made that this variable simply measures the usage of computers in companies. Furthermore, when correlating this with another question from EWCTS-2021 – namely, 'How often does your main paid job involve working with a computer, laptop, tablet, smartphone?', Spearman's correlation coefficient of 0.4187 was achieved.[67] This correlation was estimated between the two variables that result from the response to the question on computer systems being expressed in terms of a four-point Likert scale, from 'To a large extent' to 'Not at all, while responses to the question on computer usage were expressed on a five-point Liker scale, from 'Always' to 'Never'. Correlation was carried out at the individual level using EWCTS-2021 microdata.

The above correlation coefficient implies that, first, there is a connection between these two variables. This is to be expected, given that computers, smartphones, etc. are often necessary for computerised systems to exist and for AM to operate (though not always; for example, a manufacturing plant might carry out worker monitoring without such workers using any smart devices). However, it also shows that the influence of computer systems variable carries a lot of additional information beyond simple computer usage. In other words, although the correlation metric does not provide any indication of causality or robustness, the medium strength of the correlation identified adds some credence to the proposition that the question 'To what extent does a computerised system influence what

[66] See: https://publications.jrc.ec.europa.eu/repository/handle/JRC133016

[67] Spearman's correlation coefficient was calculated, rather than the more frequently used Pearson's, because the former, by its design, produces more accurate results than the latter when ordinary data are used, which was the case here. Only information from EU-27 countries was correlated.

you do in your work?' indicates the extent to which such technologies affect workers, rather than how frequently they use such technologies.

Nevertheless, given the fact that EWCTS-2021 is not an ideal metric of AM, as discussed, insights from it will be corroborated with other sources, and will not be used as a standalone source on AM usage. The study also expands on the findings presented here in subsection Hence, based purely on those sources analysed that look at the usage of AM tools among individuals with different education levels, workers with a higher level of education are more likely to be subject to algorithmic management than those with a lower level of education. One explanation for such a trend is the fact that individuals with higher education more often work in jobs using computers and other smart devices that can be used for algorithmic management. For example, according to EWCTS-2021, 74.1 % of individuals with a Bachelor's, Master's, or doctoral degree use computers 'Always', compared with only 21.1 % of individuals with upper- or lower-secondary, as well as individuals with early childhood education. This greater use of computers by those with high levels of education, in turn, creates greater opportunities for employers to use AM. An alternative explanation is that higher education tends to imply higher job complexity, which, in turn, means workers are more likely to interact with various digital tools that might involve an element of algorithmic management (e.g., workers might use email systems that are monitored by employers). Higher complexity also implies a greater likelihood that decision-making is data-driven, which may also be supported through algorithmic management.

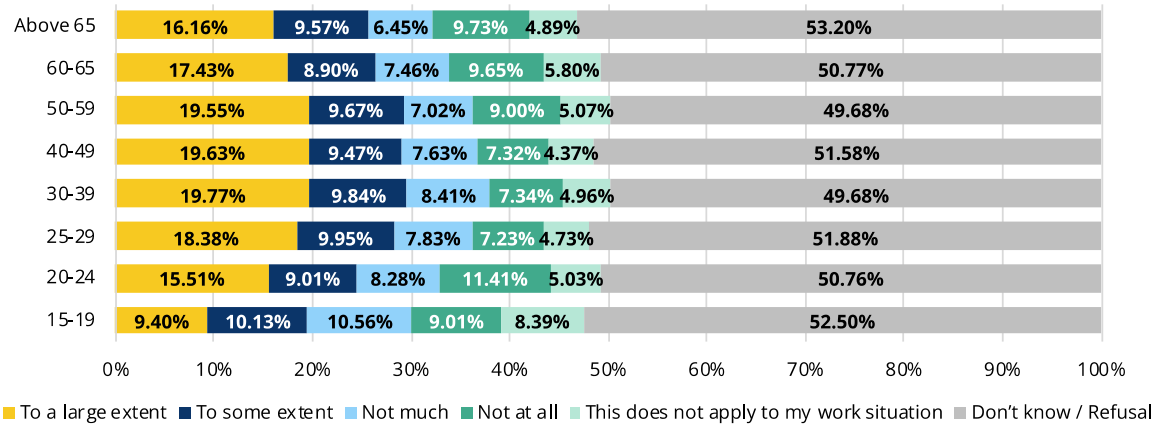
1.2.2. Evidence from the Visionary Analytics surveys, which covers the results of the survey carried out for this study on the usage of AM tools among workers.

Age group

Based on EWCTS-2021, there is no uniform relationship between the usage of AM tools in the workplace and workers' ages. For example, computer systems most frequently influence workers between the ages of 25 and 59, while the usage of such technologies is lower with respect to workers aged 60 and above, or below 25, as can be seen Figure 1-9 below. This lower percentage among younger workers in the EWCTS-2021 data, especially among 15–19-year-olds, could be explained by the fact that entry-level jobs and other jobs performed by such individuals are often lower-skilled and/or social, and therefore the use of such technologies might not be necessary. This is supported by data from the EWCTS-2021 survey[68]: out of the 1,105 individuals aged 15-19 covered in the survey, the largest shares worked in sales (20.3 %), as personal service workers (13.7 %), and as labourers in mining, construction, manufacturing, and transportation (5.4 %).

Figure 1-9: Distribution of workers in the EU-27 in 2021 by age group, based on the extent to which computer systems influence what they do in their work

[68] To ensure representativeness, validation was performed using the whole sample of countries covered in the EWCTS-2021, including the EU-28 and candidate countries, but excluding Ukraine.



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Note: N = 58,400. Data were weighted using the weight core 'weight'. Based on weighted EWCTS-2021 data, the visual represents the following numbers of workers (rounded) working in the EU-27: 3,400,000 aged 15-19; around 12,500,000 aged 20-24; around 17,610,000 aged 25-29; around 46,405,000 aged 30-39; around 51,215,000 aged 40-49; around 48,000,000 aged 50-59; around 15,250 000 aged 60-65; and around 3,420,000 older than 65. For more on weighting, see *Annex 1: Methodological annex.*
Source: Authors' elaboration based on EWCTS-2021 weighted data.

This contrasts with older workers (those aged 20+), who are distributed among many professions (e.g., the largest shares work as business and administration associate professionals (7.6 %); teaching professionals (7.4 %); and business and administration professionals (6.6 %). Meanwhile, the lower usage of AM among the older generation could be explained by their aversion to such technologies, or to change in general.[69] It is also important to flag the fact that, according to the figure below, almost half of respondents did not respond to the question asking if their work is influenced by computer systems. This implies a lack of transparency between employers and workers, which can be considered a significant issue.

Other sources also provide similar results, even though they seldom cover the 15-19 age group. Similarly, in the Trades Union Congress (TUC) survey, which surveyed around 2,100 individuals in the UK, more than half of respondents perceived that their workplace probably engages in monitoring practices. However, among 25-to-34-year-olds this percentage rises to 60 %, and among 18-to-24-year-olds, to 68 %. A significant majority of respondents (72 %) consider it at least moderately probable that some form of workplace monitoring takes place in their organisation.[70]

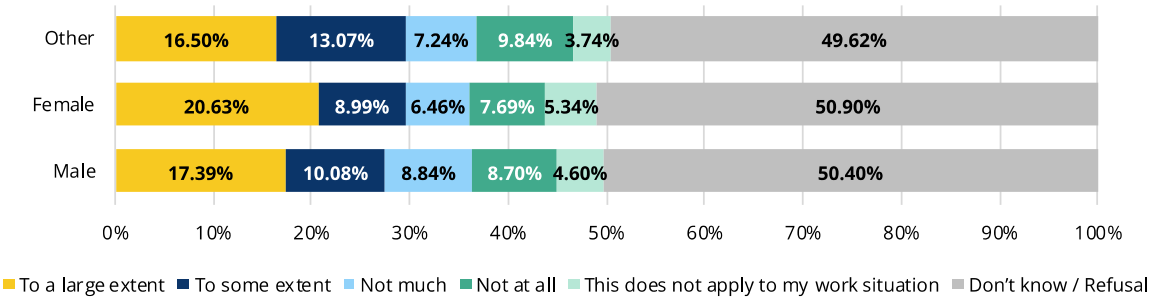
Gender

Based on EWCTS-2021 data, computer systems influence workers in a relatively similar way, irrespective of gender. Based on the figure below, for around 27-29 % of workers, computer systems influence what they do at work 'to a large degree' or 'to some extent'. However, experts who were interviewed for the present study highlighted that men are more likely to be subject to AM, given that they often work in sectors in which such tools are used relatively frequently, such as manufacturing, agriculture, and transportation and storage. Therefore, given these conflicting insights being offered by different data sources, no robust answer can be provided as to which gender is more often subject to AM.

Figure 1-10: Distribution of workers in the EU-27 in 2021 by gender based on to what extent computer systems influence what they do in their work

[69] Vaportzis, E., Giatsi Clausen, M., & Gow, A.J. (2017). Older adults perceptions of technology and barriers to interacting with tablet computers: A FOCUS Group Study. *Frontiers in Psychology*, *8*. https://doi.org/10.3389/fpsyg.2017.01687

[70] Trades Union Congress (2018). I'll be watching you: a report on workplace monitoring. London, UK: Trades Union Congress.



Note: N = 58,400. Data were weighted using the weight core 'weight'. Based on weighted EWCTS-2021 data, the visual represents the following numbers of workers (rounded) working in the EU-27: 91,100,000 women, 106,480,000 men, and 576,000 others. For more on weighting, see *Annex 1: Methodological Annex*.
Source: Authors' elaboration based on EWCTS-2021 weighted data.

Occupation

Regarding the prevalence of AM across different occupations, different data sources again provide conflicting results. For example, according to EWCTS-2021, highly skilled white-collar workers (such as managers, professionals, technicians and associate professionals)[71] and clerical support workers, who are considered low-skilled white-collar workers, are subject to AM the most (see Figure 1-11 for more details). On the other hand, according to AMPWork, clerks and professionals are those who are most subject to activity monitoring, which includes monitoring through their computers, email, or the internet. Meanwhile, different kinds of operators are the group most subject to physical monitoring, including via CCTV and the tracking of work time and entry/exit using office sensors, and vehicle tracking.

Experts interviewed for this study also expressed the opinion that blue-collar workers are more likely to be subjected to algorithmic management. The EU-OSHA (2022)[72] report agrees with this, especially when it comes to monitoring. The aforementioned report also highlights that this might be due to white-collar workers often having a stronger bargaining position than blue-collar workers, due to the greater difficulty of replacing them. This allows them to better push back against the implementation of AM tools if they believe they will infringe on their privacy or other rights. In connection with this, Adams-Prassl (2022),[73] Bodie et al. (2017),[74] Hießl (2021),[75] and others report that workers who enjoy the protection of trade unions are, in general, less susceptible to being subjected to algorithmic management, especially as the usage of such technologies becomes more prevalent. However, according to EWCTS-2021, around 67.1 % of professionals, 59.8 % of technicians, and 58.5 % of clerical workers are represented by a union in their job, compared with, for example, 48.7 % of plant and machinery operators, 49.4 % services and sales workers, and 45.4 % craft and related trade workers. These lower rates of trade union representation imply that blue-collar workers have less power to fight back against algorithmic management.

Figure 1-11: Distribution of workers in the EU-27 by occupation, based on the extent to which computer systems influence what they do in their work (2021)

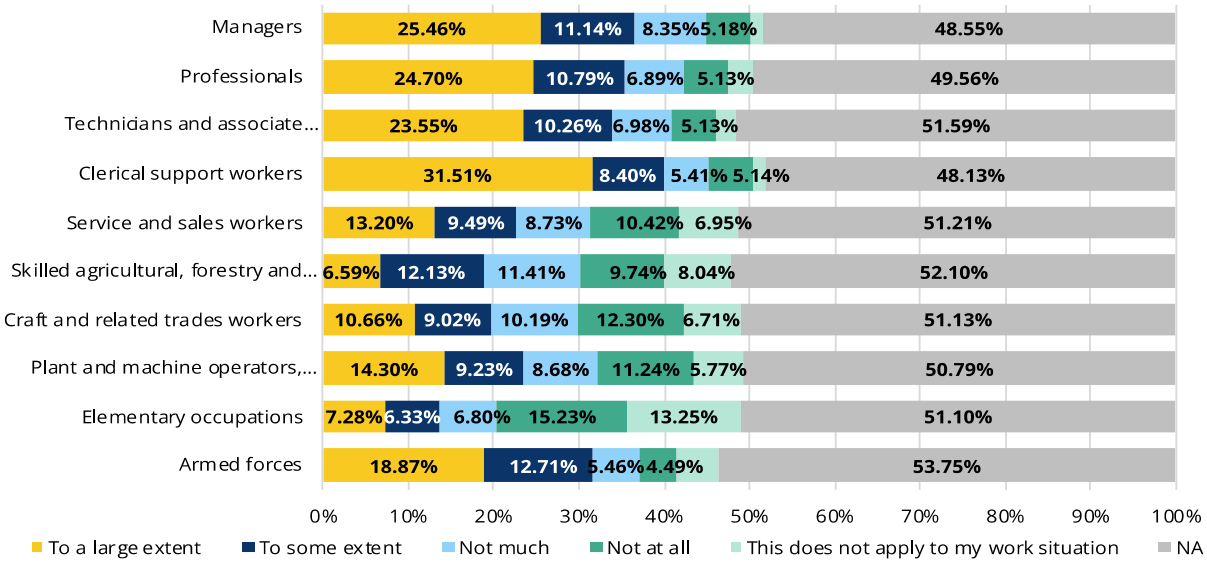
[71] Based on Eurofound classification: https://www.eurofound.europa.eu/en/coding-and-classification-standards-0

[72] Ibid.

[73] Adams-Prassl, J. (2022). Regulating algorithms at work: Lessons for a 'European approach to artificial intelligence'. European Labour Law Journal, 13(1), 30-50.

[74] Bodie, M. T., Cherry, M. A., McCormick, M. L., & Tang, J. (2017). The law and policy of people analytics. U. Colo. L. Rev., 88, 961.

[75] Hießl, C. (2021). Case law on algorithmic management at the workplace: Cross-European comparative analysis and tentative conclusions. Available at SSRN 3982735.



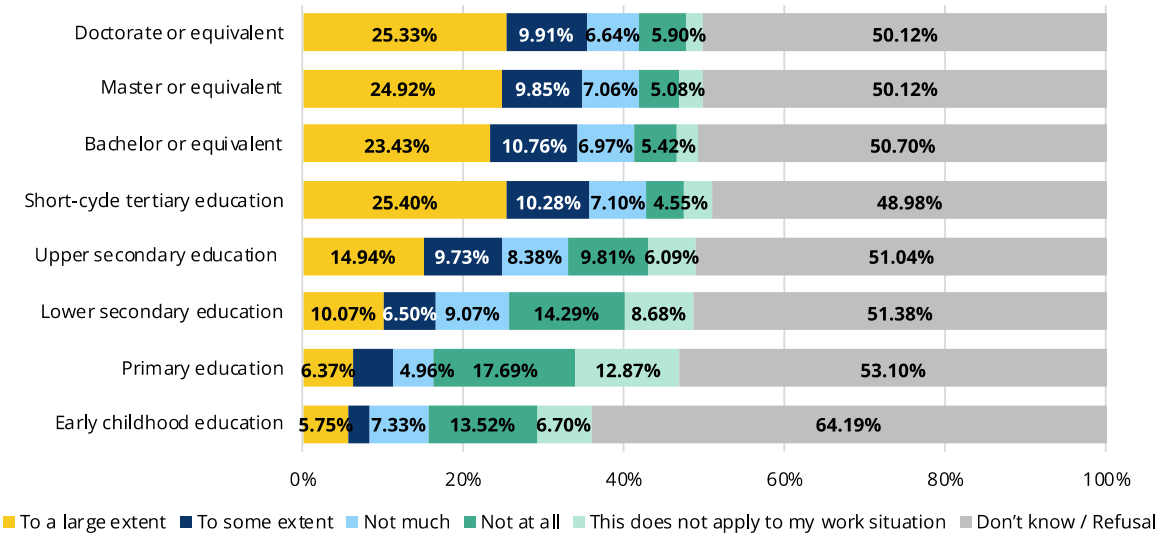
Note: N = 58,400. Data were weighted using the weight core 'weight'. Based on weighted EWCTS-2021 data, the visual represents the following numbers of workers in EU-27 (rounded): 10,000,000 managers; 42,560,000 professionals; 32,250,000 technicians and associate professionals; 19,800,000 clerical support workers; 30,950,000 service and sales workers; 5,900,000 skilled agricultural, forestry, and fishery workers; 23,100,000 craft and related trades workers; 14,950,000 plant and machine operators, and assemblers; 16,800 000 elementary occupations; and 850,000 in the armed forces. For more on weighting, see *Annex 1: Methodological annex*. *Source:* Authors' elaboration based on EWCTS-2021 weighted data.

Level of education

In terms of workers' level of education, yet again, different sources provide conflicting insights. According to EWCTS 2021, around 25 % of workers with tertiary education reported that computers influence what they do at work, compared with 6 % of those with primary and early childhood education (see Figure 1-12 below for more details).

It is interesting to note that the influence of education seems to stabilise after short-cycle tertiary education; among individuals with this or higher levels of education, computer systems influence what they do at work to a similar extent (around 35 %). This pattern therefore implies that individuals with higher levels of education are also more likely to be subject to AM. However, these results might instead indicate that people with higher education are simply more likely to do jobs that are connected to computers, and this trend is not related to AM. To examine such claims, further analysis was conducted on the workers' survey carried out for the present study.

Figure 1-12: Distribution of workers in the EU-27 in 2021 by education level, based on the extent to which computer systems influence what they do in their work



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Note: N = 58,400. Data were weighted using the weight_core 'weight'. Based on weighted EWCTS-2021 data, the visual represents the following number of workers in EU-27 (rounded): 3,934,000 workers with doctoral or equivalent degrees; 36,100,000 with Master's or equivalent; 32,530,000 with Bachelor's or equivalent; 13,540 000 with short-cycle tertiary education; 72,340,000 with upper-secondary education; 18,140,000 with lower-secondary education; 3,000,000 with primary education; and 450,000 with early childhood education. For more on weighting, see *Annex 1: Methodological annex.*

Source: Authors' elaboration based on EWCTS-2021 weighted data.

Hence, based purely on those sources analysed that look at the usage of AM tools among individuals with different education levels, workers with a higher level of education are more likely to be subject to algorithmic management than those with a lower level of education. One explanation for such a trend is the fact that individuals with higher education more often work in jobs using computers and other smart devices that can be used for algorithmic management. For example, according to EWCTS-2021, 74.1 % of individuals with a Bachelor's, Master's, or doctoral degree use computers 'Always', compared with only 21.1 % of individuals with upper- or lower-secondary, as well as individuals with early childhood education. This greater use of computers by those with high levels of education, in turn, creates greater opportunities for employers to use AM. An alternative explanation is that higher education tends to imply higher job complexity, which, in turn, means workers are more likely to interact with various digital tools that might involve an element of algorithmic management (e.g., workers might use email systems that are monitored by employers). Higher complexity also implies a greater likelihood that decision-making is data-driven, which may also be supported through algorithmic management.

1.2.2. Evidence from the Visionary Analytics surveys

This subsection of the report explores the evidence derived from the surveys of employers and workers carried out for the present study. Given that the small sample size of the employer survey includes only 155 full answers with some analysable partial ones, our focus here is predominantly on the worker survey. However, even given the larger number of responses to the workers' survey, which includes 1,270 full responses and some relevant partial responses, the sample size is still too small – especially given that different countries are represented to different extents – to make robust and granular observations about the EU as a whole. Hence, insights from this subsection should not be used as standalone evidence on the usage of AM across the EU. Rather, they should be used as additional and supporting findings to those discussed in subsection 1.2.1. Evidence from direct sources.

The remainder of this subsection describes evidence of usage from the two sources at the levels of: (i) **general and managerial function**, (ii) **employers**, and (iii) **workers**.

Usage of AM in general, and by type

Based on the VA survey of workers, around 64.2 % of the 1,286 employees in the EU-27 surveyed stated that, in their jobs, algorithms are used for at least one of the managerial functions outlined previously in

Table 1-2.[76] Those respondents who responded with 'To my best knowledge I am not aware of the usage of any of the aforementioned tools [e.g., keystroke trackers, CCTV cameras, shared email environments, etc.] in my workplace' were classified as not being subject to AM tools.[77] Out of the 247 respondents to the employers who answered this question, a similar percentage (69.2 %) indicated that AM was used in their workplace. The sample sizes here are larger than those given previously because several additional workers and employers answered this question but did not answer others after it.

Concerning the prevalence of AM tools, responses to the VA surveys differ significantly between workers and employers with regard to various managerial activities to which AM is applied, such as:

- **Recruitment** – 42.7 % of workers vs 32.6 % of employers state AM is used for worker recruitment in their organisations.
- **Work/task scheduling/distribution** – 43.5 % of workers vs 34.3 % of employers stated that algorithms are used to allocate workers' shifts.
- **Nudging/directing** – 43 % of workers vs 29.6 % of employers highlighted that AM is used to provide recommendations and instructions or, in other words, to 'nudge workers.
- **Worker monitoring/surveillance** – around 38.6 % of workers surveyed believe they are subject to worker monitoring via algorithms, compared with only 16.9 % of employers who said they do so. Meanwhile, 53.9 % of workers stated that their employers collect their personal data, while 58 % of employers stated that they do so.
- **Worker evaluation** – around 49.6 % of respondents to the VA workers' survey believe that their company uses algorithms for performance evaluation, while only 47.1 % of employers stated that they do so.
- **Talent management/training** – around 46.8 % of workers vs 38.3 % of employers stated that AM is used in their company for talent management and/or training.
- **Rewarding workers** – 42.4 % of workers said that in their job, algorithms are used to reward workers, while around 39.4 % of employers said that this is the case.
- **Worker termination/firing** – 39.8 % of respondents to the workers' survey believe that their employer uses algorithms to make firing decisions, while 18.2 % of respondents in the employers' survey stated that they do so.

As can be seen from the disparities above, while similar shares of workers and employers agree that AM is used, more workers believe AM is being used when it comes to specific AM tools. This might be explained by the fact that workers often selected several AM tools in the survey, while employers were more precise in their assessments, often focusing on only a few tools. Nevertheless, both workers and employers might misunderstand when a particular managerial function uses algorithms and when it does not. This, in turn, might explain the high estimates of AM usage from both the workers' and the employers' surveys. However, these high estimates might also reflect the possibility that those individuals who have experience with AM were more willing to fill in the survey compared with those who did not, leading to selection bias. Nevertheless, the results highlight the **complexity of the topic, which complicates an accurate assessment of how extensively such tools are used by employers and among workers.**

[76] That is, worker monitoring/surveillance, worker evaluation, work/task scheduling/distribution, nudging/direction of workers, talent management/training, recruitment, rewarding, worker termination/firing/dismissal.

[77] Because of this methodological caveat, the probabilities associated with workers that are presented in the bullet points below do not match the ones discussed in Annex 13: Factual summary of the workers' survey.

Usage by employer type

Given the relatively small sample size of the VA employer survey, the insights that can be derived from it are predominantly general. First, as discussed previously, around 69.2 % of respondents said that they use algorithms to automate at least one managerial function. Given that this statistic is much higher than other sources discussed previously – including the AMPWork survey, which was also carried out in 2023 – it is likely that there was a selection bias in the employer survey, whereby companies that use AM were more willing to fill in the survey than those that do not.

Second, according to the VA survey results, 63 % of private companies on average use algorithms to automate at least one managerial function, compared with 70 % of public companies (including public companies operating in the market, public providers of services, and public administrations), and 62.5 % of NGOs do the same. Private companies most frequently use AM to collect worker data, as stated by 51 % of these respondents. In addition, 40 % of respondents from private companies stated that they use AM to evaluate workers' performance, while 36 % highlighted that they use algorithms for rewarding workers. Public companies use AM most frequently for personal data collection, with 62.5 % of respondents from public organisations confirming that they did so; meanwhile, 40 % of public employers stated that they use AM for performance evaluations, and 37.5 % to plan worker shifts or reward workers. Similar trends can be observed among NGOs, with 62.5 % of these respondents stating that they use AM to collect personal data about workers, 43.8 % for talent management, and 40.6 % for performance evaluation. However, as these results are only based on a sample of 100 private companies, 40 public organisations, and 32 NGOs, they are not robust.

Lastly, based on the VA survey of employers, 54.2 % of micro-enterprises (1-9 employees), 57.9 % of medium-sized businesses (10-49 employees), 63.3 % of large ones (50-249 employees), and 77.1 % of large companies (250+ employees) use AM for at least one managerial function. This is consistent with the trends discussed previously. With regard to the types of AM tools used, regardless of their size, companies most frequently use AM for personal data collection, which is also consistent with the findings discussed previously. These results are based on a sample size of 24 micro-companies, 38 small businesses, 60 medium-sized companies, and 48 large ones.

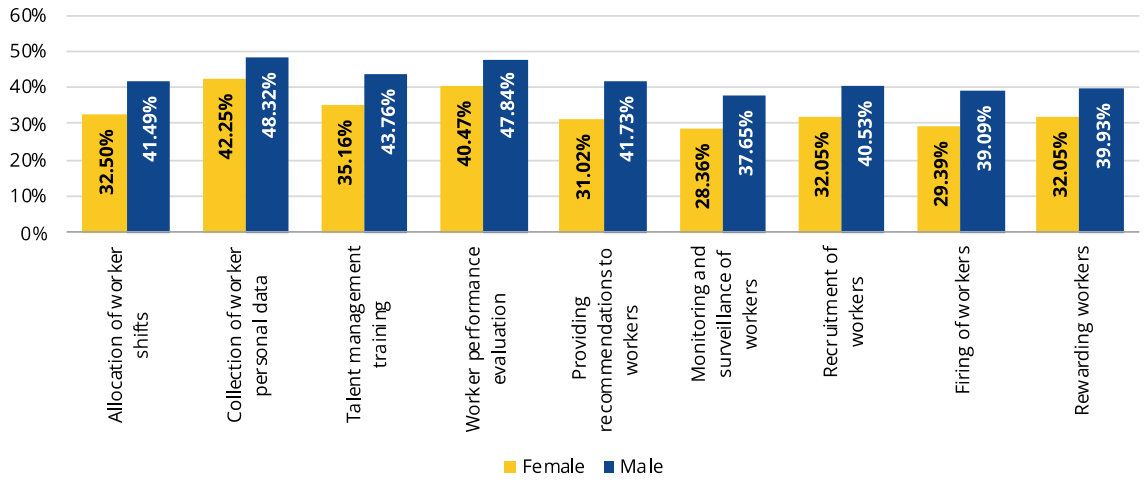
Usage by worker type

Given the larger sample size of responses to the VA workers survey, more robust insights can be derived from it. However, it still provides limited insights across the EU-27 overall, given that some countries and groups of individuals are not covered as extensively as others. Hence, the insights provided under this heading predominantly highlight possible supporting evidence rather than robust, standalone insights. The paragraphs that follow describe the usage of different AM tools among various types of workers, based on their: (i) **age group**, (ii) **gender**, (iii) **occupation**, and (iv) **level of education**.

In terms of **age**, the VA workers survey finds that individuals between ages 20-29 are subject to AM most frequently. For example, 71.6 % of 20-24-year-olds and 74.1 % of those aged 25-29 stated that they are subject to AM either 'vary rarely' or more. However, individuals between the ages of 30 and 39 are also often managed through algorithms; 62 % of this age group state that algorithms are used to manage at least one managerial function in their workplace. Above this age range, the perceived usage of AM drops rapidly, with 49.5 % of individuals aged 40-49, and 38.6 % of individuals aged 50-59 stating that AM is used in their workplace either 'rarely' or more. With regard to the types of algorithms used, they are most frequently used to collect personal data for use in workers' performance evaluations and, in the case of 20–29-year-olds, algorithms are also frequently used for the allocation of tasks.

Concerning **gender**, men appear to be subject to AM more frequently than women. Around 55.5 % of male workers stated that AM tools are used in their workplace either 'very rarely' or more, compared with 49.3 % of female workers.[78] This trend is constant across managerial functions, with male workers almost always being 5-10 % more likely to be subject to algorithms with regard to every managerial function (see Figure 1-13 below for more information). In addition, no trend can be identified whereby one gender is more subject to AM to a particular managerial function.

Figure 1-13: Distribution of workers who stated that AM is used for one or more managerial functions in their company either 'very rarely' or more, based on gender

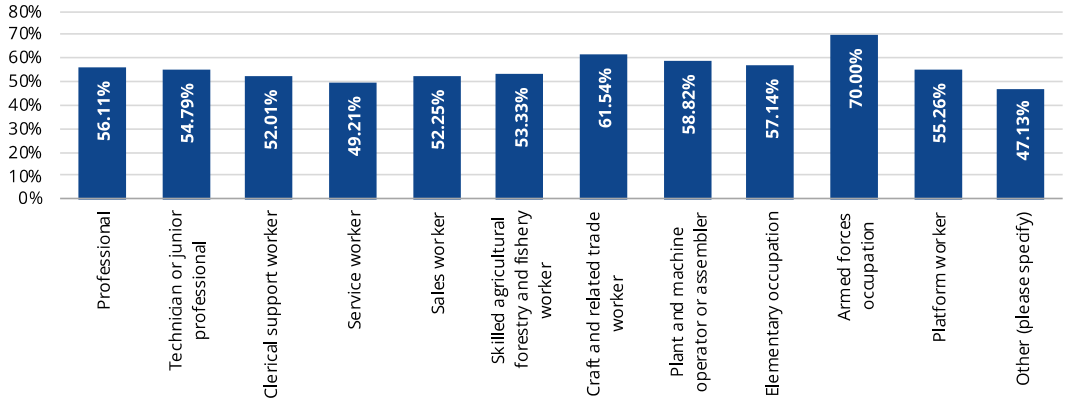


Note: N = 1,511 (female – 677; male – 834).
Source: Authors' elaboration based on worker survey carried out for the study (2023).

Concerning occupation, individuals serving in the armed forces appear to be most subject to AM (see Figure 1-14 below for details). This might be due to the large amount of personal data that is collected from workers in the armed forces, to ensure that sensitive information is not leaked. According to the data on other occupations, specific types of AM appear to be more prevalent in some than in others, but there are no large differences between the majority of them. In many cases, such differences might be due to measurement errors and noise. Hence, based on the VA survey, it appears that particular occupations, including platform work, are not subject to more AM than other types of jobs. However, given that the majority of the literature and other data sources suggest that platform work is subject to much more AM than other working arrangements, the coverage of occupations in the survey is likely to be flawed, and hence no robust conclusions should be derived from it.

[78] The 'other' category is not analysed here, given that only received three such answers were received.

Figure 1-14: Distribution of workers who stated that AM is used for particular managerial functions in their company either 'very rarely' or more, based on occupation

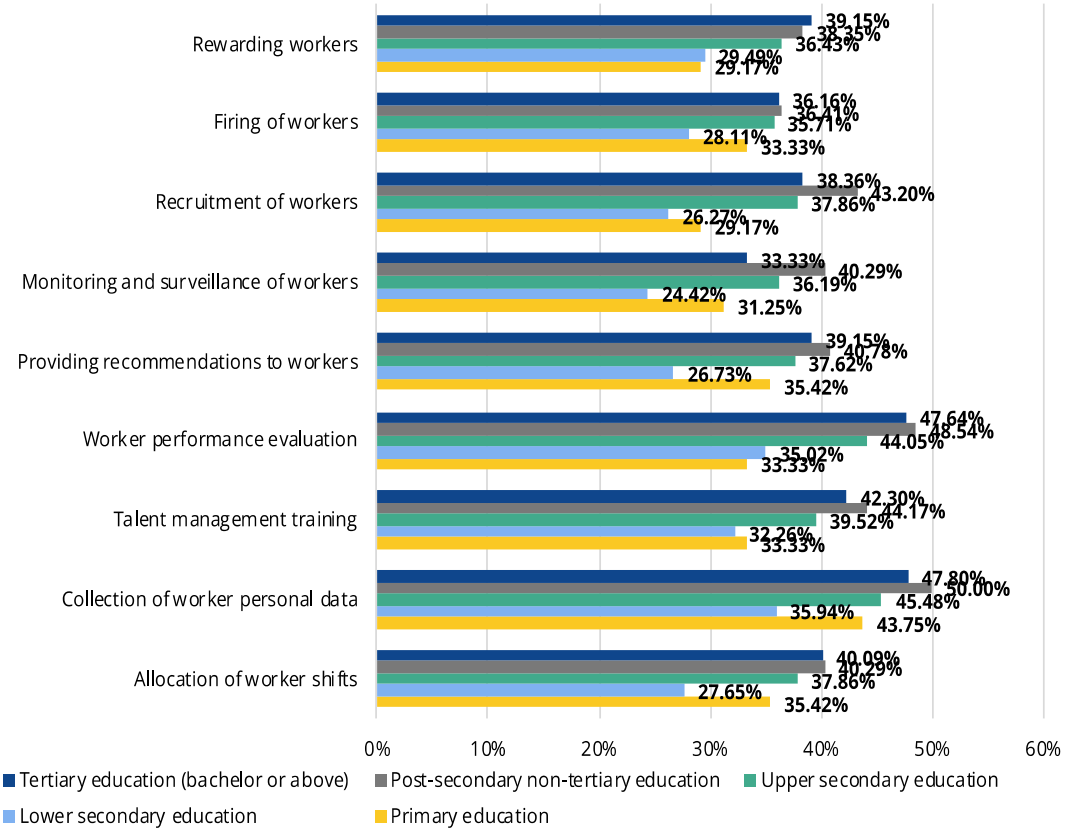


Note: N = 1,388.
Source: Authors' elaboration based on the VA survey of workers.

Level of education

Based on the VA workers' survey, individuals with lower-secondary education are, in most cases, least likely to be subject to AM (see Figure 1-15 below). Individuals with primary education are the group second least likely to be subject to AM. In some cases, however – such as with algorithms used to recruit or fire workers – individuals with lower levels of education are slightly more likely to be subject to AM. In many cases, individuals with upper-secondary; post-secondary, non-tertiary; and tertiary education are subject to similar levels of AM usage. This trend is consistent across different managerial functions. However, for some functions, the difference between higher and lower educational levels is smaller (such as in the collection of personal data and the allocation of worker shifts), as individuals with primary education are subject to such technologies almost as frequently as individuals with higher levels of education.

Figure 1-15: Distribution of workers who stated that AM is used for particular managerial functions in their company either 'very rarely' or more, by education level



Note: N = 1 225 (primary education – 39; lower-secondary education – 159, upper-secondary education – 337; post-secondary, non-tertiary education – 168; and tertiary education – 522).
Source: Authors' elaboration based on the worker survey carried out for the study (2023).

1.2.3. Evidence from the Visionary Analytics Delphi survey

As discussed previously, as part of the implementation of the present study, a Delphi survey was conducted that focused on collecting insights on the future use of AM. In addition, it contained a question about current usage, which provides some additional insights on the existing prevalence of AM. These are presented in this subsection. For more details on how the Delphi survey was carried out, including the motivation behind using this instead of other data sources, see Chapter 4. Future Trends.

First of all, the respondents to the Delphi survey were asked to indicate if they agree with a statement that around 25 % to 35 % of companies in the EU-27 currently use AM. Out of all respondents, 51.2 % agreed with this estimate (representing 109 responses); 16.1 % disagreed (34 responses); and 32.2 % (68 responses) answered 'Don't know'. Academics were the most likely to agree with this estimate, with 55.5 % of the 137 academic respondents agreeing. Workers were less willing to agree, with only 46.2 % of the 39 worker respondents agreeing. Estimates cannot be provided for other groups, as fewer than 10 responses were received.

Individuals who did not agree with the estimate were given the chance to provide their own estimate using an open question. In the majority of cases, the estimates they provided were lower. For example, five individuals indicated that they believed the usage of AM is below 5 % in workplaces. They justified this mainly by stating that because the majority of companies in the EU are SMEs, it is likely that many of them lack adequate resources to invest in AM. In addition, 11 individuals estimated that AM is used in around 10-15 % of companies in the EU, while around nine respondents stated that AM was likely to be used in between 15 % and 25 % of companies. Both groups used the justification as the previous group mentioned – namely, stating that a large number of micro- and small enterprises are

unlikely to be using AM due to resource constraints. Lastly, one individual stated that 60 % of companies use AM, but provided no justification for their estimate. The rest of those respondents who did not agree with the estimate provided failed to provide an estimate of their own.

1.2.4. Evidence from proxies

This subsection of the report explores several proxies that can provide some indirect insights into the prevalence of AM. The table below summarises the main findings from different sources, while the remainder of the subsection discusses them in depth.

The findings outlined in the table below and discussed in this subsection, while not directly connected to AM, share many indirect connections with it that might provide some insights into the future of AM. First, the concepts discussed refer to technology-driven optimisation, for which advanced technologies are used in relation to various aspects of enterprise operations. Given that AM is, to a large extent, implemented for the same purpose, an indirect connection can be seen between AM and the technologies discussed. Second, all these concepts involve various degrees of automation, and all of them involve the heavy use of algorithms. Thus, if companies employ one such type of algorithm-driven technology, it should not be too difficult for them to employ another kind. Lastly, given that cloud computing or buying in external computing power on demand were relatively novel concepts a decade ago, just as AM is today, their rapid adoption might, to some extent, be mimicked by AM in the future. For more on this, see the remainder of the subsection.

Table 1-9 – Main findings on the AM usage from proxy measures*

Evidence based on different sources
International Data Corporation (2023)[79]: The AI market in Europe had a compound growth rate of around 29.6 % annually in 2022-2026.
European enterprise survey on the adoption of AI (2020)[80]: in the EU-27, Norway, Iceland, and the UK, 42 % of enterprises use AI in at least one technology, while only 40 % say they do not use it at all.
OECD study on the AI impact (2023): 51 % of workers in Austria, 36 % in France, 34 % in Germany and 35 % in Ireland said that their workplace uses AI
DESI (2021): around 7.9 % of enterprises use AI technologies in the workplace; around 41 % of enterprises buy cloud computing services; and around 9.8 % of enterprises buy computing power to run their software.

(*) All results provided are for the EU-27 unless specified.

Recent years have seen rapid advances in AI and machine learning, which are closely linked to the growing use of AM technologies.[81] According to the European Commission (2020),[82] which references the International Data Corporation (IDC), a global provider of market intelligence, the AI market in Europe was predicted to grow by 26.5 % between 2018 and 2023, annually. However, newer data from IDC (2023), which takes into account the Ukrainian crisis and COVID-19, now estimates that the AI market in Europe will grow by around 29.6 % between 2022 and 2026.[83] However, given that these insights concern the

[79] See: https://www.idc.com/getdoc.jsp?containerId=prEUR250505223#:~:text=AI%20spending%20in%20Europe%20will,than%20%2470%20billion%20in%202026

[80] European Commission (2020). European enterprise survey on the use of technologies based on artificial intelligence: final report, Publications Office. Available from: https://op.europa.eu/en/publication-detail/-/publication/f089bbae-f0b0-11ea-991b-01aa75ed71a1/language-en

[81] Vetzo, M., & Gerards, J. (2019). Algoritme-gedreven technologieën en grondrechten. *Computerrecht*,1, 10 - 19

[82] European Commission (2020). Advanced Technologies for Industry – EU Report. Available at: https://ati.ec.europa.eu/sites/default/files/2021-01/EU%20Report%20-%20Technological%20trends%20and%20policies.pdf

[83] See: https://www.idc.com/getdoc.jsp?containerId=prEUR250505223#:~:text=AI%20spending%20in%20Europe%20will,than%20%2470%20billion%20in%202026.

growing AI market and not the usage of AM or even AI, these insights provide a very flawed proxy for algorithmic management.

Some more relevant insights can be seen by exploring the current usage of AI in the workplace. For example, according to the 2020 European enterprise survey on the adoption of AI, which surveyed 9,640 enterprises in the EU-27 as well as Norway, Iceland and the UK, 42 % of enterprises said they used AI at least in some form, while 40 % said they did not use it at all.[84] Based on this survey, Slovakia, Cyprus, and Estonia exhibit relatively low rates of AI uptake, at around 30 %, while around half of enterprises in Lithuania and Bulgaria (54 % in each), and in Austria and Luxembourg (51 % in both), have implemented at least one AI technology. Similarly, based on a recent study by the OECD, 51 % of workers in Austria, 36 % in France, 34 % in Germany and 35 % in Ireland said that their workplace uses AI.[85] This study relied on surveys in 2021 of around 750 workers in Austria, 800 in France, 846 in Germany, and 442 in Ireland working in the manufacturing or finance sector.

DESI 2021, meanwhile, paints a fairly different picture. According to these data, only around 7.9 % of enterprises in the EU-27 use AI technologies in the workplace. Companies in Denmark, Portugal and Finland use AI the most (24 %, 17 % and 16 % of companies respectively), while in Romania, only 1 % of companies use AI. In Poland, Hungary, Cyprus, Estonia, and Bulgaria, 3 % do so. However, that DESI does not cover companies from agriculture, mining, and several other sectors. It also does not cover companies with between one and nine employees. (Usage of AI in such companies can be expected to be lower than in larger companies as, according to DESI, 6.4 % of companies with 10-49 employees use AI, while 28.5 % of those with 250+ employees do so. This implies that if the micro companies were included, usage of AI across the EU-27 might be even lower.

Given the stark difference between data sources, it is important to identify the possible reasons for this. First, the differences may be due to different methodologies being applied. For example, DESI is a large-scale survey, while other surveys often focus on specific sectors and countries, which leads to biased sample sizes that may focus too heavily on companies that are more likely to use AI technologies (e.g., ICT companies). Another possible reason for inconsistencies could be that managers and workers misinterpret what AI is.[86] For example, according to the MMC Venture Report (2019), only around 1,580 (around 56 %) out of the 2,830 'AI-based start-ups'[87] surveyed and analysed from 13 EU countries[88] use AI in their day-to-day operations. In other words, companies in many cases used relatively simple algorithms that they tried to 'sell' as AI. This, in turn, means that many companies might misunderstand whether they are using AI or simple algorithms in their workplace. Though this claim largely comes from the grey literature. Some interviewees also reported that companies often misconstrue, or misreport, whether they are using AI, as saying that they do often leads to greater interest from investors.

Going beyond AI, the DESI index (see Table 1-10 below) shows that since 2014, companies have rapidly started to use cloud computing services over the internet (the percentage that do so has increased from 17.8 % in 2014 to 41 % in 2021). Similarly, from 2014 to 2021, the share of enterprises buying computing power to run their own software increased from 2.9 % to 9.8 %. Although this does not constitute a large increase in absolute terms, in relative terms it represents an increase of more than threefold in seven

[84] European Commission (2020). European enterprise survey on the use of technologies based on artificial intelligence: final report, Publications Office. Available from: https://op.europa.eu/en/publication-detail/-/publication/f089bbae-f0b0-11ea-991b-01aa75ed71a1/language-en

[85] The impact of AI on the workplace: Main findings from the OECD AI surveys of employers and workers. (2023). OECD Social, Employment and Migration Working Papers.

[86] Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek, E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work. Report prepared for EU-OSHA.

[87] This is how these companies defined themselves.

[88] Austria, Denmark, Finland, France, Germany, Ireland, Italy, the Netherlands, Norway, Portugal, Spain, Sweden and the United Kingdom.

years. This, in turn, might provide a general proxy indicating how rapidly AM usage might grow across the EU.

Table 1-10– Percentage of enterprises in the EU-27 (excluding the financial sector) that use the following

	2014	2016	2018	2020	2021
Buy cloud computing services used over the internet	17.8 %	18.9 %	23.9 %	36.1 %	41 %
Buy computing power to run the enterprise's own software	2.9 %	3.8 %	5.1 %	8.7 %	9.8 %

Source: Author's elaboration, based on DESI.

1.2.5. General summary/conclusions

Based on the evidence collected from a variety of sources, several conclusions can be derived. First, **due to the relative newness of AM, no one source fully and robustly covers the prevalence of AM in the EU**. This severely complicates the analysis, as it prevents us from painting a full picture of AM usage throughout the EU and among different types of employers and employees. Although some sources provide insights into the automation of various managerial functions, which can be aggregated, the majority of these sources provide very different results. This is true even of those that focus on similar technologies, over a similar time period, and in similar countries. In other words, **there is a wide variation in results between the different sources**. For example, according to ESENER-3 data, 8.2 % of organisations in the EU-27 use machines, systems, or computers to monitor workers' performance. Meanwhile, based on data from ECS-2019, 27 % of organisations monitor workers performance. Based on the UNI Europa and Central European Labour Studies Institute study by Holubová (2022),[89] however, around 47 % of workers surveyed believed monitoring was used in their workplace. These and many other variations can be largely explained by using different sampling strategies, survey approaches, the years in which the surveys were carried out, and the general methodologies used to collect the data. However, at the same time, this wide variation heavily complicates the analysis, as it is unclear which of these many sources is the most reliable.

To some extent, the aforementioned issues could be mitigated by running a targeted survey focusing on employers to collect up-to-date and robust information on the prevalence of AM. However, **obtaining insights about AM usage from businesses/employers is often challenging**. For example, during the implementation of the survey carried out for the present study, many difficulties were faced in reaching out to employers for interviews and surveys, despite sending emails to over 30,000 different companies. This, in turn, created a situation in which stakeholders who were interested in how the EC would tackle AM were unwilling to share their views regarding this important question.

Nevertheless, although there are a lot of issues with the existing quantitative data on AM, some conclusions can still be derived concerning the prevalence of such technologies in the EU. First, by aggregating the results, considering the different focuses of the various sources and the fact that the business landscape of the EU is largely populated by SMEs, **it can be stated that in 2023, around a quarter of enterprises in the EU-27 are using AM in some form or other**. This is based on observations that the usage of some AM technologies, such as those that monitor and evaluate workers, has grown heavily in recent years, largely prompted by the COVID-19 pandemic. The above estimate is lower than many others which cover only one AM technology, due to the fact that such sources often ignore micro-enterprises, and because they are sometimes biased towards companies that are generally more likely to use AM, such as those focusing on ICT, finance, transportation,

[89] Holubová, B. (2022). Algorithmic Management: awareness, risks, and responses of the social partners. Final Report.

etc. However, given the issues discussed previously, this estimate is not robust, and represents only our best possible estimation, based on the wide array of sources analysed.

Second, **employers use algorithms most frequently to monitor and evaluate workers; based on estimates, around one-quarter of companies in the EU-27 use these technologies for such purposes**. On the one hand, the wide prevalence of such technologies can be explained by the boost they received during the COVID-19 pandemic when many people had to telework, prompting some employers to install software solutions such as keyloggers or email screeners, which monitor what workers do on their work computers. On the other hand, algorithms and software solutions that can be used to monitor or evaluate workers are relatively old dating back to at least the 1980s,[90] and hence may be more familiar to many employers than other types of AM solutions.

Third, **large privately owned companies use AM more frequently than other types of companies**. The likely reason why larger enterprises use AM more frequently than smaller ones is because such technologies allow them to manage their large workforce more efficiently. In addition, privately owned companies might use AM more often than public companies since integrating AM can be very resource-intensive, requiring resources some public companies might not possess. The disparity between private and public companies might also be due to the different types of activities they carry out in their respective sectors. However, given that a robust and consistent trend in terms of companies from certain economic sectors making greater use of AM could not be identified, it cannot be concluded that this is the case.

Lastly, **there is little robust evidence regarding which types of workers are most subject to AM.** For example, based on the VA surveys, male workers are subject to AM more widely than female workers, across different managerial tasks. However, based on Parkes (2023),[91] 38 % of low-skilled female workers are susceptible to worker monitoring compared with 33 % of their male counterparts. Similarly, some sources indicate that white-collar workers are most subject to AM, while others highlight that the usage of AM is highest with respect to blue-collar workers. When it comes to age, many sources indicate that individuals between the ages of 20 and 35 are the most subject to AM, while other sources, such as the EWCTS (2021), provide different results.

[90] Kalischko, T. and Riedl, R. (2021). Electronic Performance Monitoring in the Digital Workplace: Conceptualization, Review of Effects and Moderators, and Future Research Opportunities. *Frontiers Psychology*, 12:633031.

[91] Parkes, H. (2023). Watching me, watching you: Worker surveillance in the UK after the pandemic, IPPR, available at: https://www.ippr.org/research/publications/worker-surveillance-after-the-pandemic

2. Opportunities and challenges

The effects of Algorithmic Management (AM) are already significant and are experienced widely in many workplaces and working arrangements. Employers view AM as a considerable opportunity to improve their business productivity as well as workers' safety and health protection. However, AM presents challenges for workers such as work intensification, invasions of privacy, and the perpetuation of bias. AM also presents challenges for employers. This chapter offers a comprehensive analysis of the opportunities arising and the challenges faced by both workers and employers. To date, much evidence about the impact of AM originates from the performance of digital platform labour. However, this study emphasises at the outset that AM is a management technique equally applicable to traditional workplaces. Thus, whilst the impacts of AM are currently most prominent in digital platform labour, AM should be conceptualised as a technological development independent of digital platforms with potential consequences across all workplaces.

The box below highlights the key findings of this analysis, while the remainder of the chapter goes into further detail.

Key findings:

- AM can simplify and reduce the workload of managers, help them allocate tasks more effectively, and improve communication. This, in turn, could lead to cost-cutting, increased productivity, and efficiency.
- AM can prevent some risks for Occupational Safety and Health (OSH) by, for example, alerting workers about dangers and hazards, identifying ergonomic risks, assessing workers' general well-being and more. These, in turn, allow improvements in workplace well-being, including identifying burnout.
- If AM tools are built using high-quality data that is devoid of as many human biases as possible, it can lead to more transparent and fair decision-making, fostering objectivity.
- AM can help workers in their career progression by identifying areas for improvement and highlighting to managers who are the best performers, and who might be ready for a new position or promotion within the company.
- AM can directly empower workers, improve collective bargaining by logging up-to-date information on wages and company actions, facilitate real-time communication between workers, and more.
- AM can negatively affect employers, not only workers.
- AM can qualitatively alter the work experience by eliminating worker autonomy over the production process, which can lead to the erosion of meaningful work and other negative psychosocial effects.
- AM can create a working environment that places substantial pressure on workers through extensive monitoring and worker evaluation, forcing them to work more and faster, which can in turn lead to work intensification.
- Heavy reliance on AM could remove the need for social interactions within the workplace, which could lead to fractured relationships between and among managers and workers.
- Some monitoring tools can lead to the invasion of workers' privacy and the erosion of transparency in decision-making, if algorithms use data or make decisions in ways that are not explained to workers.
- Some AM tools might introduce biases if they are trained using biased data. Examples include automated recruitment tools, which can continue patterns of

discrimination by preferring candidates from a specific generation, social orientation, sex, age, etc.

AM can negatively affect workers' rights by, for example, providing tools to employers that can be used to analyse large datasets of workers to potentially identify trade union activity, such as organising efforts or collective bargaining discussions.

2.1. Methodology

To structure the analysis and ensure rigour, this report is organised around several dimensions, based on a typology of the challenges and opportunities outlined in Table 2-11. This typology builds on, but is different from the one outlined and used previously in Table 1-1 in subsection 1.1.2. Functions and typologies of AM, for the following reasons:

- Different algorithms often lead to similar effects on workers and employers. For example, tools that monitor, evaluate and/or nudge workers can increase productivity and efficiency, but may also lead to stress and anxiety for workers. This means that focusing on managerial functions would result in a duplication of information.
- Different algorithms might be governed by the same legal provisions and *vice versa*. For example, the GDPR, in addition to governing data privacy, in Article 22, specify that data subjects, including workers, shall not be subjected to completely automated decisions. Given that AM tools often include automation, this implies that the majority of such tools would be subject to GDPR. Thus, focusing on managerial functions would again introduce a duplication of information. Hence, focusing on the challenges brought by AM tools avoids this duplication – although, where relevant, the effects of specific managerial functions from Table 1-1 are also mentioned at points within this chapter.
- A typology focusing on opportunities and challenges is both adaptive and expansive. This is because the opportunities and challenges that AM creates are often not limited to algorithmic management and, for this reason, the study is not limited only to very recent discussions of them. This is especially important in relation to the legal analysis, as it allows us to discuss how AM can be governed through existing regulations that do not directly relate to AM, but affect it indirectly – for example, preventing such technologies from negatively affecting workers.

Table 2-11 – Typology of the opportunities and challenges presented by AM

Dimension	Sub-dimension	Opportunities	Challenges
Working conditions	Changes in working arrangements and the nature of work	• Increased speed, efficiency and productivity • Career progression • Guidance, nudging	• Pressures around optimisation • Reduction of meaningful work • Reduction of worker autonomy • Deskilling • Fractured employment relationships • Algorithmic pay determination
	Working time and communication	• Reduction of workload • Greater efficiency in work scheduling and allocation • Increase in communication	• Workload and work intensity • Reduced choice around working time • Alienation • Pressures around optimisation
	Occupational safety and health	• Early alerts and warnings regarding OSH hazards • Real-time interventions to prevent accidents • Assessing the general well-being of workers • Interventions to reduce psychosocial risks	• Increased OSH risks • Psychosocial risks: stress, anxiety, burnout, 'bore-out', etc. • Physical risks: ergonomic, musculoskeletal issues and disorders, etc. • The psychological contract

Dimension	Sub-dimension	Opportunities	Challenges
Data protection and privacy	Data collection and storage	• Portability of data and data trusts	• Employer surveillance/monitoring • Workers' concerns over data protection
	Transparency	• Increased transparency of decision-making • Clear performance expectations	• Function creep • Secrecy in the development and usage of AM tools
	Decision-making by algorithm	• More rapid and accurate (clear and objective) decision-making • Increased accountability/oversight	• Unclear accountability • Lowered human involvement • Aggressively discipline workers
Fairness and discrimination	Bias and objectivity	• Possible reduction of bias	• AM could lead to unfair management decisions and bias
	Legality of algorithmic decision-making	• Flexibility and applicability of existing general provisions in the field	• Uncertainty regarding liability • Uncertainty regarding the applicability of presumptions
Workers' collective rights	Collective rights	• Co-determination • Co-design	• 'Black box' nature of algorithmic management (lack of transparency, opacity, intricacy)
	Collective bargaining rights	• Protecting and enhancing workers' rights • Predictive analytics regarding labour shortages and skills gaps	• Threats to collective bargaining • Lack of comprehensive regulation • Trade union busting
	Social dialogue	• Strengthening of collective bargaining • Improving individual negotiations • Fostering social dialogue and participation • Empowering workers	• Unequal access to information • Digital literacy divide • Decreasing capacity for social dialogue and workers' involvement
Effects on broader society	Effect on the public sector and social security systems	• More accurate and faster data-driven decision-making	• If AM tools are based on flawed data, they can exacerbate existing issues
	Effects on broader society	• Economic growth • Creation of tech for good that can be used beyond work	• Changing public health spending due to, for example, increased accidents at work • Social division and discrimination • Reduce overall levels of employment • Income inequality • Deskilling, increased need for upskilling
	Competitiveness	• Changes in the efficiency of companies • Opportunities for innovation	• Micro-enterprises and small companies might fall behind as AM tools might be too expensive for them to use.

Source: Authors' own elaboration, based on literature and other sources.

All the dimensions in the table are covered during the remainder of this chapter. More specifically, the first four dimensions – working conditions, data protection and privacy, fairness and discrimination, and workers' collective rights – are covered extensively in 2.2. Opportunities posed by AM for workers and employers and 2.3 Challenges posed by AM for workers and employers. The last dimension – effects on broader society and its constituent elements – is covered in Section 2.4. Effects of AM on public sector and social security

systems, Section 2.5. General effects of AM on broader society, and Section 2.6. General effects of AM on competitiveness and consumers. In addition, the legislative analysis, which can be found in 3. Legal and policy framework, is also structured in line with these dimensions, ensuring their consistency and cross-comparability.

This report's systematic analysis of challenges and opportunities in relation to AM is based on:

- **Literature review**: an extensive search and analysis of relevant literature, including academic papers, reports, evaluations, grey literature, etc. In total, 622 documents were identified, out of which around 509 were selected as being relevant for further analysis. The relevant documents were mapped by highlighting the insights derived from them. For the results of this mapping, see *Annex 2: Literature mapping*. For more information on the literature review, see *Annex 1: Methodological annex*.
- **Surveys and interviews**: these refer to surveys of workers and employers, as well as interviews with experts in the field, social partners, national representatives of employers and workers, and the creators of AM tools. Insights from these sources have helped to enrich the discussion on the challenges and opportunities of AM, in particular providing insights regarding new developments in this field.
- **Case studies**: case studies discussing the usage of AM across selected Member States, as well as case studies on specific AM tools, have also been integrated throughout the analysis. Insights from thee case studies have been used to contextualise the findings from the literature, and further enrich the discussion.

In addition, the following main difficulties were faced in this chapter:

- In some cases, it was difficult to differentiate between AM and Artificial Intelligence (AI), machine learning, and various other advanced technologies, all of which may be used in the workplace. This is especially true for topics that extend beyond the world of work, such as those covering the general effects of AM on broader society and competitiveness. Hence, in the discussion, the authors have attempted to remain as factual as possible, highlighting where a particular discussion covers a topic that is strongly related to AM, but is not necessarily a part of it (e.g. the usage of AI at work).

2.2. Opportunities posed by AM for workers and employers

The use of AM to manage workers has the potential to **cut costs,**[92] **increase productivity and efficiency,**[93] **reduce workload for managers**[94] **and workers,**[95] **improve working conditions, and more**. In addition, **there is also some agreement that AM is linked to the automation of work and tasks, or to the alternative taken to complete automation**.[96] However, there are many nuances regarding the ways and extent to which

[92] Tulieres, M., Tilley, J. & Bolz, L. (2019). Industrial robotics. Insights into the sector's future growth dynamics. Paris: McKinsey. Available at: https://www.mckinsey.com/~/media/McKinsey/Industries/Advanced%20Electronics/Our%20Insights/Growth%20dynamics%20in%20industrial%20robotics/Industrial-robotics-Insights-into-the-sectors-future-growth-dynamics.ashx

[93] Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek, E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work.

[94] Duggan, J., Sherman, U., Carbery, R., & McDonnell, A. (2020). Algorithmic management and app work in the gig economy: A research agenda for employment relations and HRM. *Human Resource Management Journal,* 30(1), 114 - 132; Griesbach, K., Reich, A., Elliott-Negri, L., & Milkman, R. (2019). Algorithmic control in platform food delivery work. *Socius: Sociological Research for a Dynamic World*, 5(4), 1-15. Note: the academic literature identifying the benefits of AM is smaller and more limited than the literature indicating the challenges.

[95] Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek, E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work

[96] De Stefano, V. (2020). Algorithmic bosses and what to do about them: automation, artificial intelligence and labour protection. In Economic and policy implications of artificial intelligence (pp. 65-86). Springer, Cham; Schaupp, S. (2022). COVID-19, economic crises and digitalisation: How algorithmic management became an alternative to automation. *New Technology, Work and Employment*; Wang, W., & Siau, K. (2019). Artificial Intelligence, Machine Learning, Automation, Robotics, Future of Work and Future of Humanity: A Review and Research Agenda. *Journal of Database Management*,

different AM tools affect workers and employers. In addition, the level of automation and human involvement involved in decision-making differs, depending on established workplace relationships and industry norms, which are themselves in a period of remodelling during the current period of recovery from COVID-19. In this context, this section outlines the arguments indicating how AM is beneficial and provides opportunities for employers and workers. It is important to highlight that, due to more data being available on the benefits AM brings to employers compared with workers, a large portion of the text focuses on these. Where possible, however, its benefits for workers are also discussed.

The remainder of the section is structured on the basis of the first four dimensions in Table 2-11 above. Relevant subdimensions from the table are also discussed in this section.

2.2.1. Working conditions

Changing working arrangements and the nature of work

One of the main hallmarks of AM is its ability to allow changes to working arrangements so as to increase speed, efficiency and scalability, in ways that are beyond human competences (see Figure 2-1 for some quantitative insights on this).[97] In other words, digital systems can reproduce changes across time and space much more quickly than humans can. When a platform company, such as Uber, makes a change to its operating procedure, it is incorporated simultaneously into a software update. This would not be feasible in the case of managerial changes that have to travel through human chains of command. This set of arguments is committed to the idea that the semi-automation of tasks is optimal for AM delivery, whereby human activities are substituted or reduced. For example, **AM allows for the creation of data-driven visualisations of such things as workflows or scheduling processes, where data 'tells its own story'** – these are seen to be more reliable than human 'cognitive predispositions', and thus, the use of data eliminates human tasks in large dataset analysis,[98] 'freeing us from sorting through multitudes of irrelevant results'.[99] Moreover, **AM tools also reduce and simplify worker accountability activities** (such as informing managers about the number of working hours spent on a particular task, working time each day, availability, sick leave, etc.), which can be automated by using algorithms (e.g. controlling workers' computers or working calendars).[100] This, in turn, is seen to be a method to increase productivity and efficiency in the company.

30(1), 61-79, http://doi.org/10.4018/JDM.2019010104; Mandl, I., Eurofound (2021). The digital age: Implications of automation, digitisation and platforms for work and employment, Challenges and prospects in the EU series, Publications Office of the European Union, Luxembourg, p. 16.

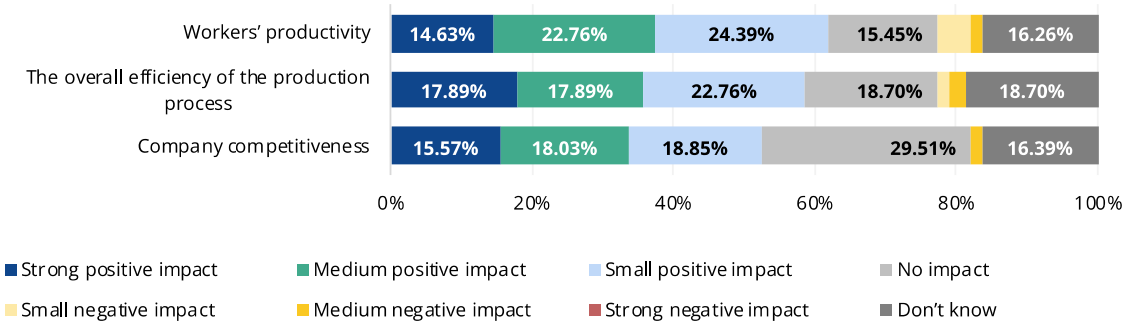
[97] Mayer-Schonberger, V., & Cukier, K. (2016). Big data: A revolution that will transform how we live, work, and think. USA: John Murray Publisher.

[98] Parry, K., Cohen, M., & Bhattacharya, S. (2016). Rise of the machines: A critical consideration of automated leadership decision making in organisations. *Group & Organization Management, 41*(5): 571-594.

[99] Spring, T. (2011). How Google, Facebook and Amazon run the internet, *PC Advisor*, available at: http://www.pcadvisor.co.uk/features/internet/3304956/how- google-facebook-and-amazon-run-the-internet/ (last visited Sept. 28, 2012).

[100] Moore, P.V. (2019). The mirror for (artificial) intelligence: In whose reflection? *Comp. Lab. L. & Pol'y J.,* 41, 47.

Figure 2-1: Effects that AM has had on surveyed organisations, based on the VA employer survey (2023)



Note: N= 123

Source: Authors' own elaboration based on the results of the employers' survey (2023).

In addition, **algorithmic determination of pay can bring some benefits to workers**. For example, AM can analyse the impact of minimum wage policies on employment rates and economic growth.[101] By examining diverse datasets and considering multiple variables, AM can provide a more nuanced understanding of the effects of minimum wages. This challenges the prevailing narrative that higher minimum wages inevitably lead to negative employment outcomes. Such insights can inform evidence-based policymaking and help to shape more effective labour rights regulations. By harnessing the potential of AM, workers and their advocates can leverage data, analytics, and predictive models to strengthen labour rights, protect workers' interests, and foster fair and equitable work environments.

Working time and communication

AM can directly empower workers through innovative services and technologies[102] by simplifying workload and enabling better quantification of work. For example, platforms such as WeClock allow employees to monitor their time and actions, helping them to track aspects such as skipped breaks, unpaid labour, long commutes, burnout, and unrealistic expectations.[103] This information empowers workers to take action and improve their overall well-being. AM can also facilitate employee self-assessment and skills development. By analysing employee data, AM algorithms can identify skills gaps and recommend training programmes or educational opportunities. Such personalised approaches to skills development enable workers to adapt to changing job requirements and enhance their employability.

Workers can benefit from AM-created support mechanisms, which reduce and simplify workload, allowing them to focus on more value-added activities. AM provides workers with more flexible schedules.[104] It can also simplify schedules, which can result in workload simplification, and may result in higher rates of productivity. This process is called 'algorithmic matching', and it occurs, for example, by matching a worker with a customer via dynamic pricing, where behavioural data also allows for personal feedback

[101] Cengiz, D., Dube, A., Lindner, A., & Zentler-Munro, D. (2022). Seeing beyond the trees: Using machine learning to estimate the impact of minimum wages on labor market outcomes. *Journal of Labor Economics*, 40(S1), S203-S247.

[102] Stacey, S. (2023). Trade unions are fighting to make AI work for the workers. *Tech Monitor*, 5 May.

[103] WeClock (2023). Give Work a Reality Check. https://weclock.it/

[104] Möhlmann, M., Zalmanson, L., Hendridsson, O., & Gregory, R.W. (2021). Algorithmic management of work on online labour platforms: when matching meets control. *MIS Q* 45(4): 1999–2022. https://doi.org/10.25300/misq/2021/15333

and productivity ratings.[105] In our survey of employers, respondents indicate an uptick in worker productivity (see Figure 2-1). According to the survey, over half of participating companies highlighted that the use of AM had produced a strong, medium, or small positive effect on working time, as it relates to worker productivity (61.8 %); to the overall efficiency of production processes (58.5 %); and to the organisation's competitiveness (52.5 %).[106] In addition, around 5 % of companies said that AM had a small or medium negative effect on these aspects, while no respondents said that it had a strong negative one. However, based on the survey of workers carried out for this study, a much smaller percentage of workers claimed that the usage of AM had a positive effect on their productivity. Around 35 % of the 970 workers surveyed said that AM had a positive impact on their productivity, while 22.2 % said it harmed it.

As part of recruitment activities, human resource and other executive suite managers historically sifted through thousands of job applications and CVs to identify suitable candidates for shortlisting. **With the help of AM, management can identify specific areas of expertise and other qualities as desired characteristics, and programmers can design algorithms that can scan through the stack of applications** (now, usually Word documents or online forms), **to identify such characteristics**. The use of AM in recruitment can therefore reduce HR managers' working time considerably.

In line with this, 8.9 % of employers surveyed said that AM had a strongly positive effect on their hiring/recruitment practices, while 12.2 % and 20.3 %, respectively, said it had a medium or a small positive impact. Of great significance here is that algorithmically aided data collection and decision-making can be accomplished in a fraction of the time that managers would traditionally spend gathering information about workers.[107] However, accuracy and effectiveness are not necessarily higher than was previously achieved without the help of such digital aids, and the use of algorithms may result in false positives and negatives, which may lead to unfair results. These potential outcomes are discussed in further detail in Section 2.3 Challenges posed by AM for workers and employers. Nonetheless, algorithms are seen to reduce human involvement in tasks relating to candidate sorting and selection, resulting in time savings for managers.

Beyond recruitment, **technology may also help managers to allocate tasks and work,**[108] including when it comes to managing large and widely dispersed groups within a specific workforce, such as taxi drivers. Such work allocation may previously have been highly tedious for switchboard operators, who would attempt to match a driver with a passenger. Now, however, AM can act as the intermediary, using geolocation to determine which driver(s) are in the vicinity of the location of a passenger. This enables a match to be made automatically, and the driver(s) notified can opt to pick up a passenger, accordingly.

Lastly, **AM allows improved communication through the use of automatic communication tools such as chatbots.** These chatbots are usually developed as computer programs and can simulate human conversation using voice or text to communicate with users, helping them to solve emerging issues, including answering their questions or providing recommendations. In relation to this, the results of the employer and worker surveys are in agreement, with **around 51.6 % of employers and 40.74 % of workers stating that AM had a positive impact on communication quality** (10.66 % of employers and 10.55 % of workers said it had a strong effect; 16.39 % and 14.68 %,

[105] Benlian, A., Wiener, M., Cram, W.A., Krasnova, H., Maedche, A., Möhlmann, M., Recker, J., & Remus, U. (2022). Algorithmic management: bright and dark sides, practical implications and research opportunities. *Bus. Inf. Syst. Eng.* 64(6): 825–839.

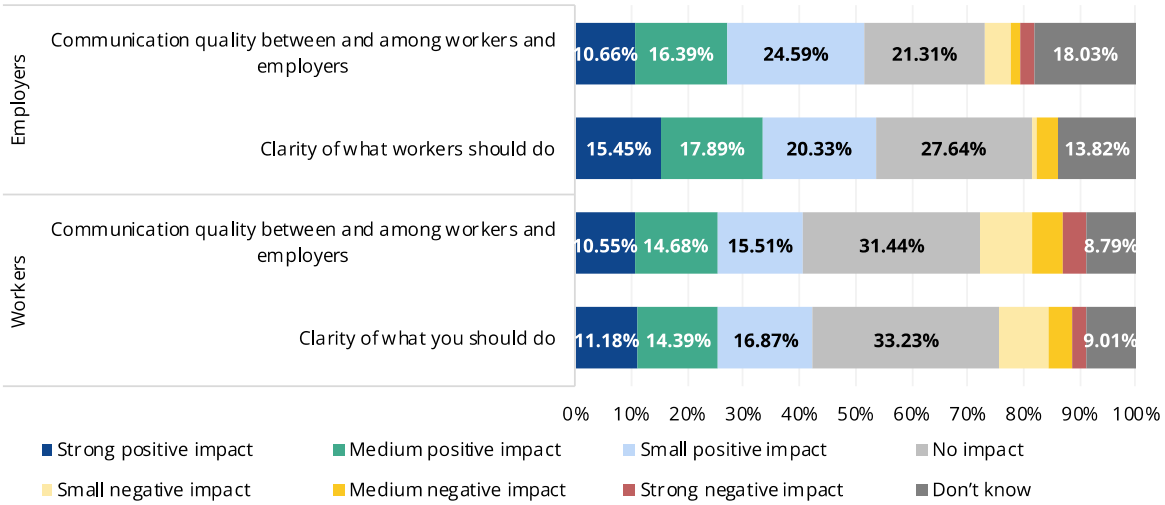
[106] The analysis was based on a sample of 122 companies.

[107] Carey, D., & Smith, M. (2016). How companies are using simulations, competitions, and analytics to hire. *Harvard Business Review.* (22 April 2016), https://hbr.org/2016/04/how-companies-are-using-simulations-competitions-and-analytics-to-hirehttps://hbr.org/2016/04/how-companies-are-using-simulations-competitions-and-analytics-to-hire.

[108] Brynjolfsoon, E., & McAfee, A. (2014). *The second machine age: Work, progress and prosperity in a time of brilliant technologies*, W.W. Norton; Duan,Y., Edwards, J.S., & Dwivedi, Y.K. (2019). Artificial intelligence for decision-making in the era of big data: evolution, challenges and research agenda. *International Journal of Information Management*, 48(3): 63-71.

respectively, said it had a medium effect; and 24.59 % and 15.51 % said it had a small effect). Meanwhile, 21.3 % of employers and 31.4 % of workers reported that it had no impact (see Figure 2-2 below for more details). It is important, however, to note that these results do not necessarily indicate an increasing usage of chatbots. Based on these data alone, exactly what kinds of algorithms improved communication cannot be accurately assessed.

Figure 2-2: Effects of AM on surveyed organisations and workers, based on the VA employer and worker surveys (2023)



Note: employers, N = 123; workers, N = 966.

Source: Authors' own elaboration based on the VA employer and workers surveys (2023).

Occupational safety and health (OSH)

AM shows some potential to improve OSH in ways that benefit workers. As highlighted by Reinhold et al. (2021),[109] AM systems can analyse large amounts of data, often in real-time, and provide **early alerts and warnings of OSH hazards, as well as analyse the efficiency of different OSH-related solutions and propose possible interventions that can improve OSH** and more. AM systems can also help workers to identify potential safety hazards and make real-time interventions to prevent accidents. Take, for instance, the ability of algorithmic software to analyse video feeds from security cameras and detect unusual activity. As the Centre for Disease Control and Prevention explains, advanced sensor technologies, which are often used to collect data inputs for workplace algorithms, can continually monitor the 'ambient work environment for hazardous substances or a worker's proximity to known hazards'.[110] Wearable technologies are also used to monitor workers' OSH and the space around them, with notifications providing up-to-date alerts that can prevent harm.[111] Some AM tools can be used to identify ergonomic risks by, for example, indicating to workers their posture is inappropriate, or recommending

[109] Reinhold K., Jarvis M., Christenko A., Jankauskaitė V., & Paliokaitė A. (2021). New forms of worker management through AI-based systems: An assessment of the implications for OSH. See: https://osha.europa.eu/en/publications/artificial-intelligence-worker-management-overview

[110] See: https://blogs.cdc.gov/niosh-science-blog/2022/09/21/algorithms-fow/

[111] Spyridopoulos, K., Kilic, L., Battaglina, M., Olausson, N., Regazzoni, P., Broughton, A., & Toro, D. (2022). Smart digital monitoring systems for occupational safety and health: workplace resources for design, implementation and use. European Agency for Safety and Health at Work, available at: https://osha.europa.eu/en/publications/smart-digital-monitoring-systems-occupational-safety-and-health-workplace-resources-design-implementation-and-use

that they take a break if they have been performing repetitive movements for a long period of time.[112]

AM can also be used to detect whether or not a worker is wearing the right protective gear, thus reducing the risk of accidents and health problems. For example, AM can detect if a worker is working at a designated height without taking adequate safety precautions (e.g. harness equipment), and warn them about this, as well as send an alarm to the control centre. As stated by the experts interviewed for the mentioned study,[113] AM can also be used to design and conduct safety training for workers by, for example, collecting and using data from the workplace to develop the best health and safety strategies. In addition, AI-based systems can assign individual risk-related profiles to workers based on the surveillance of possible health risks, a worker's current risk level, and the likelihood of future health risks by, for instance, analysing and identifying which workers are more sensitive and susceptible to specific hazards, such as noise, high/low temperatures, and similar.

Algorithms are also capable of conducting predictive analytics that can provide other safety-related benefits for workers. One example is **algorithmic sensors, which can be considered a form of algorithmic monitoring, can not only be used to scan for exposure to harmful chemicals, but can also monitor equipment and use algorithms to predict when maintenance is needed**. This reduces the risk of workers using equipment that will malfunction and result in an accident. In addition, workers with maintenance responsibilities can plan their work more efficiently and avoid unexpected downtime. Moreover, predictive analytics can help to prevent accidents outright. It has been reported that 'The Shell company now uses predictive data analytics to assess potential workplace injuries on oil rigs worldwide... Shell's computers can weigh measurable factors, such as worker engagement in safety procedures, and predict when and where [physical] injuries are likely to occur.[114]

Beyond helping with physical health, AM can also be used to assess the general well-being of workers and identify burnout, anxiety, stress, etc. For example, some AM tools that monitor workers' emails or even their tone of voice can be used to assess whether a worker is in a state of emotional distress and assess the likelihood that the worker will suffer from burnout.[115,116] In turn, such tools could also flag this issue to managers, ideally prompting a response via a notification for action. According to the surveys carried out for this study, many employers – and some workers – also believe that AM can help with safety, physical well-being, anxiety, and mental health (see Figure 2-3 below). However, employers predominantly note the positive and/or neutral aspects of such tools, while workers are more sceptical, as is discussed in Figure 2-3 below. In addition, the results of the two surveys also clearly indicate that many employers and workers are unaware of how AM can affect them, due to a high number of 'Don't know' responses, which may be explained by the newness of such tools for many of them.

[112] Reinhold, K., Järvis, M. Christenko, A, Jankauskaitė, V., Paliokaitė, A., & Riedmann, R. (2022). Artificial intelligence for worker management: implications for occupational safety and health. Prepared for EU-OSHA.

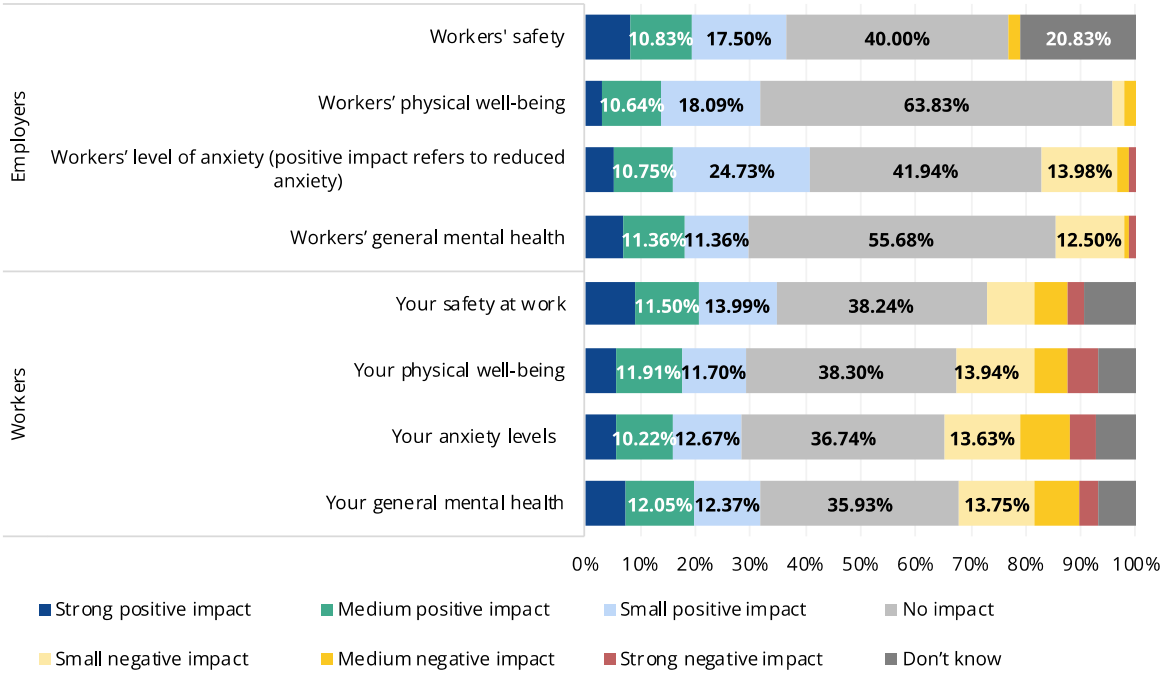
[113] Reinhold K., Jarvis M., Christenko A., Jankauskaitė V., & Paliokaitė A. (2021). New forms of worker management through AI-based systems: An assessment of the implications for OSH. See: https://osha.europa.eu/en/publications/artificial-intelligence-worker-management-overview

[114] See: https://safetymanagement.eku.edu/blog/using-predictive-analytics-to-predict-and-prevent-workplace-injuries/

[115] Cameron, G., Cameron, D., Megaw, G., Bond, R., Mulvenna, M., O'Neill, S., Armour, C., & McTear, M. (2017). Towards a chatbot for digital counselling. In: L. Hall, T. Flint, S. O'Hara, & P. Turner (Eds), *Proceedings of the 31st International BCS Human Computer Interaction Conference* (HCI 2017) (pp. 1-7), BCS Learning and Development Ltd.

[116] Grządzielewska, M. (2021). Using machine learning in burnout prediction: A survey. *Child and Adolescent Social Work Journal*, 38(2), 175-180.

Figure 2-3: Effects of AM on different aspects of work, based on the VA employers and workers survey (2023)



Note: employers, N = 123; workers, N = 966.

Source: Authors' own elaboration based on the VA employer and workers surveys.

2.2.2. Data protection and privacy

Data collection and storage

Data trusts have emerged as a potential solution to address the challenges associated with data portability and privacy,[117] which are similar concerns to those posed by AM more generally. Thus, a data trust can be seen as means to create better conditions for workers in the environment of AM. A data trust is a legal framework that allows individuals or organisations to pool their data for a specific purpose, while safeguarding privacy and ensuring the ethical use of the data.[118] Data trusts act as independent intermediaries, holding and managing data on behalf of the individuals or organisations involved. By establishing trust between data generators and data users, data trust can foster collaboration, innovation, and social benefits. They function as a means by which workers can retain control over their data and use such data in collective bargaining agreements and negotiations. Limited portability rights are provided under Article 20 of the GDPR, which are useful for supporting workers' ownership of the data within a data trust.

Transparency

AM can allow workers to better conceptualise and visualise the metrics and numeration that surround their work activities, outputs, and delivery, provided that they: 1) are given access to the data stored, and training to understand how the technology works; and 2) enjoy dialogue with management regarding what will be done with their data. One British Academy-funded study carried out by Moore (2019)[119] demonstrates a case of

[117] Delacroix, S., & Lawrence, N.D. (2019). Bottom-up data Trusts: disturbing the 'one size fits all' approach to data governance, *International Data Privacy Law*, 9(4), Nov., 236–252, https://doi.org/10.1093/idpl/ipz014

[118] NESTA (n.d.). Data trusts: A new tool for data governance (ElementAI), available at: https://hello.elementai.com/rs/024-OAQ-547/images/Data_Trusts_EN_201914.pdf

[119] Moore, P. (2019). *The Quantified Self in Precarity: Work, Technology and What Counts,* London: Routledge.

transparency and AM and identifies the ways in which this was experienced by workers and managers. As part of the Quantified Workplace study (as the organisation called it), workers were asked to wear FitBit armbands tracking their sleep, steps, and heart rate over the period of one year. All data collected from all workers' bodies were made available on a shared dashboard, rendering all data transparent. At the start of this study, workers were keen to share data transparently, not only with management, but also with their co-workers, in what they saw as a gamified environment. While this was seen as a positive advance in a progressive workplace, by the end of the study, workers became less positive, a fact that was demonstrated in one-to-one interviews with Moore for the study. Transparency of activities and the concomitant technologies fuelling AM can lead to significant benefits and opportunities, but it depends on who is being transparent, and on the surrounding employment relationship.

Another study by Schildt (2017) discusses 'computer augmented transparency', emphasising its benefits, whereby AM should allow 'a single manager to monitor and mentor a greater number of subordinates'. [120] Such arrangements are projected to benefit workers, who will be able to carry out problem-solving without their managers' involvement, which should lead to a renaissance in self-organised work. At times, however, algorithms can be seen to be inexplicable, so even where transparency is promulgated, there is no guarantee that bosses and workers will fully understand the technology. Gaudio (2022) nevertheless expresses optimism regarding tactics to 'foster transparency'[121] and make AM as democratically deployed and as beneficial as possible, where AM aids bosses. Where bosses are 'scrupulously avoiding the appearance of managerial control',[122] better transparency should mitigate issues of low understanding or lack of trust in the employment relationship.

Automated decision making

Automated decision-making, based on data collected through the use of algorithms and recognised by a company, may benefit workers' career progression. AM can help managers to identify skills gaps at both individual and wider company level and can provide recommendations regarding how to fill them.[123] This can be a double-edged sword, however: when AM exposes such gaps to the employer, the company may simply fire or demote the worker. The large volume of data collection and analysis that is intrinsic to the functioning of AM systems allows trends or associations to be identified that might be missed by human observers. This distinctive capability of algorithms could be extremely useful for understanding workers' needs, including training and upskilling benefits. Indeed, algorithms are already demonstrating the capacity to provide workers with personalised development plans in the field of education. This is supported by the workers' and employers' surveys carried out for this study as, based on the analysis provided previously, around 51.27 % of workers believe AM is used in their place of work for talent management, while 33.15 % of employers surveyed say that they do, in fact, use AM for such an activity. In addition, **during a worker's employment lifecycle, algorithms can be used to spot hidden talents** based on their productivity, efficiency, assessments of their skill level, etc. However, using AM for talent management could also result in challenges, as managers could use this information against workers, i.e. where a worker's performance is not up to a certain level, penalties might be applied. Hence, skills-gap or other performance management using AM should, like all technological advances in workplaces,

[120] Schildt, H. (2017). Big data and organisational design – the brave new world of algorithmic management and computer augmented transparency. *Organization and Management 19*(1): 23 – 30.

[121] Gaudio, G. (2022). Algorithmic bosses can't lie! How to foster transparency and limit abuses of the new algorithmic managers. *Comparative Labor Law & Policy Journal 42*(3): 707–741.

[122] Adams-Prassl, J. (2019). What if your boss was an algorithm? Economic incentives, legal challenges and the rise of artificial intelligence at work *Comparative Labor Law & Policy Journal,* 41, 123, 131-132.

[123] McKenney, M.J., & Handley, H.A. (2019). Identifying and quantifying personnel skill gaps. In: *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 63, no. 1, pp. 332-336, Los Angeles, CA: Sage Publications.

not be regarded purely as an opportunity for workers.[124] AM does not exist as an entirely automated activity, and it is usually behaviour of management in response to these tools and applications which gives it a qualitative dimension that can be judged an opportunity, benefit, challenge, or other impact.

Some AM tools might also be used to help workers directly by using data to nudge or guide them. For example, in some companies, such as Amazon, ultrasound wristbands are used that vibrate and show information on a small display, indicating where an employer can find a particular item in a warehouse.[125] This, in turn, might help workers to perform their tasks more efficiently. In addition, guidance can also go beyond direct recommendations. For example, in platform work, delivery or taxi drivers are often nudged to operate in specific high-demand locations using higher pay.[126] In regular jobs, nudging can include sending automated reminders regarding deadlines, prompts for additional training and development, as well as ensuring safety at work, which is discussed below. Based on the VA survey, it was found that around 47.59 % of workers surveyed believe that their company uses nudging, while 27.07 % acknowledge doing so.

Algorithms can be used to make decisions about workers, based on the use of data to identify and predict work disengagement and likely exit.[127] For example, by tracking workers' performance (including their productivity, personal growth indicators, and activity), algorithms can identify worker disengagement or 'quiet quitting',[128,129] as well as which workers are likely to stay the longest.[130] There are several products designed to identify patterns and trends in worker disengagement using data on voluntary turnover rates, absences, satisfaction (based on such things as surveys), as well as performance data based on metrics that determine the quality and quantity of an employee's work, their work efficiency, and organisational performance. In connection with this, algorithms may also predict potential worker burnout by tracking workers' moods (including stress levels).[131] In this way, AM tools allow companies to prepare, and take necessary action in advance to mitigate possible risks (e.g. avoiding worker burnout by reducing workload).

Related to this, **another purported benefit of AM is that it reproduces clear performance expectations in the workplace**. For example, algorithms can track worker performance on a more accurate and objective basis. They can also quickly generate

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[124] European Agency for Safety and Health at Work, Cognitive Automation: Implications for Occupational Safety and Health, European Agency for Safety and Health at Work, June 2019, https://osha.europa.eu/en/publications/cognitive-automation-implications-occupational-safety-and-health

[125] Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek, E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work, available at: https://osha.europa.eu/en/publications/artificial-intelligence-worker-management-overview

[126] Mateescu, A., & Nguyen, A. (2019). Algorithmic management in the workplace. *Data & Society,* 1-15.

[127] Moore, P.V. (2019). OSH and the Future of work: Benefits and risks of artificial intelligence tools in workplaces. EU-OSHA Discussion paper 05/07/2019. https://osha.europa.eu/en/publications/osh-and-future-work-benefits-and-risks-artificial-intelligence-tools-workplaces; Susskind, R.E., Susskind, D. (2015). *The future of the professions: How technology will transform the work of human experts.* USA: Oxford University Press.

[128] Silverman, R.E., & Waller, N. (2015). The algorithm that tells the boss who might quit, *Wall Street Journal*, https://www.wsj.com/articles/the-algorithm-that-tells-the-boss-who-might-quit-1426287935; Liu, J. (2019). This algorithm can predict when workers are about to quit – here's how. CNBC 'makeit' (10 September 2019). https://www.cnbc.com/2019/09/10/this-algorithm-can-predict-when-workers-are-about-to-quitheres-how.html; Eliot, L. (2022). AI ethics disquieted by AI getting dragged into quiet quitting mania, *Forbes*, 21 September, 2022. https://www.forbes.com/sites/lanceeliot/2022/09/21/ai-ethics-disquieted-by-ai-getting-dragged-into-quiet-quitting-mania/; Mentoring Complete (2022). Quiet quitting: causes and techniques to prevent it. https://www.mentoringcomplete.com/quiet-quitting-causes-and-techniques-to-prevent-it/

[129] Moore, P.V. (2019). OSH and the Future of work: Benefits and risks of artificial intelligence tools in workplaces. EU-OSHA Discussion paper 05/07/2019. https://osha.europa.eu/en/publications/osh-and-future-work-benefits-and-risks-artificial-intelligence-tools-workplaces; Susskind, R.E., Susskind, D. (2015). *The future of the professions: How technology will transform the work of human experts* (USA: Oxford University Press).

[130] Peck, D. (2013). They're watching you at work. *Atlantic* (Dec. 2013). http://www.theatlantic.com/magazine/archive/2013/12/theyre-watching-you-at-work/354681/.

[131] Todolí-Signes, A. (2019). Algorithms, artificial intelligence and automated decisions concerning workers and the risks of discrimination: the necessary collective governance of data protection. *Transfer: European Review of Labour and Research*, 25(4), 465-481.

evaluative outputs that are useful to workers themselves.[132] In other words, algorithmic management systems can provide workers with clear expectations and feedback about their performance, in the form of easily digestible, quantitative metrics. However, AM must be used appropriately in order for this benefit to be realised. But if this condition is met, such tools might help workers to better understand their own performance and identify key areas for improvement.

To foster such opportunities and ensure they are delivered and practised, it is essential to establish clear regulations and policies that protect workers' rights with regard to data ownership and portability, and ensure they are not left behind in the digital transformation. This includes safeguards against unfair algorithmic decision-making, ensuring access to workplace training and upskilling opportunities, as well as addressing issues such as those found in precarious work arrangements, in which a great deal of AM occurs. By prioritising workers' rights, societies can strive for a future in which the benefits of technology are shared equitably, and human dignity is upheld in the face of rapid technological advancements in AM.

2.2.3. Fairness and discrimination

Bias and objectivity

In some cases, AM can help to eliminate bias and subjectivity in management decision-making.[133] The European Agency for Fundamental Rights reports that data-supported decision-making can result in technologies 'learning to discriminate'. However, there are ways around this. Technologies can be trained to detect and avoid discriminatory behaviour.[134] This would involve specific actions and dedication to this objective on the part of unions, work councils and other representative bodies for workers, as well as the commitment of employers and government to ensuring that AM is operationalised so as to reduce discrimination. Experts repeatedly suggest the involvement of workers in making decisions about the implementation of AM at work as a way to reduce bias and discrimination. Transparent reporting of pay and discrimination in relation to this, such as is evident in the persistent gender pay gap across the EU, should involve information about how algorithms and any other technology are used to make pay decisions.[135] These types of commitments would help to better secure opportunities for ensuring bias and discrimination are not part of AM processes.

Algorithms can lead to fairer outcomes for workers, provided they are not exclusively subject to what might appear to be arbitrary decisions of human managers. Algorithms, unlike humans, make decisions explicitly based on their programming and the data they are fed. Using AM, influences such as personal opinions or emotions will ideally have less influence on decision-making processes. However, as will be discussed later, this is not always true: biased data can lead to a seemingly biased algorithm, because of the biased output/decisions that result from them. Nevertheless, if care is taken, algorithms can be less biased than human actors.[136] Where and when there is a specific intention to do so, bias can be identified and eliminated using such processes (rather than replicated and perpetuated). However, in the survey of employers carried out for the present study, 67 %

[132] Duggan, J., Sherman, U., Carbery, R., & McDonnell, A. Algorithmic management and app-work in the gig economy: A research agenda for employment relations and HRM, *Human Resource Management Journal*, 30, no. 1 (2020): 114-132.

[133] Parikh, N. Understanding Bias in AI-Enabled Hiring, Forbes Human Resources Council, 14 October 2021. https://www.forbes.com/sites/forbeshumanresourcescouncil/2021/10/14/understanding-bias-in-ai-enabled-hiring/.

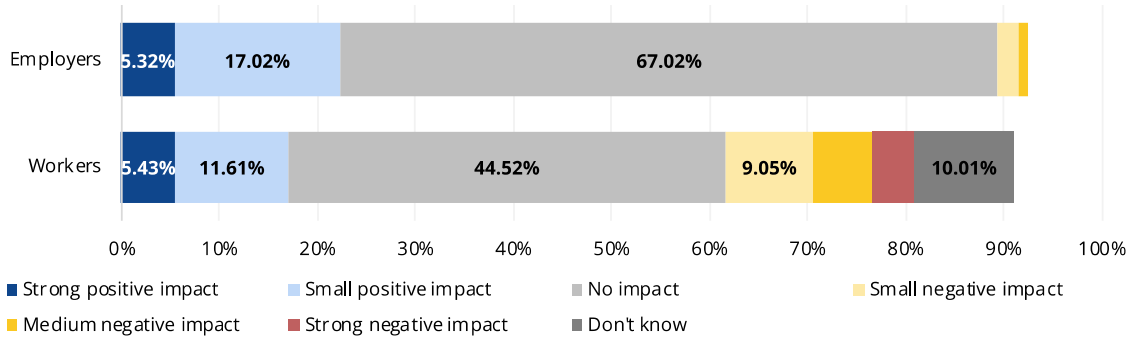
[134] European Union Agency for Fundamental Rights (2017). #Bigdata: Discrimination in data-supported decision making, FRA Focus, available at: https://fra.europa.eu/sites/default/files/fra_uploads/fra-2018-focus-big-data_en.pdf

[135] Brione, P. (2020). An ethical look at algorithmic management in the workplace, ACAS: Working for Everyone, available at: https://www.ipa-involve.com/Handlers/Download.ashx?IDMF=7129b512-2368-459a-898d-6d2b3457a039

[136] University of Montreal AI can make better clinical decisions than humans: Study. *ScienceDaily*, 10 September 2021, available at: https://www.sciencedaily.com/releases/2021/09/210910121712.htm; Bogert, E., Schecter, A. & Watson, R.T. (2021). Humans rely more on algorithms than social influence as a task becomes more difficult. *Sci. Rep.* 11, 8028, available at: https://doi.org/10.1038/s41598-021-87480-9

of organisations surveyed stated that AM has had no impact on discrimination, while 44.52 % of respondents to the workers' survey agree (see Figure 2-4 below). In addition, while employers see such tools as reducing discrimination, workers' attitudes are less clear-cut. Hence, based on the literature and interviews, it appears that while AM offers the potential to reduce discrimination, workers do not regard it as a tool that affects workplace discrimination to any great extent.

Figure 2-4: Effect that AM has had on the level of discrimination in the respondent's workplace, based on the employer and worker surveys (2023)



Note: employers N = 123; workers N = 966

Source: Authors' own elaboration based on the VA surveys of employers and workers.

It is important to highlight that the legality of algorithmic decision-making is also an essential aspect of fairness and discrimination. However, as this and many other related aspects are discussed in depth in Chapter 3. Legal and policy framework, such a discussion is omitted here.

2.2.4. Workers' collective rights

Collaboration rights

Securing workers' rights has been a struggle across industries throughout history. One route to ensuring better collective rights for workers is to include workers in the design of the same processes that will ultimately be applied to manage them, such as those used to implement AM. This requires both the existence of material collective rights, and their proper application in the context of AM.

Collaboration in the design and deployment of AM systems could be achieved by intentionally involving technically trained workers' representatives from trade unions or works councils, who should be present in negotiations any time a new technology is introduced into the workplace.[137] In particular, technically trained workers' representatives should be involved when the technologies concerned will have an impact on workers' pace of work or working time, as well as wider processes utilised for worker management.[138] Permitting workers a collaborative role in this process is very likely to create a better working environment for the deployment of AM.

Co-determination and co-design: co-determination is a critical tool for defending workers' rights, whereby having labour representatives on supervisory boards is a key ingredient ensuring a better balance of interests with investors, who tend to defend short-term

[137] Moore, P. (2020). 'First Principles and Policy Options', Data subjects, digital surveillance, AI and the future of work. Panel for the Future of Science and Technology. EPRS: STOA, European Parliament, p. 84, available at: https://www.europarl.europa.eu/RegData/etudes/STUD/2020/656305/EPRS_STU(2020)656305_EN.pdf

[138] Moore, P. (2020). 'First Principles and Policy Options', Data subjects, digital surveillance, AI and the future of work. Panel for the Future of Science and Technology. EPRS: STOA, European Parliament, p. 84, available at: https://www.europarl.europa.eu/RegData/etudes/STUD/2020/656305/EPRS_STU(2020)656305_EN.pdf

objectives rather than workers' rights.[139] Workers in most EU Member States enjoy a form of co-determination rights, including having workers' representatives on supervisory boards. Those who do not enjoy such rights include workers in Belgium, Bulgaria, Cyprus, Estonia, Italy, Latvia, Lithuania, and Romania.[140] Successful co-determination practices should be considered best practices, and companies should either adopt similar processes or implement the resulting policies uniformly across all of their branches. For example, if AM via time tracking is not allowed in the German branch of Amazon due to concerns about worker stress and psychosocial issues being raised by the works council, this type of time tracking should not be permitted in any other branch around the world.[141]

Workers' collective rights

AM is relatively new element of work management and decision-making, but it is not too early to begin assessing the likely impact it has on the exercise of labour rights across the EU, via existing tools for social dialogue and collective bargaining, and in some countries, collective determination (co-determination). This subsection looks at the potential opportunities AM provides for workers to exercise relevant labour rights via collective agreements and social dialogue between trade unions and workers' representatives, and employers.

Protecting and enhancing worker rights: machine learning algorithms can be designed and deployed in such a way as to protect and enhance worker rights.[142] Algorithms can analyse various data sources, including supply chains, industry practices and worker testimonials, to detect and prevent violations. This enables proactive interventions, the enforcement of labour laws, and the improvement of working conditions.[143] In addition, AM can help to uncover structural factors that weaken labour rights.[144] By analysing large datasets and identifying patterns, algorithms can reveal systemic issues such as precarious employment, wage theft and labour exploitation. These insights can inform policy decisions aimed at strengthening labour rights, challenging established economic orthodoxy, and promoting social justice.

Predictive analytics for labour shortages and skills gaps: AM is often discussed only as a feature of the employment relationship, but such technological processes could also be redeployed to, for example, leverage predictive analytics to anticipate and address skills and labour shortages. Such applications of AM could be instrumental to securing collective rights for workers by providing support to bargaining with analytics that demonstrates workers' rights to upskilling, fair recruitment processes, support for job transitions and the like. By analysing data about workforce demographics, industry trends and economic forecasts, AM can provide insights for employers and workers (and even policymakers) such as identifying key skills shortages. This allows proactive measures to be taken, such as implementing retraining programs, encouraging targeted recruitment efforts, or facilitating job transitions. Predictive analytics can also help to identify potential mismatches between supply and demand in the labour market, enabling early intervention to mitigate adverse impacts on workers.

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[139] Back, N., Dury, P., & Hoffman R. (2022). Co-determination at issue in Europe's top court. *Social Europe* 11 April 23, available at https://www.socialeurope.eu/co-determination-at-issue-in-europes-top-court

[140] *The Economist* (2020, 1 February). Business. Most of the world has yet to embrace co-determination, available at: https://www.economist.com/business/2020/02/01/most-of-the-world-has-yet-to-embrace-co-determination

[141] Moore, P. (2020). 'First Principles and Policy Options', Data subjects, digital surveillance, AI and the future of work. Panel for the Future of Science and Technology. EPRS: STOA, European Parliament, p. 87, available at: https://www.europarl.europa.eu/RegData/etudes/STUD/2020/656305/EPRS_STU(2020)656305_EN.pdf

[142] Mazumder, A., Ostmann, F., & Briers, M. (2021). *AI for Human Rights*. The Alan Turing Institute, available at: https://www.turing.ac.uk/ai-human-rights

[143] Gow, G. (2022). The Argument For An AI Augmented Workforce, *Forbes*, 30 October

[144] Lavelle-Hill, R., Smith, G., Mazumder, A., Landman, T., & Goulding, J. (2021). Machine learning methods for 'wicked' problems: exploring the complex drivers of modern slavery. *Humanities and Social Sciences Communications*, 8(1), 1-11.

Social dialogue

Strengthening collective bargaining: AM can enhance workers' ability to collectively bargain across a range of different national contexts. Traditionally, collective bargaining has relied on the availability of reliable and up-to-date information. AM can provide data and analytics that facilitate negotiations, empowering workers with valuable insights into market trends, company performance and comparative wage data. These insights enable workers to better negotiate working conditions and fairer wages, ultimately strengthening their collective bargaining power.[145] In addition to providing data, AM can also facilitate the coordination and communication necessary for effective collective bargaining. AM-powered platforms can connect workers, unions, and other stakeholders, enabling efficient and widespread collaboration. This fosters solidarity and enables workers to present a unified front in negotiations, increasing their leverage and improving outcomes.

Collective bargaining agreements allow the specific dimensions of the employment relationship to be determined between employers and workers' representative groups. Typical collective agreements cover the setting and release of wages and overtime; how/whether individual and/or collective dismissals proceed; promotion criteria and processes; working time; the pace of work; and what kinds of contracts are deemed acceptable. Hence, collective agreements and social dialogue should be sought to provide opportunities for workers to seek agreements in relation to the working conditions that surround AM. For example, AM might have an impact on the way in which wages are decided – which, at times, might rely on how client relations are determined. AM could also have an impact on how targets are set with clients.[146] Bargaining over wages and targets represent standard themes in trade union and works council discussions, but the ways in which algorithms work differ from other human resource discussions of this nature, making bargaining less straightforward. For example, algorithmic determination might lead to unreasonable targets being set. Therefore, new collective bargaining discussions and potential agreements should consider the nature of such algorithms – i.e. that they are expected to make better decisions than humans and may even be used to substitute decisions by human managers.

Related issues that will need to be negotiated, and preferably agreed, through collective bargaining with trade unions and works councils include the ways in which data are gathered, processed, and used. Furthermore, negotiation and bargaining should occur in relation to technical manner in which monitoring occurs, including discussions about what quantity of data are collected, from whom, when, and for what purpose, as well as how much monitoring is reasonable. Collective agreements represent an opportunity and can be used to enhance the transparency in AM, as such environments allow workers' representatives to agree on what data are collected and how they are used by the algorithm for worker management. If collective agreements are established, workers' representative groups will have access to the data collected about workers. They will also know precisely who will take responsibility for sharing data with workers (which will ideally be carried out regularly by company data protection officers), as well as what such data will be used for. Workers' representatives will also be in a position to agree or disagree to any changes in data collection and its use for AM. In this way, transparency can be achieved through collective agreements. However, one big issue with this is that AM is often employed in workplaces with limited collective bargaining.

Fostering social dialogue and participation: AM can foster new opportunities for social dialogue and participation, both among workers and between employees and employers. By leveraging AM-based platforms, employees can engage in discussions and make collective decisions on issues relating to the ethical use of technology and data in their

[145]Voss, E., & Bertossa, D. (2022). Collective Bargaining and Digitalization: A Global Survey of Union Use of Collective Bargaining to Increase Worker Control over Digitalization, *New England Journal of Public Policy,* 34(1).

[146] Aloisi, A., & Gramano, E. (2019). Artificial intelligence is watching you at work: digital surveillance, employee monitoring, and regulatory issues in the EU Context, *Comparative Labor Law & Policy Journal, 41* (1), 95-121

workplace.[147] These platforms can facilitate real-time communication, enabling workers to voice their concerns, propose solutions, and engage in meaningful dialogue with employers. This form of social dialogue can lead to the establishment of guidelines and policies for the ethical use of AM.[148] By involving workers in decision-making processes, AM can be harnessed in ways that align with labour rights and values. This participatory approach ensures that the use of AM is transparent, accountable, and respectful of workers' interests. Moreover, AM can enable workers to contribute to the design and implementation of new technology-based rights.[149] For instance, AM-powered platforms can facilitate the creation of worker-driven algorithms that ensure fair scheduling, prevent discrimination, and promote work-life balance. By actively involving workers in shaping these technologies, AM becomes a tool for enhancing labour rights and improving working conditions.[150]

Empowering workers: AM can also improve individual negotiations between employees and employers. By providing accurate and relevant information, AM algorithms can assist employees in advocating for their rights and interests. For example, AM can determine reasonable notice periods for termination, based on various factors such as industry standards, job roles and employee qualifications.[151] This ensures that employees receive fair treatment during the termination process, reducing uncertainty and potential exploitation. Furthermore, AM can assist employees in understanding their rights and entitlements, particularly in legal contexts. AM-powered chatbots or virtual assistants can provide employees with easily accessible and accurate information regarding employment laws, contractual obligations and company policies. This empowers workers to make informed decisions and to assert their rights effectively.

2.3 Challenges posed by AM for workers and employers

Alongside the opportunities outlined above, **number of drawbacks are associated with AM, for both workers and employers**. For employers, the identifiable drawbacks associated with AM include high investment costs, over-reliance on technology, as well as complexity in understanding the system. Moreover, AM can result in challenges for workers through harm to working conditions, workplace relations, a reduction in social interactions, and limited contextual understanding in decision-making. In addition, this limited understanding of contextual factors – and the fact that unions cannot properly protect workers due to the opacity of AM tools – leads to AM having the potential to exacerbate these challenges. In addition, smaller and medium-sized businesses may experience challenges differently from larger ones, and may suffer more immediately, due to having fewer resources, if, for example, AM leads to staff leaving the company. Such risks are exacerbated by the fact that some data protection and privacy laws apply predominantly to larger companies,[152] meaning that workers in smaller companies may be more vulnerable when AM is applied.

With this context in mind, the remainder of this section explores the challenges faced by employers and employees in the workplace. It is structured the same way as the section on

[147] Molina, O., Butollo, F., Makó, C., Godino, A., Holtgrewe, U., Illsoe, A., ... & Wotschack, P. (2023). It takes two to code: a comparative analysis of collective bargaining and artificial intelligence. *Transfer: European Review of Labour and Research*, 29(1), 87-104.

[148] Krämer, C., & Cazes, S. (2022). Shaping the transition: Artificial intelligence and social dialogue. *OECD Social, Employment and Migration Working Papers*.

[149] Contreras, R. (2021). Impact of digitalisation on social dialogue and collective bargaining. *European Foundation for the Improvement of Living and Working Conditions (Eurofound),* 15 December.

[150] Bernhardt, A., Suleiman, R., & Kresge, L. (2021). Data and algorithms at work: the case for worker technology rights, UC Berkeley Labor Center.

[151] Griffin, B. (2020, 9 April). Canada: Predicting Reasonable Notice Using Machine Learning. *Blue J Legal*.

[151] For example, the GPDR applies to all companies, but companies with fewer than 250 employees only have to keep records of data processing activities if data collection/processing is their company's main activity, if there is a threat to people's rights, and freedoms or if the relevant processing of data involves sensitive data or criminal data.

Opportunities, to ensure that the readers can easily find insights on the topics that interest them the most, as well as to foster comparability.

2.3.1. Working conditions

Changing working arrangements and the nature of work

Reduction in meaningful work: this challenge brings into focus the idea of meaningful work, and how the volume and availability of such work may be impacted by AM. If, on the one hand, AM acts to reduce the mundane elements of work performed by managers, it may help to elevate the meaning of their work. Indeed, by raising the average quality of management work, it may serve to make this work more attractive and meaningful. **If, on the other hand, AM takes away aspects of managers' work which they value, such as planning and human interaction, it may erode the meaning of their work.** Likewise, if AM makes the work of workers more tightly regulated and less autonomous, this can reduce their opportunities for meaningful work. AM can appear to devalue human input into work and can make for a less rewarding work experience for both managers and workers. Whether the meaningfulness of work is increased or reduced by AM will ultimately depend on whether it is designed in such a way that it respects the needs and interests of those who manage, and those who work.[153] The more priority is given to workers' and managers' shared interests and well-being, the more likely it will be that AM can add to the meaningfulness of their work.

Reduction in worker autonomy: another related consequence flowing from AM is the minimisation, if not elimination, of worker autonomy. As previously mentioned, AM alters the production process by transferring the decision-making processes from human workers (including managers) to algorithmic systems — as the latter (thanks to vast computing power) can discover more efficient ways to complete tasks. This means that some workers 'will not be trained, and, rather, be forced to implement decisions that previously they had the power to shape'.[154] Unruh et al. emphasise that there are many ways in which 'work decision algorithms' negatively impact workers' liberty.[155]

Several consequences relate to this., First, **work decision algorithms might also inadvertently reduce workers' autonomy, because they strip workers of the right to exercise their preferences in the production process**. As Sharp (2018) puts it, 'if people analytics determines that there is a better way to perform a particular job task, the managerial imperative will, of course, be to deploy that task methodology across all similar positions in the organisation', and this will inevitably 'limit the freedom of humans to work according to their styles and preferences'.[156] Second, **algorithms in the workplace can jeopardise 'decision autonomy' because 'workers lose discretion over their work, and, thereby, the ability to make, even small, decisions about their work themselves'**.[157] Thirdly, **by de-personalising the workplace, AM can be seen as reducing what could be called 'collective' or 'organisational' autonomy'.** In other words, 'if technology reduces communication with colleagues and managers…, this might reduce possibilities for workers to exchange information and to organise, and make it more difficult to contact managers, for example to receive feedback or to make complaints'.[158]

[153] Smids, J., Nyholm, S., & Berkers, H. (2020, September). Robots in the workplace: a threat to – or opportunity for – meaningful work? *Philosophy & Technology*, 33(3):503-22.

[154] Jarrahi, M.H., Newlands, G., Lee, M.K., Wolf, C.T., Kinder, E., & Sutherland, W. (2021). Algorithmic management in a work context. *Big Data & Society*, *8*(2), 20539517211020332.

[155] Unruh, C.F., Haid, C., Fottner, J., & Büthe, T. (2022). Human autonomy in algorithmic management. In: *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 753-762.

[156] Sharp, B.S. (2018). Policy implications of people analytics and the automated workplace. In *The Political Economy of Robots*, pp. 61-80. Cham: Palgrave Macmillan.

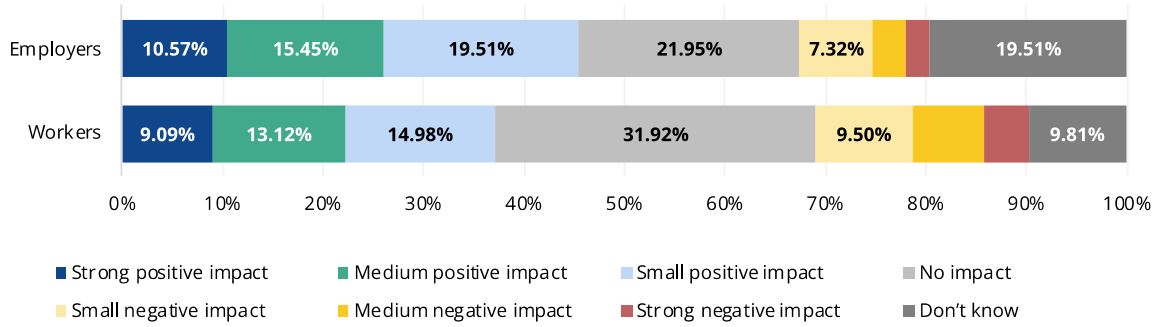
[157] Unruh, C.F., Haid, C., Johannes, F., & Büthe, T. (2022). Human autonomy in algorithmic management. In: *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 753-762.

[158] Unruh, C.F., Haid, C., Fottner, J., & Büthe, T. (2022). Human autonomy in algorithmic management. In: *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 753-762.

Ultimately, the loss of autonomy, like the loss of technical skills, is another factor contributing to worker alienation.

Although AM brings clear dangers with regard to worker autonomy, the majority of respondents to the employers' and workers' surveys carried out for this study believed that AM does not harm worker autonomy, or that it actually enhances it (see Figure 2-5). However, given that almost one in ten employers, and almost one in five workers, indicated that they 'Don't know' how AM affects worker autonomy, this implies that many may not yet be aware of what effects such tools might have. This suggests, as mentioned previously, a lack of knowledge – as well as, perhaps more importantly, the **need for educational efforts to inform employers and workers alike** regarding both the positives and the dangers that AM can bring, prior to the widespread adoption of AM within an organisation.

Figure 2-5: Effects of AM on worker autonomy, based on the VA employer and worker surveys (2023)



Note: employers N = 123; workers N = 966

Source: Authors' own elaboration based on the VA employer and workers surveys.

Deskilling: in an challenge related to the removal of autonomy, AM might also lead to the issue of deskilling – something that 'occurs when machines assume so much work previously assumed by people that workers begin to lose their acquired skill and fail to learn new ones'.[159] Added to this, **the simplification of workplace operations reduces the need for workers to develop or maintain a variety of skills**.[160] Jarrahi et al. (2021) explain that AM 'seeks to automate as many sub-components of the sociotechnical work system as possible and leave the rest to human operators', and that 'an unintended consequence of such common design approaches is to deskill humans and to relegate them to uninspiring roles'.[161] They further elaborate that in the platform economy, AM has cultivated a working environment in which 'workers are considered replaceable 'human computers', and may engage in (ghost) work that is repetitive, does not result in any new learning, and only intends to train AI systems'.[162]

Furthermore, a study by Schaupp on the state of digitalisation in the manufacturing and delivery logistics sectors found that the primary objective or function of AM in one firm

[159] Joh, E.E. (2019). The consequences of automating and deskilling the police. *UCLA L. Rev. Discourse,* 67: 133.

[160] Arguably, Adam Smith's commentary on the effects of the division of labour apply equally to algorithmic management: 'The man whose whole life is spent in performing a few simple operations, of which the effects too are, perhaps, always the same, or very nearly the same, has no occasion to exert his understanding, or to exercise his invention in finding out expedients for removing difficulties which never occur. He naturally loses, therefore, the habit of such exertion, and generally becomes as stupid and ignorant as it is possible for a human creature to become.' Smith, A. (1776/1937). *An Enquiry into the Nature and Causes of the Wealth of Nations*, New York: Random House.

[161] Jarrahi, M.H., Newlands, G., Lee, M.K., Wolf, C.T., Kinder, E., & Sutherland, W. (2021). Algorithmic management in a work context. *Big Data & Society* 8, no. 2 (2021): 20539517211020332.

[162] Jarrahi, M. et al (2021). *Algorithmic management in a work context*.

'seems to be a reduction of labour costs through deskilling'.[163] Schaupp's research revealed that for grocery delivery company Smart Shopping, 'algorithmic work control allowed for the reduction of on-the-job- training to one and a half days'. At Smart Delivery, 'algorithmic work control made it possible to replace personal training of the couriers by an online group-videoconference', which has allowed the company to adopt the principle of 'hire anyone'. In addition, one manager explained that, at Smart Electrics, AM is used to 'either to make things go faster or to enable people with less qualifications to do it'. Thereby, management wants to 'shift work to [...] less qualified, and, therefore, cheaper workers'.[164]

Another concern that has received less attention is the possibility of 'moral deskilling'. Vallor argues that 'moral skills appear just as vulnerable to disruption or devaluation by technology-driven shifts in human practices, as are professional or artisanal skills, such as machining, shoemaking, or gardening'.[165] **This is particularly evident in fields or sectors that have traditionally contained an ethical dimension in the execution of work-related tasks, such as in the military, healthcare, and education, among others**. In other words, industries that have been perceived as conveying the morals and value systems of society into aspects of life that may go beyond the workplace, may be compromised when AM is used to make decisions about workers without careful ethical consideration. The loss of moral skills is detrimental to individual workers and society at large, as such skills are 'a prerequisite to the cultivation of virtuous character'.[166]

Fractured employment relationships: In addition, **the changes in working arrangements brought by AM risk fracturing relationships between management and workers**. For example, workers can – and reportedly do – feel that the imposition of surveillance systems is invasive and violates their privacy rights.[167] This is especially relevant given that, as noted in 1.2. 1.2. Prevalence of AM in the EU, worker monitoring is one of the main applications of AM. Hence, a company's desire to install worker surveillance tools could be quickly eroded due to worker resistance, which could, in turn, lead to a workplace disruption. For example, workers might feel discouraged or demotivated when they are constantly monitored by what they perceive as overly invasive or unfair surveillance tools. In turn, these negative sentiments could contribute to declining worker morale, lower productivity, and higher turnover, and make it more difficult for firms to attract top talent.[168] All of these possibilities jeopardise the health of the business enterprise. Although more work is needed to gather evidence on these dynamics, it is important for companies to take into account the risks of certain types of AM, as well as their benefits, and weigh these against one another to decide whether or not AM should be continued. Regular risk assessments should be carried out to ensure any risks are mitigated and that the benefits continue to outweigh the risks.

Algorithmic pay determination: lastly, **algorithmic pay determination has gained attention as a method for setting employee remuneration, but it raises concerns about potential disparities in compensation among workers.** Using algorithms and data analytics, companies can automate the process of determining salaries, on the basis of various factors such as performance metrics, market rates and individual contributions. In some cases, this can in turn create a pivot – particularly among digital labour platforms – towards determining individual remuneration solely through the use of complex

[163] Schaupp, S. (2022). Cybernetic proletarianization: spirals of devaluation and conflict in digitalized production. *Capital & Class* 46, no. 1 : 11-31.

[164] Schaupp, S. (2022). Cybernetic proletarianization: spirals of devaluation and conflict in digitalized production. *Capital & Class* 46, no. 1 : 11-31.

[165] Vallor, S. (2015). Moral deskilling and upskilling in a new machine age: Reflections on the ambiguous future of character. *Philosophy & Technology,* 28(1): 107-124.

[166] Vallor, S. (2015). Moral deskilling and upskilling in a new machine age.

[167] Nuwer, R. (2022, 21 June). The employee surveillance that fuels worker distrust, *BBC Worklife*, available at: https://www.bbc.com/worklife/article/20220621-the-employee-surveillance-that-fuels-worker-distrust

[168] Bello-Perez, Y. (2021). Why You Shouldn't Spend Your Tech Budget on Employee Surveillance Software, *Unleash AI*, accessed 23 March 2023, available at: https://www.unleash.ai/hr-technology/why-you-shouldnt-spend-your-tech-budget-on-employee-surveillance-software/

mathematical models. Such models are built around a variety of factors including task complexity, demand-supply dynamics, and customer ratings.

While this approach might offer objectivity and efficiency, it also has the potential to perpetuate pay inequalities.[169] Algorithmic systems can inadvertently introduce biases or reinforce existing disparities by relying on historical data that reflects past discriminatory practices or systemic biases in pay determination, as well as other aspects relating to performance and pay.[170] In addition, the opaque nature of algorithms means that the mechanism used in setting wages can become obscured. This directly jeopardises the principles of fair and equal pay, if workers cannot understand how their pay packets are calculated. On top of this, it may be harder for workers to obtain corrective compensation in a workplace that is subject to algorithmic – as opposed to human – management. This is because such systems can perpetuate the problem of 'structural unaccountability'.[171] Lastly, the combination of the above factors can mean that workers completing almost identical work might be remunerated at different levels, which can lead to divisiveness and resentment in the workplace.

This, in turn, means that algorithmic wage discrimination can undermine the welfare of workers, and can also threaten the success of the firm. More specifically, discriminatory work environments are associated with lower workplace morale, higher rates of turnover, and exposure to legal liabilities.[172] All of these factors directly undermine a firm's competitiveness. In this light, algorithms which reproduce pay discrimination can have negative consequences for workers, companies, and the economy itself, which means that strong legal protections against it are necessary.

To mitigate this challenge, employers can closely monitor and regulate algorithmic pay determination, to ensure fairness, transparency, and accountability, and to prevent unjust wage gaps from emerging or widening among workers, based on characteristics such as gender, race or other protected attributes. Striking a balance between the benefits of algorithmic decision-making, and the need to address pay equity is essential in building a just and inclusive work environment.

Working time and communication

Workload and work intensity: constant evaluations, alongside the ever-present threat of sanction by the algorithm, can create a working environment that is characterised by substantial pressure and work intensification. For example, it has been established that there is a substantial 'correlation between the existence of [worker] monitoring and workplaces, where the pace of work is significantly determined by machines and computers'.[173] Thus, in practice, the 'centralised algorithmic control may also lead to work intensification and to worsened working time quality… [because] the number of tasks to undertake increases and/or the time to complete these tasks decreases'.[174] As Baiocco et

[169] Bernhardt, A., Suleiman, R., & Kresge, L. (2021). Case for Worker Technology Rights, UC Berkeley Labor Center, available at: https://laborcenter.berkeley.edu/data-algorithms-at-work/

[170] Vignola, E.F., Baron, S., Abreu Plasencia, E., Hussein, M., & Cohen, N. (2023). Workers' Health under Algorithmic Management: Emerging Findings and Urgent Research Questions. *Int J Environ Res Public Health*, 2023 January 10;20(2):1239. doi: 10.3390/ijerph20021239.

[171] Wedel, J.R. (2014). *Unaccountable: how elite power brokers corrupt our finances, freedom, and security*, Pegasus Books.

[172] Stamarski, C.S., & Son Hing, L.S. (2015). Gender inequalities in the workplace: the effects of organizational structures, processes, practices, and decision makers' sexism. *Front Psychol.* 2015 September 16;6:1400. doi: 10.3389/fpsyg.2015.01400.

[173] Riso, S.(2020, 9 December). Employee monitoring and surveillance: The challenges of digitalisation. Eurofound, *Working conditions and sustainable work,* available at: https://www.eurofound.europa.eu/publications/report/2020/employee-monitoring-and-surveillance-the-challenges-of-digitalisation

[174] Baiocco, S., Fernández-Macías, E., Rani, U., & Pesole, A. (2022). *The algorithmic management of work and its implications in different contexts*. No. 2022/02. JRC Working Papers Series on Labour, Education and Technology, available at: https://joint-research-centre.ec.europa.eu/publications/algorithmic-management-work-and-its-implications-different-contexts_en

al. point out, 'these aspects are defined to meet the target set centrally by the algorithms, rather than considering workers' situations or views… [so] workers must adapt their pace of work to comply with what is requested, being subject to time pressure and consequent stress'.[175]

Reduced choice regarding working time: in some cases, the distribution of jobs can be structured so as to incentivise or compel workers to meet certain algorithmically determined targets. In the case of digital labour platforms that rely on user data, such as Uber, customer ratings can play an important role in shaping how work is distributed. Previous research has demonstrated that 'workers with the best scores and the most experience tended to receive more work, due to clients' preferences, and the platforms' algorithmic ranking of workers within search results'.[176] This, in turn, can put a great deal of pressure on workers on such platforms to 'maintai[n] a high average rating and good accuracy scores'.[177] To be sure, this is a distinct algorithmically informed technique, which has a demoralising impact on workers. As Wood et al. (2019) note, **'whereas Uber's algorithmic management 'deactivates' (dismisses) workers with ratings deemed low, digital labour platforms, instead, use algorithms to filter work away from those with low ratings, thus making continuing on the platform a less viable means of making a living**'.[178]

In addition, **the number of ratings can also be a barrier to entry, or may create stress** for some workers, as some platform users may prefer someone with a large number of reviews an average rating to other individuals with small number of reviews and a high rating. Such preferences might also create a 'lock-in' situation in which individuals working on platforms might not want to leave their job, even if they are not happy with it, in order not to lose their good track record. This, in turn, can have a demoralising effect on workers, which in turn might lead to mental stress.

Alienation: AM tools may also reduce human connections in the workplace, leading workers to feel alienated and isolated. **Instead of management unfolding within social interactions between a human boss and worker, automated systems evaluate, direct and discipline workers through digital indicators**. This managerial transformation crowds out opportunities for mentoring workers, obtaining feedback from workers on the ground, and building trust in the workplace. Such outcomes deprive businesses of several key factors necessary for successful performance.[179]

Pressures around optimisation: one of the perceived benefits of AM is greater work efficiency. However, while this might be of benefit to businesses, customers, and some workers, it simplifies the production process to the potential detriment of workers. In other words, **where the purpose or function of algorithmic systems is the execution of basic and direct instructions to help them optimise, the worker may feel that they are expected to perform as well as a machine would**. In turn, this might make workers feel like 'machines' or 'robots', which might lead to psychosocial issues.[180]

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[175] Ibid.

[176] Wood, A.J., Graham, M., Lehdonvirta, V., & Hjorth, I. (2019). Good gig, bad gig: autonomy and algorithmic control in the global gig economy. *Work, Employment and Society,* 33(1): 56-75.

[177] Ibid, p. 64.

[178] Wood, A.J., Graham, M., Lehdonvirta, V., & Hjorth, I. (2019). Good gig, bad gig: autonomy and algorithmic control in the global gig economy, p. 64.

[179] Vredenburgh, K. (2022). Freedom at work: understanding, alienation, and the AI-driven workplace, *Canadian Journal of Philosophy*, 52, no. 1 (2022): 78-92.

[180] See debates in transhumanism and post-autonomous literature about a 'disciplinary society', and whether we are entering a 'control society' in which the role of machines in transforming human experiences and/or placed in competition with humans is still undetermined. For this discussion, see Foucault, M. (1977). *Discipline and Punish: The Birth of the Prison*, New York: Random House; Hardt, M., Negri A. (2000). *Empire,* Harvard University Press; Negri, A. (2008). *The Porcelain Workshop: For a New Grammar of Politics* Semiotext(e); Haraway, D. (1996). *Simians, Cyborgs, and Women: The Reinvention of Nature,* Free Association Books; Guattari, F. (1995). *Chaosmosis: An Ethico-Aesthetic Paradigm* Power Institute; Holmes, B. (1995). Guattari's Schizoanalytic Cartographies' *Escape the Overcode: Activist Art in the Control Society,* as well as https://brianholmes.wordpress.com/2009/02/27/guattaris-schizoanalytic-cartographies/

Occupational safety and health (OSH)

Increased OSH risks: evaluations carried out via algorithms, with their unique ability to provide real-time feedback without cessation, can have negative psychosocial and physical impacts on workers. Consider the application in call centres of Cogito, a form of algorithmic software that is 'used to provide both real-time and recorded activity productivity assessments that can be viewed by both workers and managers'. Roose (2019) reports how Cogito displays real-time feedback to call centre workers. Talking too fast? The program flashes up an icon of a speedometer, indicating that the user should slow down. Sound sleepy? The software displays an 'energy cue' with a picture of a coffee cup. Not empathetic enough? A heart icon pops up.[181] This tool ostensibly measures people's emotions and could be understood to help protect workers' OSH. However, without some auditing and management over their use, such tools might only be used to identify the negative externalities of overworked staff and possible consequent costs to the company, rather than to meaningfully help workers to mitigate against OSH risks.

Wood (2021) reviews systems similar to Cogito in other industrial sectors.[182] One such example is Genome*,* which is 'a passive data collection system and machine learning tool', used by the Canadian healthcare consulting firm Klick Health. This software calculates 'the average time it takes to complete a variety of tasks and alert team leaders, when projects are behind schedule… [and] notifies project leaders of outstanding and urgent things to do'.[183] Researchers have found that constant monitoring and evaluation by these kinds of AM systems can 'cause workers to behave in unnatural ways (always smiling or always being active), to achieve targets requiring great physical or psychological effort, or to be unable to interact socially with colleagues or to take breaks'.[184] Such constant pressure to satisfy extensive (if not intrusive) monitoring systems leads 'to stress and anxiety, particularly if combined with a lack of control over one's own time or in conjunction with constant reminders of such observation aimed at achieving modifications in a worker's behaviour.'[185]

Psychosocial and physical OSH risks: constant monitoring, paired with a lack of transparency regarding how algorithms are used to evaluate workers in situations of information asymmetries, may lead to emotional and physical distress at work, which are OSH concerns.[186] Hence, AM can lead to heightened levels of emotional, psychosocial and physical stress, anxiety, and sometimes burnout, due to such tools forcing workers to behave like machines, putting pressure on them perform at an optimised level. Physical issues can emerge because stress and anxiety cause people to sit in an inappropriate and ergonomically unsafe manner at their desks, leading to musculoskeletal difficulties later in life. Anxiety may be linked to a range of debilitating diseases.[187]

The psychological contract: while AM is used in a variety of management practices, it is mostly within the human resources practice of people analytics that the

[181] Roose, K. (2019). A machine may not take your job, but one could become your boss. *NY Times: The* Shift (23 June 23, 2019), available at: https://www. nytimes. com/2019/06/23/technology/artificial-intelligence-ai-workplace. html

[182] Wood, A.J. (2021). *Algorithmic management consequences for work organisation and working conditions*, No. 2021/07, JRC Working Papers Series on Labour, Education and Technology.

[183] Ibid.

[184] Todolí-Signes, A. (2021). Making algorithms safe for workers: occupational risks associated with work managed by artificial intelligence, *Transfer: European Review of Labour and Research,* 27(4): 433-452.

[185] Ibid.

[186] Aloisi, A., & De Stefano, V. (2022). *Your Boss Is an Algorithm Artificial Intelligence, Platform Work and Labour* (Bloomsbury Publishing); Cefaliello, A., Moore, P., & Donoghue, R. (2023). Making algorithmic management safe and healthy for workers: addressing psychosocial risks in new legal provisions, *European Labour Law Journal*, doi: 10.1177/20319525231167476 ; Moore, P. (2018). *The Threat of Violence and Harassment in the Digitalized World of Work,* Geneva: ACTRAV, International Labour Office, available at: https://www.ilo.org/wcmsp5/groups/public/---ed_dialogue/---actrav/documents/publication/wcms_617062.pdf

[187] Todolí-Signes, A. (2021). Making algorithms safe for workers, Making algorithms safe for workers: occupational risks associated with work managed by artificial intelligence. *Transfer: European Review of Labour and Research* 27(4): 433-452.

possibility of psychosocial risks emerges, in relation to the so-called 'psychological contract'. It refers to a set of expectations that go beyond the standard legally binding contract, whereby workers have a cognitive *perception* of what is expected of them. This also involves unwritten perceptions of the obligations that the employer has toward them. These perceptions are not always entirely explicit, but there is a range of seeming promises surrounding any workplace/space. Tomprou and Lee's research on the psychological contract emerging in AM contexts[188] identifies how the usage of decision-making technologies has an impact on the worker-employer – that is, whether algorithms have an impact on the type of organisational inducement. They find workers demonstrate lower levels of trust regarding algorithmic inducement activities, i.e. where machines are seen to have a motivating or pressuring effect on workers, based on decisions either taken by the algorithm either makes, or which the algorithm helps a manager to make. Thus, a potential challenge of AM emerges where either work candidates or workers see increased decision-making power being attributed to algorithms. Lowered trust in the employment relationship can lead to a problematic psychological contract.

2.3.2. Data protection and privacy

Data collection and storage

Employer surveillance and monitoring: in order for AM to realise its purpose of making the production process more economical, large quantities of data must be collected to serve as inputs to the algorithm. Such data primarily come from widespread surveillance and monitoring of the workplace, including of workers themselves. This aspect of AM has led to, or coincided with, an explosion in new monitoring technologies that constantly (and in real-time) observe and report on worker behaviours. **This proliferation of 'productivity monitoring applications' and wearable technologies, designed to survey workers, raises several concerns – first and foremost, the potential invasion of workers' privacy and data protection rights**.[189]

Workers' data protection concerns: although, as mentioned previously, worker monitoring is expanding, comparatively few of the employers and employees surveyed for this study believe it will harm workers' privacy. For example, 9.92 %, 3.31 % and 0.83 % of the employers surveyed believe AM has a small, medium, or large negative effect on worker privacy, respectively. Meanwhile, 11.24 %, 8.04 % and 6.08 % of the workers surveyed believe AM has a small, medium, or large negative effect, respectively. However, as can be observed, there is quite a substantial difference between the perceptions of employers and workers. Nevertheless, even though AM, and especially monitoring tools, are considered highly invasive by some academics,[190] a large number of employers and workers are not sure, or do not believe, that AM harms privacy. For more details, see Figure 2-6 below.

Transparency

Function creep: there is a risk that the data collected about an individual will be used for purposes other than the originally declared. This 'function creep' is particularly problematic

[188] Tomprou, M., & Lee, M.K. (2022). Employment relationships in algorithmic management: A psychological contract perspective, *Computers in Human Behaviour,* 126 106997 1-12.

[189] Adams-Prassl, J., Abraha, H., Kelly-Lyth, A., Silberman, M., & Rakshita, S. (2023). Regulating Algorithmic Management: A Blueprint. *European Labour Law Journal* 14.2 (2023): 124–151; Ginsburg, M. (2022). Fostering transparency and limiting the abusive use of algorithmic management practices - The view from the United States. *Comparative Labor Law & Policy Journal,* 42(): 743–755; Gaudio, G. (2022). Algorithmic bosses can't lie! How to foster transparency and limit abuses of the new algorithmic managers. *Comparative Labor Law & Policy Journal 42*(3): 707–741.

[190] For example: Berente, N., Gu, B., Recker, J., & Santhanam, R. (2021). Managing artificial intelligence. *MIS Quarterly*, 45(3): 1433-1450; Blagoycheva, H. (2021). Изкуствен интелект и корпоративна социална отговорност–сблъсък и нови възможности. Известия. *Списание на Икономически университет-Варна*, 65(1), 55-69; Gerards, J., Vetzo, M., & Nehmelman, R. (2018). Algoritmes en grondrechten.

if it is used to prevent workers from exercising their rights. As Ajunwa notes, the spread of AM will require a 'balance between the employer's pecuniary interests in monitoring productivity and the workers' privacy interests'.[191] While such a practice that is not permitted under the GDPR, it is very difficult to prove and guard against. This is because non-transparent practices may be carried out, whether by inference or clandestine action, without leaving a digital trace. The actual reasons for using a particular dataset, and for writing and deploying an algorithm, may be different from those openly expressed. For example, a set of managers may wish to reduce costs and lay off workers in the near future. Rather than informing workers that they are being tracked and monitored in order to determine who is laid off, managers might begin to look at HR data relating to which workers take the most time off. After monitoring such data over a period of time, decisions may then be made regarding who to fire and who to retain, based on data that were observed longitudinally, but were not stated as being for this purpose. In meetings with workers' representatives and/or workers themselves, managers may explain their decisions as being based on other premises, rather than on the data they have been watching. While this does not necessarily mean that an algorithm is used, the inferences drawn from patterns of data can lead to function creep in data collection, processing, and usage.

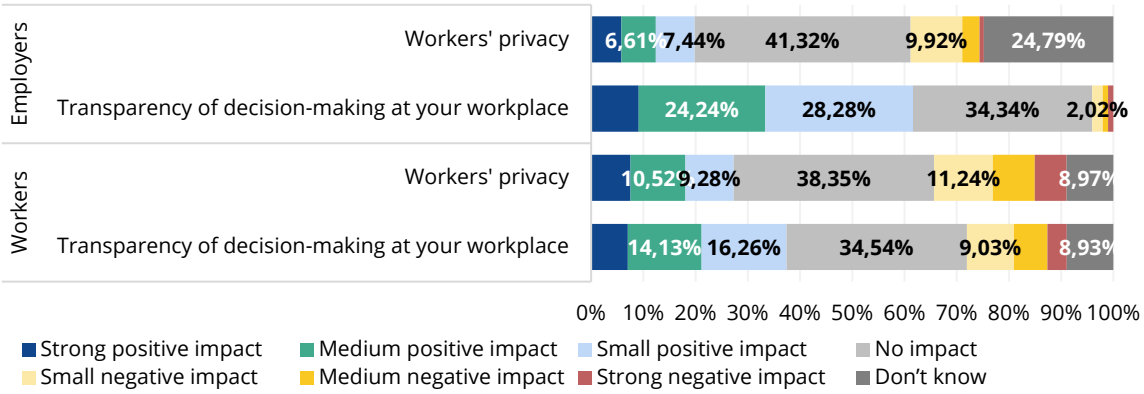
One report by UNIGlobal Union[192] outlines how workers' productivity is constantly and invasively monitored by scanners, and how social distancing was automated during the COVID-19 pandemic. While sensors indicating workers' location appeared very useful for ensuring social distancing and represented a positive development in protection from virus infection, at the same time, the company did not provide data about COVID-19 cases to workers, making the workers feel very insecure about the potential function creep of such data collection.

Based on the surveys of employers and workers carried out for the present study, many workers appear to be unaware of these issues. More specifically, only 9.03 %, 6.38 % and 3.72 % of worker respondents believe AM has a small, medium, or strong negative effect on the transparency of decision making, respectively. Meanwhile, 16.26 %, 14.13 % and 7.01 % believe that AM has a small, medium, or strong positive effect, respectively, on the transparency of decision making. The corresponding shares among employers are even more extreme: only 2.02 %, 1.01 % and 1.01 % of respondents believe that AM will have a small, medium, or strong negative effect, respectively, on privacy, while over half of employers believe that the effect of AM on transparency will be positive. These results imply that the majority of workers and employers see AM as having no effect or a positive effect in terms of privacy, perhaps without realising or considering the negatives.

[191] Ajunwa, I. (2018). Algorithms at work: productivity monitoring applications and wearable technology as the new data-centric research agenda for employment and labor law. *Louis ULJ,* 63: 21-55, p. 21.

[192] Delfanti, A., Radovac, L., & Walker, T. (n.d.). The Amazon panopticon: A guide for workers, organizers and policymakers. UNIGlobal Union, available at: https://uniglobalunion.org/wp-content/uploads/amazon_panopticon_en_final.pdf

Figure 2-6: Effects AM on privacy and transparency of decision-making, based on the VA employer and worker surveys (2023)



Note: employers N = 123; workers N = 966

Source: Authors' own elaboration based on the employer and workers surveys.

Decision-making by algorithm

Unclear accountability: a growing number of legal cases, which are discussed in detail in the legal analysis chapter of this report, have cropped up in relation to specific monitoring tactics and issues, including spying on trade union activity,[193] as well as the GPS monitoring of phones and vehicles. Moreover, wearable technologies are already being used to time bathroom breaks, track workers' walking routes around warehouses, direct workers to the correct locations for products, and even detect 'brain activity, and deliver data to workers about fatigue levels in real-time, by reading their brain waves'.[194] **Such practices are very likely to create stressful conditions for workers, who feel constantly watched., This may be exacerbated if there is insufficient communication as to why workers are being watched, and what will be done with the data collected**. Moore (2020) spoke to several workers in a range of industries, from consultancy to dentistry to content moderation, and not one of them had been informed of the reasons for data collection, although all of them knew that they were being 'watched'.[195] One worker in a property management firm was aware that the manager watched all of their activities at their desks and consoles, and, periodically, workers were called in to speak to the manager in the office. This worker informed Moore that there was no rationale for this level of tracking, nor any discussion of what would be done with the video recordings.

Lowered human involvement: algorithmic management can detract from the principle of 'human in command' when it comes to decision making. Automated systems may prioritise efficiency and cost reduction over human and employee interests. This can result in decisions that are not in line with workers' needs, desires, and rights.[196] To address this challenge, it is important to ensure that human oversight and control are maintained over

[193] Amazon has applied 'heat maps' to identify where workers are gathering, leading to fears that companies are concerned about gatherings in relation to union organising. All of these activities create suspicion regarding reasons for company data collection and tracking that can be linked to the ways in which AM may be used to sanction and penalise workers. See: Delfanti, A., Radovac, L., & Walker, T. (n.d.). The Amazon panopticon: A guide for workers, organizers and policymakers. UNIGlobal Union, available at: https://uniglobalunion.org/wp-content/uploads/amazon_panopticon_en_final.pdf

[194] Ajunwa, I. (2018). Algorithms at work, p. 38.

[195] Moore, P. (2020). *Data subjects, digital surveillance, AI and the future of work.*

[196] Köchling, A., & Wehner, M.C. (2020). Discriminated by an algorithm: a systematic review of discrimination and fairness by algorithmic decision-making in the context of HR recruitment and HR development, *Business Research*, 13(3), 795-848.

AM-driven decision-making processes. Clear lines of accountability and mechanisms for human intervention should be established to prevent undue automation, and to ensure that human rights and dignity are respected in the workplace. Ethical frameworks and guidelines can help to guide the responsible use of AM in decision making, emphasising the importance of human values and ethical considerations.

Discipline: not only are AM systems used to direct and evaluate workers, but in some cases, this information is used by employers to make decisions that lead to disciplinary measures. This raises several concerns. If disciplinary measures entail 'the punishment and reward of workers, in order to elicit cooperation and enforce compliance', there are different mechanisms by which these objectives can be achieved through platform algorithms.[197] More specifically, algorithmic platforms may facilitate an environment in which 'the fear of being considered 'abnormal' encourages individuals to make efforts to return to the disciplinary norm, under threat of sanctions'[198]. Hence, platform labour is sometimes seen as occurring within a panopticon prison structure.[199]

Beyond simple rewards, some companies – especially those employing platform workers – might use AM to dismiss workers. While the present research did not reveal a great deal of quantitative data to support this (see 1.2 1.2. Prevalence of AM in the EU), **nevertheless such use of AM might lead to such issues, which it is very important to discuss.** The most obvious method of disciplining workers is deactivation from the platform, an omnipresent possibility that can weigh on the minds of workers. Usually, digital labour platforms treat platform workers as independent contractors, not workers.[200] Therefore, many such workers lack the protection of employment legislation, such as the prohibition of arbitrary dismissal. A platform firm – such as Uber, Deliveroo, etc. – can terminate a worker's contract (i.e. access to the platform) at any time. The very real prospect of abrupt and even seemingly random termination by the algorithm has been found to motivate risky and harmful workplace dynamics. Most significantly, the 'fear of 'deactivation' from the platform that may coerce [workers] into accepting unsafe or exploitative conditions of work'.[201] Certain groups are especially susceptible to this risk, such as migrant workers, who are 'reluctant to complain for fear of deactivation by platforms…, [especially] temporary migrants who are exploited do not make claims, as they fear deportation or job loss'.[202]

Deactivation, while it is the most extreme example, is just one of the methods of discipline used by AM systems. Another involves the distribution of work itself. Platforms effectively

[197] Wood, A.J. (2021). *Algorithmic management consequences for work organisation and working conditions*. No. 2021/07. JRC Working Papers Series on Labour, Education and Technology.

[198] Galière, S. (2020). When food-delivery platform workers consent to algorithmic management: a Foucauldian perspective, *New Technology, Work and Employment,* 35(3): 357-370.

[199] A panopticon is a type of institutional building or system of control designed to maximise surveillance and minimise the possibility of escape or misconduct. The concept was first developed by the English philosopher and social theorist Jeremy Bentham in the late 18th century. The key feature of a panopticon is a central watchtower from which a single observer or guard can see into every cell or space within the institution, such as a prison, hospital or school. Individuals within the cells or spaces are unaware of when they are being watched, creating a sense of constant surveillance and the perception that they could be observed at any moment. The panopticon design was intended to promote discipline and self-regulation among the inmates or occupants. The idea is that the fear of being watched would lead people to modify their behaviour to conform to the rules and expectations of the institution, even when they were not actually being observed. While the panopticon was originally conceived as a physical architectural design, it has also been used as a metaphor for various systems of surveillance and control in modern society. It has been applied to discussions of surveillance technologies, government surveillance, and the ways in which social media and online platforms can create a feeling of being constantly watched. See: Delfanti, A., Radovac, L., & Walker, T. (n.d.). The Amazon panopticon: A guide for workers, organizers and policymakers. UNIGlobal Union, available at: https://uniglobalunion.org/wp-content/uploads/amazon_panopticon_en_final.pdf

[200] Berg, J. Income security in the on-demand economy: Findings and policy lessons from a survey of crowdworkers. International Labour Organization, available at: https://www.ilo.org/wcmsp5/groups/public/---ed_protect/---protrav/---travail/documents/publication/wcms_479693.pdf

[201] Athreya, B. (2020). Slaves to Technology: Worker control in the surveillance economy, *Anti-Trafficking Review,* 15 (2020): 82-101.

[202] Riordan, R., Robinson, R., Hoffstaedter, G. (2020). Seeking justice beyond the platform economy: migrant workers navigating precarious lives. *Journal of Sustainable Tourism* (2022): 1-18.

connect sellers of labour services with buyers in need of those services, yet 'the structure of platform food delivery work is far from the ideal type of a free marketplace'.[203] This is because platforms facilitate these market transactions 'within the context of technical control systems that shape and constrain workers' choices'. **Workers often experience these 'technical control systems' in decidedly negative ways.** On one level, the operation of these algorithmically determined systems feels opaque. In the case of platform delivery workers, this is 'because the rules of the game — the logic of pay and disbursement of orders — is black-boxed beyond workers' view, and subject to frequent change, workers often experience the algorithm as arbitrary and inscrutable…'.[204]

2.3.3. Fairness and discrimination

Bias and objectivity

AM can lead to unfair management decisions and bias: there are discussions on the risk of discrimination in filmed job interviews, which are becoming increasingly common. In 2020, it was reported that 86 % of companies asked candidates for filmed interviews.[205] While the increase in the adoption of such tactics was initially a result of the COVID-19 crisis, the process has become normalised and is set to continue. Prior to COVID-19, companies were already using AM to look for both verbal and non-verbal cues. The company HireVue allegedly collected biometric data from interviews using biometrically enabled software. This software employed facial geometry scanning and tracking to extract biometric information from job candidates. The company was taken to court for doing this without informed consent.[206] Similarly, the facial recognition company Clearview AI was taken to court for creating a database that was filled with billions of images taken from social media, which it shared with police forces without the consent of users.[207]

In addition, technologies that gather biometric data can increase biases in situations where, for example, if an interviewee's energy levels are low, a machine could detect such emotions and inform managers to have discretion. Evidence strongly shows that hiring preferences have favoured certain demographics within a population, and such patterns may be replicated in the way an algorithm teaches itself to identify specific candidates, such as in the now-famous Amazon case.[208] If the data provided to an algorithm reflect a dominant bias over time, it may scores candidates higher for 'in group'-related phrases on job applications, or for biometric data such as certain facial expressions, leading to discriminatory outcomes.[209] Where certain cue regarding sexual orientation, age and gender do not fall within this dominant category, this can lead to exclusion – particularly where a workforce is largely made up of members of the dominant characteristics pool, for example, white males, as was the case with Amazon.

[203] Griesbach, K., Reich, A., Elliott-Negri, L., & Milkman, R. (2019). Algorithmic control in platform food delivery work, *Socius,* p. 5 2378023119870041.

[204] Ibid, p. 5.

[205] Bayern, M. (2020). 86% of companies are conducting job interviews via videoconference, Tech Republic, available at: https://www.techrepublic.com/article/86-of-companies-are-conducting-job-interviews-via-video-conference/

[206] Jewett, A. (2022). HireVue Class Action Alleges Company Collects Interview Candidates' Biometric Data Without Informed Consent. *Top Class Actions*, available at: https://topclassactions.com/lawsuit-settlements/employment-labor/hirevue-class-action-alleges-company-collects-interview-candidates-biometric-data-without-informed-consent/

[207] Hart, R. (2022). Clearview AI Fined $9.4 Million In U.K. For Illegal Facial Recognition Database. *Forbes,* May 2022, available at: https://www.forbes.com/sites/roberthart/2022/05/23/clearview-ai-fined-94-million-in-uk-for-illegal-facial-recognition-database/

[208] Goodman, R. (2018). Why Amazon's Automated Hiring Tool Discriminated Against Women, ACLU News, available at: https://www.aclu.org/news/womens-rights/why-amazons-automated-hiring-tool-discriminated-against

[209] Moore, P.V. (2019). OSH and the Future of work: Benefits and risks of artificial intelligence tools in workplaces. EU-OSHA Discussion paper, 5 July 2019, available at: https://osha.europa.eu/en/publications/osh-and-future-work-benefits-and-risks-artificial-intelligence-tools-workplaces

2.3.4. Workers' collective rights

Workers' collective rights

While AM offers significant opportunities to strengthen and expand labour rights, it also presents profound challenges. *Laissez-faire* regulatory approaches, biases in AM systems, the expansion of managerial control and the automation of decision-making can all undermine labour rights and worker autonomy. To promote opportunities for social dialogue and collective agreements relating to AM, a comprehensive regulatory framework that addresses these challenges can ensure that AM technologies are used in a manner that respects and enhances labour rights.

The 'black box' nature of algorithmic management (lack of transparency, opacity, intricacy): AM often relies on extensive data collection and surveillance technologies that often operate in a 'black box' environment. One such challenge is 'algorithmic opacity',[210] whereby workers do not understand how the technology works and also do not understand how management puts the technologies into practice. This can lead to difficulties in negotiations for collective rights. Platform workers often face difficulties in understanding the algorithms' inner workings, given that algorithmic management is opaquer than its non-algorithmic counterpart.[211] Lack of transparency further leads to difficulties in organising in support of workers' collective rights.

While AM can enable organisations to monitor and optimise work processes, it also raises concerns about privacy and employee autonomy.[212] As previously noted, the widespread use of surveillance technologies can create a 'panopticon' work environment in which employees feel constantly monitored and subject to control. It is crucial to strike the right balance between using AM to achieve productivity improvements, and respecting privacy rights. Clear policies and safeguards should be put in place to protect employees' privacy and to ensure that surveillance technologies are used responsibly and ethically. Transparency, consent, and the right to be forgotten should be upheld, to preserve opportunities leading to the dignity and privacy of workers.

Collective bargaining

Current discussions surrounding the impact of AM often focus primarily on quantitative aspects, such as job loss and replacement. While it is essential to understand these quantitative effects, there is also a need to consider the qualitative impacts of AM on labour rights.[213] For example, AM technologies can lead to work intensification, whereby employees are subject to increased pressure to meet productivity targets or face continuous monitoring and surveillance. The qualitative effects of AM on work, including its impact on workers' autonomy, well-being, and job satisfaction, must be carefully examined. Policymakers, researchers, and stakeholders should engage in comprehensive studies and dialogue in order to understand the full range of effects that AM can have on labour rights.[214]

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[210] Benlian, A., Wiener, M., Cram, W.A., Krasnova, H., Maedche, A., Möhlmann, M., Recker, J., & Remus, U. (2022). Algorithmic Management: Bright and dark sides, practical implications and research opportunities. *Busi. Inf. Syst. Eng., 64*(6): 825–839.

[211] Kellogg, K.C., Valentine, M.A., & Christin. A. (2020). Algorithms at work: the new contested terrain of control. *Acad. Manag. Ann*., 14(1):366–410, available at: https://doi.org/10.5465/annals.2018.0174

[212] De Stefano, V. (2019). 'Negotiating the algorithm': Automation, artificial intelligence and labour protection. Artificial Intelligence and Labour Protection (May 16, 2018). *Comparative Labor Law & Policy Journal*, 41(1).

[213] De Stefano, V., & Taes, S. (2023). Algorithmic management and collective bargaining. *European Review of Labour and Research*, 29(1), 21-36.

[214] Cave, D. (2023) AI could diminish wage bargaining power and transform job roles, report finds, *People Management*, 12 May.

One aspect that is under threat is collecting bargaining. Collective bargaining agreements allow specific dimensions of the employment relationship to be determined between employers and workers' representative groups. Typically, collective agreements cover the setting and release of wages and overtime; how and whether individual and/or collective dismissals should proceed; criteria and processes for promotion; working time; the pace of work; and what types of contracts are deemed acceptable. Hence, collective agreements and social dialogue should be sought to provide opportunities for workers to seek agreements concerning working conditions in relation to AM. For example, AM might have an impact on how wages are decided – which is, at times, reliant on how client relationships are determined. AM can also have an impact on how targets are set with clients.[215] Bargaining over wages and targets represent standard themes in trade union and works council discussions, but **the ways in which algorithms work differ from other human resource discussions of this nature, making bargaining less straightforward**. For example, algorithmic determination may lead to unreasonable targets being set.[216] Therefore, new collective bargaining discussions and hoped-for agreements should consider the nature of algorithms, i.e. that they are expected to make better decisions than humans and may even be used to substitute decisions by managers.

Related issues that will need to be negotiated, and preferably agreed upon, through collective bargaining with trade unions and works councils include the ways in which data are gathered, processed, and utilised. Furthermore, negotiation and bargaining should occur in relation to the technical manner in which monitoring occurs, with discussions regarding what quantity of data are collected, from whom, when, and for what purpose, as well as how much monitoring is reasonable.

Lack of comprehensive regulation: despite the potential benefits of ensuring compliance, public regulation has not fully kept pace with these technological developments. Current regulatory frameworks often adopt a relatively minimal approach when it comes to protecting labour rights.[217] **This lack of comprehensive regulation leaves significant gaps, allowing companies to exploit AM technologies for managerial control and surveillance purposes without adequate oversight**. To address this challenge, robust regulatory frameworks are needed that explicitly address the impact of AM on labour rights. Such frameworks should ensure transparency, accountability and fairness in the design and use of AM systems.[218] They should also consider issues such as data privacy, consent, and algorithmic accountability. By establishing clear guidelines and standards, regulators can prevent the misuse of AM and protect workers' rights and dignity.

A factor that exacerbates problems with collective bargaining, or potentially limits its application, is the fact that the types of industries in which the use of AM is underway, such as cleaning, hotels, restaurants, and transportation, tend to have a lower density of union representation and collective bargaining. Indeed, in some industries, AM is used to explicitly limit or weaken union activities.[219] **AM can also be used to identify and reveal where trade union activity is occurring, and who is a trade union member.** The use of algorithmic management to identify trade union activity, the location of trade union activity, and trade union membership, has raised concerns regarding workers' rights and privacy. AM and the use of algorithms and data analytics to monitor and control various aspects of work, such as scheduling, performance evaluation and workforce management. While such algorithms can provide benefits in terms of efficiency and productivity, as outlined in earlier

[215] Aloisi, A., & Gramano, E. (2019). Artificial intelligence is watching you at work: digital surveillance, employee monitoring, and regulatory issues in the EU Context, *Comparative Labor Law & Policy Journa*l, 41 (1), 95-121.

[216] Ibid.

[217] Stefano, V., & Aloisi, A. (2021, 8 November). 'Artificial intelligence and workers' rights'. *Social Europe.*

[218] AlgorithmWatch (2023). Algorithmic transparency and accountability in the world of work: A mapping study into the activities of trade unions. Study conducted on behalf of the ITUC, available at: https://www.ituc-csi.org/Algorithmic-transparency-and-accountability

[219] Semuels, A. (2022, 13 October). Some companies will do just about anything to stop workers from unionising. The Future of Work. *Time Magazine*, available at: https://time.com/6221176/worker-strikes-employers-unions/; Trades Union Congress (n.d.). When AI is the Boss, available at: https://www.tuc.org.uk/resource/when-ai-boss

sections, there is a risk that they may be used to directly suppress or target trade union activities.

Trade union busting: by analysing large datasets and patterns, algorithmic systems can potentially identify indicators of trade union activity, such as organising efforts or discussions about collective bargaining.[220] This can lead to practices that victimise trade unionists, whereby workers engaging in union-related activities may face negative consequences or surveillance. In addition, algorithms can be used to track the location of trade union activity, which could enable employers to monitor and potentially interfere even with lawful organising efforts. Moreover, **AM may also be employed to infer trade union membership, based on various data points** such as social media activity or participation in specific events or groups. This can potentially violate a worker's right to privacy and freedom of association, as employers could use this information to target and discriminate against employees based on their trade union affiliations.

Social dialogue

Unequal access to information: the deployment of AM raises broader ethical and social implications. It requires a careful consideration of its impact on society, on the power dynamics between employers and employees, and its potential for exacerbating existing inequalities. Ethical frameworks and guidelines should be established to guide the design, deployment, and use of AM systems in the workplace. Furthermore, ongoing dialogue and collaboration among stakeholders, including workers, employers, regulators, and civil society organisations, could provide positive benefits that mitigate against the challenges outlined here, and ensure that the use of AM aligns with fundamental labour rights and values. However, information asymmetries can stymie the progress surrounding social dialogue.

'Social dialogue' is 'a fundamental component of the European social model. It enables social partners (representatives of management and labour) to contribute actively, including through agreements, to designing European social and employment policy'.[221] Social dialogue also involves workers' representatives playing an active role in negotiating contracts and working conditions. Elements of social dialogue include negotiations that can lead to collective bargaining agreements, employment regulations and social welfare policies; consultation, which refers to gestures made by governments to actively include both employers and worker representatives in the formulation of regulations, labour law and related matters; conflict resolution, whereby workplace conflicts can be resolved before strikes occur; collaboration, which relates to well-being, workplace rights, safety and labour standards. In this sense, social dialogue is linked to the potential for citizens and companies to play positive roles in high-level policymaking decisions, as well as in the negotiations that occur on the ground and in work environments.

The rise in the use of AM means that associations that represent workers and employers should both seek to play an explicit role in social dialogue in order to influence policymaking. This is important because, as indicated, AM is already having an impact on work. One area relating to AM that requires their explicit involvement concerns Article 22 of the GDPR, which states that decisions at work should not be made on the basis of algorithms alone. The 'Proposal for a Directive on Improving Working Conditions in Platform Work' expresses the intention of promoting 'social dialogue on algorithmic management'.[222] The right to social dialogue is a democratic right, and the use of data and the processes

[220] Maxwell, J. (2018). How a labor union is using an algorithm to predict when to organize, *VICE News*, available at: https://www.vice.com/en/article/nep5wb/how-a-labor-union-is-using-an-algorithm-to-predict-when-to-organize

[221] European Parliament, available at: https://www.europarl.europa.eu/factsheets/en/sheet/58/social-dialogue

[222] Clifford, D., Goldenfein, J., Jimenez, A., & Richardson, M. (2023). A right to social dialogue on automated decision-making: From workers' right to autonomous right, *Technology and Regulation,* available at: https://techreg.org/article/view/13258

involved in AM must respect this fundamental right, in order to realise the opportunities presented by AM.

2.4. Effects of AM on public sector and social security systems

AM and public sector workers

Research into the implications of AM in the workplace has, by and large, focused on digital platform work. But what does AM mean for workers in the public sector? On this question, the evidence is less abundant and, in some cases, even problematically scarce. This is particularly concerning, given that the use of AM systems in the public sector is expanding rapidly. AM systems are already being used for a variety of purposes in the public sector, including 'calculating of optimal routes for collecting municipal waste; analysing which buildings are more likely to catch on fire to guide which fire safety inspections should be prioritized; estimating where the chance of criminal behaviour is the highest, and subsequently, send police staff to these so-called 'hotspots'; the evaluation of teachers' performance; and, guiding physicians' behaviour through nudges based on algorithms in health care.'[223] The literature discussing public sector activity and artificial intelligence technologies (AITs) is predominantly concerned with the use of 'automated decision making' (ADM) in the delivery of public sector benefits, as opposed to how public sector workers are impacted by the use of AM systems.

Some research has documented the novel impacts of AM systems in the public sector. For example, Nagtegaal (2021)[224] conducted a study to understand police officers' perceptions of how AM systems impact the procedural fairness of their work. Nagtegaal found that 'when a decision is made by an algorithm for practices that are low in complexity, [employee perceptions of] procedural justice increases' – a positive outcome that could be enhanced by the spread of AM systems in public sector work. However, the author also discovered that 'for practices that are high in complexity, decisions involving a public manager are perceived [by workers] as higher in procedural justice compared to decisions that were made automatically by computers using algorithms.' To be sure, workers' perceptions of how AM systems impact the delivery of their services to the public is an important data point. Because public sector workers are closest to the operation of these technological tools, their perceptions of such tools ostensibly provide meaningful insights into how effective they may or may not be. Thus, further research into this issue should be pursued.

Another area of mounting concern regarding the use of AM in the public sector is how it impacts the ability of public sector workers to do their jobs in ways that best promote the common good. Some researchers have noted that 'decision-making based on the use of algorithms might impact the capacity of public employees… to organize their jobs and [exercise] autonomy'.[225] Elsewhere in this report, the negative implications of reduced autonomy at work were discussed. However, a loss of workplace autonomy poses an additional complication in the case of public sector work. The delivery of benefits and services is often a highly administrative process, wherein small errors abound (e.g. clerical mistakes such as writing the wrong date, misspellings, etc.). The offloading of tasks to AM systems, which have repeatedly demonstrated an inability to compensate for such errors, jeopardises the ability of workers to 'get fair results' as they are less 'accountable for the decisions made by those autonomous systems'.[226] In other words, a loss of autonomy

[223] Nagtegaal, R. (2021). The impact of using algorithms for managerial decisions on public employees' procedural justice. *Government Information Quarterly*, *38*(1), p.101536.

[224] Nagtegaal, R. (2021). The impact of using algorithms for managerial decisions on public employees' procedural justice. *Government Information Quarterly*, *38*(1), p.101536.

[225] Criado, J.I., Valero, J., & Villodre, J., 2020. Algorithmic transparency and bureaucratic discretion: The case of SALER early warning system, *Information Polity*, *25*(4), p. 454.

[226] Ibid.

brought about by AM may prevent workers from being able to apply the necessary discretion that best serves citizen users of welfare and public services.

Another study, of Norwegian civil servants, interestingly supports the above findings. The researchers assessed public sector workers' reactions to the digitalisation of their workplaces and work processes. The findings of the study indicated that 'public service workers are positive [towards] digitization when it promotes professional aspects of their work and that professional discretion is considered necessary to accomplish tasks of greater complexity'.[227] In short, there appears to be consistent evidence that AM and ADM systems in the public sector may be beneficial, so long as they do not strip public sector workers of the autonomy to address complex problems which automated systems lack the nuance to address.

Expansion of AM and ADM in the public sector

The previous sections have demonstrated that the increasing use of AM in the public sector has several implications. First, algorithms can be used to manage workers in these sectors, just as they are used in other industrial sectors. Simply put, a worker employed by a state welfare administration agency may be managed by an algorithm in similar ways to a worker employed by a private insurance company. Second, algorithms are increasingly used by the public sector to administer the delivery of benefits and services. The first dimension is a form of algorithmic management (AM), whereas the second is best categorised as algorithmic decision making (ADM). For workers in the public sector, the expansion of both of these dimensions impacts their experiences at work and the experiences of citizens who interact with these services. Furthermore, the logic motivating the expansion of AM and ADM are the same: to make systems of production more efficient. In addition, the lessons learned from AM can often be applied to ADM, and *vice versa*. Hence, the interlocking impacts of AM and ADM on both workers and citizens necessitates the exploration of both domains.

Due to their interlocking functions, the boundary between the AM and ADM can be fuzzy. This is because, as the ILO notes, 'The link between the concepts of automation and algorithmic management is that the latter may imply some degree of automation of management'.[228] ADM systems may be used by an organisation to automate aspects of the delivery of a service or product, such as determining which citizens are eligible to receive a certain welfare benefit. However, once ADMs are applied to management-related decisions, AM is effectively in operation in the workplace. Yet this boundary continues to be obscured by matters of degree. The ILO further explains how, 'algorithmic management involves the replacement of labour input by machine (algorithmic) input for some types of managerial tasks. If algorithms only assist human managers in their decision-making, strictly speaking there would be no automation of management, but the process still qualifies as algorithmic management'.[229] In other words, the managerial role of an algorithm can range from merely making recommendations to a human who is in control, to completely automating the process, eliminating human involvement. Generally speaking, then, 'any process of automation of production or service delivery can facilitate the adoption of algorithmic management'.[230]

It therefore comes as no surprise that governments, just like the private sector, are attracted to the potential efficiencies that can result from implementing AM systems.

[227] Busch, P.A., Henriksen, H.Z., & Sæbø, Ø., 2018. Opportunities and challenges of digitized discretionary practices: a public service worker perspective, *Government Information Quarterly*, *35*(4), pp.547-556.

[228] Baiocco, S. et al. (2022). 'The Algorithmic Management of work and its implications in different contexts,' International Labour Organisation, June 2022, https://www.ilo.org/wcmsp5/groups/public/---ed_emp/documents/publication/wcms_849220.pdf

[229] Ibid.

[230] Ibid.

The range of responsibilities shouldered by governmental agencies is vast, and the use of algorithmic decision making (ADM) is being rolled out, albeit at different speeds and relative magnitude compared to the overall size of public service provision, to serve a wide range of applications. Brown (2019) reports that ADM is already being used to make 'Decisions about where to police, whom to detain, which child maltreatment allegations to investigate, whom to grant access to permanent supportive housing, and what level of unemployment benefit to provide…'.[231] The International Social Security Association (ISSA) reports that 'There is a growing trend in social security to apply AI, particularly to improve customer services through automated 24/7 front-end support and also, more incipiently, automating back-end processes'.[232] Not only do governments confront numerous and often complex obligations, but they also suffer from ever-declining budgets to fulfil such duties, thus prompting the need to economise on administration and to manage workers using algorithms as a cost-saving strategy, as much as possible.[233] Thus, the majority of the benefits for employers and workers outlined previously, about which the empirical evidence predominantly applies to private sector workers, can also be applied to workers in the public sector.

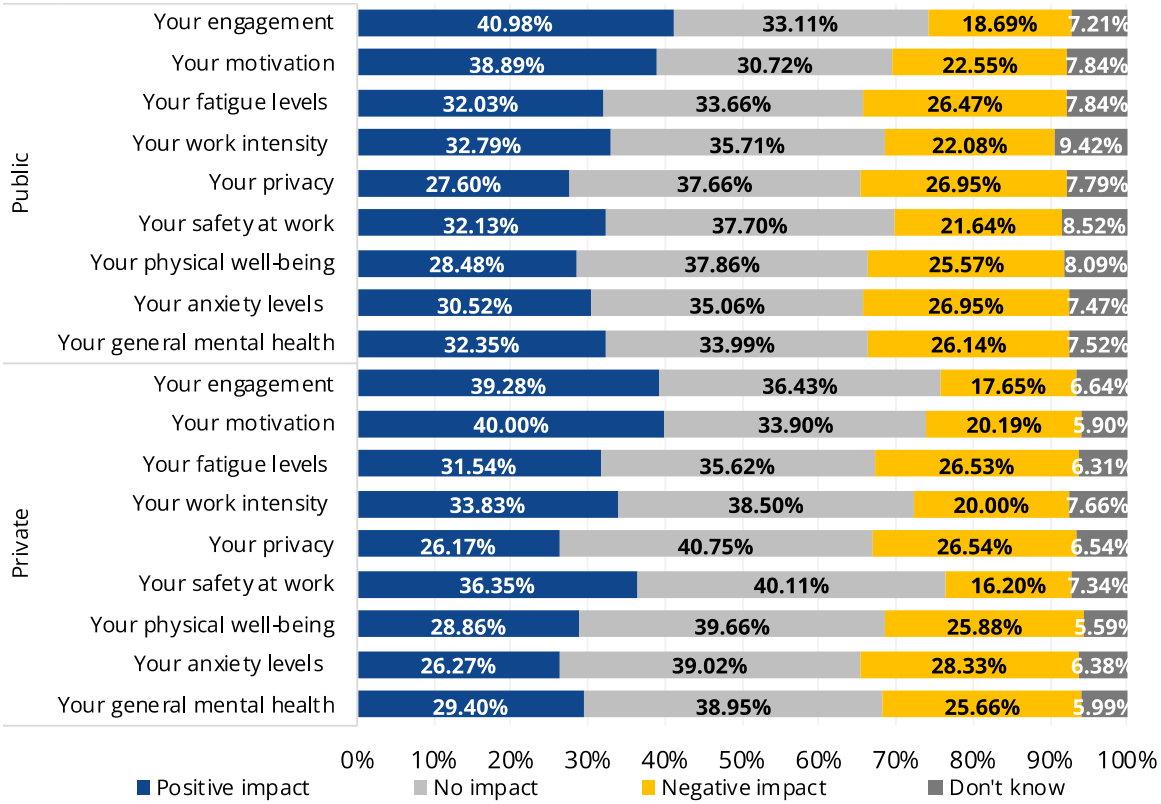
However, based on the analysis provided in Section 1.2. 1.2. Prevalence of AM in the EU, **public sector enterprises use AM less frequently than private enterprises**. Expanding on this, based on the survey of workers carried out for the present study, relevant results from which are outlined in Figure 2-7 below, workers in both public and private enterprises generally share a similar view of how such tools affect them – the differences between them are rarely more than a few percentage points, which could even be attributed to noise. The only larger differences occur in the area of safety at work, where workers in private enterprises are more likely than workers in public organisations to see AM as leading to higher safety standards. In addition, more workers in public enterprises feel that AM reduces anxiety levels and improves mental health, compared with workers in private enterprises.

[231] Brown, A., Chouldechova, A., Putnam-Hornstein, E., Tobin, A., & Vaithianathan, R. (2019, May). Toward algorithmic accountability in public services: A qualitative study of affected community perspectives on algorithmic decision-making in child welfare services. In: *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1-12.

[232] International Social Security Association (2020, 28 October). Artificial Intelligence in Social Security: Background and Experiences,, available at: https://www.issa.int/analysis/artificial-intelligence-social-security-background-and-experiences

[233] https://neweconomics.org/uploads/files/austerity-by-stealth-final.pdf

Figure 2-7: Effects of AM on workers in private and public companies, based on the VA survey of workers (2023)



Note: N = 966

Source: Authors' own elaboration based on based on the workers survey

Nevertheless, **while the turn towards algorithmic decision making (ADM) has an apparent logic administratively, cases of the implementation of ADM in the real world are already raising concerns about its ever-expanding use**. As Brown (2019) goes on to say: 'These technologies and the opaque manner in which they are deployed have drawn tremendous criticism from researchers, policymakers, civil rights groups, and the media.' Certainly, these criticisms are not based solely on theoretical speculation about the potential effects of ADMs on both private and public sector workers. In many cases, such rebukes arise from exposure to the harm incurred by communities that have interacted with those state services that have been impacted by the incorporation of ADM. The paragraphs that follow explore these concerns. Moreover, the likely rapid permeation of ADM throughout social, political, and economic institutions renders the need for an analysis of their impact even more pressing. All forms of data subjects (users, consumers, workers, governments, employers, etc.) will be subject to the consequences of this expansion of ADM into all facets of social life, including the public sector.

AM and the status of workers' social security rights

A large volume of research now exists on how AM systems are changing the world of work, in both positive and negative ways. However, just as for public sector workers, there is a notable absence of literature regarding what AM systems might mean for workers' social security rights. Social security is defined by the ILO as 'the protection that a society provides to individuals and households to ensure access to health care and to guarantee income security, particularly in cases of old age, unemployment, sickness, invalidity, work

injury, maternity or loss of a breadwinner'.[234] In particular, the following question needs to be addressed: what does the deployment of AM systems in workplaces mean for workers' access to those benefits that guarantee their health and financial security? Unfortunately, the existing literature offers little in response, as it primarily explores how AM systems threaten other types of workers' rights. For example, De Stefano and Taes (2023) investigate the more frequently discussed 'risks that new managerial technologies present for fundamental rights and principles, such as non-discrimination, freedom of association and the right to privacy', – leaving the question of social security to one side.[235]

The rise of the platform or 'gig' economy has spurred criticism over the potential loss of worker benefits. Based on exchanges with stakeholders and the analysis of relevant sources, workers employed by digital labour platforms as independent contractors forgo numerous labour standards associated with the traditional employment relationship, including minimum wage, overtime compensation, unemployment compensation, family and medical leave and employer payroll taxes, among other benefits.[236] However, this loss of rights is ascribed to workers being contracted as self-employed, not to the use of AM systems by digital labour platforms.

Researchers have thus largely been concerned with the impact of AM systems on key aspects of the workplace and production process. Specifically, they have studied how AM systems endanger the rights of workers when making hiring decisions, giving directions, evaluating performance, or exercising disciplinary power.[237] It is, therefore, imperative that future research should consider the relationship between AM and the social security rights of workers. It might be revealed that certain trends apply that are already recognised with respect to AM in workplaces. For instance, the literature has established 'a broad consensus that unions and collective bargaining contribute to a broad sharing of productivity gains'.[238] It is reasonable, therefore, to speculate that high union density and strong collective bargaining procedures will be imperative to protecting social security rights in AM-transformed workplaces.

Algorithms make mistakes

One of the most obvious problems attributed to ADM usage in social and welfare services workforces is that algorithms do not always get things right. Not only can they be programmed incorrectly, but algorithms presently lack the sophistication to assess all relevant information when making determinations relating to the administration and assessment of benefits. Of course, errors are a problem not only associated with algorithms. Human managers are capable of all kinds of shortcomings, from small clerical errors to serious acts of discrimination. However, as the examples below demonstrate, the impact of algorithmic managerial errors has the semi-unique capacity to scale rapidly across time and space. If an algorithm's programming contains one limitation, and that algorithm processes thousands or millions of automated decisions, that limitation can be replicated at incredible volume. On top of this, those who are negatively affected by the algorithm may find it difficult, if not impossible, to hold the system accountable and obtain a corrective solution.

[234] International Labour Organization. (n.d.). Facts on Social Security, ILO, available at: https://www.ilo.org/wcmsp5/groups/public/---dgreports/---dcomm/documents/publication/wcms_067588.pdf

[235] De Stefano, V., & Taes, S. (2023). Algorithmic management and collective bargaining. *Transfer: European Review of Labour and Research*, *29*(1), 21-36.

[236] Donovan, S.A., Bradley, D.H., & Shimabukuru, J.O. (2016). What does the gig economy mean for workers?, available at: https://ecommons.cornell.edu/bitstream/handle/1813/79155/CRS_What_does_the_gig_economy_mean.pdf?sequence=1

[237] Wood, A.J. (2021). Algorithmic Management. *Consequences for Work Organisation and Working Conditions*. European Commission, JRC Technical Report, available at: https://reshapingwork.net/dialogue/wp-content/uploads/sites/3/2021/06/algorithmic-management-consequences-for-work-organisation-and-working-conditions.pdf

[238] Garnero, A. (2021). The impact of collective bargaining on employment and wage inequality: Evidence from a new taxonomy of bargaining systems. *European Journal of Industrial Relations*, *27*(2), 185-202.

One example comes from Australia which, even though it lies outside the EU, provides some valuable lessons. The objective of the Robodebt scheme was 'to replace the formerly manual system of calculating overpayments and issuing debt notices to welfare recipients with an automated data-matching system'. Robodebt has since become an infamous international story of ADM use in welfare services, due to its colossal failure to accurately calculate and administer debt notices, resulting in the issuance of thousands of false statements. In the end, the deployment of Robodebt 'saw the government unlawfully raise $1.76bn in debts against 443,000 people'.[239] To understand the impact of this failure on real individuals, consider the case of 'pensioner Rosemary Gay… feeling 'sheer terror' after reading an October 2016 letter which said she owed Centrelink $64,998.17 in overpaid welfare benefits'. When she challenged CentreLink on this, she was informed there was a glitch, and her bill was reduced to over AUD 6,600. Still feeling this to be the incorrect number, she challenged the decision again, and it was reduced to just AUD 120. Nevertheless, the initial shock of the first bill is reportedly something Rosemary 'will never get over'.

A second high-profile example of errors being made by ADM systems is the Dutch government's introduction in 2014 of a new algorithm called SyRI, aimed at detecting welfare fraud. SyRI used information from a variety of different government databases to identify for further investigation individuals at high risk of committing welfare fraud. In 2018, six non-governmental organisations brought a lawsuit against the government over its use of SyRI, and in 2020 the Dutch court the Hague ruled in their favour. The court found that the algorithm violated the right to privacy under the European Convention on Human Rights. This ruling is likely to set a precedent concerning use of algorithms by governments. The court's ruling reiterated many of the concerns about algorithms long noted by academics and activists: inaccuracy, opacity, bias, and privacy.

However, it is worth emphasising that cases exist in which these artificial intelligence technologies (AITs) fulfil their objectives and improve overall outcomes. The ISSA has maintained a leading role in tracking developments surrounding the use of AITs (including ADMs) in the administration of social security benefits. Its research has indicated several instances in which these tools have been implemented with positive results. The organisation reports, for instance, on the case of the Norwegian Labour and Welfare Administration's use of a chatbot to enable a timelier response to queries during the COVID-19 crisis. This was an apparent success, as the chatbot 'responded to more than 8,000 daily inquiries, which compares to a pre-COVID number of 2,000'.[240] The Superintendence on Occupational Risks in Argentina achieved similar success with its implementation of a chatbot named 'Julieta', designed to handle questions 'about work injury benefits'. According to the ISSA, the chatbot 'fulfilled the objectives of providing automated and personalised customer services by responding not only to the most frequent questions but also by inquiring about the status of costumers' operations such as enrolment and benefit applications'. Other notable examples include the use by Employment and Social Development Canada of machine learning models to identify 'over 2000 vulnerable Canadians to be entitled to the GIS by processing more than 10 million records of unstructured text data'.

Biased data

As seen above, generating incorrect outcomes is perhaps the most impactful way in which algorithms can make mistakes. **However, algorithms or ADMs are also only as good as the data they are fed**. Evans (2021) notes that 'The issue of training is particularly

[239] Henriques-Gomes, L. (2021). Robodebt: Court Approves $1.8bn Settlement for Victims of Government's 'Shameful' Failure. *The Guardian*, Australia news section, 11 June 2021, available at: https://www.theguardian.com/australia-news/2021/jun/11/robodebt-court-approves-18bn-settlement-for-victims-of-governments-shameful-failure

[240] ISSA (2020). Artificial Intelligence in Social Security: Background and Experiences, International Social Security Association, 28 October 2020, available at: https://www.issa.int/analysis/artificial-intelligence-social-security-background-and-experiences

important when models are trying to predict outcomes involving people'.[241] One domain of public services in which this problem has been especially pronounced and the source of great worry is in policing and corrections. Bennett Moses and Chan (2018) explain that predictive policing is a 'term applied to a range of analytic tools and law enforcement practices' that are used to 'forecast where and when the next crime or series of crimes will take place'.[242]

'Dirty' data can occur for a variety of reasons, including human error, data entry mistakes, data integration issues, or technical errors. Even more worrying is that dirty data in predictive analytics is difficult to detect and correct, as the errors produced may be subtle or hidden, and therefore hard to identify. On top of the problem of detection is the reality that cleaning and validating large datasets can be a time-consuming and resource-intensive process, requiring significant amounts of time and effort. What is the likelihood that organisations will have the resources and capacity to carefully clean their datasets? While it is possible to use automated tools to detect and correct errors, it is also necessary to involve human experts to review and validate the data.

Contextual factors

A third core limitation of algorithms is that **not all aspects of social life are easy to predict, yet there is a growing interest in using computers to do so**. As IBM reports, companies, and governments today:

> *…are inundated with data from log files to images and video, and all this data resides in disparate data repositories across an organization. To gain insights from this data, data scientists use deep learning and machine learning algorithms to find patterns and make predictions about future events*.[243]

The use of predictive analytics spans all types of government services, ranging from child protective services to parole decisions, policing, adult social care, and others.

The use of machine learning algorithms to make predictions that will impact the distribution of social and welfare services raises serious ethical concerns. Glaberson (2019) argues that 'the most important risks predictive analytics present can be categorized along three axes: accuracy, fairness, and misuse'.[244] With regard to accuracy, predictive models can be led astray by 'The existence of error in the underlying data and the difficulty of identifying appropriate outcomes to measure'. This possibility is greatly exacerbated by the existence of large volumes of 'dirty data'. While the explosion of electronic monitoring has resulted in a massive expansion of data, 'it is beginning to be recognized that an inordinate proportion of data in most data sources is 'dirty'.' According to Kim et al. (2003), 'roughly speaking, dirty data means either missing data or wrong data or non-standard representations of the same data'.[245]

Another factor that threatens the accuracy of predictive algorithms is that they cannot always factor in relevant changes that happen over time. According to Glaberson (2019), 'predictive analytic algorithms tend to rely on negative risk factors, to the exclusion of protective, positive factors.' Glaberson further explains this point by way of example, noting how predictive tools used by child welfare services 'tend to project static conditions from a family's past into the future, without considering rehabilitative actions

[241] Evans, Ty. (2021). Why Do Algorithms Make Mistakes?, Psychology Today, available at: https://www.psychologytoday.com/gb/blog/trust-games/202111/why-do-algorithms-make-mistakes. Accessed 6 March 2023.

[242] Bennett Moses, L., & Chan, J. (2018). Algorithmic prediction in policing: assumptions, evaluation, and accountability, *Policing and Society*, *28*(7), 806-822.

[243] https://www.ibm.com/topics/predictive-analytics

[244] Glaberson, S.K. (2019). Coding over the cracks: Predictive analytics and child protection. *Fordham Urb. L. J.*, *46*, 307.

[245] Kim, W., Choi, B.-J., Hong, E.-K., Kim, S.-K., & Lee, D. (2003). A Taxonomy of dirty data. *Data Mining and Knowledge Discovery,* 7(81-99), available at: http://biosoft.kaist.ac.kr/bisl_2018/paperPDF/2003TaxonomyOfDirty.pdf

taken or resiliency-supporting factors that may be present'. One can easily foresee the moral injustices this might perpetuate. In this particular case, it could deny a family member who may have engaged in problematic behaviour in the past from ever regaining access to their child – even after they have engaged in meaningful efforts to reform. In addition, it can deny a child access to a key family relationship to which the child is rightfully entitled, given that the conditions of the person on the ground have changed.

The most vulnerable as primary victims

Social services systems almost always rely on historical data, which can be problematic if those data sources reflect intentional and structural prejudices and inequalities. For example, if a dataset indicates that people from certain ethnic or racial groups are less likely to access social services, an algorithm may unfairly exclude individuals from these groups in the future, thereby further marginalising and excluding vulnerable individuals. At an even more basic level lies the problem that minorities are, by definition, underrepresented in datasets covering the general population, which means that algorithms are intrinsically less well equipped to make rules or judgements about individuals from these groups. In addition, other researchers claims that social services are built around 'juked statistics' as 'police data records contain falsified crimes, planted evidence, racially biased arrests, and other actions that produce dirty data'.[246]

A second issue is that **automated decision-making systems inherently lack transparency in their operations**. This is largely because algorithms can be complex and involve multiple calculations and transformations that are difficult to understand – something often called the 'black box' problem.[247] Algorithmic complexity can be even greater in deep learning neural networks, which can have hundreds or thousands of interconnected nodes performing complex mathematical operations. Not only is it difficult, and in some cases impossible, for trained statisticians and mathematicians to understand how algorithms operate, it is effectively impossible for those who lack the proper training. This makes it difficult for vulnerable individuals to understand how decisions are made, or to appeal decisions that they believe to be incorrect or unfair. Such opacity is especially problematic for marginalised individuals who may not have the resources or knowledge to navigate the systems required to fix these errors and, where necessary, obtain compensatory justice. For these reasons, AM and ADM systems present new obstacles to workers holding employers accountable in cases of managerial malfeasance.

Third, **automated welfare systems do not consider the unique circumstances and needs of vulnerable individuals**. A system that automatically (and without human intervention) denies benefits to someone deemed 'non-eligible' might fail to consider the individual's unusual personal circumstances – something that a human could account for. Such inflexibility can result in vulnerable individuals being unfairly denied benefits or services they need to survive. ADMs eliminate human interaction, leading to a loss of empathy and understanding of the complex issues faced by those in highly precarious situations. Automated systems might use chatbots or virtual assistants to communicate with individuals, which can lack the emotional connection and empathy that a human caseworker can provide. Thus, not only do marginalised individuals face the prospect of unfairly denied benefits, but they also have no recourse to an empathetic representative to help correct errors or account for unusual or changing circumstances.

2.5. General effects of AM on broader society

Algorithms now enjoy a ubiquitous presence in modern society. Their use spans a growing number of social and technological vectors, such as social media, finance, healthcare,

[246] Richardson, R., Schultz, J.M., & Crawford, K. (2019). Dirty data, bad predictions: How civil rights violations impact police data, predictive policing systems, and justice. *NYUL Rev. Online*, *94*, 15.

[247] Ajunwa, I. (2020). The 'black box' at work. *Big Data & Society*, *7*(2), 2053951720966181.

criminal justice, and education. **Their functions are multitudinous. Algorithms are responsible for the automation of tedious tasks that consume vast amounts of labour time. They provide the infrastructure that connects people all over the world on the basis of shared attributes, interests, or goals. They are also responsible for rapidly disseminating information, such that a new finding or discovery can become common knowledge in mere seconds.**

However, **while algorithms offer potential benefits, such as greater efficiency and automation, they also raise significant questions about their impact on society, and how they transform social relationships and dynamics**. There is now wide recognition and concern about the potential for algorithms to reinforce existing biases, exacerbate social inequalities, erode privacy, undermine personal autonomy, and even disrupt democratic processes. On top of this, algorithmic decision making can be opaque and difficult to understand.

With this in mind, it is important to highlight that the analysis of the impact of algorithms on society is a notoriously complex affair that is increasingly becoming the subject of academic scrutiny. The nature of algorithms, specifically those that serve a general purpose, means they can easily be applied to new areas, and are therefore quickly spreading through our social institutions. Fortunately, researchers are reacting quickly to the scale of this phenomenon and devoting greater attention to it. Hence, there is growing awareness of this fact, and a collective witnessing of the growth of collaborative research teams and centres that bring together individuals with backgrounds in computer science, statistics, ethics, law, sociology, psychology and more. Under the headings below, the impacts of algorithms in several core domains of social life are surveyed. An overview of the growing literature is provided in the remainder of the subsection.

Indirect algorithmic management

The spread of algorithmic management is not a universal phenomenon across industrial sectors. In certain industries, workers are managed extensively (if not entirely) by algorithms. Those in the gig or platform economy are examples of this, with workers reporting minimal interaction with human managers. In other sectors, meanwhile, traditional management structures still largely exist. However, this does not mean that workers in the latter sectors are immune from the spread of algorithmic software. For many, the nature of their work is changing, due to the presence and incorporation of complementary or supportive algorithms. As explored below, workers in various industries such as the media, entertainment, education, etc., are increasingly having to take into consideration algorithmic outputs and realities when making decisions at work. This is the case for journalists, for example, the content of whose work is dictated by how it will be processed via various algorithms. Thus, while they are not explicitly told what and how to write by an algorithm in the strict sense of 'algorithmic management', they are 'indirectly' advised on what to do by the algorithms with which they interact. Hence, **this phenomenon of 'indirect algorithmic management' is having a major impact on many dimensions of society.**

To further clarify this point, consider the distinction between one's work being *controlled* by an algorithm, versus being *conditioned* by it.[248] The former situation applies when a traditional 'employment relationship' is in operation, whereby the activities of labour are contractually subordinated to the employer, granting the employer the legal entitlement of 'managerial prerogative'.[249] In the context of traditional AM, then, the algorithm is effectively

[248] Philip Pettit elaborates on how one's activity or freedoms can be 'compromised' or 'conditioned' by the nature of the social obstacles that interfere with one's pursuits. We slightly adapt this distinction to the question of management in the employment relationship, as opposed to the question of overall social freedom. For more on this, see: Pettit, P. (2006). Freedom in the Market, *Politics, Philosophy & Economics* 5, no. 2 (1 June 2006): 131–49, https://doi.org/10.1177/1470594X06064218

[249] Gaudio, G. (2022). Algorithmic bosses can't lie! How to foster transparency and limit abuses of the new algorithmic managers. *Comparative labor Law & Policy Journal, 42*(3): 707–741.

invested with the power of managerial prerogative, and therefore, controls the worker's labour. Again, the platform worker is the textbook example of this type of arrangement, wherein the platform's algorithm is responsible for administering tasks, scheduling shifts, or initiating disciplinary sanctions. However, other **workers occupy a different position, in which they are not *de jure* subordinated to an algorithm and subject to its prerogative, but *de facto* pressured to perform their labour in certain ways because of their proximity to an algorithm**. The work of a worker in this position is thus conditioned by the nature of the algorithms with which they interact. Due to this, the work of teachers, journalists and numerous other occupations are increasingly conditioned by algorithms, even if they are not controlled by them in their day-to-day work.

Work

In previous sections, the impact of AM on workers has been discussed. Here, the impact of algorithms on the world of work is emphasised in more general terms. The use of algorithms on a digital platform has been central to the rise of the gig or platform economy. This is because they enable efficient matching on online platforms of suppliers of labour with customers or clients demanding such labour. Thus, **this technology has enabled the blossoming of an ever-growing labour market characterised by short-term contracts or freelance work, where workers are paid for completing individual tasks, rather than being employed full-time by a company** – i.e., the 'gig' or 'platform' economy. While short-term work has always been around, finding people with a particular skill set to hire on a short-term basis was previously much more difficult and time-consuming. Workers would have to rely on word of mouth or personal connections to find gigs, and companies would have to go through the lengthy process of advertising, interviewing, and hiring workers.

The development of digital labour platforms, managed by algorithms, has made it easier for workers and clients to find one another, thereby reducing the 'transaction' costs traditionally attached to gig work.[250] For example, ride-sharing platforms, such Uber and Lyft, use algorithms to efficiently match drivers with passengers; freelance marketplaces, such as Upwork and Fiverr, use algorithms to match freelancers with businesses looking for their skills; and platforms such as Just Eat, or Uber Eats, connect delivery drivers with customers ordering food for home delivery. Algorithms have also helped the gig economy to scale rapidly, because digital platforms can easily move wherever there is internet infrastructure, which happens to be a large part of the world. Unlike traditional companies, digital platforms do not have to move highly immobile fixed capital, such as warehouses, storefronts, or even a large standard employed labour force, when expanding to new markets.

However, the rise of the platform economy has been a controversial political-economic development in the past 15 years. On the one hand, some argue that the gig economy produces flexible working opportunities that are highly valued by workers, as well as cheaper labour costs for those in need of task-based services.[251] However, labour movements, trade unions and some legal scholars have been sounding the alarm about the impact of the platform economy on working conditions. Vocal representatives of these groups see the gig economy as a threat to the victories of previous struggles to improve workers' rights and build a standard employment relationship within employment law. Based on exchanges with stakeholders and the analysis of relevant sources, the platform economy involves the casualisation of work, which means that traditional employment protections, such as minimum wage, health insurance and retirement benefits are no longer secured or guaranteed for such workers. In short, workers in the gig economy can be often underpaid and subject to excessive work intensity, all while lacking traditional benefits and job security.

[250] Hansen Henten, A., & Maria Windekilde, I. (2016). Transaction costs and the sharing economy. *info*, *18*(1), 1-15.

[251] See: https://www.heritage.org/jobs-and-labor/report/the-rise-the-gig-economy-good-workers-and-consumers

Media content and consumption

Algorithms are rapidly transforming how media content is produced, distributed, and consumed. The economic imperative for media producers – or even 'content creators', to use a more voguish term – today, is arguably not the same as it was for producers in the past. The most notable source of difference is that 'attention' has displaced 'sales' as the primary measure of success in the digital and algorithmic media environment.[252] Attention has become the currency of digital platforms because it is a valuable resource for advertisers and platform owners. Digital platforms, such as social media, search engines and streaming services, rely on user attention to generate revenue through advertising and subscriptions. The more time users spend on these platforms, the more opportunities advertisers have to reach their target audiences, and the more revenue the platform can generate.

The imperative of attracting attention dramatically *conditions* how businesses and workers operate, and what is expected of workers in the media and entertainment industries. Algorithmic management is, thus, used in these industries to ensure the creation and delivery of content designed to capture and retain as much user attention as possible. Take, for instance, the process of selecting stories in the media. Algorithms now play an increasing role in determining which stories will be covered, and how they will be covered for. The widespread availability of audience and engagement analytics means algorithms are helping journalists and editors to make editorial decisions by identifying topics and trends that are popular among readers. This 'data-driven' approach means that news and media outlets are guided ever more by the interests and preferences of their customers or viewers, and less so by other potential journalistic values.

There are other issues that media producers must take into consideration, due to the proliferation of algorithms – and this again represents a form of 'indirect algorithmic management'. While social media algorithms don't 'control' journalists or video makers directly, they do heavily condition their work because failure to comply with certain principles will result in poor audience engagement and interaction. Consider, for example, how algorithmic management conditions the ways in which the outputs of media and content creators are distributed. Because their work is increasingly released onto digital platforms that are themselves governed by algorithms, work and editorial-related decisions have to be made on the basis of what will succeed on those platforms. Journalists, for instance, must take actions that will optimise their visibility in digital ecosystems, such as considering presentation aesthetics, keywords, formatting, and headlines. These choices will be directed towards boosting their engagement and visibility.

Another example of this kind of 'indirect algorithmic management' is the extent to which content creators have to personalise content, optimise user engagement, and create addictive user experiences. Algorithms analyse user data to understand their interests and preferences, allowing media producers to create and curate content that is personalised to the needs and tastes of certain individuals and audiences.[253] This personalised approach ensures that users are presented with content that is more likely to capture and maintain their attention. Companies are finding new ways to identify the most engaging content for users, such as A/B testing.[254]

Education

[252] Bhargava, V.R., & Velasquez, M. (2021). Ethics of the attention economy: The problem of social media addiction. *Business Ethics Quarterly*, *31*(3), 321-359.

[253] Mittelstadt, B. (2016). Automation, algorithms, and politics| auditing for transparency in content personalization systems. *International Journal of Communication*, *10*, 12.

[254] Gomez-Uribe, C.A., & Hunt, N. (2015). The Netflix recommender system: Algorithms, business value, and innovation. *ACM Transactions on Management Information Systems (TMIS)*, *6*(4), 1-19.

Algorithms are rapidly changing the provision of education in several ways. These range from how students learn, and instructors teach, to the ways in which educational institutions are managed and outcomes are evaluated. Algorithms are the force behind the onset of 'data-driven instruction', which functions as a form of 'indirect algorithmic management'. That is, teachers and educators are now the beneficiaries of substantial information and insights about their students, in the form of algorithmic outputs. These outputs may not come in the form of direct managerial instruction. However, the incorporation of these insights is increasingly important and relevant for student success, and this incorporation process conditions certain workplace choices and discretions of workers in the educational sector – just as journalists cannot ignore algorithmic functions and outputs in their sector.

'Data-driven instruction' unfolds in a variety of different ways within the educational sector. For example, the algorithms used in education systems are capable of ushering in 'personalised' or 'adaptive' learning that involves adjusting the difficulty level and content of learning material, based on a student's performance, learning outcomes, and other relevant markers. After such analysis, the algorithms direct teachers towards the most suitable learning material and even help to adjust it to the student's needs. For instance, the algorithm might select certain multimedia content that best matches the student's learning style and interests, at a level of difficulty that is challenging, but not overwhelming, to the student. **Adaptive learning is further reinforced by the capability of educational algorithms to function as 'recommendation engines' that can pair students with the materials that will be most helpful in their learning.** These recommendation engines condition the work of educators, by leading them to make choices that may not be autonomously derived.

Another example of the involvement of algorithms in education is the phenomenon of 'intelligent tutoring'. Algorithms are now integrated into intelligent tutoring systems and can generate useful feedback for educators to act upon. The algorithms embedded into these systems play a key role in assessing students' answers and providing feedback that teachers can draw upon and incorporate into their educational approaches, thereby conditioning how their work is carried out. Such technology relies on the potential of predictive analytics, whereby advanced statistical techniques are used to analyse student data and generate predictions about how they will perform in the future. The significance of such analytics is that they can help schools and teachers identify which students are at risk of under-performing, failing, or even dropping out. Such a possibility offers schools the capacity to take preventive steps to promote the students' best interests, as opposed to only reactive measures.[255] In other word, if predictive analytics flags that a student is at risk of falling behind, the school can direct additional support and resources to that student, with the hope of making a positive intervention. There is growing evidence that the use of predictive analytics in education does improve student outcomes.[256]

2.6. General effects of AM on competitiveness and consumers

The literature on the relationship between AM as a technique for managing workplaces and consumers is thin. This scarcity can partly be explained by the newness of the subject, and partly by researchers focusing more on the impacts of general AI algorithms, which encompass broader aspects of this AI-consumer relationship. This is also possibly because AM is not a customer-facing technique: its central purpose is to automate managerial responsibilities regarding workers. However, it is possible to discern some potential impacts of AM on consumer interactions and experiences.

Most notably, aside from the purpose of managing their workers, organisations might employ AM to directly or indirectly affect customers' experiences and improve their

[255] Ekowo, M., & Palmer, I. (2017). Predictive analytics in higher education. New America.

[256] Foster, C., & Francis, P. (2020). A systematic review on the deployment and effectiveness of data analytics in higher education to improve student outcomes. *Assessment & Evaluation in Higher Education*, *45*(6), 822-841.

satisfaction with provided services. For example, **AM can render business operations more efficient by streamlining administrative processes, which could have indirect effects on customers**. Specifically, improved business efficiency might translate into customer-friendly outcomes such as faster delivery times, lower costs, and increased convenience. Insights from previous research confirm this. Specifically, delivery workers, predominantly those working through a platform, acknowledge that AM exerts positive effects, by improving deliverers' service efficiency, work quality, reliability, and customer-oriented behaviour, hence providing better service to consumers.[257]

The growing impact of AM is also evident in its effect on the competitiveness of organisations. **In a similar manner to general AI, research indicates that AM can have a positive influence on innovation, efficiency, and productivity.** From a broader perspective, Cockburn et al. (2018) argue that innovations not only have the potential to influence both the production and the characteristics of products and services, but also to affect the process of organisational innovation itself.[258] Moreover, according to Wirtz, the emergence of technologies such as AI, machine learning, the Internet of Things, mobile technology, virtual reality, geo-tagging, speech recognition and biometrics provide significant opportunities for extensive service innovations. These innovations, AM included, have the potential to enhance customer experience, service quality and productivity. Overall, the literature suggests that the adoption of AI-based tools in work processes, including AM tools, enhances the innovativeness of companies and pushes them to develop new and more efficient solutions to respond to a changing competitive market.[259] However, **it is worth noting that the majority of literature focuses on researching AI and other emerging technologies and their effect on the competitiveness of organisations, rather than separating out the specific effects of AM tools.**

[257] Wu, X., Liu, Q., Qu, H., & Wang, J. (2023). The effect of algorithmic management and workers' coping behaviour: An exploratory qualitative research of Chinese food-delivery platform. *Tourism Management*, 96, 104716.

[258] Cockburn, I.M., Henderson, R., & Stern, S. (2018). The impact of artificial intelligence on innovation: An exploratory analysis. In: *The economics of artificial intelligence: An agenda*, pp. 115-146. University of Chicago Press.

[259] Abrardi, L., Cambini, C., & Rondi, L. (2022). Artificial intelligence, firms and consumer behavior: A survey. *Journal of Economic Surveys*, 36(4), 969-991.

3. Legal and policy framework

This chapter aims to conduct a comprehensive mapping and a robust analysis of legislation at the European Union (EU) and Member State (MS) level (both legislations already in force and regulatory interventions that are planned), as well as **social partner agreements or initiatives**, in relation to the challenges and opportunities arising from AM systems at work. In addition, this examination will encompass a **selection of case law** at both national and supranational levels, along with relevant **decisions and opinions issued by enforcement authorities**.

The chapter flags which aspects relating to AM are covered in the current legal framework and which are not, in line with the typology in Chapter 2. Opportunities and challenges. **Its overarching goal is to assess how relevant and effective the regulatory tools are in addressing AM systems**, and any gaps remain.

Key findings:

- AM systems and AM-driven processes intersect with a wide range of legal domains within the EU. Although many laws in the EU *acquis* can, in principle, be applied to various facets and operational phases of AM, no legislation exists that is tailored specifically to the nuances of AM, with the exception of Chapter III of the proposed PWD. Instead, various items of legislation tend to address the broader situations under which AM falls, leading to a more indirect connection. As a result, legal issues relating to AM may require interpretation, based on legislation that was not expressly designed for it, given the general nature of the legislation.
- AM systems function in ways that diverge significantly from the way in which managerial functions are exercised in conventional work relationships in terms of volume, velocity and variety. Their influence is readily apparent within organizational frameworks. This means that the effectiveness of the various relevant legal instruments can vary considerably.
- The gaps in the current legal frameworks arise for two major reasons. First, although AM systems perform traditional managerial roles, their method of operation contrasts sharply with the previous model in conventional work relationships. Second, some existing legal structures have inherent limitations that do not relate exclusively to AM, but which are exacerbated when matched against AM systems' capabilities to perform or support key managerial tasks. While some current legislative gaps might appear minor in the traditional context, they can have amplified consequences when confronted by AM.
- The prevailing sentiment in much of the literature, especially that concerning the effectiveness of current legal structures in the context of AM, leans towards a sceptical outlook. This may reflect concerns about how algorithms are transforming labour dynamics, or the intricacies of ensuring appropriate protections in a rapidly evolving landscape. Nevertheless, the present analytical exercise strives for balance, seeking to understand both the potential benefits and the challenges that AM brings to social institutions.
- Several legal instruments, such as the Working Time Directive (WTD), the Transparent and Predictable Working Conditions Directive (TPWCD) and the Work-Life Balance Directive (WLBD) offer workers protections in relation to AM, ensuring their health and safety, reducing the unpredictability of work patterns and facilitating flexibility for parents and caregivers. In the area of working time legislation, some structural challenges that may be relevant to AM systems concern the binary distinction between working time and rest time, the lack of indications as to how to compensate for productive and unproductive working time, and insufficient protection against underemployment. The Occupational Safety and Health (OSH) Framework Directive addresses challenges such as the routinisation and standardisation of work,

diminished worker decision-making autonomy, heightened health and safety hazards and deteriorating work-related health.

- The legal frameworks governing workers' privacy and data protection stand out as one of the most apt and relevant when it comes to regulating AM systems. By and large, they provide responses to handle the intricate challenges posed by such systems in terms of grounds for lawful processing, transparency and safeguards in the case of solely automated decision-making. Despite the limited presence of provisions that are tailored specifically to work environments and the use of broad and general terms, the General Data Protection Regulation (GDPR) provides a robust foundation, allowing more focused interventions, in alignment with its core principles and requirements. At national level, however, emerging regulatory initiatives may potentially result in legal fragmentation.
- EU equality instruments in principle can ensure discrimination-free professional environments, even with the adoption of AM systems. However, the constitutive features of algorithmic discrimination are barely compatible with the 'traditional' notions of discriminatory practices, thus reducing the effectiveness of the existing rules and redress mechanisms. AM systems design individualised instructions, treatments and requirements for workers by aggregating legitimately collected data, which are not necessarily aligned with the set of characteristics protected under EU law. Moreover, the datasets used in AM play a crucial role and, even if accurate, can result in discriminatory outcomes when they reproduce unfair realities. Some gaps in addressing algorithmic discrimination stem both from its unique nature and from structural constraints within the equality framework, such as a limited set of protected grounds given the legal basis, as well as exceptions and justifications in the case of indirect discrimination, and the obstacles that stand in the way of gathering evidence.
- Collective rights are eminently relevant to the context of AM systems and functions. Effective information, consultation and social dialogue can improve the reception of AM and the outcomes of its decisions. More specifically, the role of social partners and worker representatives is recognised in various pieces of EU legislation, including the GDPR and the proposed Platform Work Directive (PWD). Furthermore, collective agreements are beginning to include terms regarding AM systems, typically addressing impacts on occupational safety and health, workers' data protection and work performance.

3.1. Methodology

This section analyses how the legal framework at EU and national levels responds to the emerging challenges and opportunities presented by AM tools. To do so, legislation, case law and decisions by enforcement authorities are relied upon. This analysis also identifies gaps in the existing legal framework. By adopting an **analytical and doctrinal approach**, the section will address, but will not be limited to, the following questions:

- How relevant and effective are the current legal frameworks at the EU and national levels in addressing the challenges and opportunities posed by the use of AM systems?
- To what extent does the current legal framework suffer from gaps, and in what areas is there a need for action?

The analysis relies on the following sources:

- A **mapping of developments at EU and MS level**, covering legislation directly or indirectly relevant to AM, as well as other policy initiatives (both those in force and, to the extent that they are known, any other planned policy initiatives). The mapping also includes legislative and judicial developments, collective agreements, and any other types of initiatives, be they planned or in force. Relevant national debates either directly or indirectly related to AM are considered. The methods used for the mapping include desk research carried out by the core team along with national experts covering all

EU-27 MS, as well as interviews with experts and national stakeholders. The mapping also involves the identification of relevant regulatory developments in a selection of the EU's main partners. This section draws on this mapping and is complemented by a deep examination of the academic literature.

- The **analysis of five selected court decisions** that touch upon critical aspects of AM systems and practices.
- The **results of the analysis on challenges and opportunities** provided in Chapter 2. Opportunities and challenges, including the literature review, case studies on AM tools, surveys of workers and employers, etc. The analysis of the regulatory framework is informed by the typology of challenges and opportunities proposed in Chapter 2 (see Table 2-11 above). This classification considers thematic areas, and aggregates legislation based on its potential ability to address a series of central dimensions of AM.
- **Results of the three thematic workshops**, collaboratively conducted with experts and stakeholders, enriched by a **validation workshop** that included prominent academics highly knowledgeable in the relevant subjects. During this culminating event, the external experts validated the findings of the legal analysis.

In this section, certain methodological issues can be identified:

- The methodology adhered to in conducting this analysis employs a **thematic and hierarchical approach to structure the examination**. However, the nature of the legislation concerned is, by default, interconnected and interdependent. This means that various laws and regulations are often related to each other in complex ways, whereby one piece of legislation may influence or rely upon another. To accurately represent these multifarious relationships within the legislative framework, this section will not simply follow a linear or isolated examination of each part. Instead, it will include numerous cross-references. These serve as connections between the different components of the legislation, highlighting areas in which they interact or influence each other. By highlighting these interrelationships, this analysis aims to offer a more informed and integrated perspective, reflecting the true nature of legislative practice and its inherent complexities.
- Since AM represents a novel and unprecedented phenomenon within the legal landscape, several of the pieces of legislation associated with this area are in uncharted territory, having not been examined or applied in real-world practice or case law. This **lack of precedent** is challenging for legal professionals, lawmakers, and scholars. Without existing judgments or practical applications to refer to, the interpretation and understanding of the relevant laws must rely on theoretical examination, doctrinal analysis, and speculative reasoning. This approach requires the careful consideration of existing legal principles, the intended purpose of the legislation, and how it might be applied in hypothetical scenarios. Such analysis also presents a degree of uncertainty, as the real-world application of these laws may yet yield unexpected outcomes or interpretations.
- Another methodological limitation pertains to the generalisation of some of the findings related to **artificial intelligence (AI)** and, to a lesser extent, **platform work** to the context of AM. Given the close relationship between AI and AM systems, the analysis includes specific comments relating to AI, which in many cases is the underlying technology supporting AM systems. Similarly, where relevant, the study draws insights from the narrower field of platform work, given that much of the literature exploring the legal and factual implications of AM has primarily emerged in this area. The present analysis recognises that while some of the findings from research into platform work could be applied or extended to AM systems in 'conventional' work settings, there may be differences in the way algorithms operate in platform work compared with other sectors.
- The temporal scope of this analysis extends to end of **September 2023**. In addition, the primary focus of our examination of proposed legislation currently under discussion is on

the proposal as originally presented by the Commission. Alternative versions of such proposed legislation are only referenced when they have introduced significant elements that warrant consideration in this analysis. Nevertheless, where appropriate, major developments were added during the last stretch of revisions. Between the submission of the Final Report and the publication of this study, a political agreement has been reached on the Artificial Intelligence Act and the Platform Work Directive. These developments do not substantially change the analysis presented here.

The box above highlights the key findings of the analysis, while the remainder of the chapter discusses them extensively and comprehensively. Each section begins with a summary of key findings, enabling readers interested in specific aspects of the analysis to quickly grasp the main points.

3.2. Working conditions

Key findings:

- The Working Time Directive (WTD), the Transparent and Predictable Working Conditions Directive (TPWCD), the Work-Life Balance Directive (WLBD), and the Occupational Safety and Health (OSH) Framework Directive offer safeguards to address some of the core challenges relating to AM. The overarching framework of these legal tools is well equipped to tackle underlying aspects of the various issues related to working conditions (e.g. work intensity, transparency and predictability, and health and safety), whether workers are engaged in conventional or algorithm-driven work settings.
- The OSH Framework Directive underscores employers' duties regarding safety and health in all work-related aspects. It mandates preventive and protective measures, proactive adaptation to changing circumstances, informing, and training workers about safety and health risks, and involving them by means of consultation. These safeguards are crucial to managing the physical and psychosocial risks associated with use of AM systems.
- One of the core structural limitations of the EU *acquis* relates to the scope of the directives mentioned above, which leave self-employed workers unprotected. Other challenges that may be relevant to AM systems around working time include the binary distinction between working time and rest time, the lack of indications as to how to compensate for 'productive' and 'unproductive' working time, and insufficient protection against underemployment. Admittedly, these concerns are not unique to AM, but represent broader challenges.
- The risk-based approach adopted in existing and proposed instruments, such as the OSH Framework Directive, the AI Act and the proposed Platform Work Directive, is in principle suitable for addressing some of AM-related OSH risks, such as the routinisation and standardisation of work, diminished worker decision-making autonomy, heightened health and safety hazards and deteriorating work-related health provided that the assessment and mitigation processes are carried out iteratively throughout all phases of AM.

3.2.1. Working time, and the changing nature of work and working arrangements

3.2.1.1. EU legal framework

A. Workload, work intensity and working time qualification

The automation of work processes has the potential to allow **increased flexibility and job autonomy, as well as reduced workload and work intensity**. Algorithms can be programmed in a way that considers both the managerial needs and workers'

preferences.[260] AM systems can optimise work performance, streamlining and prioritising tasks more efficiently on the basis of deadlines and priorities. They can reduce the need for long hours and overtime and increase the predictability of work schedules. Moreover, AM can increase workers' autonomy, ultimately improving their well-being in the workplace.[261] However, AM systems can also have the reverse consequences, namely **work intensification and deterioration in the quality of working time**.[262] In some segments of the labour market, AM systems incentivise workers to work according to the goals set by algorithmic tools.[263] Various algorithmic control mechanisms are deployed to make workers increase their working pace and the number of tasks they undertake. Some AM tools may **nudge** workers to extend their availability[264] or to work long and/or anti-social hours.[265] For example, workers with greater availability may have a higher assessment, access to more rewarding tasks, or get a bonus for work performed during a peak period as in the model of platform work. Failure to follow such nudges and pre-determined goals might have negative consequences, such as lower performance rankings, which could lead to a decrease in income or disciplinary sanctions. Moreover, algorithmic control systems, especially if they are insufficiently transparent, generate a feeling of job insecurity, which pushes workers to work longer hours in order to achieve their algorithmically set targets.[266] Another AM-related factor that may lead to work intensification and increased workload is connected with the use of **performance metrics** to create team ratings indicating the top performers, which results in a highly competitive work environment.[267]

Although there are currently **no EU instruments that address these issues directly, some overarching legal principles and provisions can curb excessive workload and work intensity**.

A worker's right to working conditions that respect his or her health, safety, and dignity, as well as the right to a limitation on maximum working hours, daily and weekly rest periods, and an annual period of paid leave, are enshrined in Art. 31 of the Charter of Fundamental

[260] Zhang, A., Boltz, A., Wang, C.-W., & Lee, M.K. (2022) Algorithmic Management Reimagined for Workers and By Workers: Centering Worker Well-Being in Gig Work; Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '22); New Orleans, LA, USA. 29 April–5 May 2022; Parent-Rocheleau, X., & Parker, S.K. (2022). Algorithms as Work Designers: How Algorithmic Management Influences the Design of Jobs. *Hum. Resour. Manag. Rev.* 2022;32:100838; Lippert, I., Kirchner, K., & Saunders, C. (2023). The Dynamic Relationships between Algorithmic Management and Workers' Occupational Well-being: A Job Demands-Resources Perspective. Proceedings of the ECIS 2023 Research Papers, available at https://backend.orbit.dtu.dk/The_Dynamic_Relationships_between_Algorithmic_Management_and_Work.pdf

[261] Kinowska, H., & Sienkiewicz, Ł. J. (2023). Influence of algorithmic management practices on workplace well-being – evidence from European organisations. *Information Technology & People* Vol. 36 No. 8, pp. 21-42, Emerald Publishing Limited.

[262] Baiocco, S. et al. (2022). 'The algorithmic management of work and its implications in different contexts' (2022). JRC Working Papers Series on Labour, Education and Technology 2022/02, available at https://www.econstor.eu/handle/10419/262292; Piasna, A. (2023). Algorithms of time: how algorithmic management changes the temporalities of work and prospects for working time reduction. *Cambridge Journal of Economics.*

[263] Parent-Rocheleau, X. & Parker, S.K. (2022). Algorithms as work designers: How algorithmic management influences the design of jobs, Human Resource Management Review 32(3) 100838, p. 6, which contains references to warehouses, the cleaning sector, and digital labour platforms. For evidence on the link between the deployment of algorithmic management and the determination of the pace of work in regular workplaces, see: Eurofound (2020). Employee monitoring and surveillance: The challenges of digitalisation, Publications Office of the European Union, Luxembourg (as referred to in Baiocco, S. et al. (2022), p. 26.

[264] Piasna, A. (2023). Algorithms of time: how algorithmic management changes the temporalities of work and prospects for working time reduction. *Cambridge Journal of Economics.*

[265] Mostly evidenced in the gig economy sector. See, for example, Wood, A.J., Graham, M., Lehdonvirta, V., & Hjorth, I. (2019a). Good gig, bad gig: autonomy and algorithmic control in the global gig economy, *Work, Employment and Society*, 33(1):56-75.; Work in the Planetary Labour Market, Fairwork Cloudwork Ratings 2021, available at https://fair.work/wp-content/uploads/sites/17/2021/06/Fairwork-cloudwork-2021-report.pdf. Similarly, for accounts from Amazon warehouses, see Delfanti, A. (2021) Machinic dispossession and augmented despotism: Digital work in an Amazon warehouse, *New Media & Society* 23(1) 39-55, p. 50.

[266] Urzí Brancati, C., & Curtarelli, M. (2021). Digital tools for worker management and psychosocial risks in the workplace: evidence from the ESENER survey, JRC125714, p. 6, Seville: European Commission.

[267] Levy, K.E. (2015). The contexts of control: Information, power, and truck-driving work, *The Information Society*, 31 (2), pp. 160-174. See also: Parent-Rocheleau, X., & Parker, S.K. (2022), p. 7, for further references.

Rights of the European Union. The **Working Time Directive (WTD)**[268] gives substance to this provision, **setting a minimum standard for the organisation of working time to protect workers' safety and health**. Among other provisions, this Directive stipulates that rest periods must be adequate, i.e. sufficiently long and continuous. The daily rest period of at least 11 consecutive hours within a 24-hour period must be observed (Art. 3). Every worker is entitled to a rest break if the daily working time exceeds six hours (Art. 4). Art. 5 WTD determines the right to a **weekly rest period,** stating that every worker should have an uninterrupted rest period of 24 hours plus the daily rest period of 11 hours per seven-day period.[269] In addition, the average weekly working time may not exceed 48 hours, including overtime (Art. 6 lit. b). Art. 6 WTD provides for the **maximum weekly working time**. MS must ensure that the period of weekly working time is limited, and the average working time for each seven-day period does not exceed 48 hours.[270] The WTD does not specify whether its provisions set absolute limits in the case of concurrent contracts with one or more employer(s), or if they apply to each employment relationship separately. This is a **broader question that pertains to the interpretation and application of the Directive**. However, in light of the Directive's objective to improve the health and safety of workers, the limits on average weekly working time, and daily and weekly rest, should, as far as possible, apply per worker.[271] As clarified on several occasions by the Court of Justice of the European Union (CJEU), given the fundamental nature of the right to health and safety of workers, the application of this right should not be made subordinate to purely economic considerations.[272]

The benchmarks in the WTD, although **established in relation to traditional work environments**, **are also applicable to workplaces that apply AM systems.** The aforementioned provisions setting minimum rest periods, maximum working time, etc., are equally relevant to workers whose schedule is set algorithmically as to those who work in more conventional settings.

However, **some specificities of AM might challenge the effectiveness of the provisions of the WTD in AM-driven contexts.** The archetypal working relationship, to which the existing EU legal *acquis* is predominantly tailored, is based on a 'time-centred' model with regular, linear, continuous and predictable working days.[273] The WTD provides for a binary divide between 'working time' and a 'rest period'.[274] Some AM systems, not only in the platform economy,[275] increase working time fragmentation by dividing work assignments into ever-smaller units of precisely measured time, which signifies a departure from the prevalent linear paradigm.[276] While the problems with delineating working time

[268] Directive 2003/88/EC of the European Parliament and of the Council of 4 November 2003 concerning certain aspects of the organisation of working time.

[269] According to Art. 16 WTD, the Member States may determine a reference period of 14 days for the weekly rest period. This period may be reduced to 24 hours only if there are objective, technical, or work organisation circumstances. MS may provide for a reference period of up to four months (Art. 16(b)).

[270] An opt-out from this provision is possible upon the completion of requirements set out in Art. 22 of the WTD.

[271] In judgment C-585/19, Academia, the CJEU stated that when workers have several employment contracts with the same employer, the minimum daily rest period applies to those contracts taken as a whole. See the Report on the implementation by Member States of Directive 2003/88/EC concerning certain aspects of the organisation of working time {SWD(2023) 40 final}. Note, however, that MS practices vary considerably on this point.

[272] Judgment of the Court, 14 May 2019, Federación de Servicios de Comisiones Obreras (CCOO) v Deutsche Bank, C-55/18, paragraph 66; Judgment of the Court, 9 September 2003, Jaeger, C-151/02, EU:C:2003:437, paragraphs 66 and 67; Judgment of the Court, 26 June 2001, BECTU, C-173/99, EU:C:2001:356, paragraph 59.

[273] Piasna, A. (2023). Algorithms of time: how algorithmic management changes the temporalities of work and prospects for working time reduction. *Cambridge Journal of Economics.*

[274] The WTD defines working time as 'any period during which the worker is working, at the employer's disposal and carrying out his activity or duties, in accordance with national laws and/or practice' (Art. 2(1)). Conversely, the term 'rest period' is defined as any period that is not working time (Art. 2(2)).

[275] Piasna, A. (2023). Algorithms of time: how algorithmic management changes the temporalities of work and prospects for working time reduction. *Cambridge Journal of Economics, p. 7,* referring to on-location platform work, freelance and microwork platforms, as well as service sectors including retail, warehouses, care and hospitality.

[276] In the words of Piasna, working time can become 'a patchwork of ever-shorter units of paid working time scheduled in irregular and discontinuous patterns'. See Piasna, A. (2023). Algorithms of time: how algorithmic management changes the temporalities of work and prospects for working time reduction. *Cambridge Journal of Economics.*

from rest periods have long existed in traditional contexts,[277] **the adoption of AM at work gives new prominence to the question of the classification of stand-by time, potentially posing novel interpretative challenges**. AM systems can lead to an 'artificial' separation between 'productive' and 'non-productive' segments of time (these two definitions are contentious too), increasing waiting time between units of working time devoted to the actual time performance.[278] Moreover, work-related activities executed offline may be mistakenly classified as 'idle time' by systems that track workers' online progress at work. [279]

This problem with **delineating the difference between working time and rest time** became prominent in the context of the platform economy, and its **implications could reverberate in an AM-driven workplace as well**. The scholarly debate on how to calculate the time that platform workers (e.g. drivers or riders) spend on a platform ready to take up assignments, without actively working, is unsettled. Some scholars have argued that such time should be fully considered as working time under the WTD, given that workers are significantly constrained in using their stand-by time for other purposes.[280] This was also expressed by the UK Supreme Court in the *Uber vs. Aslam* judgement.[281] At the same time, the literature warns against making decisive statements regarding the classification of the stand-by time of platform workers, flagging up that it might turn out to be 'the next [legal] battleground'.[282] In some contexts, the entire work system is based on being in a **stand-by, 'on-call' mode**,[283] which is different from a typical scenario in which a worker has a regular work schedule and, in addition, has to be reachable during some stand-by periods.[284] The problem stemming from this legal uncertainty has been recognised by the European Parliament, which called on the Commission and the MS 'to ensure that waiting time and being available on the platform is considered as working time for platform workers in an employment relationship'.[285] This volatile model prevailing in the platform economy could be replicated in other sectors, including for blue-collar workers who become engaged in distinct, individual tasks rather than working within a regular and linear timespan. To date, little is known about the classification of working time in algorithmically managed settings outside the platform economy.

A related question is whether **stand-by time needs to be remunerated on the same basis as 'active' or 'productive' working time**. As argued by Katsabian and Davidov in a reflection on platform work, while this should be the case under the current legal framework,[286] the remuneration of all those who make themselves available to work (e.g. by

[277] There is a rich supply of jurisprudence from the CJEU on the notion of 'stand-by time'. As found in *Matzak,* Judgment of the Court (Fifth Chamber) of 21 February 2018 in Case C-518/15 *Ville de Nivelles v Rudy Matzak*, stand-by time counts as working time if the worker is obliged to remain at a place specified by the employer, and if this limits his/her personal and social interests. Subsequent judgments of the CJEU further clarified that stand-by time can be considered working time only if the constraints imposed on the worker during that period objectively and *very significantly* reduce their freedom to manage the time during which their professional services are not required, and to devote that time to pursuing his interests. See *RJ v Stadt Offenbach am Main*, 9 March 2021; Judgment of 11 November 2021 in Case C-214/20, *MG v Dublin City Council*.

[278] Piasna, A. (2023). Algorithms of time: how algorithmic management changes the temporalities of work and prospects for working time reduction. *Cambridge Journal of Economics.*

[279] De Stefano, V., & Taes, S. (2023). Algorithmic management and collective bargaining, *Transfer: European Review of Labour and Research* 29(1), p. 27.

[280] Katsabian, T., & Davidov, G. (2023). Flexibility, choice, and labour law: The challenge of on-demand platforms. *University of Toronto Law Journal*, *73*(3), 348-379.

[281] *Uber BV v Aslam*, [2021] UKSC 5.

[282] Katsabian, T., & Davidov, G. (2023). Flexibility, choice, and labour law: The challenge of on-demand platforms, *University of Toronto Law Journal* 73(3), p. 350.

[283] Davies, A.C.L. (2020). Wages and Working Time in the 'Gig Economy', *King's Law Journal,* 31:2, 250-259.

[284] Katsabian, T., & Davidov, G. (2023). Flexibility, choice, and labour law: The challenge of on-demand platforms, *University of Toronto Law Journal* 73(3), p. 350, p. 363.

[285] European Parliament resolution of 16 September 2021 on fair working conditions, rights, and social protection for platform workers – new forms of employment linked to digital development (2019/2186(INI)), para. 28.

[286] See also, to this effect, Uber BV v Aslam, (2021), UKSC 5, para. 134.

being logged into the app or platform), but who are not actively working, could render the current on-demand system unsustainable.[287] This statement could arguably be extended to other algorithmically managed business models that are reliant on a large pool of workers who are remunerated only for the time they actively spend on task performance. However, automation tools can be employed to pinpoint areas for work process optimisation; therefore, **the negative implications often associated with excessive stand-by time in the context of platform work may not necessarily be replicated across all traditional sectors.**

Due to its legal basis and to the limited competences in this area (Art. 153(5) TFEU), the WTD does not regulate the issue of the **remuneration** of workers for working time, including stand-by and waiting time,[288] and the CJEU has not yet ruled on this issue.[289] The problem of **income instability** links back to the salient issue of work intensification: workers are often pushed to take up extra assignments and extend their work availability for purely economic reasons.[290] Given that technology is used to optimise work performance, leading to a reduction in the time actively spent on individual tasks, **the questions on payment for the 'unproductive' working time may become even more relevant in the future**.

Additional safeguards are provided under the proposed **Platform Work Directive (PWD)**, Art. 7(1) of which requires digital labour platforms to 'regularly monitor and evaluate the impact of individual decisions taken or supported by automated monitoring and decision-making systems/ on working conditions, including working time. In addition, Art. 7(2) stipulates that **automated monitoring and decision-making systems should not be used by platforms in a way that puts 'undue pressure' on platform workers, or otherwise puts at risk their physical and mental health**.[291] Platforms should ensure sufficient human resources for monitoring the impact of AM on working time. Thus, platform workers with the status of employee will benefit from protection under WTD, and will enjoy an additional layer of protection against the potentially negative effects of AM.

Another EU legal instrument that can be indirectly relevant for the **mitigation of the negative effect of extended availability for work and increased workload** is the **Work-Life Balance Directive (WLBD)**.[292] Algorithmically distributed time slots may not always fit in neatly with family commitments, rendering it difficult to fulfil responsibilities as caregivers.[293] In this regard, the WLBD, which has as one of its main aims to better **uphold work-life balance for parents and carers**, provides meaningful safeguards. Among other provisions, workers with children up to a specified age, as well as carers, have the right to request **flexible working arrangements for caring purposes**. Employers should consider and respond to requests within a reasonable period of time, taking into account the needs of both the employer and the worker (Art. 9). Reasons for any refusal of such a request or any postponement of such arrangements should be provided. Less favourable treatment of workers on the ground that they have applied for or taken flexible work arrangements, should be prohibited (Art. 14). These provisions promote **work-life reconciliation and flexible work patterns in both traditional and AM-managed settings**.

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[287] See Katsabian, T., & Davidov, G. (2023).

[288] The only exception to this is the provision of paid annual leave. As laid down by Article 153 (5) TFEU, pay is an area that is excluded from the EU's competences.

[289] In *Yodel*, the CJEU was asked about the methods used to calculate the working time of couriers who are not required to work fixed hours but are free to determine their working hours within certain parameters. What appeared problematic for the referring court was that couriers may, during the time spent on parcel delivery spend time on different (even personal) activities, and simultaneously provide their services to other undertakings. However, the CJEU remained silent on this issue, focusing instead only on the consideration of whether such couriers qualify as 'workers' under the WTD. See *Order of the Court (Eighth Chamber) of 22 April 2020. B v Yodel Delivery Network Ltd.*

[290] Piasna, A. (2023). Algorithms of time: how algorithmic management changes the temporalities of work and prospects for working time reduction. *Cambridge Journal of Economics*).

[291] For a more detailed discussion of this provision, see subsection on 3.2.2. Occupational safety and health.

[292] Directive (EU) 2019/1158 of the European Parliament and of the Council of 20 June 2019 on work-life balance for parents and carers and repealing Council Directive 2010/18/EU.

[293] Warren, T. (2021). Work–life balance and gig work: 'Where are we now' and 'where to next' with the work–life balance agenda?' *Journal of Industrial Relations*, 63(4) 522.

Lastly, relevant safeguards against the potential negative impacts of extended availability, to which AM can contribute, can also be found in the **OSH Framework Directive.** As further detailed in the section that follows, this instrument obliges employers to **assess all risks, including emerging risks brought about by new technologies at work, to take the necessary measures to avoid risks and protect workers, and to inform and/or consult with workers and their representatives**. Thus, the OSH Framework Directive contributes to the prevention of psychosocial risks related to the algorithmic organisation of working time, including excessive workload and extended availability.[294]

Given the requirements stemming from Art. 153 TFEU, the instruments discussed above apply only to **workers** in an employment contract or relationship.[295] Considering that the CJEU still largely relies on the expansive 'Lawrie Blum' formula[296] of 'worker', many casual workers and the self-employed, whose work can also be algorithmically managed, may be left unprotected.[297] This is all the more problematic, given that AM systems may in principle facilitate the process of **contracting work outside the standard employment relationship,** since they provide various tools for direct and indirect control (e.g. nudges and ratings) that can ensure a smooth workflow even without the engagement of subordinate workers.[298] While the exercise of algorithmic control via nudging tools can constitute an indicia of an employment relationship,[299] not all work arrangements relying on AM systems will qualify as an employment relationship.[300] Externally contracted workers whose schedules are algorithmically driven can be especially affected by the unpredictability of work schedules, as they are subject to on-demand calls from companies to a greater extent.[301] **The use of AM systems for the governance of nominally self-employed workers poses serious issues with regard to the applicability of EU legal instruments to such arrangements**. The *Yodel* case,[302] which concerned the notion of 'worker' under the WTD in the context of an on-demand business model, illustrates this challenge. The CJEU noted that the requirement to provide a parcel delivery service during specific time slots[303] is 'inherent to the very nature of that service since compliance with

[294] Visionary Analytics (2023). Study exploring the social, economic and legal context and trends of telework and the right to disconnect, in the context of digitalisation and the future of work, during and beyond the COVID-19 pandemic, forthcoming, p. 98.

[295] The WTD speaks of 'workers' without explicitly defining them. The directives on atypical employment apply to workers hired under an employment contract or relationship defined in law, collective agreements, or practice in each MS (Clause 2(1)). The TPWCD applies to every worker 'who has an employment contract or employment relationship, as defined by the law, collective agreements, or practice in force in each Member State with consideration to the case-law of the Court of Justice' (Art. 1 (2)). The same wording is used in the Work-Life Balance Directive and the proposed Platform Work Directive.

[296] The 'Lawrie-Blum formula' or 'Lawrie-Blum doctrine' refers to the essential criteria defining the notion of a worker, formulated in the *Deborah Lawrie-Blum v Land Baden-Württemberg* judgment of 3 July 1986 (Case 66/85). These crucial elements of the employment relationship are that 'for a certain period of time a person performs services for and under the direction of another person in return for which he receives remuneration'. See Kountouris, N. (2018). The concept of 'worker' in European labour law: fragmentation, autonomy and scope, *Industrial Law Journal* 47, no. 2: 192-225.

[297] Georgiou, D. (2022). The new EU Directive on Transparent and Predictable Working Conditions in the context of new forms of employment, *European Journal of Industrial Relations*, 28(2: 193, accessed 24 May 2022, p. 201.

[298] See: Piasna (2023), p. 8; also Wood, A.J. (2021). Algorithmic Management Consequences for Work Organisation and Working Conditions, Seville: European Commission, who notes that 'a potential consequence of the wider of adoption of algorithmic management beyond platform work may be an acceleration and expansion of fissured employment relations', p. 9.

[299] See the case law on digital labour platforms, mostly in the delivery and transport sector, but also on-location crowdwork, e.g., decision 9 AZR 102/20 of the Federal Labour Court. For an overview of case law reclassifying platform workers as employees, see Christina Hiessl, 'The Classification of Platform Workers in Case Law: A Cross-European Comparative Analysis Collective Bargaining for the Self-Employed' (2021) 42 Comparative Labor Law & Policy Journal 465.

[300] For example, freelancers working through online platforms such as Upwork are commonly considered to be genuinely independent, even though they are subject to algorithmic control. See e.g., Jarrahi, M.H., Sutherland, W., Nelson, S.B. *et al.* (2020), Platformic Management, Boundary Resources for Gig Work, and Worker Autonomy. *Comput Supported Coop Work* 29, 153–189; Bucher, E.L. et al. (2020). 'Pacifying the algorithm – Anticipatory compliance in the face of algorithmic management in the gig economy.' *Organization* 28: 44 - 67.

[301] Baiocco, S., Fernández-Macías, E., Rani, U., & Pesole, A. (2022). *The algorithmic management of work and its implications in different contexts*. No. 2022/02. JRC Working Papers Series on Labour, Education and Technology, p. 22.

[302] Order of the Court (Eighth Chamber) of 22 April 2020. *B v Yodel Delivery Network Ltd*.

those time slots appears essential to ensure the proper performance of that service'.[304] This **significant discretion over working time** was one of the factors that led the Court to suggest that the courier enjoys 'a great deal of latitude in relation to his putative employer',[305] and that his independence does not appear to be fictitious.[306] Unless this independence is merely notional and the existence of a subordinate relationship can be established, a self-employed person could be precluded from being classified as a 'worker' under the WTD.[307] On the other hand, algorithmic control mechanisms are increasingly being recognised by national courts across Europe as a sign of control and ultimately of an employment relationship.[308]

B. Unpredictability and lack of transparency over timeslot allocation

AM tools allow the **optimisation of working time** by generating work schedules based on the **workers' availability and real-time predictions of customer demands and market fluctuations.** Such 'just-in-time' systems are increasingly applied in conventional workplaces across different sectors.[309] On the one hand, when programmed to incorporate workers' preferences and capacities, they may contribute to workers' physical and mental well-being.[310] On the other hand, however, frequent changes in work schedules and their announcement at short notice make the organisation of working time **unpredictable**.[311] Schedules can be erratic and volatile, leaving workers with **little 'time sovereignty' and control over the work pace and labour process**. This may further lead to poor work-life balance and income instability.[312] Moreover, the mechanisms governing the allocation of tasks to workers may not always be fully transparent, which renders negotiations over working hours challenging for workers and their representatives.[313]

Relevant safeguards against some of these challenges are provided under the **Transparent and Predictable Working Conditions Directive (TPWCD)**. The

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[303] The service agreement specified that parcels must be delivered between 07.30 and 21.00. Within this time window, couriers could organise the order and the timing of the deliveries, tailoring their working time to their personal needs and preferences, rather than solely according to the interests of the parcel delivery company.

[304] Ibid., para 42.

[305] Order of the Court (Eighth Chamber) of 22 April 2020, *B v Yodel Delivery Network Ltd*, para 35.

[306] Ibid., para 43. The other key factors taken into account by the Court were: the freedom to use subcontractors or substitutes to perform the service; the freedom to accept or reject the tasks offered by the company and to decide about the maximum number of those tasks; and the possibility to provide services to a third party, including the employers' competitors.

[307] Ibid., para 36.

[308] This is most visible in the growing body of case law on the employment status of platform workers. For a comprehensive overview of judicial and administrative decision on the employment classification of platform workers in Europe, see for example, Hiessl, C. (2022). The classification of platform workers in case law: A cross-European comparative analysis. *Comparative Labour Law & Policy Journal,* vol. 42, no. 2, 465–518; Potocka-Sionek, N. (2023). Easier done than said? An Empirical Analysis of Case Law on Platform Work in the EU'. *Hungarian Labour Law e-Journal, 2023/*1: 46.

[309] For example, in the building security sector, data collected through video cameras and building sensors is used to automate decisions about the times at which to deploy security guards, while in customer service, AM systems measure in-store customer traffic, estimate sales productivity scores for each worker, and generate work schedules on that basis. Bernhardt, A., Kresge, L., & Suleiman, R. (2023). The Data-Driven Workplace and the Case for Worker Technology Rights, *ILR Review* 76(1), p. 7-8, with further references. Examples of such systems can also be found in the retail, food and hospitality industries, as documented in Loggins, K. (2020, 24 February). Here's What Happens When an Algorithm Determines Your Work Schedule, *VICE,* available at: https://www.vice.com/en/heres-what-happens-when-an-algorithm-determines-your-work-schedule, accessed 28 October 2023. See also Kantor, J. (2014). Working Anything but 9 to 5, *New York Times*, 13 August 2014, U.S. section, available at: https://www.nytimes.com/interactive/ 2014/08/13/us/starbucks-workers-schedulinghours.html, as referred to in Unruh, C.F. et al. (2022). Algorithmic Scheduling in Industry: Technical and Ethical Aspects.

[310] Unruh, C.F. et al. (2022). Algorithmic Scheduling in Industry: Technical and Ethical Aspects, p.5.

[311] Adams-Prassl, J. (2022). Regulating algorithms at work: Lessons for a 'European approach to artificial intelligence'. *European Labour Law Journal*, *13*(1), 30-50.

[312] Jarrahi, M.H. et al. (2021). Algorithmic management in a work context, Big Data & Society, 8(2) 205395172110203, p. 4.

[313] De Stefano, V., & Taes, S. (2023). Algorithmic management and collective bargaining. *European Review of Labour and Research*, 29(1), 21-36.

effectiveness of the Directive has not been assessed within the framework of AM systems, and there have been no rulings to validate its application in this specific context.[314] Nonetheless, it holds the potential to introduce **meaningful and effective rules that can help to minimise the lack of predictability and enhance the transparency of scheduling and programming**. By and large, the goal of this instrument is to improve working conditions by **promoting more transparent and predictable employment patterns**, while ensuring labour market adaptability (Art. 1(1)). The Directive provides workers with a set of **information rights**, concerning 'essential aspects of the employment relationship', including how workers will be paid for the additional hours worked, when exactly their assignments will start, and within what timeframe a call can be cancelled. Its fundamental premise is that **increased awareness serves as a catalyst for enhancing working conditions**, a notion that remains highly pertinent, particularly within the framework of AI-driven settings.

The TPWCD provides additional information rights for workers whose work patterns are 'entirely or mostly unpredictable'. These kinds of work arrangements could be particularly relevant in the context of AM, given their prevalence in 'just-in-time' models. In such cases, Art. 4(2)(m) provides that workers should be informed of the very fact that the work schedule is variable; of the number of guaranteed paid hours and the remuneration for work performed in addition to those guaranteed hours; of the reference hours and days within which the worker may be required to work; of the minimum notice period to which the worker is entitled before the start of a work assignment and, where applicable, the deadline for cancellation. Moreover, under the Directive, the length of a worker's standard working day or week, any arrangements for overtime and its remuneration and, where applicable, any arrangements for shift changes, must be communicated to workers with an entirely or mostly unpredictable schedule (Art. 4(2)(l)). An important limitation of this catalogue is the lack of inclusion of **data collection** as one of the 'essential aspects of the employment relationships' about which workers must be informed.[315] This issue could be addressed by applying these rules in conjunction with the GDPR provisions.

In addition to the information rights laid down in Art. 4(2), Art. 10 of the TPWCD sets out rules for **minimum predictability of work patterns that are entirely or mostly unpredictable**. According to this provision, a worker whose work pattern is entirely or mostly unpredictable cannot be required to work unless the work takes place within the predetermined reference hours, and the worker is informed of a work assignment within a reasonable notice period. Art. 11 TPWCD, in turn, provides that MS may decide to **limit the use and duration of on-demand or similar employment contracts**, to introduce a rebuttable presumption of an employment contract, with a minimum amount of paid hours calculated on the basis of the average hours worked during a given period, or to take 'other equivalent measures' to prevent the abuse of on-demand contracts.[316] Moreover, according to Art. 12 TPWCD, workers who have worked for at least six months with a given employer can request a transition to 'more predictable and secure' employment, if available. The employer does not have to accept this request but is only obliged to provide a 'reasoned' reply. All these provisions afford important protective mechanisms against working time insecurity, to which AM systems can contribute.

[314] In particular, the question remains open as to whether the TPWCD can oblige an employer to inform workers about the existence and logic of at least those algorithms which are covered by the catalogue of information rights under Art. 4 of the Directive. As suggested by Hiessl, this could provide workers with an alternative path to claim information rights to that set out under the GDPR, in case such information rights cannot be based on the latter instrument. See Hiessl, C. (2022). Algorithmic Management in the Workplace: Taking Stock of Case Law and Litigation in Europe, Hungarian Labour Law e-Journal 2022/2, p. 20.

[315] Adams-Prassl, J. (2022). Regulating algorithms at work: Lessons for a 'European approach to artificial intelligence', *European Labour Law Journal* 13(1): 30.

[316] Bednarowicz, B. (2019). Delivering on the European pillar of social rights: the new directive on transparent and predictable working conditions in the European Union. *Industrial Law Journal*, *48*(4), 604-623.

The TPWCD can be seen as an instrument complementary to the WTD, integrating with and advancing its provisions,[317] and offering reliable safeguards for all workers with short, variable, and unpredictable work hours, including algorithmically managed ones. Importantly, **the TPWCD provides a set of redress mechanisms to stimulate the enforcement of its provisions**.[318]

Moving on to **platform work arrangements**, the **proposed PWD** guarantees persons performing platform work the right to be informed by the platforms on which they work about automated decision-making systems used to take or support decisions that significantly affect their working conditions (Art. 6 (1)(b)). **Working time is explicitly mentioned** as one of the aspects covered by this important provision. Such information should cover the fact that such AM systems are used or are in the process of being introduced, the categories of decisions taken or supported by these systems, the main parameters taken into account by these systems, and their relative importance (Art. 6(2)). Importantly, the right to information in this regard applies to all persons performing platform work, regardless of their employment status (Art. 10(1)). Moreover, as per Art. 8, platform workers are entitled to an explanation from the platform with regard to a decision made or supported by an automated decision-making system that significantly affects their working time. They should have access to a contact person designated by the platform to discuss and clarify the facts, circumstances and reasons that have led to the decision. **These provisions foster the transparency of AM systems with regard to task allocation and shift scheduling in this specific context, making it possible to mitigate potential bias and challenge decisions that may have negative effects on workers**.

Transparency in this regard is also fundamental to ensuring that workers' entitlements under other EU instruments, such as the WTD and WLBD, are realised. However, **the proposed PWD does not mandate transparency of working conditions as such.** Unlike the TPWCD, it does not give platform workers the right to information about their work schedules. Consequently, only those platform workers who fall within the 'worker' definition provided under the TPWCD will benefit from information rights about working conditions, including working time scheduling under that instrument. Thus, not all persons performing platform work are protected against working hour insecurity.

C. Communication

AM systems can also be used to **enable remote collaboration**, allowing employees to work from anywhere. In terms of faster and better communication, AM tools ensure that information is readily accessible to all and provide insights into work progress. These data can, in turn, be used to ensure the equitable distribution of workload and offer constructive feedback for improvement. **Team dynamics and communication can be enhanced by AI-powered tools such as virtual meeting platforms, chatbots or real-time communication infrastructure**, whether in traditional or remote work settings. Information can be shared more effectively, thus enhancing the productivity of workers, even if they work in a highly flexible way, outside the office premises.

Against this background, AM systems and AI-based tools or platforms have been hailed as great facilitators and **catalysts of remote work and telework**.[319] In this regard, the 2002

[317] Aloisi, A., Rainone, S., Countouris, N. (2023), *An unfinished task? Matching the Platform Work Directive with the EU and international 'social acquis'*, ILO Working Paper (Geneva, ILO).

[318] MS must ensure that workers, including those whose employment relationship has ended, have access to effective and impartial dispute resolution and a right to redress in the case of infringements of their rights arising from this Directive (Art. 16). Moreover, adequate measures must be ensured to protect workers against any adverse treatment by the employer and any adverse consequences resulting from a complaint lodged with the employer or resulting from any proceedings initiated with the aim of enforcing compliance with the rights provided for in this Directive (Art. 17). Art. 18 provides workers with further safeguards against dismissal (or its equivalents) on the grounds that they have exercised the rights provided by the Directive.

[319] See, for example, https://www.forbes.com/sites/glebtsipursky/2023/05/09/the-ai-revolution-transforming-hybrid-and-remote-work-and-the-return-to-office; https://virtualspace.ai/blogs/the-future-of-remote-work-6-top-ai-use-cases-and-tools

social partners' **Framework Agreement on Telework**[320] needs to be analysed. This defines telework as a voluntary 'form of organising and/or performing work, using information technology, in the context of an employment contract/relationship, where work, that could also be performed at the employer's premises, is carried out away from those premises on a regular basis'. Notably, **the Agreement establishes a wide-ranging principle of non-discrimination regarding employment conditions between teleworkers and comparable workers at the employer's premises** (Art. 4). The workload and performance standards of the teleworker need to be the same as those of comparable workers at the employer's premises. The employer must guarantee data protection (Art. 5) and implement training procedures for teleworkers (Art. 10). In addition, employers are in charge of supplying, setting up and maintaining the essential equipment necessary for ongoing telework, unless the teleworker opts to use their own equipment (Art. 7). Moreover, it is the duty of the employer to adopt measures preventing the teleworker from being isolated from the rest of their co-workers (Art. 9). These general principles can be **implemented via complementary collective and/or individual agreements**. All the provisions of the Agreement are also relevant to algorithmically managed teleworkers, even if though were tailored to another 'generation' of technologically driven work arrangements and are **not AM-specific**. AM tools can foster the application of these provisions – for instance, by improving communication and thus mitigating social isolation from other co-workers. However, the Agreement does not directly address such issues as algorithmic transparency, the right to disconnect or other measures that might protect algorithmically managed teleworkers from heightened levels of exposure to communication. The true effectiveness of the Framework Agreement on Telework can only be assessed by examining its **implementation** at national level, a process that is heavily contingent on the quality and strength of industrial relations within each MS.

The EU cross-industry social partners Work Programme for the social dialogue for 2022-24, signed in June 2022 between the European Trade Union Confederation and European business associations, foresaw the **review and update of the 2002 Framework Agreement on Telework**, with an attempt to address the right to disconnect. The objective of the social partners was to agree on a new, binding agreement to be turned into EU law.[321] That notwithstanding, in November 2023, social partners closed their negotiations without an agreement on a new text.

At the same time, the **Autonomous Framework Agreement on Digitalisation**,[322] signed in June 2020 aims to address the issues around the modalities of connecting and disconnecting and sets out measures that aim to facilitate the respect of working time rules and teleworking and mobile work rules. The national-level implementation of this Agreement is underway, exhibiting varying degrees of effectiveness.

Another potential impact of AM on communication at the workplace is **worker alienation**.[323] The risk is that 'workers can become mere parts of a production chain',[324] and that the social interactions will become minimal. This problem has been salient in the platform

[320] See https://resourcecentre.etuc.org/sites/default/files/2020-09/Telework%202002_Framework%20Agreement%20-%20EN.pdf

[321] European Social Dialogue Work Programme 2022–2024, available at https://www.businesseurope.eu/reports_and_studies/2022-06-28_european_social_dialogue_programme_22-24_0.pdf. See also the European Parliament resolution of 21 January 2021 with recommendations to the Commission on the right to disconnect (2019/2181(INL)): 'The right to disconnect refers to workers' right not to engage in work related activities or communications outside working time, by means of digital tools, such as phone calls, emails, or other messages. The right to disconnect should entitle workers to switch off work-related tools and not to respond to employers' requests outside working time, with no risk of adverse consequences, such as dismissal or other retaliatory measures. Conversely, employers should not require workers to work outside working time. Employers should not promote an 'always on' work culture in which workers who waive their right to disconnect are clearly favoured over those who do not. Workers reporting situations of non-compliance with the right to disconnect in the workplace should not be penalised. The right to disconnect should apply to all workers and all sectors, both public and private'.

[322] See: https://resourcecentre.etuc.org/sites/default/files/2020-09/Telework%202002_Framework%20Agreement%20-%20EN.pdf

[323] Unruh, C.F., Haid, C., Fottner, J., & Büthe, T. (2022). Human autonomy in algorithmic management. In: *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 753-762.

[324] Todolí-Signes, A. (2021). Making algorithms safe for workers: occupational risks associated with work managed by artificial *intelligence, Transfer: European Review of Labour and Research*, 27(4): 433, p. 439.

economy,[325] but it is not clear whether this issue will automatically translate into all AM-driven settings. The proposed PWD provides safeguards in this regard by obliging platforms to create the **possibility for platform workers to contact and communicate with each other, and to be contacted by platform workers' representatives** (Art. 15). This should be made possible through the platforms' digital infrastructure 'or similarly effective means'. In addition, MS are required to ensure that digital platforms do not access or monitor such communications. Apart from this proposed instrument, which directly addresses the problem of the alienation of platform workers, no other specific instruments are tailored to this issue.

3.2.1.2. Selected national legislation and policies

A. EU countries

There are to date **no national laws specifically addressing the challenges relating to AM and working time**. At country level, the general labour law framework implementing the WTD lays down limits for maximum working hours, minimum rest periods and overtime regulations.[326] In some countries, legislative reforms to reduce maximum working time are underway.[327] The implementation of the TPWCD into national legal frameworks has brought (or, in countries that have yet to transpose this Directive, *will* bring),[328] further **safeguards concerning the predictability of working time schemes, which are also relevant in the AM context**.

Some countries – **the Netherlands** being a prime example – provide special rules concerning remuneration for on-call work. Under Art. 7:628a of the Dutch Civil Code, where a period of less than 15 working hours per week has been contracted and working times have not been fixed, or where the amount of working hours has not been fixed, the employee is, for every period of less than three hours in which he/she has performed work, entitled to the wages to which he/she would have been entitled if he/she had performed work for three hours. Furthermore, in some countries, there is a **right to minimum working hours**. In Belgium, national law regulates minimum daily working time, providing that except for derogations (laid down by royal decree or collective agreement), the length of each work session cannot be less than three hours.[329]

Moreover, risks posed by digital technologies, such as **permanent availability, work intensification and irregular work schedules, are addressed by regulations concerning work-life balance and telework**. For instance, in the **Netherlands**, companies are required to minimise the psychological stress that results from working with visual display units under the Working Conditions Decree.[330] As reported in the Commission's study on telework and the right to disconnect, specific legislation on the right to disconnect has been passed in **Belgium, Croatia, France, Greece, Ireland, Italy,**

[325] In the context of platform work, see Glavin, P., Bierman, A., & Schieman, S. (2021). Über-Alienated: Powerless and Alone in the Gig Economy. *Work and Occupations*, *48*(4), 399-431; Wood, A.J. et al. (2019). Good Gig, Bad Gig: Autonomy and Algorithmic Control in the Global Gig Economy *Work, Employment and Society*, 33(1): 56, accessed 3 November 2020; Seetharaman, B., Pal, J., & Hui, J. (2021). Delivery Work and the Experience of Social Isolation. *Proc. ACM Hum. Comput. Interact.,*5, 1–17; Duggan, J. et al. (2021). Boundaryless careers and algorithmic constraints in the gig economy, *International Journal of Human Resource Management,* 1.

[326] See the Report on the implementation by Member States of Directive 2003/88/EC concerning certain aspects of the organisation of working time {SWD(2023) 40 final}.

[327] See the Draft Bill in the UK, proposing to reduce the maximum working time from 48 to 32 hours. Available at https://publications.parliament.uk/pa/bills/cbill/58-03/0164/220164.pdf

[328] As of June 2023, the Directive has not been implemented in Luxembourg, Slovenia and Austria. See https://iuslaboris.com/insights/implementing-the-transparent-and-predictable-working-conditions-directive-across-europe/.

[329] The Labour Act of 16 March 1971. See the commentary at https://bit.ly/47n8Kai

[330] See Visionary Analytics (2023). Study exploring the social, economic and legal context and trends of telework and the right to disconnect, in the context of digitalisation and the future of work, during and beyond the COVID-19 pandemic, forthcoming, p. 160.

Luxembourg, Portugal, Slovakia and Spain.[331] However, these regulations do not contain rules regarding AM specifically.

B. Non-EU countries

Examples of regulations that specifically targeting the issue of algorithmically driven work intensification can be found in the US. In **California**, Assembly Bill 701 (AB 701, in force since January 2022) is designed to regulate the use of quotas within logistics warehouses.[332] Certain large companies in the industry are required to disclose allocations and pace-of-work standards to each employee upon hiring. Moreover, employers must share a 'written description of each quota to which the employee is subject, including the number of tasks to be performed or materials to be produced or handled, within a defined period', In addition, 'Any potential adverse employment action that may result from failure to meet the quota must also be shared. Employees can disregard any quotas that prevent compliance with meal or rest periods, the use of bathroom facilities, or health and safety laws.[333]

In a related context, the **State of New York** has advanced a targeted Warehouse Worker Protection Act (enacted on 19 June 2023). The key purpose of this is to make productivity quotas more transparent within warehouse distribution centres[334] – that is, workers (and regulators) must be provided with documentation concerning any such quotas and must be notified about any change in the company's expectations with regard to the pace of work and deliverables. The Act applies only to warehouses using quotas or monitoring workers' speed data, without obliging them to put such a mechanism in place. **Quotas that are too stringent,** thereby preventing workers from taking breaks for personal reasons, **are banned.** Workers must receive information about their quotas within the first month of being hired and have the right to access both current and previous productivity data (up to a maximum of three months). In addition, they are empowered to pursue a court injunction against any detrimental practices, while a rate of injury that is too high will trigger the possibility of a state investigation. This ensures protection against retaliation for workers who make a complaint.

3.1.1.3. Social partner agreements and initiatives

Even though, historically, **working time has been one of the major concerns for trade unions**,[335] the impact of AM on working time has so far not been directly addressed in many collective bargaining agreements. More generally, some collective agreements provide that the introduction of a new technology that 'has a significant impact on employment, work organisation, or working conditions' requires notice of at least three months. Moreover, the reasons justifying its introduction and the social consequences related to it must be explained, and workers' representatives must be consulted on the social consequences of introducing the new technology.[336]

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[331] Visionary Analytics (2023). Study exploring the social, economic and legal context and trends of telework and the right to disconnect, in the context of digitalisation and the future of work, during and beyond the COVID-19 pandemic, forthcoming, p. 78.

[332] As referred to in Aloisi, A., & De Stefano, V. (2023). Between risk mitigation and labour rights enforcement: Assessing the transatlantic race to govern AI-driven decision-making through a comparative lens, *European Labour Law Journal*, *14*(2), 283–307.

[333] De Stefano, V., & Aloisi, A. (2023). Between risk mitigation and labour rights enforcement: Assessing the transatlantic race to govern AI-driven decision-making through a comparative lens. *European Labour Law Journal,* 14 (2), 283-307. Available at: https://www.natlawreview.com/article/california-passes-legislation-regarding-job-performance-quotas-large-warehouse.

[334] This legislation covers any employer that directly or indirectly employs or has control over the wages, hours or working conditions of 100 or more employees at a single warehouse distribution centre, or 1,000 or more employees at one or more warehouse distribution centres within the state. See https://dol.ny.gov/WWPA.

[335] Keune, M. (2006). Collective bargaining and working time in Europe: an overview. *2006) Collective bargaining on working time: recent European experiences, Brussels: ETUI-REHS*, 9-29.

[336] Collective agreements in the private sector in Belgium, as referred to in: European Trade Union Confederation (2022). Neue Ansätze für Kollektivvertragsverhandlungen: kurze Handlungsempfehlungen zu zukunftsrelevanten Themen, p. 8.

An example of a specific instrument is the Spanish **national collective bargaining agreement for the sector of travel agencies** (2019-2022),[337] which set up a committee composed of business and workers representatives, to establish a 'productivity measurement system appropriate to the sectoral circumstances and the level of the normal productivity index or base period for comparisons'. The committee was tasked with 'studying issues related to technological innovation',[338] considering *inter alia* 'adequate working conditions in terms of repetitive tasks, health and hygiene', and retraining those workers affected by these digital innovations. At the time of writing, no further information is available on the results of this initiative.

Several social partner initiatives concerning **platform workers** are also worth mentioning, since they include some AM-related provisions regarding the topics discussed above. For example, during a strike in July 2017, Deliveroo riders in Spain demanded at least two deliveries per hour in order to reduce salary uncertainty stemming from the lack of predictability of algorithmically managed schedules, as well as a minimum working time of 20 hours per week.[339] Several collective agreements have granted platform workers the right to disconnect,[340] or set a maximum allowed number of deliveries per hour, with a view to preventing work intensification.[341] Moreover, some collective bargaining agreements have stipulated a minimum number of work hours. For instance, a collective agreement signed by Takeaway Express (Just Eat) in Italy guaranteed the minimum duration of daily working time and a shift lasting at least two consecutive hours, and explicitly provided that waiting time is to be remunerated. Overall, however, **the definition of working time and the criteria for pay determination have proven to be among the most controversial issues during negotiations regarding the working conditions of platform workers**.[342] In more standard settings, it could be easier to establish these terms and conditions as they are not as contentious as in the platform economy.

3.1.1.4. Other relevant initiatives

A. EU countries

The number of initiatives referring specifically to challenges related to AM and working time remains limited. In **Poland**, the 'Position on the impact of algorithmic management on the working environment', issued by the political party Razem, highlighted the risks that AM poses in relation to working time, such as the arbitrary setting of working hours and pay, or the violation of the right to disconnect from work when at home. It called for legal regulations to ensure full disclosure to employees regarding the collection and processing of data, including the methods used to collect and use such data, and the use of applications in the area of AM.[343]

B. Non-EU countries

In the **UK**, the TUC issued a manifesto, 'Dignity at work and the AI revolution', which included proposals to ensure that AI and ADM systems are properly regulated in the workplace, to make sure that they do not have an adverse impact on workers, including with regard to **work intensification**.[344]

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[337] Resolución de 29 de diciembre de 2021, de la Dirección General de Trabajo, por la que se registra y publica el Convenio colectivo laboral de ámbito estatal para el sector de agencias de viajes, para el período 2019-2022, available (in Spanish) at https://www.boe.es/eli/es/res/2021/12/29/(10)

[338] Art. 58.

[339] https://apps.eurofound.europa.eu/platformeconomydb/riders-x-derechos-riders-for-rights-103036

[340] For example, the collective agreement between Just Eat Spain and the Spanish unions CCOO and UGT, effective as of January 2022. See https://uniglobalunion.org/news/historic-victories-for-just-eat-and-deliveroo-workers-in-europe/

[341] Lamannis, M. (2023). Collective Bargaining in the Platform Economy: A Mapping Exercise of Existing Initiatives. *ETUI Research Paper-Report*, *2* p. 45.

[342] Ibid, p. 42.

[343] See: https://partiarazem.pl/stanowiska/2021/11/02/stanowisko-wplywu-zarzadzania-algorytmicznego-na-srodowisko-pracy

[344] Creagh, M. (2023). Work Intensification. The impact on workers and trade union strategies to tackle work intensification, TUC. Available at https://www.tuc.org.uk/sites/default/files/2023-07/WorkIntensificationrReportJuly2023.pdf

In **Ontario**, the 'Fair Workplaces, Better Jobs Act', introduced in 2017, provided that employers must pay a minimum of three hours for an on-call 'shift' when this is not considered working time and the employee is not called for work.[345] However, this law was repealed before it entered into force.[346]

The Chinese 'Internet Information Service Algorithmic Recommendation Management Provisions', adopted in 2022 by the Cyberspace Administration of China, stated that the providers of algorithmic recommendation services who coordinate workers' services must protect workers' rights, including the right to receive a salary, to rest and vacation. Moreover, they should establish algorithms related to assigning orders, salary composition and payment, work times, rewards, and penalties.[347]

3.1.1.5. Selected case law

Amazon case, VIII Pa 135/19, District Court in Poznań, 10 June 2020

This case concerned the **dismissal of a warehouse worker** at an Amazon fulfilment centre in Poland. The worker was dismissed **for having failed to meet minimum performance outcomes**. The assessment of workers' achievements at the centre was measured according to the number of items prepared per hour, for which required minimum target values were set by the management (the so-called 'indicators'). These indicators against which the employee's performance was compared were established every four weeks on the basis of data from other employees. Employees who made more than 120 % of the required target value received praise; those who reached a score of 100-120 % received 'basic feedback', while employees with scores below 100 % were requested to improve their performance, e.g. through 'coaching sessions'. According to the company's rules, workers could be dismissed if they received six negative evaluations over 12 months. Three criteria were considered when assessing employees: productivity, quality of results, and attendance. The minimum threshold corresponded to the result achieved by 90 % of the employees. For employees who did not reach this minimum, the training and disciplinary procedure described above were to be implemented.

In the present case, Amazon claimed that the worker had failed to achieve the required weekly standard 13 times throughout his contract, having received six negative reviews in total. The worker demanded a declaration that the termination of his employment contract was ineffective. His claim was successful. In its judgment of 6 February 2019,[348] the Regional Court in Poznań held that Amazon's internal rules were detrimental to employees, and incompatible with the **Polish Labour Code**.[349] The evaluation system assumed that not all workers would reach the required threshold. It was practically impossible for all workers to reach the norm, even if, objectively, they exercised their work with maximum diligence. In the words of the Court, this system led to the creation of a **'system akin to a race' among workers, who had to compete against each other** so as not to miss the target.[350] The assessment criteria were, according to the Court, not objective, as they were not based on any verifiable findings as to the amount of work that an employee can do per day. This system 'distorted the nature of an employment relationship'[351], the Court said, since the least productive employees were dismissed simply because their performance was 'worse' than 90 % of the workforce as a whole, without considering objectively what performance the employee could achieve by performing his or her duties conscientiously and diligently.

[345] *Fair Workplaces, Better Jobs Act*, SO 2017, adding s 21.4 to the *Ontario Employment Standards Act*, SO, 2000, c 41 [*Ontario Employment Standards Act*].

[346] *Making Ontario Open for Business Act*, SO 2018, c 14, s 27(5).

[347] See: https://www.chinalawtranslate.com/en/algorithms/#.

[348] V P 713/16, Regional Court in Poznań, 6 February 2019.

[349] Art. 9 (2) of the Polish Labour Code states that the provisions of collective bargaining agreements and collective agreements, as well as regulations and statutes, may not be less favourable to employees than the provisions of the Labour Code.

[350] V P 713/16, p. 15, reaffirmed by the District Court, VIII Pa 135/19, p. 12.

[351] V P 713/16, p. 15, reaffirmed by the District Court, VIII Pa 135/19, p. 13.

Moreover, in the case at hand, the court observed that the worker's results fell below the 90 % minimum for only eight weeks during the whole period of employment, and that the termination notice was issued not when the worker's results were lowest, but when they had improved, reaching above the minimum level.

In its ruling of 10 June 2020, the District Court confirmed the position and argumentation of the Regional Court. The Court, which held that the evaluation system was geared towards the expulsion of employees, and its essence was therefore **contrary to the basic principles of labour law and incompatible with the Polish Labour Code**. As stated by the Court, Amazon 'abused its position as an employer' by creating 'competition' among the employees'.[352] From the perspective of an objective observer, it was unattainable for all workers to meet the standard, even when exercising the utmost care and efficiency in the execution of their tasks. In other words, the system assumed that at least 10 % of workers, regardless of whether they were objectively diligent and conscientious, would receive a negative evaluation and suffer the consequences. The Court emphasised that standards and minima were not fixed; they depended on the work not only of the employee in question, but also of other employees. As these minima shifted over time, employees were left uncertain as to their ability to meet these benchmarks. The evaluation standards were not only skewed against the workers, but also unverifiable for them. Thus, the Court concluded that the worker's dismissal was unsubstantiated and ineffective.

3.1.1.6. Conclusions

There are currently **no specific rules under the EU regulatory framework that directly address AM-related changes to the organisation of work**. However, as discussed above, the existing EU legal instruments provide **meaningful safeguards against some of the key challenges related to the impacts of AM on the organisation of working time**. A composite set of provisions addressing these concerns can be pieced together by considering elements from various existing and proposed instruments. Admittedly, many of the issues posed by AM present a continuation of the broader trend of the flexibilization of working time and the growing individualisation of work patterns.[353] The core challenges, namely work intensification, blurred boundaries between work and private life, extended availability, social isolation, a heightened level of communication and a lack of predictability regarding work schedules, are **neither novel nor limited to AM**. They have previously been examined in the contexts of remote work, telework and casual work, among others. Therefore, **the debate on the challenges and opportunities of AM and the effectiveness of the existing regulatory framework to respond to them needs to be put into this broader context**. For example, the increasing fragmentation of working time facilitated by some AM systems in platform work and beyond[354] makes it necessary to revisit the long-standing debate over the 'intermediate' time between the period of working time and rest time. Although the CJEU case law shows an expansive interpretation of the notion of 'working time', the issue of the calculation and remuneration of 'stand-by' time in the on-demand economy is far from settled.

The existing legal framework is in principle suited to addressing the foundational aspects of a range of issues related to workers' working conditions, encompassing aspects such as work intensity, transparency, predictability, and communication. This applies equally to workers in traditional employment and to those operating in algorithm-driven work environments, highlighting the inherent capacity of existing tools to address recurrent concerns in the context of the AM-driven workplace. However, the adequacy of the current regulatory framework in addressing the **distinctive challenges introduced by**

[352] VIII Pa 135/19, District Court in Poznań, 10 June 2020, p. 13.

[353] Murray, J. (1999). Normalising temporary work. *Indus. L. J.* 28: 269, in: Vosko, L.F. (2011). *Managing the margins: Gender, citizenship, and the international regulation of precarious employment*. Oxford: OUP.

[354] For evidence of the fragmentation of working time beyond platform work, see in particular Piasna, A. (2023). Algorithms of time: how algorithmic management changes the temporalities of work and prospects for working time reduction. *Cambridge Journal of Economics;* Devetter, F. X., & Valentin, J. (2023). Long day for few hours: impact of working time fragmentation on low wages in France. *Cambridge Journal of Economics*, bead048.

AM systems remains uncertain. While existing legal instruments touch upon certain concerns related to the impact of AM on working time, a notable lack of specific rules tailored to the organizational changes brought about by this set of technologies is evident. As AM continues to reshape work patterns with mixed and conflicting results, questions linger about whether the current regulatory approach is sufficiently comprehensive to address evolving issues and safeguard the well-being of workers and competitiveness of businesses.

Some limitations of the existing legal framework stem from the mismatch between the design of certain legal instruments and the operational dynamics of AM systems. This disconnect arises because some legal tools were conceived on the basis of **archetypal types of work organisation that are transformed by the application of AM systems**. These misalignments can lead to challenges in applying the law effectively within the complex and rapidly evolving landscape of modern workplaces. Recognising and addressing these issues is essential in order to ensure that the legal *acquis* remains relevant and adaptable to the ever-changing nature of work. Another systemic limitation of the legal *acquis* relates to the **personal scope** of the directives, which leaves self-employed persons unprotected. This lack of safeguards for those who do not qualify as workers, but are subject to algorithmic management and control, is a salient legal gap. It remains to be seen how the scope of the recently adopted directives (TPWCD, WLBD) or proposed instruments (PWD), which define the concept of worker on the basis of a 'hybrid' formula, will be implemented into national legal systems.[355]

AM systems have the potential to enhance transparency, verifiability, and programmability in various aspects of employment. They can provide practical solutions to the broader challenges listed in this subsection and can be used to enforce the provisions of existing instruments. For example, monitoring systems could be put in place to encourage or require workers to take a break in due time, in compliance with the limits set out in the Working Time Directive. Workers could be automatically notified about the time when a break is needed, as per applicable legislation.[356] Moreover, AM tools can be used to substantially reduce and simplify worker accountability activities, such as informing managers about the number of working hours spent on a particular task, working time each day and availability. **The enforcement of WTD in the algorithmic setting can be further facilitated** by applications that track workers' presence at work as well as their working time. It is worth pointing out that, in the realm of the platform economy, the proposed PWD provides that labour inspectorates, social protection bodies and other relevant authorities should be provided with information regarding the general terms and conditions applicable to platform-mediated contractual relationships, including on working time.[357] This could increase the transparency of platform work and – more broadly – non-standard forms of employment, including in cross-border contexts, and stimulate the enforcement of the existing provisions.

More research is needed on the impact of algorithmic scheduling on workers. Scholars argue for human-centred form of work organisation that consider employee preferences in task allocation and scheduling.[358] Algorithms should be not only driven by

[355] In the context of the Transparent and Predictable Working Conditions Directive, see Bednarowicz, B. (2019). Delivering on the European pillar of social rights: the new directive on transparent and predictable working conditions in the European Union. *Industrial Law Journal*, *48*(4), 604-623; in the context of the proposed PWD, see Rosin, A. (2022). Towards a European Employment Status: The EU Proposal for a Directive on Improving Working Conditions in Platform Work. *Industrial Law Journal* 51(2): 478–493; on the complex interplay between the personal coverage of PWD and TPWCD. See also Durri, I. (2023). Casual work arrangements and platform-based work: The casual work agenda as a way to enhance the labour protection of platform workers (Dissertation, *Bulletin of Comparative Labour relations,* volume 115), p. 206, Wolters Kluwer.

[356] As mentioned in the context of the gig economy in Davies, A.C.L. (2020). Wages and Working Time in the 'Gig Economy. *King's Law Journal* 31(2) 250, p. 259.

[357] Art. 12 (1) (b). See also Recital 42 of the PWD, with direct reference to information on working time. Note, however, that under the Commission's Proposal this information right is limited to cases when platforms unilaterally determine these conditions and apply them to a large number of contractual relationships.

[358] Unruh, C.F., Haid, C., Fottner, J., & Büthe, T. (2022). Human autonomy in algorithmic management. In: *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 753-762.

the needs of customers and businesses but should also consider workers' well-being and capacities. They should ensure data protection and the transparency of the algorithms driving the work organisation, providing workers with collective rights in this regard. Ensuring the protection of personal data, is crucial to achieving this goal.[359]

As discussed in the subsections that follow, several important responses to the above issues can be found in the EU **privacy and data protection frameworks.** These provide a set of safeguards against the excessive use of the personal data that fuel these metrics. By acting at the level of data collection and processing, instruments such as the GDPR (further analysed under the relevant heading) can limit the negative uses of AM systems that affect working conditions, including working time. Important responses can be also deployed at the collective level, for instance by leveraging **information and consultation rights** or engaging in **social dialogue**, as elaborated upon in more detail below.

3.2.2. Occupational safety and health

3.2.2.1. EU legal framework

As discussed in Chapter 2. Opportunities and challenges of this report, **deployment of AM systems at work may have both positive and negative implications for the occupational health and safety of workers**.

Algorithm-driven applications in the field of OSH include descriptive, diagnostic and predictive systems, as well as prescriptive systems that support decision-making or even take actions themselves (i.e. semi-autonomous systems).[360] These systems provide numerous opportunities in terms of the assessment of the general health and well-being of workers, the facilitation of early alerts and warnings of OSH hazards, the removal of humans from dangerous tasks and materials, and accident prevention (e.g. human-machine collisions in hybrid environments).[361] Therefore, **AM systems can allow for the effective reduction of physical risks** (e.g. ergonomic and musculoskeletal disorders)[362] **and psychosocial risks** (e.g. 'tech-fatigue', work intensification, work-related stress or burnout).[363] Some AM technologies can also be useful in determining when a worker is able to return to work after an injury.[364] Moreover, AM tools can monitor and evaluate workers' adherence to workplace safety measures (as occurred during the COVID-19 pandemic),[365] thus **improving the enforcement of preventive and protective mechanisms**. Strengthening the health and safety of workers is reported as one of the motivations for adopting AI systems at work, along with the need to improve worker performance, reduce staff costs and address skill shortages.[366]

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[359] Ibid, p. 6.

[360] For the typology of AM-driven OSH applications, see Howard, J. (2022). Algorithms and the future of work. *American Journal of Industrial Medicine,* 65(12) 943, p. 945.

[361] See e.g. Baldassarre, A., Mucci, N., Lecca, L.I., Tomasini, E., Parcias do Rosario, M.J., Pereira, C.T., Arcangeli, G., & Oliveira, P.A.B. (2020_. Biosensors in Occupational Safety and Health Management: A Narrative Review. *Int. J. Environ. Res. Public Health,* 17, e2461; Arana-Landín, G., Laskurain-Iturbe, I., Iturrate, M., & Landeta-Manzano, B. (2023). Assessing the Influence of Industry 4.0 Technologies on Occupational Health and Safety. *Heliyon*, *9*, e13720; Ramos, D., Cotrim, T., Arezes, P., Baptista, J., Rodrigues, M., Leitão, J. (2022). Frontiers in Occupational Health and Safety Management. *Int. J. Environ. Res. Public Health*, *19*, 10759.

[362] For example, in the construction sector, the use of AI has proven to lead to a decrease in musculoskeletal disorders. See Mustapha, Z., & Akomah, B.B. (2023). Artificial Intelligence in Digitalised Workplaces, *International Journal of Innovative Science and Research Technology*, (7) 8: 2361.

[363] According to the EU-OSHA, 'psychosocial risks arise from poor work design, organisation and management, as well as a poor social context of work, and they may result in negative psychological, physical and social outcomes such as work-related stress, burnout or depression'. See https://europa.eu/!3jh8mP

[364] Ajunwa, I. (2019). Algorithms at work: productivity monitoring applications and wearable technology as the new data-centric research agenda for employment and labor law. *St Louis U Law J* 2019;63:21.

[365] Howard, J. (2022). Algorithms and the future of work. *American Journal of Industrial Medicine,* 65(12) 943, p. 945.

[366] According to the OECD study, nearly 40 % of managers in the manufacturing sector and approximately one-quarter of managers in the finance and insurance sector were motivated to adopt AI by the need to improve OSH at the workplace. Meanwhile, 32 % and 39 % of employers that adopted AI in finance and manufacturing, respectively, said that the effect of AI on managers' ability to measure worker performance had been positive. See: Broecke, S., Lane, M., & Williams, M.

At the same time, **AM systems may expose workers to physical and psychosocial health and safety risks**.[367] In particular, automated and algorithm-driven tools used for worker monitoring may increase psychosocial risks, which in turn can lead to work-related stress and other mental health concerns.[368] Work intensification and the routinisation of tasks, which may be spurred by some AM tools, are also recognised as non-negligible psychosocial risks.[369] The deployment of certain technologies, such as robots, may cause some as-yet-unknown risks (e.g. miscommunication with workers), potentially leading to workplace accidents.[370] The paragraphs that follow examine the existing and proposed legal instruments **relevant to governing these AM-related OSH risks**.

The general principles concerning the prevention and protection of workers against occupational accidents and diseases are comprehensively set out in the Occupational Safety and Health (OSH) Framework Directive.[371] This instrument provides for a general duty on the part of the employer to ensure workers' safety and health 'in every aspect related to the work' (Art. 5(1)). Even though this Directive does not address the challenges related to AM explicitly, **its general principles are applicable to AM-driven work settings.** This flexibility is reflected in the open-textured framing of the directive, which allows for its principles to be adapted to encompass emerging challenges such as those relating to AM, rather than serving as a limitation.

The OSH Framework Directive obliges employers to take the **measures necessary for the protection of workers' safety and health**, including the prevention of occupational risks, the provision of information and training, and the adoption of the necessary organisation and means (Art. 6(1)). Importantly, the Directive mandates a **proactive risk-based approach**,[372] **as the employer is required to adjust the preventive and protective measures to the changing circumstances** (Art. 6(1)). Employers are required to develop a coherent prevention policy covering technology, the organisation of work, working conditions, social relationships and the influence of factors related to the working environment (Art. 6 (2)). **These general provisions provide a relevant set of crucial safeguards with regard to the OSH risks related to AM systems**. They mandate that the introduction of AM tools should always be preceded by the evaluation of the extent to which such tools will impact workers' health and safety. This implies that, when integrating software at work, an employer will need to consider all the risks identified by the provider during the risk assessment and management phases and communicated to them according to Art. 13(3)(iii) AI Act (more on this below).[373]

In line with Art. 6(2)(d) of the OSH Framework Directive, the use of AM should be **adjusted to workers** and, in particular, **monotonous work and work at a predetermined work rate**

(2023). *The impact of AI on the workplace: Main findings from the OECD AI surveys of employers and workers.* OECD Social, Employment and Migration Working Papers No. 288, p. 33.

[367] Cefaliello, A., Moore, P.V., & Donoghue, R. (2023). Making algorithmic management safe and healthy for workers: Addressing psychosocial risks in new legal provisions. *European Labour Law Journal*, 14(2) 192.

[368] Urzí Brancati, C., & Curtarelli, M. (2021). Digital tools for worker management and psychosocial risks in the workplace: evidence from the ESENER survey, JRC125714, Seville: European Commission.

[369] High work intensity and monotonous work are recognised as some of the most common psychosocial risks at work, also beyond the AM context. See Eurofound & the European Agency for Safety and Health at Work (2014). Psychosocial Risks in Europe: Prevalence and strategies for prevention, Luxembourg: Publications Office of the European Union. For a discussion on this issue in the context of AM, see Todolí-Signes, A. (2021). Making algorithms safe for workers: occupational risks associated with work managed by artificial intelligence, *Transfer: European Review of Labour and Research,* 27(4) 433.

[370] Jarota, M. (2023). Artificial intelligence in the work process. A reflection on the proposed European Union regulations on artificial intelligence from an occupational health and safety perspective, *Computer Law & Security Review,* 49 105825, p. 2.

[371] Council Directive of 12 June 1989 on the introduction of measures to encourage improvements in the safety and health of workers at work (89/391/EEC).

[372] Paper on digitalisation and the use of machinery and robotics using artificial intelligence, prepared for the Senior Labour Inspectors Committee (SLIC) and SLIC working group MACHEX, p. 38.

[373] Cefaliello, A., & Kullmann, M. (2022). Offering false security: How the draft artificial intelligence act undermines fundamental workers rights, *European Labour Law Journal*, 13(4) 542, p. 559.

must be mitigated.[374] This is crucial in the context of AM systems, which often break down tasks into simple, fragmented processes that can be easily monitored and reassembled. Such practices may have detrimental effects not only on workers' health and safety, but also on their skill levels,[375] job satisfaction and the mutual trust between workers and employers.[376] Through Art. 6(2)(d) of the Framework Directive, the EU legal framework is equipped to address and limit these risks.

Furthermore, the OSH Framework Directive provides important **information and consultation rights**, which are of paramount relevance in the context of AM at work. According to Art. 6(3)(c), employers must ensure that the planning and introduction of new technologies are subject to consultation with workers and/or their representatives as regards the consequences for workers' safety and health of the choice of equipment, the working conditions and the working environment.[377] **Workers and/or their representatives must be informed by the employer about all safety and health risks and protective and preventive measures put in place** (Art. 10). They must be consulted by the employer and should be allowed to take part in discussions regarding all questions relating to safety and health at work (Art. 11(1)). Moreover, they shall have the **right to appeal** to the authority responsible for OSH protection (Art. 11(6)) and must receive health surveillance appropriate to the health and safety risks to which they are exposed at work (Art. 14(1)). The importance of these provisions cannot be underestimated in the context of AM. As is often emphasised in literature and by stakeholders, it is vital to ensure the involvement of worker representatives in aspects relating to the introduction of AM in a workplace. **Collective negotiation enables the '*ex-ante'* governance** of AM systems, which is more effective than '*ex-post'* measures put in place after an OSH hazard has materialised.[378] The **participatory model of AM**[379] is key to ensuring the fair and human-centred use of AM and the mitigation of OSH risks that may be posed AM systems. By adopting a 'worker-centric approach',[380] these provisions of the OSH Framework Directive enable workers' representatives to shape the use of algorithms at work and to design adequate preventive and protective strategies. Moreover, the direct involvement of workers in decision-making may mitigate the stresses related to their exposure to AM systems, as it shifts the locus of control to them.[381] Such involvement also increases **trust in the technologies** introduced[382] and gives workers a **sense of agency over the work organisation**, improving their work attitude.[383]

Another crucial right guaranteed under the OSH Framework Directive is the **right to adequate safety and health training,** which should be provided, among other situations, **when new work equipment or new technology is introduced** (Art. 12(1)). In the context of AM at work, safeguarding this right is key to raising awareness among workers about

[374] Ibid, p. 559.

[375] Wagner, B. (2019). Liable, but Not in Control? Ensuring Meaningful Human Agency in Automated Decision-Making Systems, *Policy & Internet*, Volume 11(1) 104, p. 117.

[376] For OSH risks relating to the 'micromanagement', see Todolí-Signes, A. (2021), p. 439.

[377] Jarota, M. (2023). Artificial intelligence in the work process. A reflection on the proposed European Union regulations on artificial intelligence from an occupational health and safety perspective. *Computer Law & Security Review*, *49*, 105825.

[378] De Stefano, V., Taes, S. (2023). Algorithmic management and collective bargaining, *Transfer: European Review of Labour and Research*, 29(1), 21.

[379] Lee, M.K. et al. (2021). Participatory Algorithmic Management: Elicitation Methods for Worker Well-Being Models, AAAI/ACM Conference on AI, Ethics, and Society.

[380] CCefaliello, A., & Kullmann, M. (2022). Offering false security: How the draft artificial intelligence act undermines fundamental workers rights, *European Labour Law Journal*, 13(4) 542, p. 559.

[381] Urzí Brancati, C., & Curtarelli, M. (2021). Digital tools for worker management and psychosocial risks in the workplace: evidence from the ESENER survey, JRC125714, Seville: European Commission, p. 8.

[382] Ibid.

[383] Jarota, M. (2023). Artificial intelligence in the work process. A reflection on the proposed European Union regulations on artificial intelligence from an occupational health and safety perspective. *Computer Law & Security Review*, *49*, 105825, p. 11.

potential risks and ensuring that workers are kept 'in the loop'.[384] By and large, this provision can be considered a 'frontline activity' in countering the OSH risks stemming from AM.[385]

All in all, the **risk management, information and consultation, and safety and health training provided for under the OSH Framework Directive are a relevant procedural sequence that is well suited to addressing the risks emerging in the context of the adoption of AM at work**, despite the OSH Framework Directive not being a tool designed exclusively for AM.

The OSH Framework Directive is complemented by several **'daughter' directives focusing on specific or sectoral aspects of safety and health.** These individual directives tailor the general principles of the OSH Framework Directive to specific tasks, definite workplace hazards, specific workplaces and sectors, and specific groups of workers or work-related aspects. **Some of these directives may be particularly relevant to the challenges posed by AM for some categories of workers, or for those involved in some specific work environments**.

i. Directive 90/270/EEC on **display screen equipment** obliges employers to analyse workstations to assess their safety and health conditions.[386] Specifically, possible risks to eyesight, physical problems, and problems of mental stress should be assessed (Art. 3(1)). The Directive states that workers must be informed about all aspects of safety and health relating to their workstation (Art. 6). Moreover, no 'quantitative or qualitative checking facility' may be used without workers' knowledge. When deciding about the use of the display screen equipment, the employer must consider whether the software used is suitable for the task, easy to use and, where appropriate, adaptable to the operator's level of knowledge or experience. The Directive also mandates that the principles of software ergonomics must be applied, in particular, to human data processing.[387]

ii. Another instrument worth mentioning is Directive 2009/104/EC on **work equipment**, which lays down the employer's duty to ensure that work equipment is safe.[388] It sets forth several basic requirements, including training for workers on how to engage with this work equipment, and consultation with worker representatives. This Directive is of particular relevance to deployment in the workplace of new management technologies, including AM and AI tools.

Specific AM-related risks to OSH are explicitly targeted in the **proposed Platform Work Directive (PWD)**. This draft instrument clarifies the existing EU OSH legal *acquis* in the context of platform work, wherein the health and safety risks are particularly well-evidenced and pronounced,[389] by expressly stating the OSH obligations of digital labour platforms. First, platforms must **inform persons performing platform work** about automated decision-making systems that are used to take or support decisions that significantly affect their occupational safety and health (Art. 6(1)(b)). Second, platforms must regularly **monitor and evaluate the impact of individual decisions taken or supported by**

[384] See Wagner, B. (2019); also Moore, P.V. (2018). Artificial Intelligence: Occupational Safety and Health and the Future of Work, WZB Social Science Centre Berlin, Weizenbaum Institute for the Networked Society, European Agency for Safety and Health at Work (EU-OSHA).

[385] Todolí-Signes, A. (2021), p. 443.

[386] Council Directive 90/270/EEC of 29 May 1990 on the minimum safety and health requirements for work with display screen equipment (fifth individual Directive within the meaning of Article 16 (1) of Directive 89/391/EEC).

[387] Annex, point 3.

[388] Directive 2009/104/EC of the European Parliament and of the Council of 16 September 2009 concerning the minimum safety and health requirements for the use of work equipment by workers at work (second individual Directive within the meaning of Article 16(1) of Directive 89/391/EEC.

[389] Digital platform work and occupational safety and health: a review Safety and health at work EU-OSHA, available at https://osha.europa.eu/en/publications/digital-platform-work-and-occupational-safety-and-health-review, accessed on 13 March 2022; Garben, S. (2017). Protecting workers in the online platform economy an overview of regulatory and policy developments in the EU, Luxembourg: Publications Office of the European Union, accessed 6 April 2020; Williams, P., McDonald, P., & Mayes, R. (2022). The Growing 'Gig Economy'. Implications for the Health and Safety of Digital Platform Workers. In: *Handbook on Management and Employment Practices*, Cham: Springer.

automated monitoring and decision-making systems (i.e. AM systems) on the working conditions (including OSH), of persons performing platform work (Art. 7(1)). Importantly, these provisions apply to persons working through platforms, regardless of their employment status (Art. 10(1)). This is a crucial step forward, given the applicability of the OSH Directives only to 'workers'. Art. 7(2) further specifies platforms' obligations regarding OSH, although the personal scope of this provision is limited to those platform workers who enjoy employment status. First, it requires platforms to assess the risks posed by AM systems to the safety and health of platform workers, with regard to possible risks of work-related accidents, as well as psychosocial and ergonomic risks. Second, it obliges platforms to evaluate whether the safeguards in place are suitable for the risks identified. Third, platforms are responsible for introducing appropriate preventive and protective measures. Lastly, platforms must not use AM systems 'in any manner that puts undue pressure on platform workers, or otherwise puts at risk the physical and mental health of platform workers'. Importantly, Art. 7(2) of the proposed PWD is the first one to explicitly recognise the impact of AM on workers' health and safety and to expressly mention psychosocial risks. As can be inferred from Art. 7(1), the evaluation of AM and its impact on OSH should be carried out regularly, in a dynamic way.[390] Platforms must ensure **sufficient human resources for monitoring the impact of AM on working conditions, including on OSH**. This function should be performed by people with the necessary competence, training, and authority (Art. 7(3)). In summary, the regulation of OSH-related risks in the proposed PWD is an important regulatory intervention, given the well-evidenced OSH risks prevailing in the platform economy.

To date, **there has been no social partner framework agreement dedicated specifically to AM-related OSH risks**.[391] However, some broader instruments are worth mentioning in this context:

i. In the case of AM systems applied to **remote workers and teleworkers**, the 2002 Social Partners' Framework Agreement on Telework is relevant. It restates and specifies the principles set out in the OSH Framework Directive, stating that the employer bears responsibility for the protection of the OSH of the teleworker, and for informing them of the company's policy on OSH, in particular requirements regarding visual display units. The teleworker should apply these safety and health policies correctly.

ii. The **Autonomous Framework Agreement on Digitalisation** of June 2020 by EU cross-industry social partners, which addresses opportunities and challenges in the world of work resulting from digital transformation, dedicates some of its provisions to safety and health. The agreement recognises that while monitoring systems and other technologies offer opportunities to safeguard health and safety, as well as improving efficiency, some risks arise from it in the realm of working conditions. The agreement highlights that the technologies introduced need to be safe; namely, they should prevent harm and apply the **'human-in-control' principle**. Moreover, a risk assessment system, 'including opportunities to improve safety and prevent harm such as for human physical integrity, psychological safety, confirmation bias or cognitive fatigue', should be put in place.

Moreover, specific references to challenges related to the digitalisation of the world of work are made in the **EU Strategic Framework on Health and Safety at Work 2021-2027**, adopted by the European Commission in 2021.[392] In particular, this Framework highlights two drivers of risks posed to OSH by new technologies: (i) growing irregularity in the performance of work, and (ii) the risks related to new work tools and machinery.[393] Although

[390] Cefaliello, A., Moore, P.V., & Donoghue, R. (2023). Making algorithmic management safe and healthy for workers: Addressing psychosocial risks in new legal provisions. *European Labour Law Journal*, 14(2) 192, p. 204.

[391] For a discussion of such an option, see Todolí-Signes, A. (2021).

[392] Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, EU Strategic Framework on health and safety at work 2021-2027 Occupational safety and health in a changing world of work, COM/2021/323 final.

[393] Ibid, p. 6.

it does not mention AM systems directly, it refers broadly to the challenges related to the spread of remote work and the increased use of mobile devices.[394] The Commission committed itself to modernising the OSH legislative framework by reviewing the Workplaces Directive and the Display Screen Equipment.[395]

Another crucial piece of regulation relevant to risks related to AM, including those concerning OSH, is the draft **Artificial Intelligence Act (AI Act)**.[396] This instrument allows for the use of AI systems at work, as long as they comply with the conditions set out under the Act.[397] At the same time, it acknowledges their potentially adverse impact on health and safety[398] and envisages mechanisms aimed at offsetting these risks. AI systems are defined broadly as software developed that use one or more techniques enumerated in Annex I of the AI Act, which can 'for a given set of human-defined objectives, generate outputs such as content, predictions, recommendations, or decisions influencing the environments they interact with' (Art. 3 (1)).

Some categories of AI systems, including some used at work, are classified as 'high risk' under this proposed legislation. According to Art. 6(1), this is the case for AI systems intended to be used as safety components of products that are subject to third-party *ex-ante* conformity assessment. Moreover, 'standalone' AI systems explicitly referred to in Annex III of the draft AI Act must also be considered high-risk. Annex III (3) refers specifically to some AI systems in the area of employment, worker management, and – notably – access to self-employment. More precisely, it classifies **two subcategories of AI systems as high risk**: (i) those intended for recruitment purposes, and (ii) those intended for deciding on promotion and the termination of work contracts, for task allocation, and for monitoring and evaluating the performance and behaviour of persons in such cases.

Moreover, a limitation of the catalogue of high-risk AI systems in Annex III is that the **classification depends on the provider's *intended* purpose**,[399] which may in some cases differ from the actual use of such systems. In simple terms, a system might not be intended to monitor or evaluate workers, but it may be nevertheless suitable for such purpose and eventually be used to this end by the employer.[400]

AI systems classified as high-risk must comply with a detailed set of requirements laid down in Title III of the draft AI Act. A risk management system should be put in place, consisting of 'a **continuous iterative process** run throughout the entire lifecycle of a high-risk AI system' (Art. 9(2). Providers should identify and mitigate the known and foreseeable risks associated with such systems (Art. 9(2)(a)). They should also estimate and assess the risks that may emerge when the high-risk AI system is deployed 'in accordance with its intended purpose and under conditions of reasonably foreseeable misuse' (Art. 9 (2)(b)) and adapt suitable risk management measures (Art. 9(2)(d). When

[394] Ibid, p. 7

[395] Ibid, p. 9.

[396] Unless otherwise specified, reference is made to the original proposal of the Artificial Intelligence Act (AI Act) presented in April 2021. Proposal for a Regulation of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain union legislative acts, COM/2021/206 final.

[397] Kelly-Lyth, A. (2022). 'The AI Act and Algorithmic Management, *Comparative Labor Law & Policy Journal Dispatches*; De Stefano, V., & Wouters, M. (2022). AI and digital tools in workplace management and evaluation. An assessment of the EU's legal framework, European Parliamentary Research Service Scientific Foresight, p. 51.

[398] Recital 28 states that 'AI systems could produce adverse outcomes to health and safety of persons, in particular when such systems operate as components of products.' For a broader discussion, see Section 3.3. Data protection and privacy.

[399] This is defined in Art. 3 (12) as 'the use for which an AI system is intended by the provider, including the specific context and conditions of use, as specified in the information supplied by the provider in the instructions for use, promotional or sales materials and statements, as well as in the technical documentation'. See, also, the Explanatory Memorandum stating that 'the classification as high-risk does not only depend on the function performed by the AI system, but also on the specific purpose and modalities for which that system is used.'

[400] For a criticism of this approach, see, for example, Klengel, E., & Wenckebach, J. (2021), p. 165. Note that if the user substantially modifies the intended purpose of a high-risk AI system that has already been introduced, he will be considered a provider for the purposes of the AI Act, rather than the initial provider (Art. 28(1)(b) AI Act). In the employment context, this means that an employer who uses the AI system in a way that is substantially different from that intended by provider bears all the responsibility, instead of the provider (e.g. the designer) of the system.

designing the software, the provider should thus consider in the risk management system the impacts on OSH of AI-driven work processes.[401] For instance, AI systems should be designed to alleviate monotonous work and work at a predetermined work rate, and to reduce their potentially negative effect on health and safety. To give a concrete example, task allocation and target setting could be connected with the analysis of heart rate and other signals that indicate when a worker is tired and should be adjusted accordingly.[402]

Thus, **if adequately implemented, this set of provisions under the draft AI Act holds great potential to make providers contribute to the fair application of AM at work and ensure the effective realisation of workers' rights under the OSH Framework Directive and other relevant instruments** (e.g. the Working Time Directive). To prevent OSH risks in algorithmic-driven settings, AI systems need to be adaptive to workers' capacities and take account of their limitations and concrete circumstances. Researchers have long emphasised the need to program algorithms in a way that takes account of foreseeable OSH risks and minimises those risks as part of the mandatory risk management performed by the programmers.[403] Elimination of flaws and biases in the training data and the design of the algorithms is pivotal to ensuring that AI systems do not create or exacerbate health risks. In technical terms, controlling 'upstream' factors, such as the input data and algorithm design, prevents 'downstream' risks.[404] Obliging the provider of an AI system to put in place a risk management system is thus crucial to achieving this goal.[405] Given that the nature of AM tools is dynamic and subject to evolution, it is important to emphasise that the provider's obligations to assess and mitigate risks should go beyond the *ex-ante* design phase.[406] Rather, the elimination or reduction of the risks should also be ensured **throughout the entire life-cycle of the AI system**, i.e. also during its deployment phases (Art. 9(4)).[407]

Employers will most typically qualify as 'users' under the draft AI Act.[408] As such, they need to be informed by the provider of the risks of the AI system (Art. 9(4) AI Act)) and be given instructions for the use of high-risk AI systems. Any known or foreseeable circumstance that might lead to risks to health and safety or to fundamental rights needs to be disclosed (Art. 13(3)(b)(iii)). Employers must use the AI tools according to their instructions for use (Art. 29(1)) and, to the extent that they control the input data, they must ensure that input data are relevant, given the intended purpose of the AI tool (Art. 29(3)). They must keep records of any automatically generated events ('logs') to the extent that these logs are under their control (Art. 29(5)). Moreover, they must monitor the operation of AI systems on the basis of the instructions for use and report serious incidents or malfunctions to the provider or distributor. Under Art. 53(3) of the AI Act, any significant risks to health and safety identified during the development and testing of such systems must result in immediate mitigation. If such mitigation fails to take place, the development and testing process must be suspended.

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[401] Cefaliello, A., & Kullmann, M. (2022). Offering false security: How the draft artificial intelligence act undermines fundamental workers rights, *European Labour Law Journal*, 13(4) 542.

[402] Ibid, p. 560.

[403] Todolí-Signes, A. (2021). Making algorithms safe for workers: occupational risks associated with work managed by artificial intelligence, *Transfer: European Review of Labour and Research,* 27(4): 433-452.

[404] Howard, J. (2022). Algorithms and the future of work. *American Journal of Industrial Medicine,* 65(12) 943, p. 947.

[405] Kelly-Lyth, A. (2021). Dispatch No. 39 – European Union, The AI act and algorithmic management, p. 9.

[406] Cefaliello, A., Moore, P.V., & Donoghue (2023); Cebulla, A., Szpak, Z., & Knight, G. (2023). Preparing to work with artificial intelligence: assessing WHS when using AI in the workplace, *IJWHM*, 16(4) 294.

[407] See also Art. 12(1) AI Act, according to which 'the logging capabilities shall ensure a level of traceability of the AI system's functioning throughout its lifecycle'.

[408] Unless they introduce the AI system under their name or trademark, modify the intended purpose of a high-risk AI system already put into service, or substantially change the high-risk AI system. In such cases, according to Art. 28 (1), they become providers and are subject to all obligations relating to a provider.

Even though employers are not direct addressees of most of the provisions of the draft AI Act,[409] they remain responsible for **ensuring safe and healthy working conditions when AI systems are in use in the workplace**.[410] The information obtained from the provider of an AI system should be used to evaluate the potential impact of the AI system at work and to design appropriate preventive and protective measures.[411] Art. 29(2) of the AI Act makes clear that the employer's obligations to use AI systems in accordance with the instructions received from the provider are without prejudice to their obligations stemming from other EU or national regulations. Moreover, employers have full discretion over organising how the human oversight measures indicated by the provider will be implemented. For example, they could implement procedures for internal control by workers and/or their supervisors to ensure an effective oversight system. Likewise, they could establish a company dialogue on emerging AI-related OSH risks.[412]

Workers do not qualify as 'users' of a high-risk AI system within the meaning of the AI Act.[413] Even so, high-risk AI systems under the Act include human-machine interface tools that a natural person (e.g. a worker) oversees while they are in use (Art. 14(1)). The AI Act does not establish specific individual or collective rights for workers in the context of AI use. For example, it does not make explicit reference to the right of worker representatives to access the training data.[414]

Overall, **risk management provisions under the draft AI Act provide a potent mechanism to offset the potentially negative effects of AI systems on OSH**. Nevertheless, some **structural features of this regulation need to be highlighted that might impair its effectiveness**. Most fundamentally, the draft regulation provides for a 'regulatory ceiling'. This element risks undermining potentially stronger and more specific national-level regulations or collective instruments, as discussed below in this legal analysis.[415] Moreover, the evaluation of the conformity of the AI system with the risk management system is subject only to the provider's self-assessment.[416]

Lastly, Regulation (EU) 2023/1230 (the **Machinery Regulation**), published on 29 June 2023[417] and replacing the Machinery Directive 2006/42/EC, can be **relevant to some of the AM and AI-related challenges to OSH**. This Regulation responds to the challenges related to new technologies, including machine learning systems, mobile machinery, and autonomous mobile machinery, some of which may also be used in workplaces. It states specifically that 'Member States may lay down requirements to ensure that persons, *including workers*, are protected when installing or using machinery or related products, provided that such rules do not allow for the modification of machinery or a related product in a way that is not compatible with this Regulation' (Art. 5, emphasis added). Art. 8 provides that machinery or related products must only be made available on the market or put into service if they meet the essential health and safety requirements set out in Annex

[409] Kelly-Lyth, A. (2021). Dispatch No. 39 – European Union, The AI act and algorithmic management. Dispatch No. 39– European Union. *Comparative Labor Law & Policy Journal*.

[410] The only exceptions to this are extraordinary and unforeseeable circumstances beyond the employer's control, or exceptional causes that could not have been avoided despite the exercise of due diligence (Art. 5 (4) of the OSH Framework Directive).

[411] Cefaliello, A., & Kullmann, M. (2022), p. 559.

[412] Jarota, M. (2023). Artificial intelligence in the work process. A reflection on the proposed European Union regulations on artificial intelligence from an occupational health and safety perspective, *Computer Law & Security Review,* 49 105825, p. 5.

[413] Art. 3 (4) provides that only those 'using an AI system under its authority' are to be considered users.

[414] Klengel, E., & Wenckebach, J. (2021). Artificial intelligence, work, power imbalance and democracy – why co-determination is essential, *ILLeJ,* 14(2) 157.

[415] De Stefano, V., & Wouters, M. (2022). AI and digital tools in workplace management and evaluation. An assessment of the EU's legal framework, European Parliamentary Research Service Scientific Foresight.

[416] For a critical assessment of this aspect, see De Stefano, V., & Taes, S. (2023), in which the authors argue for the introduction of an *ex-ante* assessment by third parties instead.

[417] Regulation (EU) 2023/1230 of the European Parliament and of the Council of 14 June 2023 on machinery and repealing Directive 2006/42/EC of the European Parliament and of the Council and Council Directive 73/361/EEC.

III. **It is the duty of manufacturers to ensure that machinery has been designed and constructed in accordance with these essential health and safety requirements** (Art. 10). The manufacturer must carry out a risk assessment to determine the essential health and safety requirements that apply to the machinery or related product. The machinery or related product must then be designed and constructed so as to eliminate hazards or, if that is not possible, to minimise all relevant risks. Importantly, the **risk assessment and risk reduction** must 'include hazards that might arise during the lifecycle of the machinery or related product that are foreseeable at the time of placing the machinery or related product on the market as an intended evolution of its fully or partially self-evolving behaviour or logic as a result of the machinery or related product designed to operate with varying levels of autonomy' (Annex III, Part B, 1). Users must be informed of the risks and of the potential need to carry out any training. Moreover, the Regulation explicitly states that 'the discomfort, fatigue and physical and psychological stress faced by the operator shall be eliminated or reduced to the minimum possible' (Annex III, Part B, 1.1.6).

3.2.2.2. Selected national legislation and policies

A. EU countries

To date, at national level, there are **no regulations specifically addressing AM-related occupational safety and health risks to workers**. General legislative frameworks on OSH apply to AM tools.

Some regulations are targeted specifically at OSH risks in the context of the **platform economy**. Art. 71 (1) of the **Greek** Law 4808/2021 states that platforms should have the same health and safety obligations towards service providers who work through them as independent contractors, as they would have towards them if they were with them in an employment relationship. A law regulating health and safety aspects of platform work was also introduced in the Italian Region of **Lazio**. This regional law explicitly specified that platforms should account for health and safety at work by providing protective devices and training, and that they are obliged to insure workers against accidents at work and occupational diseases.[418]

In several countries, policy documents addressing a wide range of AM-related challenges draw attention to OSH risks. To give one example, in **Malta**, the Parliamentary Secretary for Financial Services, Digital Economy, and Innovation issued a policy document, 'Towards Ethical and Trustworthy AI', which aims to ensure that AM systems used at work are accompanied by safeguards preventing overconfidence or overreliance on the AI system. It specifically mentions that the allocation of tasks between human and automated processes should enhance or augment human capabilities.[419]

3.2.2.3. Other relevant initiatives

Some standards issued by the **International Organisation for Standardisation (ISO)**, a worldwide federation of national standards bodies establishing a set of best practices for businesses, are relevant to the challenges arising from AM. For example, ISO 9241-210:2019 on Human-centred Design for Interactive Systems provides requirements and recommendations for human-centred design principles and activities throughout the life cycle of interactive computer-based systems. It is intended to be used by those managing design processes and is concerned with the ways in which both hardware and software components of interactive systems can enhance human–system interaction. It provides an overview of human-centred design activities but does not tackle all aspects of project management.[420] However, ISO standards are not free and are developed by business, without following tripartite approach (workers, employers, and governments).

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[418] This text is available at https://www.consiglio.regione.lazio.it/consiglio-regionale/?vw=leggiregionalidettaglio&id=9349&sv=vigente

[419] See: https://malta.ai/wp-content/uploads/2019/10/Malta_Towards_Ethical_and_Trustworthy_AI_vFINAL.pdf

[420] See: https://www.iso.org/standard/77520.html.

There have also been some bottom-up, non-binding initiatives addressing OSH risks, including those relating to AM in the context of the platform economy. An interesting example, which presents a comprehensive approach to OSH risk prevention and management, is the 'Charter of Fundamental Rights of Digital Labour in the Urban Context' (the so-called '**Bologna Charter**'). It required the undersigning digital labour platforms to develop an OSH management system and to adopt all appropriate measures to assess, prevent and reduce risks to OSH. It also provided workers' right to refuse tasks without repercussions when faced with extraordinary weather conditions.[421]

3.2.2.4. Conclusions

The existing and proposed instruments at EU level reviewed in this subsection are, in principle, suitable for addressing the OSH risks emerging from AM systems, even if none of them lays down explicit provisions in this regard. The general prevention principles and the worker-centric approach adopted under the OSH Framework Directive are as valid as they were over 30 years ago when it was conceived.

Although the OSH Framework Directive does not directly address AM systems, many of the risks they pose are adequately covered by its provisions. Both physical and psychosocial risks, including those related to AM, need to be included in organisational health and safety strategies. While psychosocial risks are often perceived as more complex and less 'tangible', and thus harder to address,[422] the OSH Framework Directive also provides, in principle. a 'suitable frame'[423] for combating these risks. Even though the AI Act does not address the specificities of the employment context (i.e. it does not provide for individual or collective rights for workers), **its risk-based model is a suitable anticipatory method for assessing and mitigating OSH risks**. In the context of the platform economy, in which AM-related challenges to OSH are particularly prevalent, OSH risks are explicitly addressed in the proposed PWD. This concretisation of platforms' obligations in the context of OSH – notably, regardless of the employment status with regard to some of the platforms' duties – is an important clarification of the legal framework and platform workers' rights in this regard.

The OSH Framework Directive emphasises the employer's responsibility for all work-related aspects of safety and health. The Directive requires employers to take **preventive measures**, provide **information** and **training,** and **adapt to changing circumstances proactively**. It also stipulates that employers must inform workers about safety and health risks and preventive and protective measures, involve them in safety discussions, allow redress to OSH authorities, and provide appropriate health surveillance. These provisions offer vital safeguards for managing the occupational safety and health risks associated with AI systems. Involving worker representatives promotes proactive governance and mitigates OSH risks, fostering a worker-centric approach, reducing monitoring-related stress, building trust in technology, and enhancing workers' sense of agency.

Some concerns voiced in the literature regarding the suitability of the existing EU regulatory framework revolve around the **lack of a specific focus on AM-related risks in the work environment in the OSH Framework Directive and in the draft AI Act**. Against this background, some scholars advocate for a dedicated instrument addressing the risks to OSH posed by AM.[424] However, there is no consensus as to how best to overcome the

[421] For an analysis of this initiatives, see, for example, https://osha.europa.eu/sites/default/files/2022-02/A_national_local_answer_challenges_platform_economy_Italy.pdf.

[422] Cefaliello, A., Moore, P.V., & Donoghue, R. (2023); Jespersen, A.H., Hasle, P., & Nielsen, K.T. (2016). The Wicked Character of Psychosocial Risks: Implications for Regulation, *NJWLS*, 6(3) 23.

[423] Walters, D., & Wadsworth, E. (2014). Contexts and Determinants of the Management of Occupational Safety and Health in European Workplaces, *Policy and Practice in Health and Safety*, 12:2, 109-130, as cited in Dahler-Larsen, P. (2020). Can occupational health and safety management systems address psychosocial risk factors? An empirical study, *Safety science*, Volume: 130, p. 2.

[424] Cefaliello, A., Moore, P.V., & Donoghue, R. (2023) advocate for a more comprehensive AM in Workplaces Directive that would include OSH-specific provisions; Todolí-Signes, A. (2021) suggests an individual OSH directive on AM work; see

limitations of the current or proposed regulations. Given the novelty of the challenges concerned, empirical evidence on the effectiveness of the existing regulatory models in an AM context is lacking. In addition, it is important to contextualise those risks that may arise also, but not only, in an AM setting (e.g. psychosocial risks such as work-related stress or work intensification). Studies suggest a correlation between psychosocial risks and management technologies,[425] while at the same time noting that some of these psychosocial risks stem from other organisational factors involved in work and social relations.[426]

Lastly, while this section has analysed instruments that address the AM-related risks to OSH most directly, it should be stressed that the risks relating to AM may result from aspects such as privacy violation[427] or discrimination.[428] A lack of transparency regarding algorithmic monitoring and decision-making may cause anxiety and lead to psychological distress.[429] Even the very awareness of employers' access to sensitive data can be considered a psychosocial risk factor.[430] Hence, **the analysis of the EU instruments on OSH cannot be detached from instruments such as the General Data Protection Regulation, anti-discrimination legislation and other instruments targeting algorithmic transparency, which can contribute greatly to combating OSH risks related to AM.** These instruments will be examined in detail in other sections of this report.

AM systems provide both opportunities and challenges in terms of the enforcement of the legal *acquis*. In the context of AM, the **enforcement of these regulations by national labour inspectorates** through conventional means might prove more difficult than in a traditional context.[431] Conventional methods of labour inspection focus mainly on physical, chemical, biological and psychological hazards that arise within a given geographical and physical space. Moreover, the cross-border character of AM-related risks poses challenges to traditional methods of labour inspection, suggesting the potential need for a new way of approaching these challenges.[432] Enforcement may be further complicated by the fact that some of the OSH risks posed by AM are not easily visible and detectable.[433] Another critical issue is the need for the improvement of national reporting data on OSH. Such reporting is pivotal to the development of estimates and prevention strategies, including in the context of AM. This reflects the general issues related to the monitoring of OSH data, and specifically data relating to psychosocial risks, both at European[434] and international level.

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also Jarota, M. (2023); Ponce Del Castillo, A. (2021). The AI Regulation: entering an AI regulatory winter? Why an ad hoc directive on AI in employment is required, ETUI Policy Brief.

[425] Urzí Brancati, C., & Curtarelli, M. (2021). Digital tools for worker management and psychosocial risks in the workplace: evidence from the ESENER survey, JRC125714, Seville: European Commission, p. 14.

[426] Dahler-Larsen, Peter, Anna Sundby, and Adiilah Boodhoo. 'Can occupational health and safety management systems address psychosocial risk factors? An empirical study', *Safety Science* 130 (2020): 104878.

[427] For example, on privacy concerns related to wearables used for self-tracking see Schall, M.C., Sesek, R.F., & Cavuoto, L.A. (2018). Barriers to the Adoption of Wearable Sensors in the Workplace: A Survey of Occupational Safety and Health Professionals, *Human Factors*, 60 (3) pp. 351–362; Ball, K. (2021). Electronic monitoring and Surveillance in the Workplace. Literature Review and Policy Recommendations, JRC125716, p. 25.

[428] Cefaliello, A., Moore, P.V., & Donoghue, R. (2023); Todolí-Signes, A. (2021)

[429] As identified in Chapter 2. Opportunities and challenges.

[430] Jarota, M. (2023). Artificial intelligence in the work process. A reflection on the proposed European Union regulations on artificial intelligence from an occupational health and safety perspective, *Computer Law & Security Review,* 49 105825, p. 3.

[431] Nordic Future of Work Group (2020). Work today and in the future. Perspectives on Occupational Safety and Health challenges and opportunities for the Nordic labour inspectorates, p. 63. Available at: https://julkaisut.valtioneuvosto.fi/bitstream/handle/10024/162419/STM_2020_30_rap.pdf

[432] Ibid.

[433] Jarota, M. (2023). Artificial intelligence in the work process. A reflection on the proposed European Union regulations on artificial intelligence from an occupational health and safety perspective, *Computer Law & Security Review,* 49 105825, p. 10.

[434] According to the assessment of the European Commission from 2017, only 56 % of the respondents in a pan-European survey had sufficient information on how to include psychosocial risk factors in their risk assessment. *See* European Commission (2017). Ex-post evaluation of the European Union occupational safety and health Directives, Brussels: European Commission, as referred to in Dahler-Larsen, P., Sundby, A. & Boodhoo, A. 'Can occupational health and safety management systems address psychosocial risk factors? An empirical study' *Safety Science* 130 (2020): 104878.

In this regard, **AM systems could be potentially mobilised to further improve data collection and to streamline bureaucratic procedures**. Moreover, they could ensure a quicker, if not real-time, and more efficient reporting of injuries and diseases.

3.3. Data protection and privacy

Key findings:

- AM systems present increased difficulties for data subjects and data controllers attempting to exercise their rights and fulfil their obligations. This is due to the novelty of the technology, a lack of awareness regarding the collection and processing of personal data, the potential tension between the operations of the AM system and some key principles of the GDPR, among other factors.
- There is no single EU legal act in the area of privacy and data protection that is specifically tailored to the use of AM, with the notable exception of Chapter III of the proposed PWD. Despite this, the GDPR covers several relevant aspects, such as data processing, transparency and access, and solely automated decision-making, all of which are relevant to AM systems. Hence, many direct and explicit legal responses to the challenges at stake can be found in the GDPR and the PWD.
- More specifically, the GDPR is one of the most effective frameworks regulating certain AM functions. It offers solutions to address the intricate challenges posed by such systems, encompassing issues relating to lawful processing and transparency, and providing safeguards in cases of solely automated decision-making. Significant developments have been witnessed in this respect, thanks to strategic litigation, collective agreements, company protocols and enforcement.
- Emerging case law reveals the GDPR's forward-looking approach to the challenges posed by AM, while also highlighting certain interpretive gaps in its application. In particular, the GDPR is limited by its horizontal scope, its lack of specific provisions tailored to the workplace environment and its use of broad terminology.
- While comprehensive in its approach, the GDPR offers flexibility tailored to different contexts, especially when it comes to the workplace. Specifically, Article 88 GDPR allows for the adoption of 'more specific rules to ensure the protection of the rights and freedoms in respect of the processing of workers' personal data in the employment context'. Such rules must 'include suitable and specific measures to safeguard the data subject's human dignity, legitimate interests and fundamental rights'. This is especially pertinent when considering the transparency of data processing, the sharing of personal data among business groups or entities involved in joint economic activities, and monitoring systems in the workplace. As a consequence, while the regulation employs general terms or possesses a predominantly 'individualised' character, its application can be adapted to address the intricate nuances of AM in work environments.
- The GDPR can also be used as a starting point to establishing workers' data trusts, fostering trust and empowering workers with greater control over their personal data in the face of the advanced challenges posed by AM.

The international, EU and national principles and rules on data protection and privacy are **a direct and suitable way to tackle certain aspects of AM** and their underlying resource, namely personal data. Indeed, for AM to operate efficiently, large swathes of data must be collected and processed. If big data represent the backbone of algorithm- or AI-based systems, 'personal data' are their 'fuel and by-product'.[435]

AM tools and practices pose a **combination of old and new challenges** in this respect, as detailed in Chapter 2. Opportunities and challenges. The workplace is the archetypal

[435] Mitrou, L. (2018). Data protection, artificial intelligence, and cognitive services: is the general data protection regulation (GDPR) 'artificial intelligence-proof?' Tilburg: TILT Law & Technology Working Paper Series.

context in which workers' privacy comes into tension with the interests of employers which enjoy wide managerial latitude in running their businesses. Notably, **the employer–employee relationship presents a rich set of specificities**. Employment law may impose legal obligations requiring personal data processing.[436] For instance, employers need to aggregate and process troves of employees' personal data for personnel administration purposes.[437] This is evident during the phase of hiring, when employers collect information about all aspects they deem important in order to vet candidates, as well as throughout the entire cycle of the employment relationship.[438] Data generated by workers are crucial to assessing their performance, to rewarding or sanctioning them and, more broadly, to organising production. Using workers' data to inform decision-making is aimed at making more rapid, accurate, and objective managerial decisions.

It is undisputed that advanced technologies such as smartphones, desktops, tablets, vehicles, and wearables can be used to monitor and control the behaviour of individuals.[439] As a consequence, the proliferation of interconnected systems has expanded employers' monitoring capabilities. New technologies alter **power dynamics** and the **information imbalance** between contractual parties, largely in favour of data holders. Moreover, digital devices and software allow the production, collection, storage, and processing of an unprecedented volume of data springing from multiple sources. These data lay the groundwork for AM applications which, in turn, generate new data. The combination of these shifts also raises new questions. Therefore, research is ballooning into how to attune the existing frameworks in this area with new AM practices.[440]

The scope, variety, and velocity of the 'datafication'[441] of the workplace represent a partial novelty that is **putting to a test the suitability and effectiveness of existing legal frameworks**.[442] First, the principles regarding employee monitoring were conceived for work settings in which authority was exercised directly and immediately. Due to the spike in the adoption of remote and flexible or hybrid work schemes, the boundaries between professional and private lives – and between home and work premises – are increasingly being blurred. Consider, for example, the use of personal devices for work-related purposes, and *vice versa*. As will be demonstrated in this section, data protection rules are **particularly relevant when it comes to regulating data collection and processing**, but **only partially effective as regards some functions of AM**. Despite this, there is significant room for manoeuvre in establishing more specific regulations where necessary,

[436] For example, Art. 6 of Directive 89/391/EEC on the introduction of measures to encourage improvements in the safety and health of workers at work establishes that an employer must collect data and information on workers in order to maintain adequate levels of health and safety. Art. 6(3) requires the employer, following a risk assessment, to 'ensure that the planning and introduction of new technologies are the subject of consultation with the workers and/or their representatives, as regards the consequences of the choice of equipment, the working conditions and the working environment for the safety and health of workers'.

[437] Hendrickx, F. (2019). Privacy 4.0 at work: regulating employment, technology and automation, *Comp. Lab. L. & Pol'y J.*, *41*, 147.

[438] The Article 29 Working Party (hereinafter Art. 29 WP), Opinion 08/2001 on the processing of personal data in the employment context, WP 48 (mentioning the following data as possible relevant data in the employment context: application forms and work references; payroll and tax information-tax and social benefits information; sickness records; annual leave records; unpaid leave/special leave records; annual appraisal/assessment records; records relating to promoting, transfer; training, disciplinary matters; records relating to accident at work; information generated by computer systems; attendance records; family members; reimbursement of expenses, e.g. travel). Art. 29 WP is the advisory body replaced by the European Data Protection Board (EDPB) on 25 May 2018. The opinions of the Art. 29 WP nevertheless remain relevant and valid now that it has been replaced by the EDPB.

[439] European Fundamental Rights Agency (2010). Data Protection in the European Union: The Role of National Data Protection Authorities. Luxembourg: Publications Office of the European Union, p. 37.

[440] Abraha, H. (2023). Regulating algorithmic employment decisions through data protection law, *European Labour Law Journal*, *14*(2), 172–191.

[441] Adler-Bell, S., & Miller, M. (2018). The datafication of employment: how surveillance and capitalism are shaping workers' futures without their knowledge. *The Century Foundation.*

[442] According to Article Art. 29 WP, the adoption of technologies 'that enable more systematic processing of employees' personal data at work' is 'creating significant challenges to privacy and data protection'. Art. 29 WP, Opinion 2/2017 on data processing at work WP 249.

as long as these regulations align with fundamental data processing principles and other relevant requirements.

3.3.1. EU legal framework

3.3.1.1. Data collection and storage, processing and portability

Privacy is conceived as a **fundamental right** whose ambit of application includes 'everyone'.[443] Needless to say, this extends to relationships in work contexts as well. However, workers are found to be in a situation of information asymmetry, a structural condition of the labour market. This **asymmetry complicates the exercise of their right to privacy**, making it challenging for workers to fully assert and protect this fundamental entitlement.

When discussing the relevance and suitability of the *acquis* in this area, it is crucial to consider its 'complex, perhaps intricate, **multi-source regulatory architecture**',[444] which comprises international legal provisions, the General Data Protection Regulation (Regulation (EU) 2016/679) (GDPR), and domestic measures with regard to employee monitoring and data processing at work. This corpus of law is supplemented by a growing body of guidance and decisions in individual cases issued by **data protection authorities** (DPAs).[445] To this must be added newly proposed instruments such as the proposed Platform Work Directive (PWD) and the draft Artificial Intelligence (AI) Act. This multi-layered legal system, though designed to be comprehensive, can sometimes lead to confusion, potentially making companies hesitant to embrace new technologies. However, it is vital to recognise that the GDPR's enforcement model emphasises proximity to data subjects, and consistent application is fortified by the European Data Protection Board, as outlined in Art. 60, and following the GDPR.

In principle, the GDPR is relevant and suited to addressing at least some of the challenges related to certain practices executed in the context of AM systems, such as lawful data processing and solely automated decision-making.[446] In the case of hiring, promotion, wage setting and termination – to highlight only a few of the phases of an employment relationship – decisions may only be made in reliance on personal data collected and processed if they follow both the substantive and the procedural rules that will be discussed in the paragraphs that follow. **The number of provisions of the GDPR that are in principle relevant to regulating some AM functions is noteworthy. Nevertheless, the novel nature of AM systems places strain on the existing GDPR rules, highlighting potential areas for improvement within the current legal framework.**

A. Employee monitoring

Art. 88(1) GDPR specifically references 'the context of employment', and permits the establishment of 'more specific rules', through laws or collective agreements, to safeguard the rights and freedoms related to processing employees' personal data. This provision acknowledges that employers may process employees' personal data for a variety of purposes, such as recruitment, the performance of the contract of employment, and the management of the workforce. A comprehensive set of cases is considered with reference to the organisation of work, equality and diversity in the workplace, safety and

[443] Article 12, Universal Declaration of Human Rights; Article 8, European Convention of Human Rights; and Article 7, European Charter of Fundamental Rights.

[444] Aloisi, A., & De Stefano, V. (2022). Your Boss is an Algorithm: Artificial Intelligence, Platform Work and Labour, Hart Publishing. See also: European Declaration on Digital Rights and the Principles of the Digital Decade adopted by the European Parliament, the Council, and the Commission on 15 December 2022

[445] Hießl, C. (2023). Jurisprudence of national courts in Europe on algorithmic management at the workplace, European Centre of Expertise in the field of labour law, employment, and labour market policies (ECE).

[446] Kellogg, K.C., Valentine, M.A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control, *Academy of Management Annals*, *14*(1), 366-410.

health at work, the protection of the employer's or a customer's property, the exercise of rights and benefits related to employment and the termination of the employment relationship. It is recognised that this situation can give rise to risks to the fundamental rights and freedoms of employees, such as their right to privacy. Moreover, Art. 88(2) GDPR stipulates that the rules laid down under Art. 88(1) include '**suitable and specific measures to safeguard the data subject's human dignity, legitimate interests and fundamental rights, with particular regard to the transparency of processing**, sharing personal data among business groups or entities involved in joint economic activities, **and monitoring systems at the workplace**'. The empowerment of MS and social partners to introduce more focused provisions suggests that the GDPR recognises the **special nature of monitoring and personal data processing in the context of employment**, and the distinct challenges they present. In essence, the GDPR provides a robust foundation, while allowing for more focused interventions in alignment with its provisions.

As Abraha remarks, 'the extent to which [MS] can utilise the facultative clause under Article 88 GDPR, and the yardstick to measure whether national legislation meets the minimum requirements of the clause remain unclear'.[447] The very first important **benchmark to evaluate the interaction between national legislation and EU law in the field of data protection** was provided by the CJEU in March 2023, in the *Hauptpersonalrat der Lehrerinnen* case.[448] This decision concerned the compatibility of some provisions of the Hessian Law on Data Protection and Freedom of Information and the Hessian Civil Service Act with the conditions laid down in Art. 88 GDPR. In short, although MS have the discretion to establish more specific rules governing the processing of personal data, their freedom in doing so is not unfettered. More detailed rules introduced on the basis of Art 88 GDPR need to **satisfy the conditions of both paragraphs 1 and 2, which pertain to specific and suitable safeguards to protect employees' rights as data subjects. If this is not the case, the national court must disregard these 'more specific rules'**.[449] The interpretation of domestic rules should consider the goal of the GDPR to attain a harmonised, consistent and uniform application of data protection rules throughout the EU. In doing so, such specific rules 'must have a normative content…, which is distinct from the general rules of [the GDPR]'.[450] Merely restating the GDPR principles and rules within national legislation would not serve this objective, given that the GDPR is directly enforceable in each MS. As a result, a national provision that duplicates or refers to relevant GDPR provisions does not fulfil the criteria outlined in Art. 88.

Art. 88 GDPR should be seen as an enticing **opportunity** for regulating AM instruments and practices through domestic law and/or national collective agreements.[451] This provision lays the foundation for both national lawmakers and social partners to play an important role. Experts agree that responsive and participatory methods can uphold adaptation to the changing world of work while reducing uncertainty (see the discussion in Section 3.5 Workers' collective rights). In this respect, the *Hauptpersonalrat der Lehrerinnen* case offers valuable guidance on how to most appropriately implement the optional specialisation clause included in Art. 88 GDPR.

B. Workers' privacy and data protection

In conjunction with a mosaic of domestic labour-related schemes, **the GDPR affords meaningful protection by regimenting the process by which data can be collected**

[447] Abraha, H. (2023). C-34/21 - Hauptpersonalrat v Wiesbaden: Article 88 GDPR and the Interplay between EU and Member State Employee Data Protection Rules, *Modern Law Review*, available at SSRN: https://ssrn.com/abstract=4446107

[448] Case C-34/21, *Hauptpersonalrat der Lehrerinnen und Lehrer beim Hessischen Kultusministerium v Minister des Hessischen Kultusministeriums*, ECLI:EU:C:2023:270

[449] See, in particular, paras 74-75 and 82 of the judgement.

[450] Ibid at para 61.

[451] De Stefano, V., & Taes, S. (2023). Algorithmic management and collective bargaining. *Transfer: European Review of Labour and Research*, *29*(1), 21-36.

and otherwise processed, limiting the subtlest applications of AM, and rendering data-driven business practices accountable. The enforcement of rights, such as information, notification, access, rectification, and human intervention is complex, but could constitute an advantageous way of designing and implementing AM systems that perform well and are privacy compliant.

The **GDPR** applies across the EU. Its personal, material, procedural and territorial[452] scope is wide-ranging and applicable to both employees and self-employed persons.[453] This aspect overcomes the limitations of the binary divide between employment and self-employment. A similarly broad personal scope can be found in Chapter III of the proposed **Platform Work Directive** (**PWD**), labelled 'algorithmic management', which would mandate strong information and disclosure rights.[454]

Given the ample definition of **personal data**,[455] which covers most of the information relevant for HR purposes,[456] the **GDPR is eminently relevant in the case of most AM practices**.[457] The notion of personal data includes subjective evaluations of individuals, regardless of whether the information is true, correct or demonstrated. Data that have undergone pseudonymisation are also covered.[458] Under the GDPR, **inferred personal data**, meaning data based on assumptions about data subjects, are protected as personal data. Treating inferred personal data as **'pure' personal data** addresses important questions inherent to the functioning of AM applications. In the case of automated determinations, both input and output data can be considered 'personal'. No consensus exists as to whether processes of so-called 'inferential analytics', namely the process by which algorithms extract recurring patterns from large amounts of data and advance real-time predictions,[459] fall within the scope of the GDPR. Despite this, it can be admitted that it would be unreasonable to limit the collection of data that is based on concrete facts, without limiting the inferences drawn from such data.[460] An example of this is the case of workers who are 'classified', 'clustered' or 'rated' on the basis of new information that is generated from data that they have shared in the first instance. Art. 29 WP considers both verifiable and unverifiable inferences to be classified as personal data. However, it does not conclusively determine whether the reasoning and methods behind these inferences should be treated in the same way. The CJEU has not yet reached a definitive stance on these matters.[461]

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[452] The law extends its reach to any company processing the data of EU citizens. See Bradford, A. (2012). *The Brussels Effect*, NW. U. L. REV. 107:1.

[453] As stressed by Art. 29 WP, '[w]here the word 'employee' is used in this Opinion [on data processing at work], WP29 does not intend to restrict the scope of this term merely to persons with an employment contract recognised as such under applicable labour laws. Over the past decades, new business models served by different types of labour relationships, and in particular employment on a freelance basis, have become more commonplace'. See Art. 29 WP, Opinion 2/2017 on data processing at work.

[454] Proposal for a Directive of the European Parliament and of the Council on improving working conditions in platform work, COM (2021) 762 final (9 December 2021). Arts. 6, 7(1) and (3), and 8 apply to all persons performing platform work irrespective of their employment status.

[455] Defined in Art. 4 GDPR as 'any information relating to an *identified or identifiable natural person* ('data subject'); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural, or social identity of that natural person' (emphasis added).

[456] De Stefano, V., & Wouters, M. (2022). AI and digital tools in workplace management and evaluation: An assessment of the EU's legal framework.

[457] Purtova, N. (2018). The law of everything. Broad concept of personal data and future of EU data protection law. *Law, Innovation and Technology*, *10*(1), 40-81; Gellert, R.M. (2021). Personal data's ever-expanding scope in smart environments and possible path (s) for regulating emerging digital technologies.

[458] Recital 26.

[459] Bodie, M.T., Cherry, M.A., McCormick, M.L., & Tang, J. (2017). The law and policy of people analytics. *U. Colo. L. Rev.*, *88*, 961; Otto, M. (2018). Workforce analytics v fundamental rights protection in the EU in the age of big data. *Comp. Lab. L. & Pol'y J.*, *40*, 389.

[460] Aloisi, A., & De Stefano, V. (2022). *Your boss is an algorithm: artificial intelligence, platform work and labour*. Bloomsbury Publishing, p. 47-48.

[461] Wachter, S., & Mittelstadt, B. (2019). A right to reasonable inferences: re-thinking data protection law in the age of big data and AI, *Colum. Bus. L. Rev.*, 494.

The GDPR principles govern the design of law-abiding data processing systems and the trustworthy implementation of related operations.[462] However, **the broad framing of its overarching principles and provisions does not ensure they are effectively applied in practice in the context of AM systems used in work settings**. As mentioned, though, the GDPR also foresees the introduction of **more specific rules to address such nuances**. In many countries, DPAs have adopted many 'context-specific' guidelines and recommendations, as well as orders and fines issued. A constant interpretive effort has been made by companies to master and adapt internal procedures in consideration of the DPAs' guidelines and decisions.

a. General principles

Art. 5 GDPR lays down a catalogue of **paramount principles** relating to the processing of personal data that must be complied with. These principles are especially relevant to AM systems. Personal data must (a) be processed in a lawful, fair and transparent manner; (b) collected for specified, explicit and legitimate purposes, and not further processed in a manner that is incompatible with those initial purposes; (c) must be adequate, relevant, and limited to what is necessary in relation to the purposes for which they are processed; (d) must be accurate and, where necessary, kept up to date; (e) kept in a form which permits identification of data subjects for no longer than is necessary for the purposes for which the personal data are processed; (f) processed in a manner that ensures appropriate security of the personal data, including protection against unauthorised or unlawful processing, and against accidental loss, destruction, or damage, using appropriate technical or organisational measures.

Lawfulness, fairness and transparency, purpose limitation, data minimisation, accuracy, storage limitation, integrity and confidentiality are the keystones of the legitimate processing of personal data that are used to fuel AM systems. These principles are expected to prevent the most intrusive applications of AM systems. To give an example, they do not allow employers to install devices that capture and process 'as much data as possible', or to feed them into software that can use them for purposes other than the original goal(s) for which such data were collected. This is particularly pertinent, given the multi-use, evolvable and versatile nature of AM systems: the GDPR principles could limit the operation of the most advanced and problematic AM applications, and may simultaneously reduce the risks of such systems being used improperly. The principles of purpose limitation and data minimisation are particularly prominent. Upon further examination, they sit uneasily with predictive and prescriptive AM systems. These principles address the processing for purposes that are 'unspecific, imprecise, or inexplicit'[463] and limit the problem of 'function creep', which involves the progressive widening of the use of AM systems beyond the purposes for which they were originally intended.[464]

Under Art. 6(4), the GDPR sets out broadly conceived yardsticks that can support the implementation of the **purpose limitation principle** by helping the data controller ascertain whether the processing of data for another purpose is compatible with the purpose for which the personal data were initially collected.[465] Among these, one can find: (i) any link between the purposes for which the personal data have been collected and the purposes of the intended further processing; (ii) the context in which the data have been collected, in particular regarding the relationship between data subjects and the controller; (iii) the nature of the personal data, in particular whether special categories of personal data are processed or whether personal data related to criminal convictions and offences are processed; (iv) the possible consequences of the intended further processing for data subjects; and (v) the existence of appropriate safeguards, which may include encryption or

[462] Waas, B. (2022). *Artificial Intelligence and Labour Law*, 127-128.

[463] Butterworth, M. (2018). The ICO and artificial intelligence: The role of fairness in the GDPR framework. *Computer Law & Security Review*, *34*(2), 257-268.

[464] For a discussion on the challenge of function creep, see Chapter 2 of this report.

[465] See also: Art. 29 WP, Opinion 03/2013 on purpose limitation.

pseudonymisation. Sartor and Lagioia remark that, in the case of personal data used to draw individualised inferences and predictions, the legitimacy of the repurposing activity must be assessed by considering several criteria, such as the proximity between the new purpose and the original purpose, the consistency of the new purpose with the data subjects' expectations, and the controller's 'guard rails' to minimise risk and prevent nefarious consequences.[466]

The impact of the principle of **data minimisation** on some AM systems is particularly noteworthy. Only those data that are necessary to pursue the intended goal can be collected, processed, and kept only for the period that is necessary, considering the goal. This conflicts with some 'scraping' or 'crawling' functions of AM tools, which extract data from multiple sources, and end up including information that is not relevant to the specified objective. One example is hiring or CV-screening software, which is programmed to evaluate candidates by mapping and processing available information from social media,[467] such as pictures (a worker who is smoking or attending a given event, for instance), from which it is easy to infer not only relationships, but also possible illnesses, economic conditions and political affiliations.[468] This AM-based HR function is, in principle, not permitted under the GDPR. However, the existence in the market of software that purportedly incorporates such advanced capabilities begs several questions as to the effectiveness of these very broadly conceived principles.

Art. 29 WP provides a useful illustration of the application of the principles mentioned above.[469] A tool for managing mobile devices (e.g. wearables or GPS-equipped phones) can be used by employers 'to locate devices remotely, deploy specific configurations and/or applications, and delete data on demand'.[470] It can also be used for real-time tracking of the device.[471] Upon carrying out a **data protection impact assessment** to evaluate compliance with the principles of proportionality and subsidiarity, further discussed below, employers (or data controllers, following the GDPR terminology),[472] 'must ensure that the data collected as part of this remote location capability is processed for a specified purpose and does not, and could not, form part of a wider programme enabling ongoing monitoring of employees'.[473] This example can be extended to other circumstances. In general, **the GDPR is suitable for preventing the use of AM tools to capture, accumulate and use data for purposes other than those clearly specified**. On paper, the principles of purpose limitation and data minimisation are effective in limiting the collection of unrelated and overabundant data that not even 'the data subject considered or could have even imagined' (sometimes referred to as 'garbage-in, garbage-out').[474] However, **due to the functioning of AM tools, compliance with the principles of purpose limitation and data minimisation could affect the accuracy and appropriateness of the AM functions**, as the final decision could be based on a smaller amount of data. This highlights the tension arising between certain GDPR principles when confronted with the reality of AM systems.

Implementing the principle of **accuracy** is a way to prevent errors, unintended consequences, and lack of trust. When datasets are considered inaccurate, the worker has

[466] Sartor, G. & Lagioia, F. (2020). *The impact of the General Data Protection Regulation (GDPR) on artificial intelligence*, pp. 45-46. See also Art. 6(4) GDPR.

[467] Art. 29 WP, Opinion 2/2017 on data processing at work, p. 11.

[468] Aloisi, A. & De Stefano, V. (2022). p. 51.

[469] Art. 29 WP, Opinion 2/2017 on data processing at work, p. 17.

[470] Ibid.

[471] Ibid.

[472] EDPB, Guidelines 07/2020 on the concepts of controller and processor in the GDPR Version 1.0 Adopted on 02 September 2020.

[473] Ibid.

[474] EDPS, Opinion 3/2018 Opinion on online manipulation and personal data, p.15.

grounds to 'obtain from the controller restriction of processing'.[475] As per the draft **AI Act**,[476] datasets must be of sufficiently high quality, meaning that the training, validation, and testing must be relevant, representative, free from error, and complete. The potential presence of errors in the dataset may deter employers from using AM tools.

b. Legal bases for lawful processing

Identifying the most **appropriate legal basis for lawful processing** is a precondition that must be fulfilled before processing personal data used by AM systems. Art. 6 GDPR enumerates several conditions of lawfulness for processing: consent given by the data subject; necessity in the context of a contract; compliance with a legal obligation; protection of vital interest of the data subject; performance of a task carried out in the public interest or in the exercise of official authority vested in the controller; and legitimate interest, as long as the interests or fundamental rights and freedom of the data subject are not endangered. It is expected that the basis of 'contractual necessity' would be commonly used in this context of AM at work.

Given the unbalanced positions of the contractual parties, '**consent**' (conditions for legitimacy of which are further specified under Art. 7 GDPR), a widely used legal basis in commercial transactions, **is deemed not to be a solid legal ground for lawful processing in the employment context**.[477] As stated by Art. 29 WP, 'employees are seldom in a position to freely give, refuse, or revoke consent', given the consequences they may face and the fear of retaliation. Since the use of consent to justify the adoption of AM systems is highly questionable, employers are expected to rely on another legal ground. Moreover, according to the European Data Protection Board (EDPB),[478] obtaining consent does not reduce the controller's obligations to observe the GDPR principles on processing, especially those regarding fairness, necessity, proportionality and data quality.

Typically, another legal ground could be the '**necessity**' of such processing '**for the performance of a contract'** to which the data subject is party, or to take steps at the request of the data subject before entering a contract' (Article 6(1)(b)). For example, AM tools can be used to process payslips and other administrative documents. Using the basis of contractual necessity could be a way to introduce AM systems. Far from being regarded as 'safe conduct', in order for this legal ground to be valid, the 'necessity' of the processing is subject to a balancing test that considers whether the data collection or processing is required to achieve a legitimate purpose; whether it outweighs the right to data protection; as well as the measures that have to be taken to ensure that infringements of the rights to private life and secrecy of communications are limited to a minimum. As De Stefano and Wouters remark, while some extent of managerial control is necessary for the performance of an employment contract, it does not mean that the implementation of intrusive AI mechanisms is strictly necessary.[479] In other words, **it is not a given that all AM systems are strictly 'needed' in order to fulfil contractual obligations**. Rather, less privacy-infringing tools and models could be available and could be favoured by companies. Employers are encouraged to evaluate between available alternatives and to select privacy-enhancing AM applications.

Similar considerations apply to the legal ground of **legitimate interests**,[480] another basis for uses that exceed those included in the notion of contractual necessity. This basis could be

[475] Art. 18 GDPR.

[476] Proposal for a Regulation of the European Parliament and of the Council Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act) and Amending Certain Union Legislative Acts, COM (2021) 206 final (22 April 2021) (AI Act).

[477] Niezna, M., & Davidov, G. (2023). Consent in Contracts of Employment. *The Modern Law Review*.

[478] EDPB, Guidelines 05/2020 on consent under Regulation 2016/679 Version 1.1 Adopted on 4 May 2020.

[479] De Stefano, V., & Wouters, M. (2022), p. 36.

[480] Article 6(1)(f)). On a related matter, see the interpretation of the GDPR in the context of data processing by an online social network operator. Case C-252/21, *Meta Platforms Inc., formerly Facebook Inc., Meta Platforms Ireland Ltd, formerly Facebook Ireland Ltd, Facebook Deutschland GmbH v Bundeskartellamt*, ECLI:EU:C:2023:537 (on the criteria for lawful data processing under Article 6(1) of the GDPR, focusing in particular on two points: (b) and (f)).

adopted to justify worker monitoring to protect their safety, secure company assets, track productivity or ensure the smooth operation of the business. In the case of AM systems introduced in reliance on this legal basis, workers can exercise their right to object to data processing (Art. 21 GDPR). As explained by the Art. 29 WP, the legitimate interests basis 'leaves much room for interpretation and has sometimes […] led to lack of predictability and lack of legal certainty'.[481] However, such a concept must not be interpreted in an 'absolute' sense, but as 'a requirement to apply the principle of proportionality, balancing the employer's interests in data processing against those of the employee concerned'.[482] In short, if 'an application provides benefits that are outweighed by the disadvantages imposed on the data subjects, […] the application fails to have a basis according to Article 6(1)(f)'.[483]

According to preliminary analyses of the legal provisions applying to **AI-driven applications**, the legal grounds of contractual necessity, legitimate interests, and the necessity for compliance with a legal obligation could be in principle used to justify the adoption of AM tools. Despite this, these grounds can **hardly be invoked to support or legitimise the most intrusive and privacy-infringing applications of AM**, unless this can also be justified in light of the benefits that they would bring to workers upon successfully passing the 'balance of competing interests' test. To be more specific, a movement-tracking device could be introduced upon justification of its necessity to foster compliance with occupational safety and health requirements in a warehouse, where mechanical harms are at risk of collision with workers. After being installed and used for such purposes, the data collected by the instrument should not be used to assess the quality of a given worker's performance, nor can it be used to fuel a model based on incentives and disincentives. While these principles appear in theory to address many aspects relevant to AM, in practice, their effectiveness faces challenges due to the mixed-use and self-reinforcing characteristics of AM.

c. Sensitive data

The processing of categories of **sensitive personal data** 'revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership' are prohibited under Art. 9(1) GDPR. The same prohibition applies to the processing of 'genetic data, biometric data […], data concerning health or data concerning a natural person's sex life or sexual orientation'. According to the CJEU, the 'strengthened protection regime' regarding sensitive personal data extends to the processing of personal data that could indirectly reveal sensitive information concerning a natural person.[484]

This rule may **restrict the deployment of some AM functionalities** that are driven by such sensitive personal data. However, this general limitation does not apply when such processing 'is necessary for the purposes of carrying out the **obligations and exercising specific rights of the controller or of the data subject in the field of employment and social security and social protection law** in so far as it is authorised by Union or Member State law or a collective agreement providing for appropriate safeguards' (Art. 9(2)(b)). Moreover, sensitive personal data can be processed when such processing is necessary for the purposes of preventive or occupational medicine, for the assessment of the working capacity of the employee, for medical diagnosis, for the provision of health or social care or treatment, or for the management of health or social care systems and services (Art. 9 (2)(h)). The general prohibition does not apply when sensitive data are manifestly made

[481] Article 29 Working Party Opinion 06/2014 on the notion of legitimate interests of the data controller under Article 7 of Directive 95/46/EC (9 April 2014) WP217.

[482] Bundesarbeitsgericht [BAG] [Federal Labor Court] Nov. 17, 2016, 2 AZR 730/15, *Neue Zeitschrift für Arbeitsrecht*,394, 2017 (Ger.); Kort, M., *Der Beschäftigtendatenschutz gem. § 26 BDSG-neu, Ist die Ausfüllung der Öffnungsklausel des Article 88 DS-GVO geglückt? Zeitschrift für Datenschutz,* 319, 320 (2017); Seifert, A. (2019). GDPR, *Article 88, margin 57,* in: *Datenschutzrecht,* S. Simitis, G. Hornung, I. Spiecker gen. Döhmann (eds.), 1st ed, Nomos.

[483] Sartor, G., & Lagioia, F. (2020). The impact of the General Data Protection Regulation (GDPR) on artificial intelligence, p. 22

[484] See Case C-184/20, *Vyriausioji tarnybinės etikos komisija*, 1 August 2022, ECLI:EU:C:2022:601 (inferred special categories of data are special categories of data as defined in Article 9 of the GDPR).

public by the data subject (Art. 9(2)(e)). This general ban is relevant to the field of **biometric data** that can be collected by wearables (e.g. step counters, sensors measuring gait, heart rate or menstrual cycles, and devices that enable sleep analysis). For instance, some AM applications could be programmed to combine health data that are collected by relying on the 'contractual necessity' ground in the context of hiring procedures, with publicly available data (such as posts on social media, such as an online review of a medical establishment), in order to infer information about the worker's current health status, personality traits or future behaviour. AM functionalities based on such a combination of data and inferences would not be deemed legitimate. Under the GDPR, it is always and only the responsibility of the data controller to comply with, and be able to demonstrate compliance with, the GDPR rules. However, **important challenges may arise at the level of enforcement**.[485]

The GDPR framework is complemented by the proposed Platform Work Directive (PWD). **Art 6 PWD** prohibits the processing of data that is not strictly necessary for the performance of the contract, such as personal, emotional, or psychological data, or health data. This also applies to personal data with regard to private conversations, including communication with platform workers' representatives under Art. 6(5)(c) of the PWD. The PWD also prohibits the collection of personal data while the worker is not offering or performing platform work (Art. 6(5)(d)). These provisions increase the legal certainty regarding the application of GDPR in the context of platform work.

d. Technical and procedural obligations

Data controllers must be able to demonstrate that they have appropriately followed the key principles under Art. 5 GDPR and fulfilled their obligations, including those laid down in the provisions on **'privacy by design'** and **'privacy by default'**.[486] Considering the 'nature, scope, context, and purposes of the processing, as well as the risks of varying likelihood and severity for the rights and freedoms of natural persons', Art. 24(1) GDPR imposes on the data controller the duty to 'implement appropriate technical and organisational measures to ensure and to be able to demonstrate' that the data processing is performed in accordance with the key principles. In practice, this could involve the drafting of accurate privacy notice documentation that is shared with the workers. The implementation of 'appropriate technical and organisational measures for ensuring that, by default, only personal data, which are necessary for each specific purpose of the processing are processed' (Article 25(2) GDPR) can be relevant and effective in **strengthening the accountability, legitimacy, and trustworthiness of AM applications**. These 'meta-obligations'[487] facilitate conformity on the part of data controllers by imposing procedural steps that pave the way to adherence to the GDPR principles.[488] The need to fulfil such general requirements serves as a **compliance-enhancing method**; an opportunity that may reduce the risks borne by companies in terms of their vulnerability to contestation and litigation. Related to this, Article 25(1) GDPR fleshes out an employer's duty to effectively 'implement appropriate technical and organisational measures', and to 'integrate the necessary safeguards into the processing in order to meet the requirements of [the GDPR] and protect the rights of data subjects'.

[485] Presthus, W., & Sønslien, K.F. (2021). An analysis of violations and sanctions following the GDPR, *International Journal of Information Systems and Project Management*, *9*(1), 38-53.

[486] Michelakaki, C., & Vale, S.B. (2023, May). Unlocking Data Protection By Design & By Default: Lessons from the Enforcement of Article 25 GDPR. FPF: Future of Privacy Forum.

[487] Quelle, C. (2018). Enhancing compliance under the general data protection regulation: the risky upshot of the accountability-and risk-based approach. *European Journal of Risk Regulation*, *9*(3), 502-526. See also Coglianese, C., & Mendelson, E. (2010). Meta-Regulation and Self-Regulation', in: R. Baldwin, M. Cave, & M. Lodge (eds), *The Oxford Handbook of Regulation,* Oxford University Press, p. 146–168.

[488] Binns, R. (2017). Data protection impact assessments: a meta-regulatory approach, *International Data Privacy Law*, *7*(1), 22-35.

The GDPR addresses potential 'transparency deficits', thanks to participatory instruments, such as the **Data Protection Impact Assessment** (DPIA),[489] premised upon the prospect of involving workers and DPAs (Art. 36 GDPR).[490] When data processing using new technologies 'is likely to result in a high risk to the rights and freedom of natural persons',[491] a DPIA must be drafted (Art. 35 GDPR).[492] Art. 35(3)(a) GDPR **requires a DPIA in case of a 'systematic and extensive evaluation of personal aspects relating to natural persons which is based on automated processing, including profiling, and on which decisions are based that produce legal effects concerning the natural person or similarly significantly affect the natural person'**. This exercise must also be completed by the data controller to demonstrate compliance with the principles under Arts. 5 and 6 GDPR. This should not be a one-off exercise, and must be conducted prior to the implementation of AM, and updated throughout its deployment. The assessment must include a systematic description of the operations and the purposes of the processing, a clarification of its necessity and proportionality, the risks faced by workers, and measures to address those risks and demonstrate compliance with the GDPR (Art. 35(7)).[493] Significantly, the systematic description of AM systems could be disclosed to workers in a plain and accessible form, thanks to the DPIA. **The requirement to carry out a DPIA might not be applicable to every AM system**. For example, it is not mandatory in situations of automated decisions and profiling unless the decisions rooted in such profiling lead to legal consequences or have comparably significant impacts on data subjects.

By implementing this risk-based method, the DPIA offers collective and process-oriented governance tools that provide a protective shield for individual rights. Instruments such as the DPIA or **codes of conduct** (Art. 40 GDPR)[494] can prompt the redesign of internal practices to ensure full compliance with the law and, at the same time, lay the foundations for the exercise of due-process rights. Besides **contributing to the prevention of unfairness, inaccuracy and discrimination**, this technical instrument can serve to establish the substance for redress mechanisms by data subjects.[495] The DPIA procedure also presupposes an employer's ability to 'trace back' automated decisions and to demonstrate that personal data processing is informed by all of the principles laid down in the GDPR, and that the employees have received appropriate information regarding the methods and purposes of the processing. One consequence for employers is the need to scrutinise software provided by third parties to ensure they have appropriate technical understanding of its implications. In short, the DPIA emphasises risk avoidance, and ensures more collegial data processing.

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[489] Gellert, R. (2015). Data protection: a risk regulation? Between the risk management of everything and the precautionary alternative, *International Data Privacy Law*, *5*(1), 3.

[490] 'The controller must 'seek the views of data subjects or their representatives' (Art. 35(9)) where appropriate. Art. 29 WP considers that 'those views could be sought through a variety of means, depending on the context (e.g. a generic study related to the purpose and means of the processing operation, a question to the staff representatives, or usual surveys sent to the data controller's future customers) […]; if the data controller's final decision differs from the views of the data subjects, its reasons for going ahead or not should be documented', Art. 29 WP, Opinion 2/2017 on data processing at work.

[491] See Recital 75.

[492] Janssen, H.L. (2020). An approach for a fundamental rights impact assessment to automated decision-making. *International Data Privacy Law*, *10*(1), 76-106.

[493] Drafting the document satisfies the controller's duty to implement, review and update 'appropriate technical and organisational measures to ensure and to be able to demonstrate' that the processing is compliant with the Regulation (Art. 24(1) of the GDPR). ADMS, profiling based on 'aspects concerning the data subject's performance at work', systematic monitoring and processing involving vulnerable data subjects all fall within the definition of high risk. See Art. 29 WP, Guidelines on Data Protection Impact Assessment (DPIA) (wp248rev.01) (using the example of '[a] company systematically monitoring its employees' activities, including […] employees' workstation, internet activity, etc.' as a case in which the DPIA is specifically required).

[494] Johnston, H., & Silberman, M. (2020). Using GDPR to improve legal clarity and working conditions on digital labour platforms: Can a code of conduct as provided for by Article 40 of the General Data Protection Regulation (GDPR) help workers and socially responsible platforms? ETUI Research Paper-Working Paper.

[495] See also Art. 7, Proposal for a Directive of the European Parliament and of the Council on improving working conditions in platform work, COM(2021) 762 final (introducing a *sui generis* algorithmic impact assessment of the risks of automated monitoring and decision-making systems to the safety and health of platform workers).

Taken together, these provisions trigger a **multidimensional process** whose ownership, in terms of its application, resides prominently with employers, who enjoy a margin of appreciation and manoeuvre regarding the concrete steps to ensure appropriate technical and organisational measures.[496]

In the context of **platform work**, the proposed PWD introduces a duty to 'monitor and evaluate the impact of individual decisions taken or supported by automated monitoring and decision-making systems' on the working conditions of persons performing platform work (Art. 7(1)). It involves the **constant assessment of the suitability of the safeguards introduced to address such risks, as well as the introduction of appropriate preventive and protective measures**. In other words, platforms can be required by MS to introduce a constant and evolvable plan of action that identifies and addresses risks arising from digital surveillance and ADMS. Little is said, however, about the documentation of the outcomes of the risk assessment exercise or the notification of workers' representatives and national labour authorities.[497]

C. Data portability

To whom do data belong? The general answer, in the employment context, is not unequivocal, as data generated by workers can be bundled together with company data, including those protected as commercial secrets or under intellectual property law. Art. 20(1) GDPR states that **workers have the right to receive personal data provided to the employer 'in a structured, commonly used, and machine-readable format'**. Workers have also the 'right to transmit those data to another controller (read: employer) without hindrance from the controller to which the personal data have been provided'. **Personal data** must be considered included in the provision, as confirmed by Art. 29 WP.[498] This aspect corroborates the significance of data portability in the context of AM systems at work.

Only processing operations based on the worker's consent or a contractual necessity between the worker and the employers fall under the scope of data portability. The employer shall inform the workers about the existence of this right, at the time when personal data are obtained. Albeit less ample than the right to access, which applies regardless of the legal basis, the right to data portability is relevant to AM at work, as most data collection and processing practices are likely to be based on a contractual necessity ground. As concluded by Art. 29 WP, while some processing operations in the realm of HR management, such as recruitment or pay services, will be covered by the right to data portability, in some other instances a case-by-case approach will be needed to verify whether this will be the situation.[499]

Another important limitation lies in the fact that the data portability rights only cover personal data that have been **'provided by'** the data subject (i.e. the worker) to the controller (i.e. the employer). This also covers, according to Art. 29 WP, **'observed data'** provided by the data subject, e.g. by virtue of the use of a device. However, **'inferred data' and 'derived data'** which are created by the data controller on the basis of the data 'provided by the data subject' are not covered by Article 20(1) GDPR. This means that

[496] Quelle, C. (2018). Enhancing Compliance under the General Data Protection Regulation: The Risky Upshot of the Accountability- and Risk-based Approach, European Journal of Risk Regulation 9, 502. See also Coglianese, C., & Mendelson, E. (2010). 'Meta-Regulation and Self-Regulation', in: R. Baldwin, M. Cave & M. Lodge (eds.), *The Oxford Handbook of Regulation,* Oxford University Press, pp. 146–168.

[497] Abraha, H., Adams-Prassl, J., & Kelly-Lyth, A. (2022). Finetuning the EU's Platform Work Directive. Available at: https://bit.ly/37V9A4a.

[498] Art. 29 WP, Guidelines on the right to data portability. Adopted on 13 December 2016, as last revised and adopted on 5 April 2017 WP 242 rev.01. See also: Janal, R. (2021, March). Data portability under the GDPR: A blueprint for access rights? In *Data Access, Consumer Interests and Public Welfare,* Nomos Verlag, pp. 319-342.

[499] Ibid, p. 11

assessments or profiles created by the employer by analysing other personal data collected in the process of work performance will arguably fall out of current data portability rights.

Art. 20(2) places obligations on data controllers for transferring the portable data directly to other data controllers **'when technically feasible'**. Yet there is no obligation on controllers 'to adopt or maintain processing systems, which are technically compatible'.[500] While compatibility is not made obligatory, Art. 29 WP recommends technical solutions that promote interoperability and standardisation, such as Application Programming Interfaces (APIs).

The main challenges pertaining to this right lie in its **limited scope** (only personal data provided by the data subject, and processed for the performance of a contract or based on consent), **practical limitation** (often reduced to copies of historical data), **and technical complexity** (further discussed below).[501] These difficulties, further aggravated by practical issues, such as delays in responding to requests, incomplete files and a lack of machine-readable formats, could be the reason for the limited popularity of data portability practices among workers' representatives and trade unions.

In summary, **workers may face some obstacles when attempting to move their data in the context of AM**. First, many systems often rely on complex and proprietary algorithms, making it arduous to standardise formats and ensure compatibility between different providers and users. Second, the vast amount of data generated and processed by advanced AM systems complicates data portability. These systems generate data in real time, creating massive datasets that might be difficult to handle and transfer efficiently. In addition, the effectiveness of data portability is undermined by a lack of awareness among workers, who may find it cumbersome to exercise this right due to a lack of user-friendly tools and clear, accessible guidelines. Furthermore, privacy concerns play a role in hindering effective data portability. AM systems may include and process sensitive personal information and ensuring that these data remain secure during the portability process is of the utmost importance. Companies may be reluctant to allow seamless data transfers for fear of data breaches or unauthorised access.

Only by overcoming these legal, practical, and technical barriers can data portability truly deliver its intended **benefits**, fostering innovation, competition, informational self-determination and worker empowerment. The right to portability pursues the goal of 'empowering' data subjects and giving them more control over their data. **Commentators propose to use this right in order to transfer data to worker 'data trusts'**[502] – a type of secure system and independent legal entity that manages data on behalf and in the interests of workers. This concept originates in common law countries but could be facilitated by the so-called **Data Governance Act**,[503] which promotes data-sharing services that must comply with fiduciary duties. This Act could stimulate more responsible business models for data sharing,[504] and could bolster the ability of workers and works councils to aggregate their bargaining power over data. The adoption of data governance models for **data stewardship**, such as data trusts, data collectives and data cooperatives, has been suggested by European social partners,[505] with the aim of providing workers with access to

[500] Recital 68 GDPR.

[501] Commission Staff Working Document, Impact Assessment Report Accompanying the document Proposal for a Regulation of the European Parliament and of the Council on harmonised rules on fair access to and use of data (Data Act) SWD/2022/34 final.

[502] Calacci, D., & Stein, J. (2023). From access to understanding: Collective data governance for workers. *European Labour Law Journal*, *14*(2), 253–282.

[503] Regulation (EU) 2022/868 of the European Parliament and of the Council of 30 May 2022 on European data governance and amending Regulation (EU) 2018/1724 (Data Governance Act) (Text with EEA relevance).

[504] Nogarede, J. (2021). No Digitalisation Without Representation. An Analysis of Policies to Empower Labour in The Digital Workplace, *FEPS Policy Study, 34*. See also Fostering a digital ecosystem for AI (OECD AI Principle 2.2).

[505] ETUC (2020). Resolution on the European strategies on artificial intelligence and data, https://bit.ly/ETUCres. See also: Ada Lovelace Institute and UK AI Council (2021). Exploring legal mechanisms for data stewardship, available at: https://understandingpatientdata.org.uk/accountability-transparency.

and rights over the collection, analysis and storage of data concerning them.[506] An advantage of such models would be **promoting trustworthy and benign use of the data that is collected or used by AM applications**.[507]

One legal constraint on the mobility of data is the inclusion of data from other (non-consenting) data subjects in a dataset of personal information. Data in the workplace are often relational.[508] Also, the **technical challenges** associated with the strategy are even more considerable. The inability to transfer or display records relating to their careers or to other achievements would increase workers' dependence on the employer. Data migration can be impractical and cause 'lock-in' effects – that is, a situation of heavy dependence on a particular system, making it complicated to switch to an alternative option, given the technical or economic costs (which could be disproportionate, especially for small companies). Lock-in effects can hurt competition and innovation.

3.3.1.2. Transparency

Transparency is a baseline right, when it comes to the adoption and deployment of AM tools and practices. The way in which such a right is ensured and exercised is key to avoiding transparency-enhancing provisions being turned into empty promises. The complexity of AM systems and the opacity of the so-called 'black boxes' involved constitute a threat to the effectiveness of rights to information and legibility. At the same time, **transparency must be seen as a means to ensure the trustworthiness of AM technologies, rather than a goal in itself**. Transparency helps in understanding contextual factors and can be used to enable a set of rights from multiple legal domains. Therefore, applying transparency does not mean overwhelming employers with excessive paperwork – an act that would be burdensome and time-consuming for companies. The sharing of readable, clear, accessible, and unsophisticated information is aimed at placing the onus on employers to **deploy AM processes that are not only efficient from an organisational perspective, but also reasonable and reportable**.[509] At the same time, such practice confers rights on data subjects. Broadly conceived notions of transparency are indeed essential to the field of AM, for a variety of reasons. Transparency rights and obligations represent an opportunity to increase workers' performance,[510] promote their trust in management, and reduce employee turnover. Conversely, secrecy, covert data collection and processing, and lack of communication can exacerbate a perception of AM as being punitive.[511]

A full reconstruction of the inner workings of AM systems is complicated. This does not mean, however, that employers who consciously adopt AM tools to pursue goals that could be achieved through less intrusive means, introducing key commands into the algorithm, or

[506] Colclough, C. (2020). Towards Workers' Data Collectives, https://projects.itforchange.net/digital-new-deal/towards-workers-data-collectives/.

[507] Salvi del Pero, A., Wyckoff, P. & Vourc'h, A. (2022). Using Artificial Intelligence in the workplace: What are the main ethical risks*? OECD Publishing;* Edwards, L., & Veale, M. (2017). Slave to the algorithm? Why a 'right to an explanation' is probably not the remedy you are looking for. *Duke L. & Tech. Rev.*, *16*, 18; Veale, M., & Edwards, L. (2018). Clarity, surprises, and further questions in the Article 29 Working Party draft guidance on automated decision-making and profiling. *Computer Law & Security Review*, *34*(2), 398-404.; Kaminski, M.E. (2019). The right to explanation, explained. *Berkeley Technology Law Journal*, *34*(1), 189-218. See also: Ananny, M., & Crawford, K. (2018). Seeing without knowing: Limitations of the transparency ideal and its application to algorithmic accountability. *New Media & Society*, *20*(3), 973-989; Adadi, A., & Berrada, M. (2018). Peeking inside the black box: a survey on explainable artificial intelligence (XAI). *IEEE access*, *6*, 52138-52160.

[508] Viljoen, S. (2021), 'Democratic data: A relational theory for data governance', *Yale Law J,* 13: 573-654.

[509] According to Art. 29 WP, while not indispensable, 'a complex mathematical explanation about how algorithms or machine-learning work' should also be provided if it is necessary to allow experts to further scrutinize how the decision-making process runs. Art. 29 WP, Guidelines on Automated individual decision-making and profiling 2016/679 (wp251rev.01). See also Hildebrandt, M. (2008). Profiling and the identity of the European citizen. In *Profiling the European citizen: Cross-disciplinary perspectives*, Dordrecht: Springer Netherlands, 303-343.

[510] McNall, L.A., & Roch, S.G. (2007). Effects of electronic monitoring types on perceptions of procedural justice, interpersonal justice, and privacy. *Journal of Applied Social Psychology*, 37 pp. 658-682.

[511] Ball, K. (2021). Electronic monitoring and surveillance in the workplace. European Commission Joint Research Centre, p. 17.

validating the original datasets,[512] can disclaim their responsibility by hiding behind the technical complexity of the machines involved.[513] There are fundamental **concepts in labour law that are designed to address the lack of shared information**, such as the reversal of the burden of proof on to the employer[514] (discussed under the heading of 3.4. Fairness and discrimination), the establishment of presumptions in favour of workers, the delivery of multi-stakeholder risk assessment and management exercises, or the attribution of ample powers of discovery and access to evidence to judiciaries or administrative bodies.[515] All of these mean that transparency rights and obligations extend beyond the provisions of the GDPR.

A. Transparency of decision-making

To begin with, Art. 13 and 14 GDPR require that the **data subject is informed about the purposes of the processing for which the personal data are intended, as well as the legal basis for the processing**. This duty can be fulfilled by sharing well-drafted privacy notices with workers.[516] The GDPR requires information to be provided to the worker 'in a concise, transparent, intelligible, and easily accessible form, using clear and plain language' (Article 12 GDPR). These rights promote workers' **'informational self-determination'**. In a similar vein, Art. 15 GDPR establishes the right of the data subject to obtain from the controller confirmation as to whether or not personal data concerning him or her are being processed. Where such processing is taking place, the data subject has a right of **access to the personal data** and to information regarding, among other things, the purposes of the processing; the categories of personal data concerned; the recipients or categories of recipient to whom the personal data have been or will be disclosed; where possible, the envisaged period for which the personal data will be stored, or, if this not possible, the criteria used to determine that period; the existence of the right to request from the controller rectification or erasure of personal data, or the restriction of the processing of personal data concerning the data subject, or to object to such processing; and the right to lodge a complaint with a supervisory authority. These provisions are especially pertinent when considering AM systems in the workplace. Moreover, as stressed by the EDPB, they 'should not be seen in isolation', but are instead closely interlinked with other provisions of the GDPR, in particular with data protection principles including the fairness and lawfulness of processing and the controller's transparency obligation. [517]

Crucially, Art. 13(2)(f) and Art. 14(2)(g) provide for an obligation to **notify data subjects that their data are involved in 'automated decision-making, including profiling**, referred to in Art. 22(1) and (4) GDPR'. In addition, 'meaningful information about the logic involved, as well as the significance and the envisaged consequences of such processing for the data subject', must be offered when personal data are collected from the data subject or a third party, respectively. Art. 15(1)(h) uses the same wording to establish the **individual's right to access**, whereby the data controller is required, upon request, to share meaningful information about the logic, significance, and consequences of the data processing. These provisions offer a channel for examining the lawfulness of processing

[512] Aloisi, A. (2024). Regulating Algorithmic Management at Work in the European Union: Data Protection, Non-discrimination, and Collective Rights. *International Journal of Comparative Labour Law and Industrial Relations* 40, no. 1.

[513] Todolí-Signes, A. (2019). Algorithms, artificial intelligence, and automated decisions concerning workers and the risks of discrimination: The necessary collective governance of data protection. *Transfer: European Review of Labour and Research*, *25*(4), 465-481.

[514] Art. 88(2) GDPR.

[515] See, for instance, Art. 16(2) PWD, providing that 'national courts or competent authorities [shall be] able to order the digital labour platform to disclose any relevant evidence which lies in their control'.

[516] Recital 60 GDPR.

[517] EDPB, Guidelines 01/2022 on data subject rights - Right of access Version 2.0 Adopted on 28 March 2023, which contain a comprehensive set of guidelines and examples, including one on dismissal, on p. 10.

conducted via AM, and for triggering legal remedies.[518] However, **their effectiveness depends heavily on the workers' legal literacy, and is particularly affected by the fact that such entitlements are often considered to apply to individuals, rather than to groups or communities**.

Another pressing question relates to the **technical complexity** inherent in the exercise of transparency rights. Some AM tools based on AI and machine learning systems are non-deterministic. Developers might not have a comprehensive understanding of the exact logic these systems follow, or the specific criteria they apply to make decisions. As a result, reconstructing the decision-making chain could be an exceedingly challenging task. In turn, this lack of transparency could result in unjust outcomes for certain workers or groups, reinforcing inequalities and undermining the principles of equal treatment (see the section on 3.4. Fairness and discrimination).

However, the Art. 29 WP states that employers using automated decision-making (ADM) systems are required to find **simple ways to inform the data subject about the rationale or criteria behind an automated decision**.[519] Thus, what is necessary is the sharing of **meaningful information about its logic** (including the factors and their weights), to allow the worker to understand and, possibly, to challenge the decision, or adapt their behaviour in the future to achieve different outcomes. However, it is not straightforwardly clear what type(s) and amount of information would meet the criterion of meaningfulness. A functional assessment can be carried out by considering the ultimate purpose of transparency rights. **Their goal is to allow workers to understand how their data are processed, and to offer them resources to evaluate the lawfulness of the processing and its compliance with key principles**. As reiterated by the EDPB, workers must receive 'sufficient, transparent, and easily accessible information about the processing of their personal data so that they can be aware of and verify the lawfulness of the processing and the accuracy of the processed data'.[520] Moreover, information under Art. 15(1)(h) has to be more specific in relation to reasoning that leads to specific decisions concerning the data subject who has asked for access.[521] From a business perspective, this entails the creation of an appropriate and data subject-friendly channel that allows workers to effectively assert their rights, and for managers to be entirely in control of processes.[522] In concrete terms, this requires efforts on the part of the developers, who must strive to incorporate **interpretability into AM** and machine learning models, allowing a clear understanding of the decision-making process. Emerging methods, such as Explainable AI (XAI)[523] can allow workers to gain insights into how their data are being used, and to comprehend the factors influencing automated decisions.

Information and access rights risk being of little practical help when workers lack the means to exercise them. The GDPR includes **anticipatory and collaborative measures**, such as the DPIA (discussed above),[524] aimed at streamlining compliance and its documentation. This is relevant to AM systems, as it could allow for the deployment of systems that are more accurate and less dysfunctional. In general, these requirements can increase 'data autonomy',[525] fostering the confidence of both employers and workers in such tools.

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[518] Custers, B., & Heijne, A.S. (2022). The right of access in automated decision-making: The scope of Article 15(1)(h) GDPR in theory and practice. *Computer Law & Security Review*, *46*, 105727.

[519] Guidelines on Automated individual decision-making and Profiling for the purposes of Regulation 2016/679, 25.

[520] Guidelines 01/2022 on data subject rights - Right of access Version 1.0 Adopted on 18 January 2022

[521] EDPB, Guidelines 01/2022 on data subject rights - Right of access Version 2.0 Adopted on 28 March 2023, point 121.

[522] De Stefano, V., & Wouters, M (2022). See also Arts. 17 and 18 GDPR.

[523] Gunning, D., Stefik, M., Choi, J., Miller, T., Stumpf, S., & Yang, G.-Z. (2019). XAI—Explainable artificial intelligence, *Science Robotics* 4, no. 37: eaay7120.

[524] Gellert, R. (2015). Data protection: a risk regulation? Between the risk management of everything and the precautionary alternative. *International Data Privacy Law*, *5*(1), 3.

[525] Calacci, D., & Stein, J. (2023). From access to understanding: Collective data governance for workers. *European Labour Law Journal*, 281

Transparency rights apply also to **self-employed persons,** under the Regulation on promoting fairness and transparency for business users of online intermediation services (**P2B Regulation**).[526] According to Art. 3 of this Regulation, providers of online intermediation services must ensure that their terms and conditions: (a) are written in clear and easily understandable language; (b) are readily available to business users throughout their relationship with the online intermediation service provider, including during the pre-contractual stage; (c) specify the reasons for decisions to suspend, terminate, or impose any other restrictions on the provision of their online intermediation services to business users; (d) provide information about any additional distribution channels and potential affiliate programmes, through which the online intermediation service providers may promote goods and services offered by business users; (e) include general information about the impact of the terms and conditions on the ownership and control of intellectual property rights belonging to business users. In the case of decisions involving the restriction, suspension, and/or termination of a given business user by a provider of online intermediation services, the business user concerned must be provided with a statement of reasons for that decision (Art. 4(1)). The provider of online intermediation services must allow the self-employed worker to clarify the facts and circumstances within the framework of the internal complaint-handling process. Art. 5 grants self-employed workers the right to be informed about 'the main parameters determining the rankings and the reasons for the relative importance of those main parameters as opposed to other parameters'.

Chapter III of the proposed **PWD** introduces robust **transparency rights** in relation to AM. Persons performing platform work need to be informed about the existence and specific scope of '(a) automated monitoring systems, which are used to monitor, supervise, or evaluate the work performance of platform workers through electronic means' and '(b) automated decision-making systems, which are used to take or support decisions that significantly affect those platform workers' working conditions' (Art. 6 PWD). Regarding the latter category, employed and self-employed platform workers must be informed about the criteria used to make a decision, the 'weight' of these criteria, and 'the grounds for decisions to restrict, suspend, or terminate the platform worker's account, to refuse the remuneration for work, on the platform worker's contractual status', and 'any decision with similar effects'. These rules are particularly suited to addressing **AM systems in the platform economy**.

Annex III of the draft **AI Act** is relevant to AM at work. Tools and software adopted in the context of '**employment, workers management and access to self-employment**' (point 4) are included among '**high-risk**' AI systems.[527] In particular, reference is made to AI systems that are intended to be applied for the recruitment or selection of natural persons, for deciding about promotion and the termination of work-related contractual relationships, for task allocation, and for monitoring and evaluating the performance and behaviour of persons in such relationships. The Annex captures some of the most common uses of AM in the work context, i.e. human resource management practices that 'may appreciably impact future career prospects and livelihoods of these persons' (Recital 36). The draft **AI Act defines procedural steps with which providers must comply in order to ensure that AI systems conform to certain requirements**. Providers must put in place assessment procedures prior to putting such systems on the market or into service.

According to Art. 9 AI Act, the providers of high-risk AI systems must establish, implement, document, and maintain a **risk management system throughout the lifecycle of a given AI system**. High-risk AI systems are allowed so long as a 'continuous iterative process' is put in place and systematically updated (Article 9(2)). The providers are required to identify and analyse the 'known and foreseeable risks', and to evaluate the risks that may emerge when the high-risk AI system is used in accordance with its intended purpose and under conditions of reasonably foreseeable misuse'. Moreover, they must evaluate other 'risks

[526] Regulation (EU) 2019/1150 on promoting fairness and transparency for business users of online intermediation services (Text with relevance to the EEA).

[527] Article 6(2) AI Act.

based on the analysis of data gathered from the post-market monitoring system' and, notably, adopt 'suitable risk management measures'. Within this margin of appreciation afforded to them, providers can adopt risk mitigation strategies and deploy internal controls. Such controls should be based on the verification of the quality management system, and examination of the information included in the technical documentation.[528] There should be evidence that the design and development process of the AI system is consistent with the technical documentation provided.[529]

Importantly, Art. 29 of the AI Act imposes obligations on users of high-risk AI systems. **Employers, who are likely to be users, must use the AI systems in accordance with the instructions for use that accompany such systems.** Moreover, to the extent that users exercise control over the input data, they must ensure that the input data are relevant, given the intended purpose of the high-risk AI system. Users must also ensure that high-risk AI systems are monitored. Interestingly, users of high-risk AI systems must use the information provided under Art. 13 to comply with their obligation to carry out a data protection impact assessment under Art. 35 GDPR, where applicable.

This regulatory framework highlights strong complementarities, alongside some overlaps and discrepancies. Since data are the underlying element enabling the efficient functioning of AI systems, it remains to be seen how legal tools, such as the AI Act and the GDPR, will operate in a synergetic way.[530] Even though they are based on different criteria depending on the relevant legal instrument (the AI Act list includes also activities which do not necessarily include processing of personal data, and under Art. 35 GDPR there are specific criteria in light of which each processing operation has to be assessed on a case-by-case basis). In short, the 'high-risk' classifications could produce mutual feedback effects.

3.3.1.3. Decision-making by algorithms

The use of automated decision-making systems (**ADMS**) can have 'several advantages'.[531] When employed appropriately, these systems can **improve efficiency and the quality of decisions**, as they have the capability to eliminate human errors and expedite processes. According to some commentators, incorporating AM systems at some stages of the recruitment process could prove advantageous for businesses – if not essential, in the case of companies considering large numbers of applicants.[532] ADMS is accompanied by inherent risks in the area of privacy, such as the collection and processing of sensitive information about workers. **Technical failures, algorithmic opacity, high costs and legal compliance issues** are among the other risks associated with ADMS based on algorithmic systems. Striking a balance between the benefits and risks of AI requires thoughtful planning, continuous monitoring and a commitment to lawful, sustainable, and responsible practices that ensure fair and effective processes at work.

Art. 22 GDPR is highly relevant to AM systems in workplaces, as it firmly enshrines the '**right not to be subject to a decision based solely on automated processing, including profiling**, which produces legal effects concerning him or her or similarly significantly affects him or her'.[533] To give an example of automated decision-making

[528] Veale, M., & Zuiderveen Borgesius, F. (2021). Demystifying the Draft EU Artificial Intelligence Act—Analysing the good, the bad, and the unclear elements of the proposed approach. *Computer Law Review International*, *22*(4), 97-112.

[529] AI Act, Annex VI.

[530] Bogucki, A., Engler, A., Perarnaud, C., & Renda, A. (2022). The AI Act and emerging EU digital acquis. See also: Ponce, A., & Naranjo, D. (2022). Regulating Algorithmic Management: An Assessment of The EC's Draft Directive on Improving Working Conditions in Platform Work. ETUI Research Paper-Policy Brief, *8*.

[531] Lukács, A., & Váradi, S. (2023). GDPR-compliant AI-based automated decision-making in the world of work. *Computer Law & Security Review*, *50*, 105848.

[532] Ibid.

[533] Art. 29 WP, 'Guidelines on Automated Individual Decision-Making and Profiling for the Purposes of Regulation 2016/679' (22 August 2018) WP 251 rev.01, 19; Bygrave, L.A. (2019). 'Minding the Machine v2.0. The EU General Data Protection Regulation and Automated Decision-Making', in: K. Yeung and M. Lodge (Eds.), *Algorithmic Regulation,* Oxford University Press.

(ADM), Recital 71 of the GDPR refers to decisions such as 'e-recruiting practices without any human intervention'.[534] Art. 22 GDPR refers to 'profiling', which consists of 'any form of automated processing of personal data evaluating the personal aspects relating to a natural person, in particular, to analyse or predict aspects concerning the *data subject's performance at work*, economic situation, health, personal preferences or interests, reliability or behaviour, location, or movements' (emphasis added).[535] AM systems can perform these tasks. However, for the time being, it appears that fully automated decision-making AI is not widespread.[536]

Art. 22 GDPR regulates some of the practices that are included in the definition of AM, namely e-recruiting without any human intervention, and the fully automated administration of employment-related issues. The provision, though relevant in principle, is subject to uncertainties in the interpretation of its components, which can pose challenges for those seeking to adhere to data protection rules and principles. Recently, the key components of this article, such as the notion of 'solely automated processing',[537] the inclusion of profiling, and the concept of the legal or 'similarly significant' effects, have been scrutinised by experts, DPAs and courts.[538] The variety of interpretations given to such phrases could have a **considerable impact on the effectiveness of the general ban on ADM**. Ambiguities regarding the interpretation of Art. 22 have been addressed in Art. 29 WP guidelines.[539] At the workplace level, most decisions could fulfil the 'legal impact' criterion for consideration, in light of the circumstances, of the behaviour, choice of individuals concerned, their prolonged effect, and the ultimate result. Examples of this can be fully automated choices about wage increases, promotions, bonus allocations, and termination. The determination of what constitutes a 'similarly significant' influence on the data subject is often left to **case-by-case interpretation**, as this must exceed a certain level and be notably great – a criterion that, as per Art. 29 WP, varies depending on the specific circumstances of each case.

With regard to **semi-automated systems**, Art. 29 WP has clarified that **nominal human interventions** that are entirely subservient to automatic decisions do not prevent the application of Art. 22. In the field of AM systems that require cooperation between humans and machines, **actions limited to 'rubber-stamping' or the inclusion of a 'token gesture' between the human input and a complex, fully automated decision-making process** do not weaken the application of the general prohibition.[540] This means that for a decision to qualify as being made by humans, genuine human input would require a manager to have decisional authority and flexibility to change a negative computer-generated employee evaluation, to give an example, thereby overriding the output of the AM without suffering negative consequences. In the Advocate General's Opinion in AG opinion in C-634/21, it is argued that the concept of 'decision' 'based solely on automated processing' ought to be interpreted broadly and 'can include a number of acts capable of affecting the data subject in many ways'. The AG concluded that a probability value output by an algorithm represents a 'decision' for the purposes of Art. 22 GDPR, where a (human)

[534] Del Pero, A.S., Wyckoff, P., & Vourc'h, A. (2022). Using Artificial Intelligence in the workplace: What are the main ethical risks? OECD Social, Employment and Migration Working Papers, No. 273, OECD Publishing. See also: Drożdż, A. (2020). *Protection of natural persons with regard to automated individual decision-making in the GDPR*. Kluwer Law International BV.

[535] Art. 4(4) GDPR.

[536] Leyer, M., & Schneider, S. (2021). Decision augmentation and automation with artificial intelligence: Threat or opportunity for managers? Business Horizons 64: 715.

[537] Sancho, D. (2020). 'Automated Decision-Making and Article 22 GDPR: Towards a more substantial regime for solely automatic decision-making', in: M. Ebers & S. Navas (Eds.), *Algorithms and Law*, Cambridge University Press, pp. 142–144.

[538] Bygrave, L.A. (2020). 'Article 22', in: C. Kuner et al. (Eds.), *The EU General Data Protection Regulation (GDPR): A Commentary*, New York: Oxford University Press, p. 522.

[539] Art. 29 WP, Guidelines on Automated individual decision-making, and Profiling for the purposes of Regulation 2016/679 (WP251rev.01), as last revised and adopted on 6 February 2018, p. 19 (endorsed by the EDPB), p. 21.

[540] See, for instance, *Garante per la protezione dei dati personali, Ordinanza ingiunzione nei confronti di Foodinho s.r.l.* – 10 June 2021, available at https://www.garanteprivacy.it/web/ guest/home/docweb/-/docweb-display/docweb/9675440

third party controller strongly drew on such a value for their decision on the establishment, implementation, or termination of a contractual relationship with the data subject.[541]

The general 'prohibition' of ADM[542] is subject to some **exceptions** that are **highly relevant to AM**.[543] The ban does not apply when the automated processing is: (a) '**necessary for entering into, or performance of, a contract** between the data subject and a data controller'; (b) when it is **authorised** by Union or Member State law, provided that suitable measures to safeguard the data subject's rights and freedoms and legitimate interests are laid down; and (c) when it is based on the data subject's explicit consent (Art. 22(2) GDPR). **Contractual necessity** may provide a solid justification for employers who wish to adopt ADM systems, but it is questionable whether any kind of AM tool could be presented as a legitimate and proportionate way of pursuing the aforementioned goal. As commentators argue, 'Article 22 is usually not even applicable to certain use-cases of AI-based decision-making, […when] partly automated decision-making takes place, where the AI assists and does not replace the human decision-maker'.[544] Relatedly, **consent cannot be considered a robust legal exception**, by analogy with the opinion of the Art. 29 WP, as the conditions for its validity are not met within a work relationship.[545] When automated individual decision-making without human intervention is not based on the data subject's consent or on a contract, it must be based on Union or Member State law (including the required safeguards), as stipulated under Art 22(2) GDPR.

Art. 22(3) GDPR is a central provision relevant to the use of AM in the context of employment. It states that when the ban does not apply, based on grounds of **contractual necessity** or consent grounds, 'the data controller shall implement suitable measures to safeguard the data subject's rights and freedoms and legitimate interests, *at least* the right to obtain human intervention on the part of the controller, to express his or her point of view, and to contest the decision'. These **rights to human intervention, dissent and objection can thus empower workers and encourage employers to redesign their internal processes**. In addition, a possible interpretation of Art. 22, read in conjunction with Recital 71,[546] could militate in favour of the existence of the right 'to obtain an **explanation** of the decision'.[547] However, explaining a decision made by algorithms can be particularly complex. Indeed, a socio-technical discussion is currently taking place as to the feasibility providing an explanation of automated decisions:[548] As discussed under the previous heading of 3.3.1.2. Transparency, AM users should find accessible and unsophisticated ways to match conducts and effects (e.g. Art 13 and 14 GDPR).[549] This would ensure that

[541] Opinion of Advocate General Pikamäe delivered on 16 March 2023, Case C-634/21, OQ v Land Hesse, Joined party: SCHUFA Holding AG, ECLI:EU:C:2023:220. While case C-634/21 concerns credit scoring, the AG considerations seem relevant in the AM context.

[542] Art. 29 WP 251/2017, 19

[543] Parviainen, H. (2022). Can algorithmic recruitment systems lawfully utilise automated decision-making in the EU? *European Labour Law Journal*, *13*(2), 225-248.

[544] Lukács, A., & Váradi, S. (2023). GDPR-compliant AI-based automated decision-making in the world of work. *Computer Law & Security Review*, *50*, 105848, p. 7.

[545] Art. 29 WP, Opinion 2/2017 on data processing at work.

[546] For an overview, see Casey, B., Farhangi, A., & Vogl, R. (2019). Rethinking Explainable Machines, *Berkeley Technology Law Journal*, *34*(1), 143-188; Goodman, B., & Flaxman, S. (2017). European Union regulations on algorithmic decision-making and a 'right to explanation', *AI magazine*, *38*(3), 50-57.

[547] While it is well known that recitals are not legally binding, they can 'cast light on the interpretation to be given to a legal rule'. Case 215/88 *Casa Fleischhandels-GmbH v Bundesanstalt für landwirtschaftliche Marktordnung* of 13 July 1989 ECLI:EU:C:1989:331. See Klimas, T., & Vaičiukaitė, J. (2008). The law of recitals in European Community legislation. *ILSA J. Int'l & Comp. L.*, *15*, 61. For an overview, see Casey, B., Farhangi, A., & Vogl, R. (2019). Rethinking Explainable Machines. *Berkeley Technology Law Journal*, *34*(1), 143-188; and Goodman, B., & Flaxman, S. (2017). European Union regulations on algorithmic decision-making and a 'right to explanation' *AI magazine*, *38*(3), 50-57.

[548] Hacker, P., & Passoth, J.H. (2020). Varieties of AI Explanations Under the Law. From the GDPR to the AIA, and Beyond. In: *International Workshop on Extending Explainable AI Beyond Deep Models and Classifiers,* Cham: Springer International Publishing, pp. 343-373.

[549] Malgieri, G., & Comandé, G. (2017). Why a right to legibility of automated decision-making exists in the general data protection regulation. *International Data Privacy Law*, *7*(4), 243-265. See also Wachter, S., Mittelstadt, B., & Floridi, L. (2017). Why a right to explanation of automated decision-making does not exist in the general data protection regulation. *International Data Privacy Law*, *7*(2), 76-99.

accountability is increased, and errors minimised. In general, this would lead to the prioritising of technologies over which management and labour have oversight, rather than 'off-the-shelf' AM tools designed and sold by third parties. Art. 22(4) GDPR introduces a prohibition on fully automated decisions based on special categories of personal data, with a set of exceptions.[550]

In this regard, Art. 88 GDPR could play a significant role in concretising Art. 22 GDPR in safeguarding employees' rights at the national level. As discussed above, Art. 88 GDPR grants Member States and social partners the flexibility to establish more precise regulations tailored to the context of employment. Currently, this opportunity remains largely unexplored. However, at national level, several labour-related provisions regarding disciplinary procedures and individual dismissals have been designed in such a way as to **'proceduralise' the exercise of power by employers**. To curb the risk of arbitrariness,[551] stringent due process and procedural fairness requirements must be followed. In the example of a lawful disciplinary dismissal,[552] a worker must be notified about the motives and offered an opportunity to appeal against the disciplinary decision by responding to the evidence offered in support of the dismissal. These guardrails are hardly consistent with solely automated processing.

Similarly, under **Art. 8(1) PWD**, persons performing platform work have the right to receive a **written explanation** of how decisions were reached that were *taken* or *supported* by an automated decision-making system, and which significantly affect their working conditions. This provision covers decisions on, for example, their access to work tasks, their income, occupational health and safety, working time, the possibility of promotion, contractual status, as well as the restriction, suspension or termination of their account (Art. 6(1)). The reference to the ancillary role of algorithmic tools gives the provision a broader scope than Article 22 GDPR.[553] This definition is more ample than the GDPR notion of 'decisions based solely on automated processing', as it includes semi-automated decisions that have critical effects in areas such as restriction, account suspension or termination, denial of remuneration for work done, and contractual status. Moreover, persons performing platform work can **request that a decision is revised, if the explanation provided does not prove persuasive, or if they feel hindered in relation to their rights**. The platform is obliged to provide a detailed reply within a reasonable timeframe, 'without undue delay and in any event within one week of receipt of the request' (Art. 8(2)). Given the broader scope of these provisions, the adoption of the proposed PWD could lead to a situation in which persons performing platform work would, at least on paper, and in the context of automated monitoring and decision-making, be protected to a greater extent than workers and the self-employed in traditional sectors.[554]

At international level, in 2018, the Council of Europe (CoE) adopted a Protocol to modernise the 1981 Convention for the Protection of Individuals with regard to Automatic Processing of Personal Data, which was adopted to protect human rights at a higher level. The updated **Convention 108+** sets out the right of every individual 'not to be subject to a decision significantly affecting him or her based solely on an automated processing of data without having his or her views taken into consideration' (Art. 9). This advance is particularly relevant to the context of AM, especially if read through the lens of the case law of the European Court of Human Rights, which has interpreted privacy rights as a means to

[550] Points (a) or (g) of Art. 9(2).

[551] Atkinson, J., & Collins, P.M. (2023). Artificial Intelligence and Human Rights at Work. In: *Artificial Intelligence and Human Rights*, Oxford University Press.

[552] Collins P. (2021), Automated Dismissal Decisions, Data Protection and The Law of Unfair Dismissal, in: UK Labour Law Blog, available at: https://uklabourlawblog.com/2021/10/19/automated-dismissal-decisions-data-protection-and-the-law-of-unfair-dismissal-by-philippa-collins/.

[553] Art. 29 WP 21. See also Veale, M., & Edwards, L. (2018). Clarity, Surprises, and Further Questions in the Article 29 Working Party Draft Guidance on Automated Decision-making and Profiling, *Computer Law & Security Review*, 34, 398.

[554] Aloisi, A., & Potocka-Sionek, N. (2022). De-gigging the labour market? An analysis of the 'algorithmic management' provisions in the proposed Platform Work Directive, *Italian Labour Law e-Journal*, 15(1) 29.

safeguard workers' 'personal autonomy' in workplaces.[555] **The wording of Convention 108+ is similar to that of the GDPR**. Both instruments emphasise similar principles and standards, ensuring a comparable level of safeguarding regarding data protection. According to the CoE, legislation must establish a robust and reliable framework for review that can rectify human rights violations effectively. Moreover, workers, in their capacity as data subjects, should have the right to comprehend the rationale behind data processing, including the basis for any conclusions reached in the case of ADMS.[556]

3.3.2. Selected national legislation and policies

A. EU countries

At national level, specific rules regarding monitoring tools and employment-related data collection and processing had been adopted using statutory provisions in national legislation (examples of which include the French Labour Code and Italian Workers' Statute), or through collective agreements tailored to the context of employment. Several national rules on the introduction of tools with monitoring capabilities predate the adoption of the GDPR, and differ between the EU MS. A shared characteristic of the national attitudes to surveillance devices, which clearly precede the emergence of AM, lies in their **technologically neutral approach: the emphasis is on the key function (monitoring), rather than on the source**.[557] These provisions are relevant to the case of AM tools. Their key goal is to restrain employers' monitoring powers in the context of employment. The rationale for this is the protection of human integrity and dignity, balanced against the employer's interests in ensuring the protection of its property and the smooth operation of the company.[558] In some cases, national rules allow the engagement of workers' representatives and data protection authorities.[559]

In most EU MS, **no specific regulations directly address AM at work. General labour codes and data protection regulations are nevertheless relevant to AM systems.** While the detailed analysis of such national frameworks is beyond the scope of the present study, 15 MS have either maintained or introduced more specific rules with regard to employee data processing, following the entry into force of the GDPR. Slovenia is to date the only MS that had not yet aligned its legislation with Art. 88 GDPR,[560] but a new proposal for a private data protection law is pending.[561] Finland, meanwhile, has adopted comprehensive and freestanding data protection rules applying exclusively to employment relations.[562] In Germany, Section 26 of the Federal Data Protection Act (*Bundesdatenschutzgesetz, BDSG*) lays down specific purposes for which employees' data may be processed, strict rules on obtaining consent, and conditions for the processing of

[555] See *Bărbulescu v. Romania* in the section 3.3.5. Selected case law [GC], 2017, para. 73.

[556] Council of Europe Convention 108 +, the Convention for the protection of individuals with regard to the processing of personal data, Commentary to Article 9 para. 75, available at: https://rm.coe.int/convention-108-convention-for-the-protection-of-individuals-with-regar/16808b36f1

[557] Art. L. 1222-4 of the French Labour Code; Art. 4 of the Italian Workers' Statute.

[558] *López Ribalda and Others v. Spain* [GC] - 1874/13 and 8567/13 Judgment 17.10.2019 [GC]

[559] The Italian DPA stated that compliance with domestic provisions concerning employee monitoring must be considered a precondition for meeting the requirements of the principle of lawful processing (Art 5. GDPR). See: *Ordinanza ingiunzione nei confronti di Foodinho s.r.l.* – 10 giugno 2021 [9675440]

[560] Abraha, H.H. (2022). A pragmatic compromise? The role of Article 88 GDPR in upholding privacy in the workplace. *International Data Privacy Law*, *12*(4), 276-296.

[561] *Zakon o varstvu osebnih podatkov.* This specifies, for example, conditions regarding video-control, and also conditions for implementation of such control. It also speaks about biometric data and specifies conditions for their collection and use. Available at: https://e-uprava.gov.si/drzava-in-druzba/e-demokracija/predlogi-predpisov/predlog-predpisa.html?id=10208.

[562] Section 3 of the Act on the Protection of Privacy in Working Life provides that 'the employer is only allowed to process personal data directly necessary for the employee's employment relationship, which is connected with managing the rights and obligations of the parties to the employment relationship or with the benefits provided by the employer for the employee or which arises from the special nature of the work concerned.' (759/2004; amendments up to 347/2019 included), available at: https://www.finlex.fi/fi/laki/kaannokset/2004/en20040759.pdf) .

sensitive data.[563] In some countries, the implementation of Art. 88 GDPR in national legislation has been turned into an opportunity to update previous rules and make them compatible with the GDPR, as in the cases of Poland,[564] Romania[565] and Hungary.[566]

In addition, some countries have enacted regulations specifically targeting AM at work. The **Spanish Law** 12/2021 (the so-called **'Riders' Law'**) is a prime example of this. Aside from introducing a presumption of employment for riders working through digital labour platforms, it established important data rights concerning all employers, as well as in traditional sectors. A point (d) was added to Article 64.4 of the Workers' Statute,[567] providing the works council with the right to be informed by the company about 'the parameters, rules, and instructions on which the algorithms or artificial intelligence systems are based, which used for decision-making practices, including profiling, that may affect working conditions, access, and maintenance of employment'. This law partly overcame the problem of the lack of a collective dimension of transparency rights under the GDPR.[568] The revised Workers' Statute now allows workers to learn about AM systems' parameters, rules, and instructions, but does not introduce the right to co-determine such metrics.[569] Systems for objection and redress are not included in the law.[570]

Another example of a national regulation addressing the data protection dimension of AM at work is the **Italian** Legislative Decree no. 104 of June 27, 2022 (the so-called '**Transparency Decree'**), which entered into force on 13 August 2022. This Decree transposed the Directive on Transparent and Predictable Working Conditions[571] into the Italian legal order, introducing notification obligations in the case of the use of automated decision-making or monitoring systems. Under Art. 4(1)(b) of the Transparency Decree[572], **the employer is required to inform workers of the use of (fully) automated decision-making and monitoring systems that are used to provide information for the purposes of hiring, managing, or terminating the relationship or for assigning tasks**. In addition, information must be disclosed on the purposes, logic and functioning of automated systems; the categories of data and the main parameters used to program or train such systems; the control measures adopted for the decision processes, their level of accuracy, robustness, and cybersecurity; the metrics used, as well as their potentially discriminatory impacts. A similar catalogue of information rights is provided under Art. 6 PWD. The reference to 'fully' automated decisions was introduced in 2023.[573] This could

[563] Section 26 BDSG provides that 'personal data of employees may be processed for employment-related purposes where necessary for hiring decisions or, after hiring, for carrying out or terminating the employment contract or to exercise or s*atisfy rights and obligations of employees' representation laid down by law or by collective agreements or* other *agreements between the employer and works council'*.

[564] Act of 10 May 2018 on the Protection of Personal Data, Dz. U. 2018 poz. 1000.

[565] Law No. 190/2018 Implementing the General Data Protection Regulation (Regulation (EU) 2016/679).

[566] Act CXII of 2011 on the Right to Informational Self-Determination and Freedom of Information (as amended by Act XXXVIII of 2018 to implement the General Data Protection Regulation (Regulation (EU) 2016/679).

[567] Ley 12/2021, de 28 de septiembre, por la que se modifica el texto refundido de la Ley del Estatuto de los Trabajadores, aprobado por el Real Decreto Legislativo 2/2015, de 23 de octubre, para garantizar los derechos laborales de las personas dedicadas al reparto en el ámbito de plataformas digitales, https://www.boe.es/eli/es/l/2021/09/28/12. See also the 'Guide to corporate obligations on the use of algorithmic information in the workplace and instrument for practical application', May 2022, available at: https://prensa.mites.gob.es/WebPrensa/noticias/laboral/detalle/4125..

[568] Todolí-Signes, A. (2019). Algorithms, artificial intelligence, and automated decisions concerning workers and the risks of discrimination: The necessary collective governance of data protection. *Transfer: European Review of Labour and Research*, *25*(4), 465-481; Todoli-Signes, A. (2021). Spanish riders law and the right to be informed about the algorithm. *European Labour Law Journal*, *12*(3), 399-402.

[569] Aloisi, A. (2022). Platform work in Europe: Lessons learned, legal developments and challenges ahead. *European Labour Law Journal*, 13(1), 4-29.

[570] Aranguiz, A. (2021). Spain's Platform Workers Win Algorithm Transparency. *Social Europe*. Retrieved from: https://socialeurope.eu/spains-platform-workers-win-algorithm-transparency.

[571] Directive (EU) 2019/1152 of the European Parliament and of the Council of 20 June 2019 on transparent and predictable working conditions in the European Union.

[572] This provision introduced the new Art. 1 bis to Legislative Decree no. 152 of 1997.

[573] D.L. 4 maggio 2023, n. 48 (c.d. Decreto lavoro). Another intervention in the same Legislative Decree rewrote paragraph 8 to the effect that these information obligations 'do not apply to systems protected by industrial and commercial secrecy'.

impact the effectiveness of the newly introduced provisions, but according to scholars, this specification does not affect the logic and rationale of the provision, which could be soon tested in court.[574]

Furthermore, Portuguese Law 13/2023 (the '**Decent Work Agenda**'), in force since May 2023, is relevant to AM. It provides that collective bargaining agreements can regulate the use of algorithms, AI and associated technologies only in that ways that are more advantageous to employees.[575] Employers are obliged to inform employees about the 'parameters, rules, and instructions on which algorithmic or AI systems are based'.[576]

Moreover, several regulations have addressed AM at work only in the specific context of **platform work.** For example, in Italy, the Cape V-bis to the Legislative Decree No 81/2015, addressed only to self-employed delivery couriers, reaffirmed the protection of privacy, in accordance with the GDPR and Legislative Decree No 196/2003.[577]

In some EU MS, workers must be informed about the introduction and usage of data-gathering technologies that can perform AM functions in workplaces.[578] To give an example, disclosure is needed in the case where an employer is contemplating introducing a CCTV camera, a phone with GPS, or an RFID wristband to access the work premises. Workers and their representatives should learn that such tools could be used to gather data.

B. Non-EU countries

Prominent regulatory initiatives regarding employee data protection can be found also in non-EU countries. For example, **in the US,** the Illinois Artificial Intelligence Video Interview Act has, since January 2020, imposed duties of disclosure, transparency and erasure on employers that use AI systems to analyse interviews for recruitment purposes. Similarly, since October 2020 in Maryland, employers have been prohibited from using facial recognition technology during pre-employment job interviews without first obtaining the candidate's consent.[579] Employers must inform applicants that AI is being used, and must disclose how the system works, and what variables are under scrutiny. Another important development is the California Workplace Technology Accountability Act (AB 1651),[580] which confers on workers the right to be informed of, review, correct and secure data collected from them by an employer. It enshrines the principles of data minimisation and purpose limitation, thereby limiting the ability of employers to use data beyond the purposes indicated. The Act also requires employers that adopt hiring or management technologies to draft and publish a related impact assessment document. Moreover, it mandates that human reviewers should be given sufficient authority, discretion, resources, and time to verify the outputs of AM.

In 2022, in **Canada**, the Canadian Federal Government tabled Bill C-27, which includes legislation concerning consumer privacy protection (the CPPA), as well as AI and data (the AIDA). Section 23 of this proposal provides that 'an organisation may collect, use, or

[574] Recchia, G.A. (2023). Condizioni di lavoro trasparenti, prevedibili e giustiziabili: quando il diritto di informazione sui sistemi automatizzati diventa uno strumento di tutela collettiva, *Labour & Law Issues*, *9*(1), R.32-R.57. See also: Garante per la protezione dei dati personali, Questioni interpretative e applicative in materia di protezione dei dati connesse all'entrata in vigore del d. lgs. 27 giugno 2022, n. 104 in materia di condizioni di lavoro trasparenti e prevedibili (c.d. 'Decreto trasparenza'), available at: https://www.garanteprivacy.it/web/guest/home/docweb/-/docweb-display/docweb/9844960.

[575] Article 3 (3).

[576] Art. 106(3)(iv).

[577] Article 47-sexies.

[578] This principle has been reaffirmed in Article 9 of the proposed Platform Work Directive.

[579] Aloisi, A., & De Stefano, V. (2023). Between risk mitigation and labour rights enforcement: Assessing the transatlantic race to govern AI-driven decision-making through a comparative lens. *European Labour Law Journal*, *14*(2), 283-307. See also Del Pero, A. S., Wyckoff, P., & Vourc'h, A. (2022). Using Artificial Intelligence in the workplace: What are the main ethical risks?. OECD Publishing.

[580] Algorithmic Accountability Act of 2022, HR 6580, 117th Cong (2022). For a more detailed analysis see Hilliard, A., Kazim, E., Kemp, T., & Bageire, K. (2023). Overview and commentary of the California Workplace Technology Accountability Act, *International Review of Law, Computers & Technology*, *37*(1), 91-109.

disclose an individual's personal information without their knowledge or consent if it was produced by the individual in the course of their employment, business or profession and the collection, use, or disclosure is consistent with the purposes for which the information was produced'. Organisations are allowed to 'use or disclose an individual's personal information' without their consent if 'the collection, use, or disclosure is necessary to establish, manage, or terminate an employment relationship between the organisation and the individual', provided that 'the organisation has informed the individual that the personal information will be or may be collected, used, or disclosed for those purposes' (Section 34). Consequently, in federally regulated work (which includes, among others, the banking, mail, radio, and television broadcasting sectors), employers would have the authority to introduce and use electronic and AM and monitoring systems, with employees having no recourse if they have been informed about the existence and use of such systems.

The 2018 General Data Protection Law of **Brazil**[581] safeguards the interests of data rights holders, including both individual workers and groups. Associations, including trade unions, have the authority to bring collective claims on behalf of workers whose data rights have been violated. Notably, the law provides the option to shift the burden of proof on to the employer if the judge deems the workers' allegations to be plausible.

3.3.3. Social partners' agreements and initiatives

Works councils and trade unions are well positioned to exchange information, raise awareness, and bring claims to prevent the improper use of AM systems.[582] Workers and their representatives have become aware of the potential of data protection rules in exercising their rights to transparency and access to information. Collective agreements are starting to include sections on the adoption and regulation of AM tools. **These rarely focus exclusively on AM systems, but rather deal more generally with the impact of AM tools and practices in connection with occupational health and safety, privacy, and the evaluation of work performance**. Workers and businesses can seize the opportunity to negotiate AM systems using collective agreements that can be 'tailored to the context of specific firms or industries'.[583] The main challenge in this respect materialises in cases where employers buy AM software from third-party companies. While it might be far-fetched for workers' representatives to influence the inner workings of such software, they can guide their managers towards the procurement and adoption of more benign systems.

In 2020, an agreement was reached at IBM in **Germany** concerning the deployment of AI systems. This was the result of a collaborative venture initiated in 2019 that involved the works council, human resources delegates, and AI specialists. The text defines standards regarding transparency, interpretability, non-discrimination, and quality control for AI systems. It posits that all AI systems at IBM should follow the fundamental rule that decision-making should always remain with humans. Lastly, the agreement provides for the creation of an AI Ethics Council at IBM Germany to supervise and continue to refine the agreement.[584]

An increasing number of collective agreements are regulating data privacy issues in **platform work.**[585] Interesting provisions in this regard have been included *inter alia* in the Just Eat agreement in Spain from December 2021. Apart from addressing the right to

[581] *Lei Geral de Proteção de Dados Pessoais*, or LGPD; Lei 13709/2018.

[583] Collins, P., & Atkinson, J. (2023). Worker voice and algorithmic management in post-Brexit Britain. *Transfer: European Review of Labour and Research*, *29*(1), 37-52. See also Dagnino, E., & Armaroli, I. (2019). A seat at the table: Negotiating data processing in the workplace: A national case study and comparative insights. *Comp. Lab. L. & Pol'y J.*, *41*, 173.; De Stefano, V. (2019). 'Negotiating the Algorithm': Automation, Artificial Intelligence, and Labor Protection. *Comp. Lab. L. & Pol'y J.*, *41*, 15.

[584] Herrmann, C. (2022). KI in den Diensten des Menschen. *Arbeitsrecht im Betrieb* 3: 43–44.; Krzywdzinski, M., Gerst, D., & Butollo, F. (2023). Promoting human-centred AI in the workplace. Trade unions and their strategies for regulating the use of AI in Germany. *Transfer: European Review of Labour and Research* 29, no. 1: 63

[585] e.g., the LaConsegna agreement; Montegrappa agreement; Tadan agreement; the agreement on riders' rights concluded in Italy by the Tuscany Region and CGIL, CISL, UIL. For an overview, see Lamannis, M. (2023, February). 'Collective bargaining in the platform economy: a mapping exercise of existing initiatives' (ETUI, Brussels).

digitally disconnect, and the right to privacy from the use of digital surveillance,[586] this agreement also created an 'algorithmic committee', with an equal number of representatives drawn from workers and the employer, tasked with ensuring that transparency obligations and human oversight are complied with.[587] The agreement also mandates that the company must inform trade unions and workers in advance before implementing any video or sound recording systems. Other bottom-up, social partner initiatives in the realm of platform work are also noteworthy. The 'Charter of Fundamental Rights of Digital Labour in the Urban Context' (the so-called 'Bologna Charter') guarantees platform workers the right to data protection, including a right to oppose the processing of personal data for control and monitoring, and the right to be informed about the existence and effect of reputational rating mechanisms. Given the signatory parties to date, the practical relevance of this Charter is limited to app-based food delivery.[588]

Outside of platform work, collective agreements addressing the data-related challenges of AM are less common, although some sectoral initiatives should be mentioned. In Spain, the **XXIV collective agreement in the banking sector 2019/2023** regulates 'new tools based on algorithms [that] can make management more efficient, facilitating the improvement of management practices'. [589] In line with this, 'employees have the right not to be subject to decisions based solely and exclusively on automated [systems, with no human intervention], except in those cases provided for by law, as well as the right to non-discrimination in relation to decisions and processing, when both are based solely on algorithms'. In such cases, workers can request the assistance and intervention of the persons designated for this purpose by the company, if and when they notice a discrepancy between the final decision and what they expected, based on their knowledge. Companies must inform workers' representatives about the use of data analytics or artificial intelligence systems when human resources and labour relations decision-making processes are based exclusively on digital models without human intervention. Such information must at least cover the data feeding the algorithms, their logic, and the evaluation of the results. The agreement also lays down workers' rights to privacy and disconnection.

Some collective agreements covering data-driven practices have been signed.[590] In **Belgium,** video surveillance in the workplace was regulated as early as in the Collective Labour Agreement No 68 of June 1998. Under Art. 6 of this agreement, video surveillance is permitted only if one of the following objectives is pursued: security and health; protection of the commodities of the firm; control of the production process; and control of the employees' labour (more specifically, measurement and control with a view to assessing wages or implications with regard to the rights and duties of the supervisory staff).[591] Moreover, in April 2002, a national collective agreement on the protection of the private lives of employees with respect to controls on electronic online communications data (Collective Agreement No. 81) was signed by social partner representatives on Belgium's National Labour Council. This agreement aims to ensure a balance between the protection of employees' private lives of and legitimate control by the employer. The control of data may not infringe the private life of the employee, and if it does, this must be kept to a

[586] Wray, B. (2021). Gig Economy Project – Just Eat signs 'pioneer' labour agreement with Spanish unions. *Brave New Europe*. Available at: https://braveneweurope.com/gig-economy-project-just-eat-signs-pioneer-labour-agreement-with-spanish-unions

[587] https://www.eurofound.europa.eu/lt/data/platform-economy/initiatives/collective-agreement-for-platform-workers-between-just-eat-ccoo-and-ugt-in-spain

[588] Waeyaert, W., Hauben, H., Lenaerts, K., & Gillis, D. (2022). Italy: A national and local answer to the challenges of the platform economy. *Policy case study*, EU-OSHA 2022.

[589] XXIV Convenio Colectivo de Banca 2019/2023 (XXIV Collective Banking Agreement 2019/2023). Available at https://www.federacionfine.es/documentos/20210129_Texto_XXIV_CONVENIO_DE_BANCA_2019_2023_1_.pdf. See Navarrete, C. M. (2021). Duelo al sol (digital). ¿Un algoritmo controla mi trabajo? Sí; a tu empresa también. Revista de Trabajo y Seguridad Social. CEF, 5-21.

[590] Dagnino, E., & Armaroli, I. (2019). A seat at the table: Negotiating data processing in the workplace: A national case study and comparative insights. *Comp. Lab. L. & Pol'y J.*, *41*, 173.

[591] An unofficial English translation of this collective agreement is available at: http://leewhiteconsultants.com/Libraries/Articles/CLA68EN.pdf

minimum. Control over electronic communications data is allowed only in pursuit of the objectives listed in the collective agreement.[592] The employer is obliged to inform the workers and works councils of all aspects of the control introduced by a control system.

In **Italy**, the major trade unions, CGIL, CISL and UIL, negotiated an agreement with a food delivery platform which addresses the issue of AM practices. The same unions also struck a deal with employers' associations in the cooperative sector. This agreement outlines rights to information when AI-based systems are being implemented in the workplace and mandates that human supervision over technology is maintained by restricting the use of automated decision-making systems.[593]

3.3.4. Other relevant initiatives

A. EU countries

Other initiatives addressing data privacy in the context of AM at work are emerging. These include mostly practical documents and **recommendations for worker representatives,**[594] as well as clarifications issued by **national authoritative bodies** to explain and specify some aspects of the existing legal framework. To give some examples, the Finnish Data Protection Ombudsman's Office defined profiling as the automated processing of personal data for the evaluation of personal aspects of an individual. In particular, profiling refers to the analysis or prediction of aspects concerning the data subject's performance at work, their economic situation, health, personal preferences or interests, reliability or behaviour, location, or movements.[595] In **Austria**, the context of Article 15 GDPR (right of access) was clarified by the Austrian Data Protection Authority,[596] which held that the scope of the right to access also includes the right to be informed about all relevant elements of an automated decision. In **France**, the National Commission for Information Technology and Civil Liberties (*Commission Nationale de l'informatique et des libertés, CNIL*), has adopted several decisions relating to AI and algorithms. It condemned the use of any biometric device to manage employees' schedules, unless 'it can be established that there are exceptional circumstances based on a specific security requirement'.[597] Moreover, the use of any biometric device for the purposes of access control is only allowed to control access to a limited number of devices, IT applications and premises. It is up to the data controller to demonstrate the need for such a biometric device, in particular by detailing 'the specific context requiring a high level of protection', and 'the reasons justifying the use of biometrics, rather than another technology', with supporting documents.[598]

In **Germany**, the Federal Ministry of Labour and Social Affairs established an interdisciplinary council to examine whether standalone legislation should be enacted on employee data protection. On 17 January 2022, the council released its detailed final report,[599] concluding that a separate law on the protection of employee data is necessary, within the framework of the possibility opened up by Article 88 GDPR. In February 2022, the German Trade Union Confederation (DGB) published a draft Employee Data Protection

[592] These objectives include: 'the prevention of illegal or defamatory acts, acts that are contrary to good ethics or which can damage the dignity of another person; the protection of the economic, commercial and financial interests of the company; the security and good operation of the company's information technology network systems; and the faithful observance of the principles and rules applicable in the company for the use of on-line technologies.'

[593] De Stefano, V., & Taes, S. (2023). Algorithmic management and collective bargaining. *Transfer: European Review of Labour and Research*, 29(1), 28.

[594] See also: TUC (2021). When AI is the boss: An introduction for union reps; Prospect (2020). Data Protection Impact Assessments: A Union Guide; Lighthouse (2021). A Guide to Good Data Stewardship for Trade Unions; UNI Europa (2020). Algorithmic Management – A Trade Union Guide.

[595] See at https://tietosuoja.fi/en/automated-decision-making-and-profiling

[596] As of 8.9.2020, DSB 2020-0.436.002. See https://www.globallegalpost.com/lawoverborders/artificial-intelligence-1272919708/austria-197245367

[597] Deliberation No. 2018-009 of 6 September 2018.

[598] Deliberation No. 2019-001 of 10 January 2019.

[599] Report of the independent interdisciplinary council on employee data protection, available at https://www.denkfabrik-bmas.de/Report_of_the_independent_interdisciplinary_Employee_Data_Protection_Advisory_Committee.pdf

Act.[600] This draft addresses several issues including the use of AI systems, behavioural or performance monitoring, and video surveillance. On 4 May 2022, the German Conference of Federal and State Data Protection Authorities (DSK) published a call for the creation of an Employee Data Protection Act. The DSK issued a resolution unambiguously declaring its view regarding the timing of an Employee Data Protection Act.[601] According to media reports, the German Federal Ministry of Labour is currently working on an **employee data protection law**.[602]

In addition to developments at national level, an interesting initiative which shows the potential of cross-border cooperation is the 'Declaration on AI in the Nordic-Baltic Region', signed by the Ministers responsible for digital development from Denmark, Estonia, Finland, the Faroe Islands, Iceland, Latvia, Lithuania, Norway, Sweden and the Åland Islands. The signing parties agreed to collaborate, to enhance access to data, and to develop ethical and transparent guidelines that enable privacy, security, and trust. Meanwhile, the Visegrad countries issued a position paper on AI, emphasising the need to ensure privacy and data security, and calling for 'ethical artificial intelligence'.[603]

Another notable **transnational initiative concerning data rights** is the request by Amazon workers from Germany, Italy, Poland, Slovakia, and the UK, filed in March 2022, with the assistance of the UNI Global Union and privacy advocates NOYB, demanding the right to be informed about the processing of their personal data under Art. 15 GDPR. NOYB and UNI found Amazon's response insufficient, *inter alia* with regard to the unclear purposes and legal basis for the processing of workers' data.[604]

B. Non-EU countries

At **international level**, in the US, the White House Office of Science and Technology Policy published a Blueprint for an **AI Bill of Rights**. The proposed AI Bill of Rights is a white paper, the outcome of an input-gathering process involving stakeholders. This means that the rights included are not legally enforceable and are mostly addressed to the Federal Government. The key principles of the Bill are the right to scrutiny over how data are used; the right to opt out of automated decision-making; the right to live free from ineffective or unsafe algorithms; the right to be informed of the fact that AI is making a decision about them; and the right to not be discriminated against by algorithms. General Counsel Jennifer Abruzzo published a memo on behalf of the National Labor Relations Board (NLRB),[605] announcing the intention to focus on the electronic monitoring of workers, which results in chilling effects on the exercise of collective bargaining rights.[606] To the agency's lawyers, who prosecute unfair labour practice cases, the document stresses the importance of enforcement, and the need to adopt a new framework 'to adapt the [NLRA] to changing patterns of industrial life'.[607] Significantly, given their pervasiveness both at work and in private lives, AM tools and surveillance technologies are identified as systems that potentially impair the exercise of collective rights.

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[600] https://www.dgb.de/uber-uns/dgb-heute/recht/++co++82a3178c-88c4-11ec-b434-001a4a160123

[601] https://www.datenschutz-berlin.de/infothek-und-service/publikationen/beschluesse-datenschutzkonferenz

[602] https://www.sueddeutsche.de/wirtschaft/computer-gesetzentwurf-zu-ki-im-arbeitsleben-geplant-dpa.urn-newsml-dpa-com-20090101-230512-99-658344

[603] Visegrad 4 countries' thoughts on the Artificial Intelligence and maximising its benefits ahead of release of the European Commission's Communication on the topic. Available at https://www.gov.pl/web/cyfryzacja/stanowisko-grupy-wyszehradzkiej-dotyczace-sztucznej-inteligencji

[604] https://uniglobalunion.org/news/gdpr_amazon/. See also https://uniglobalunion.org/wp-content/uploads/Algorithmic-Management-Opportunities-for-Collective-Action.pdf

[605] The NLRB is an independent federal agency established in 1935 to protect employees from unfair labour practices and ensure the right of private sector employees to join together, with or without a union, to improve wages, benefits and working conditions.

[606] Section 7 of the National Labor Relations Act (NLRA) on employees' ability to engage in protected activity. See: Office of Public Affairs (2022, 31 October). 'NLRB General Counsel Issues Memo on Unlawful Electronic Surveillance and Automated Management Practices', *National Labor Relations Board*, https://www.nlrb.gov/news-outreach/news-story/nlrb-general-counsel-issues-memo-on-unlawful-electronic-surveillance-and, accessed 28 November 2022.

[607] *NLRB v J Weingarten, Inc* (1975) 420 US 251, 266.

On 30 October 2023, **President Biden issued an Executive Order aimed at seizing the promising and manage the risks of artificial intelligence (AI)**.[608] The Executive Order establishes new standards for AI safety and security, protects Americans' privacy, advances equity and civil rights, stands up for consumers and workers, and promotes innovation and competition. It makes direct reference to the impact of AI on workers and their rights, notably stating that it is essential to ensure collective bargaining rights. Biden's Administration is also committed to seeking to adapt job training and education to support a diverse workforce and help provide access to opportunities that AI creates. Moreover, the Order provides that the use of AI should not undermine rights, worsen job quality, encourage undue worker surveillance, lessen market competition, introduce new health and safety risks, or cause harmful labour-force disruptions.[609] Within six months, the Chairman of the Council of Economic Advisers shall submit a report to the President on the labour-market effects of AI, and the Secretary of Labor shall submit to the President a report analysing the abilities of agencies to support workers displaced by the adoption of AI and other technological advancements.[610]

Importantly, in October 2023, the leaders of the **Group of Seven (G7)** welcomed the Hiroshima Process International Guiding Principles for Organizations Developing Advanced AI Systems[611] and the Hiroshima Process International Code of Conduct for Organizations Developing Advanced AI Systems.[612] These documents recognise the need to manage risks and to protect individuals, society and key principles including the rule of law and democratic values, keeping humankind at the centre. The leaders of the G7 have also stressed the importance of inclusive governance for AI, based on participatory models such as multi-stakeholder engagement. Joint efforts through the Hiroshima AI Process are expected to 'foster an open and enabling environment where safe, secure, and trustworthy AI systems are designed, developed, deployed, and used to maximize the benefits of the technology while mitigating its risks, for the common good worldwide, including in developing and emerging economies to close digital divides and achieve digital inclusion'.[613]

In a similar vein, **the UK Parliament introduced the Artificial Intelligence (Regulation) Bill** in November 2023.[614] This Bill outlines the creation of an AI Authority and delineates clear principles for the AI Authority to consider in regulating AI. These principles include 'safety, security, and robustness; appropriate transparency and explainability; fairness; accountability and governance; contestability and redress'. Moreover, the Bill requires businesses engaged in AI to maintain transparency, conduct comprehensive testing of their AI systems, and adhere to existing laws, encompassing those pertaining to data protection and intellectual property. The legislation also underscores the importance of ensuring non-discrimination in AI applications, accommodating diverse socio-economic groups, the elderly, and individuals with disabilities. The Bill mandates businesses to designate an AI officer, responsible for 'the safe, ethical, unbiased and non-discriminatory use of AI by the business'.

[608] The White House, Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence, available at https://www.whitehouse.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/

[609] Section 2 (c).

[610] Section 6

[611] See: https://www.mofa.go.jp/files/100573471.pdf

[612] See: https://www.mofa.go.jp/files/100573473.pdf

[613] See: https://digital-strategy.ec.europa.eu/en/library/g7-leaders-statement-hiroshima-ai-process

[614] See: Artificial Intelligence (Regulation) Bill [HL] (parliament.uk)

3.3.5. Selected case law

Bărbulescu v. Romania – European Court of Human Rights, 5 September 2017

Mr. Bărbulescu was dismissed from his job for using a Yahoo Messenger account – which was intended exclusively for work purposes (interaction with customers) – for personal communications. Mr. Bărbulescu exchanged messages with his brother and his fiancée relating to personal matters. Such use was strictly prohibited by the company's rules of conduct, previously signed by the employee. This improper behaviour was revealed after real-time monitoring of the Messenger account was introduced. Mr. Bărbulescu accused the employer of a criminal offence; namely, breaching the secrecy of (private) correspondence as enshrined in **Article 8 of the Convention for the Protection of Human Rights and Fundamental Freedoms** (henceforth, 'the Convention'), and of an unlawful disciplinary dismissal. Having lost the case in national proceedings, Mr. Bărbulescu brought his claim before the European Court of Human Rights (ECtHR), arguing that Romania had failed to comply with its positive obligation to protect his rights under Art. 8 of the Convention.

The ECtHR confirmed the applicability of Art. 8 to the case. It interpreted Art. 8 as guaranteeing 'the right to private life, including the right to lead a 'private social life', that is, the possibility to develop one's social identity'.[615] While the Court left the question of whether Mr. Bărbulescu had a 'reasonable expectation of privacy' open, it held that 'an employer's instructions cannot reduce private social life in the workplace to zero'.[616] According to the Grand Chamber of the ECtHR, the Romanian courts' **failed to determine whether Mr. Bărbulescu was notified in advance that his communications would be monitored, or about the nature and extent of the monitoring**. Moreover, the national tribunal did not examine the degree of the intrusion of monitoring into his private life and correspondence. The tribunal had also failed to determine the specific **reasons justifying the introduction of the monitoring; whether the employer could have used less intrusive measures**, and whether the communications might have been accessed without his knowledge.[617]

The ECtHR found that 'domestic authorities should ensure that the introduction by an employer of measures to monitor correspondence and other communications, irrespective of the extent and duration of such measures, is accompanied by adequate and sufficient safeguards against abuse'.[618] Thus, the majority of judges of the Grand Chamber of the ECtHR (11 votes to six) reversed the decision of the Fourth Section on 12 January 2016, finding that there has been a violation of Art. 8 of the Convention.

In the *Bărbulescu* case, the ECtHR established a ground-breaking precedent by recognising that Article 8 imposes **positive obligations on states to implement safeguards against excessive workplace monitoring by companies**. The Court outlined essential principles that domestic legal systems must follow when balancing these interests.[619] A key takeaway of *Bărbulescu* is that Art. 8 ECHR safeguards cover the workplace, necessitating that employees must be adequately informed in advance about the extent and nature of surveillance. Such a right to privacy at work does not depend on an

[615] Case *Bărbulescu v. Romania* (Application no. 61496/08, 5 September 2017), Para 70. 'The Court acknowledges that everyone has the right to live privately, away from unwanted attention […]. It also considers that it would be too restrictive to limit the notion of 'private life' to an 'inner circle' in which the individual may live his or her own personal life as he or she chooses, thus excluding entirely the outside world not encompassed within that circle […]. This is in line with the previous case law of the ECHR. See, e.g., Case Niemietz v. Germany (Application no. 13710/88, 16 December 1992), in which the ECtHR held that there are no reasons to exclude 'activities of a professional or business nature' from the scope of the notion of 'private life''.

[616] ECtHR, Bărbulescu v Romania, Appl No 61496/08 Judgement of 5 September 2017, Para 80.

[617] Para 140.

[618] Para 120 and 121.

[619] Hendrickx, F., & Van Bever, A. (2013). Article 8 ECHR: judicial patterns of employment privacy protection, in: F. Dorssemont, K. Lörcher & L. Schömann (Eds.), *The European Convention on Human Rights and the employment relation*, Bloomsbury Publishing. See also: Hendrickx, F. (2019). Article 8–Protection of Personal Data, in: F. Dorssemont, K. Lorcher, S. Clauwaert, & M. Schmitt (Eds.), *The Charter of Fundamental Rights of the European Union and the Employment Relation*, Bloomsbury Publishing, p. 249.

employee's reasonable expectations and cannot be removed by internal company policies. Employers should provide valid justifications for the measures implemented and strive to minimise surveillance whenever feasible by opting for less intrusive methods. It is crucial to maintain a **balance between the employer's legitimate interests and the privacy rights of employees** which, by means of interpretation, holds true for several types of surveillance, not limited to the monitoring of personal communications in the workplace.[620] The 'right to a social identity' is also relevant in the case of AM systems, which may exacerbate the intrusion of employers into workers' spheres. In particular, the criteria for safeguarding the right to a private life in respect of workplace monitoring established in this judgement are relevant to AM, as they inform the activity of legitimate monitoring – a key function that has been increasingly delegated to AM tools. However, the ruling in the *Bărbulescu* case does not fully elucidate the factors that trigger Art. 8 or specify when there is no reasonable expectation of privacy, leaving the scope of this right uncertain. In addition, the impact of the decision on unfair dismissal cases involving disproportionate monitoring remains unclear.

Uber case – Amsterdam Court of Appeal, 4 April 2023[621]

This case concerned a claim by Uber drivers, who challenged the temporary block imposed on their accounts by the platform for allegedly fraudulent behaviour. In the first instance,[622] the Amsterdam Civil Court found there had been no violation of **Art. 22 GDPR**. First, the court held that claimants had not disputed Uber's argument that its decision was undertaken with the meaningful human involvement of a 'risk team', in charge of handling such cases. Thus, Uber's decision was considered a not *fully* automated one. Second, the temporary deactivation of the drivers' accounts was seen as lacking long-term or permanent effects on the drivers, which meant that the requirement of legal consequences or similarly significant effects under Art. 22 (1) was not met. The drivers' claim was partially successful only with regard to access to personal data under Art. 15 (1) GDPR. Uber was required to clarify which specific fraudulent action resulted in the deactivation of drivers' accounts, giving drivers access to data in such a way that they could verify the correctness and lawfulness of the processing of their personal data.[623]

The Court of Appeal overturned the first instance ruling, recognising the drivers' information rights under Art. 15(1)(h) GDPR. It first assessed the scope of Art. 15(1)(h) of the GDPR, determining that the information right it enshrines relates not to *any* form of automated decision-making, as provided for under the Dutch legislation implementing the GDPR, but only to automated decision-making referred to in Article 22 paragraphs 1 and 4 of the GDPR.[624] Thus, Art. 15(1)(h) GDPR guarantees the right to information, as long as the controller 'subjects' the data subject to 'a decision based solely on automated processing, including profiling, which produces legal effects for him or otherwise significantly affects him'.[625] It was therefore crucial to establish whether the **automated deactivation of drivers' accounts met the criteria of Art. 22 GDPR**. The court found this to be the case, as the drivers provided sufficient evidence that deactivation was based solely on automated processing, and that Uber had not plausibly disputed this fact.[626] The court reasoned that: (i) the privacy statement stated that Uber makes automated decisions when deactivating users; (ii) profiling was used when making the deactivation decisions; (iii) the decisions addressed to the drivers were worded in a very general, vague and abstract way, without

[620] For a detailed analysis of this case see, for example, Jervis, C. E. (2018). Barbulescu v Romania: Why There is no Room for Complacency When it Comes to Privacy Rights in the Workplace. *Industrial Law Journal*, *47*(3), 440-453; Keane, K. (2018). The GDPR and Employee's Privacy: Much Ado but Nothing New, *King's Law Journal*, Volume 29 Issue 3, 354-363; Atkinson, J. (2018). Workplace monitoring and the right to private life at work. *Modern Law Review*, 81 4, 688-700.

[621] Case number 295.742/01.

[622] Rb. Amsterdam - C/13/692003/HA RK 20-302.

[623] This was found in respect of two of the four applicants.

[624] Para 3.16.

[625] Para 3.17.

[626] Para 3.20.

an explanation of any concrete conduct.[627] The company did not show that its employees, based in Cracow, took anything more than a 'purely symbolic act', or that they had taken into consideration all relevant data, without having heard the drivers. Thus, the human intervention concerned was considered insufficient.[628] Moreover, the deactivation of the workers' accounts affected drivers 'to at least a considerable extent', as the company's decision meant that the drivers were no longer able to earn income through the transport app.[629]

Uber's claim for refusing the requested information based on Art. 15(4) (i.e. that it contained trade secrets regarding its anti-fraud processes) was found to be unsubstantiated. Complete rejection of information requests would not be a proportionate and necessary means of protecting trade secrets under the Dutch law implementing the GDPR. Moreover, Uber's argument that it had provided drivers with sufficient information in the Community Guidelines and the Terms and Services Agreements was rejected. According to the court, Uber failed to **explain sufficiently and concretely the way in which it provides information regarding the factors considered in the decision-making process, and their respective weighting at an aggregate level**. Nor had it otherwise been shown that the information was complete enough for drivers to understand the reasons for the deactivation decision taken against them. It was not sufficient for Uber to explain in such information what actions might lead to deactivation. Uber should have explained what **factors formed the basis of its decision, and the weighting of those factors**. It should also have provided drivers with other information necessary to understand the reasons for the decision. Uber was ordered to fulfil the information requests, based on Article 15(1)(h) GDPR, subject to a penalty in the amount of EUR 4,000 for each day that it failed to comply with this order.

Ola case – Amsterdam Court of Appeal, 4 April 2023

Much as in the Uber case above, Ola drivers, supported by the App Drivers & Couriers Union and Worker Info Exchange, demanded access to personal data processed by the platform, based on **Art. 15(1) GDPR.** They requested access to such categories of data as customer transactions, booking cancellation history, booking acceptance history, ratings, GPS data for each trip, and device data. In the first instance, the District Court of Amsterdam[630] obliged Ola to provide access to certain categories of personal data, such as ratings data (while observing the privacy rights of passengers, thus anonymised), and how its 'fraud probability score' was generated. Personal data regarding the drivers' 'earning profile', based on which bonuses were awarded to certain drivers, and data that had been processed by Ola's 'Guardian system', which was used to detect irregularities, also had to be made transparent to drivers. The court did not consider Ola's system an automated decision-making process within the meaning of Art. 22 (1) GDPR. For example, while the 'fraud probability score' was qualified by the court to be 'profiling' within the meaning of Art. 4 (4) GDPR, the drivers had not shown that automated decisions had been taken on the basis of this risk profile. As for the drivers' 'earning profile', the court held that the possibility of obtaining a bonus had not been proven to have a legal or significant effect on drivers.

The Court of Appeal arrived at the opposite conclusion in this regard. It considered that the 'fraud probability score' and the 'earning profile', as well as the allocation of journeys, significantly affected the drivers,[631] and were based solely on automated processing, fulfilling the criteria of Art. 22(1) GDPR.[632] Only the Guardian system did not qualify as such, since Ola demonstrated sufficiently that its staff would contact the passenger or driver after a signal was issued in the case of an anomalous pattern in location.

[627] Para 3.21.

[628] Para 3.24.

[629] Para 3.19. In addition, the court considered that the allegation might have serious criminal or other repercussions.

[630] Rb. Amsterdam - C/13/689705/HA RK 20-258.

[631] Paras 3.37-3.40.

[632] Para 3.45.

The Court of Appeal further expanded **the scope of the data to which drivers must be given access**. It did so, *inter alia*, with regard to 'device data', i.e. data on the mobile phones used by the drivers to carry out their journeys and to operate the Ola Driver App. According to the court, this kind of data constitutes personal data, and Ola failed to demonstrate that it did not contain information about drivers. At the same time, it refuted the drivers' claim that *all* of their personal data in the categories of 'customer transactions, booking cancellation history, and booking acceptance history', including passenger data, needed to be provided. The Appellate Court upheld the District Court's view that, in the event of such a broad request for access to personal data concerning thousands of passengers, the controller cannot be required to ascertain which interest prevails for each transaction or cancellation.[633] For example, access should be granted to the drivers' own location, while no direct or indirect access to the location data of individual passengers should be given. In other words, such a right cannot infringe the rights and freedoms of passengers or other third parties.

Both the Uber and Ola cases are prime examples of successful strategic litigation that has led to the **mobilisation of GDPR data subject rights *vis-à-vis* digital labour platforms.**[634] Drivers were successful in enforcing the right to learn the logic behind automated decision-making, as well as the specific weighting criteria used in the model. These cases confirmed a broad scope of personal data rights under Art. 4 GDPR. They also made clear that platforms' claims that the requested information contains trade secrets about their business model does not justify the complete rejection of the access to information, as this would not be proportionate and necessary, as required by Art. 41(1)(i) GDPR. At the same time, these cases revealed salient difficulties in deciding whether a decision is made solely on automated processing, and whether such a decision produces legal effects or similarly significant effects on the data subject. As noted by Hiessl, these rulings imply that data controllers do not even need to inform subjects about the existence of algorithms which, according to them, are not covered by Art. 22(1) GDPR, as long they comply with the information obligations under Arts. 13-15 GDPR.[635]

Amazon case – Hannover Administrative Court, 9 February 2023

In its judgment of 9 February 2023, the Hannover Administrative Court examined the use of a **system employed in an Amazon warehouse to continuously collect and process qualitative and quantitative data concerning the work performed by warehouse workers**. The data were collected **in real time by means of hand-held scanners and evaluated using performance management software** (the so-called 'Fulfilment Centre Labour Management' (FCLM)). Based on these data, an application called 'Associate Development and Performance Tracking' (ADAPT) generated feedback proposals that served as the basis for a discussion with supervisors. The top 10 % of workers were given 'recognition feedback', while the lowest performing 5 % of workers were given 'development feedback'. Performance data were taken into account when measuring 'individual process paths', and to shift employees between different work areas to ensure the smooth operation of the logistics centre. Moreover, such data were used for training purposes and personnel decisions, including the termination of employment.

The use of the applications FCLM and ADAPT was prohibited by the Data Protection Commissioner in Lower Saxony on 28 October 2020, with the justification that it was not necessary for the achievement of the interests pursued by Amazon, namely: (i) control of logistics processes[636]; (ii) control of the employees' qualification; and (iii) creating objective

[633] The drivers claimed that Ola must show, pursuant to Article 41 UAVG, with regard to testing each data subject individually, and thus on a case-by-case basis, whether a refusal is justified by invoking the rights and freedoms of others, under Article 15(4) AVG.

[634] Gaudio, G. (2024). Litigating the Algorithmic Boss in the EU: A (Legally) Feasible and (Strategically) Attractive Option for Trade Unions?, *International Journal of Comparative Labour Law and Industrial Relations*, *40*(1)

[635] Hiessl, C. (2023). Jurisprudence of national courts in Europe on algorithmic management at the workplace., p. 38. Available at: https://ssrn.com/abstract=3982735.

[636] In particular, the exact calculation of the location of goods, in order to meet the delivery date guaranteed to a customer.

bases for assessment for the purposes of feedback and personnel decisions. Therefore, the Data Commissioner found such practices to be in violation of Section 26 of the (German) Federal Data Protection Act (BDSG), which specifies **Art. 6 GDPR**.[637] Upon Amazon's appeal against this order, the court needed to assess whether this ban was justified, i.e. whether the system of processing employees' data was lawful under Section 26 of the BDSG, in connection with **Art. 88(1) GDPR**.

The court held that Amazon had a substantial **legitimate interest** in the collection of the data, and that such data collection was suitable and necessary to protect these interests.[638] According to the court, an equally effective yet less employee privacy-detrimental means was not available. The data processing was assessed as proportionate and transparent,[639] and its consequences were considered 'not excessively serious'.[640] This was justified in light of the high number of workers and the complexity of their effective management, also taking into account that some workers had voiced a preference for an 'objective', transparently operating algorithmic mechanism for human surveillance and decision-making by the managerial personnel. In summary, the court saw no breach of Art. 5(1), Art. 6(4), Art. 88(1) or Art. 22 (1) of the GDPR in this case. Thus, the Data Protection Commissioner's order was held to be substantively unlawful, as the data processing was lawfully based on Art. 88(1) of the GDPR, in conjunction with Section 26 of the BDSG. The Court's position is highly contentious,[641] and, arguably, inconsistent with previous jurisprudence of German courts.[642] This ruling is under appeal.[643]

3.3.6. Conclusions

AM tools and practices are relatively novel, albeit that the purposes they accomplish are as old as the notion of management. In this respect, **the data protection framework is one of the fundamental sets of operating rules targeting various AM tools, functions, and practices**.[644] The direct applicability of the GDPR is advantageous, **it compels providers of AM systems and their adopters (i.e. employers) to meet the general principles and other requirements for processing of personal data contained therein**. The key existing principles and rules, such as the grounds for lawful processing of personal data, the principles of purpose limitation and data minimisation, the prohibition of solely automated processing, and the required safeguards **make data protection rules a primary, yet not exclusive, building block for the effective regulation of AM at work**.

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[637] According to Section 26 (1) BDSG, the personal data of employees may be processed for purposes of the employment relationship if this is necessary for the decision on the establishment of an employment relationship or, after the establishment of the employment relationship, for its implementation or termination or for the exercise or fulfilment of the rights and obligations of employee representation.

[638] Three essential purposes pursued by Amazon were: 'the control of logistics processes, the control of the individual qualification of employees, and the creation of objective assessment bases for individual feedback and personnel decisions'. See para. 62 of the judgment. For the detailed reasoning as to why the use of FCLM and ADAPT was suitable and necessary, see paras. 66-91 of the judgment.

[639] See para. 101 of the judgment.

[640] See para. 108 of the judgment.

[641] See, for example, https://lfd.niedersachsen.de/startseite/infothek/presseinformationen/thiel-das-allgemeine-personlichkeitsrecht-der-mitarbeiterinnen-und-mitarbeiter-uberwiegt-unternehmerische-interessen-219596.html.

[642] See, for example, the BAG ruling of 27 July 2017, ref. 2 AZR 681/16, stating that the use of key logger software is unlawful under the German privacy law, unless an employer has a concrete, reasonable suspicion of a criminal offence, or any other serious breach of duty by an employee. See also the decision of the Regional Administrative Court of Wiesbaden of 17 January 2022, ref. 6 K 1164/21, which considered that the GPS tracking of vehicles in the logistics sector is generally not permitted unless it is done in real time only, and with proper data protection measures in place, including a data protection impact assessment. Live tracking of logistics vehicles is considered a more privacy-friendly approach than storing GPS location data, and is therefore required by data protection laws. The logistics company had processed data for tracking its company vehicles via a SaaS cloud solution. The software used also enabled the measurement of fuel consumption and management of the tachograph.

[643] Appeal pending before the OVG Lüneburg under case no. 11 LC 105/23.

[644] See the case of generative AI systems: https://www.wired.com/story/italy-ban-chatgpt-privacy-gdpr/. See also: Lynskey, O. (2014). Deconstructing data protection: the 'added-value' of a right to data protection in the EU legal order. *International & Comparative Law Quarterly*, *63*(3), 569, 570.

More specifically, the GDPR is **one of the most effective frameworks for regulating some AM functions**. Significant developments have been witnessed in this respect, thanks to strategic litigation, collective agreements, company protocols and enforcement. The growing body of case law and DPAs' decisions proves the potential of this far-reaching instrument,[645] and the enforcement of data protection regulations will gain importance in the years to come.[646] The **role played by DPAs is crucial**, as they have competence to facilitate, monitor and enforce compliance with the GDPR and national laws implementing it.[647] It will be important to develop stronger collaboration with labour inspectorates and between countries, especially when dealing with companies that operate in several countries, and with providers of AM tools that offer their software in the EU market.

However, as a general framework, the GDPR rules do not address *all* of the challenges emerging in the field of workplace-related AM. **Not all AM functionalities are comprehensively tackled by existing regulatory tools**. A case in point is a decision based on semi-automated processing, which raises similar issues to a decision based on solely automated practices, and yet is not fully captured by the GDPR. For example, Articles 13(2)(f), 14(2)(g) and 15(1)(h) of the GDPR do not extend their protections to semi-automated decision-making. **Unless a decision is based on entirely automated processing of personal data**, employees do not possess the right to receive information about the existence of the decision-making system, or about the reasoning, importance, and implications of the decision-making process. Although the GDPR focuses on individual decisions and how they have been made, this limitation represents a concern, given that many AM systems which impact individuals combine both automated and (genuine) human-driven components. **Semi-automated processes still pose significant risks to data subjects**, such as potential bias, inaccuracy, or unfair outcomes. Excluding semi-automated decision-making from the protections afforded by the GDPR might leave data subjects vulnerable in scenarios where human intervention, although it is effective and meaningful, does not necessarily mitigate the risks inherent in automated processes. While its inclusion might expand the list of duties required of some data controllers, it would do so with the objective of better protecting data subjects in an evolving digital landscape.

The inherent limitations of the current frameworks pertain to the fact that the GDPR was not originally designed to address AM systems at work. This aspect could undermine the strength of the data protection framework with respect to AM-related issues emerging from the context of employment; however, it does not preclude the implementation of other, more targeted, rules.

Moreover, the **general nature** of many GDPR provisions represents a shortcoming in the field of AM used at work, as it leads to an increasing number of interpretive problems.[648] This issue could slow down the adoption of AM tools, due to a lack of clarity as regards the legal framework and limited trust. Some provisions and their defining features (e.g. 'solely automated' and 'significant effects' in Art. 22 GDPR; 'provided by the data subject' in Art. 20 GDPR, etc.) may read as open-textured. **The availability of carve-outs in important provisions** means that the lawfulness of certain managerial decisions is still 'ambiguous'.[649] Moreover, the interpretation and application of such provisions requires a **delicate balancing of rival interests, which leads to different interpretations being made on a case-by-case basis**. Such balancing involves an assessment of the interests at

[645] Demetzou, K., Zanfir-Fortuna, G., & Vale, S. B. (2023). The thin red line: refocusing data protection law on ADM, a global perspective with lessons from case-law. *Computer Law & Security Review*, *49*, 105806.

[646] Ibid, p. 12.

[647] Jóri, A. (2015). Shaping vs applying data protection law: two core functions of data protection authorities. *International Data Privacy Law*, *5*(2), 133.

[648] See, for example, Sartor, G., & Lagioia, F. (2020). The impact of the General Data Protection Regulation (GDPR) on artificial intelligence, whose authors observe that 'the uncertainties involved in applying indeterminate concepts and balancing competing interests are aggravated by the novelty of the technologies, their complexities, the broad scope of their individual, and social effects', p. 78.

[649] Adams-Prassl, J., Abraha, H., Kelly-Lyth, A., Silberman, M.S., & Rakshita, S. (2023). Regulating algorithmic management: A blueprint. *European Labour Law Journal*, *14*(2), 124-151, p.142.

stake, and may be influenced by a consideration for the state of the art. It cannot be denied that the openness of the GDPR and its use of terms of a general nature pose considerable challenges for legal practitioners. Despite this, in a rapidly evolving context, **openness cultivates both flexibility and resilience**.

Other limitations pertain to the primarily **'individualised'** character of the GDPR, which is geared around the standardised category of the 'data subject', making it somewhat inadequate for effectively dealing with the complexities inherent in AM. Coupled with a lack of awareness, the GDPR's **scarcely developed collective dimension** has been identified as an area for concern,[650] given that in workplaces, information is unevenly distributed and data are eminently relational. The concern is that employees may frequently hesitate to assert their rights while in the course of their employment. Furthermore, some issues arise concerning **access to evidence**, particularly when the exercise of a redress mechanism could result in negative effects for employees.

In recent years, several EU Member States have **drawn upon the GDPR's provisions, introducing national regulations that provide more specific tools for application in the context of employment**.[651] These examples demonstrate that the EU data protection framework **can be expanded and specified to better address AM-related challenges**. The result for both workers and businesses, however, is replete with potential uncertainties caused by this **fragmentation** into national schemes. Companies operating across countries may find themselves dealing with a variety of schemes, thereby experiencing an increase in their compliance costs, which could be all the more burdensome for small and medium-sized enterprises. Given their newness, such national developments ought to be monitored to assess their performance in terms of enforcement and impact.

Given the peculiarities of AM, the **adoption of a distinct set of protections not currently detailed in the GDPR** has been suggested.[652] This issue is not novel or justified solely by the emergence of AM systems. Appeals to complement the data protection framework with 'sectoral employment-specific' legislation[653] have been advanced ever since the adoption of the 1995 Data Protection Directive.[654] The distinct characteristics of the employer-employee relationship, such as information asymmetry and the pre-existing imbalanced contractual power structures, may call for **more specific rules at the workplace level**.

3.4. Fairness and discrimination

Key findings:

- EU equality instruments play a pivotal role in ensuring that professional environments remain free from discrimination, including when AM systems are adopted. As the use of AM systems gains traction, the relevance of EU equality instruments becomes even more pronounced.
- 'Algorithmic discrimination' significantly differs from traditional discriminatory behaviours, for which existing anti-discrimination rules are primarily tailored. For instance, AM

[650] Mahieu, R.L., & Ausloos, J. (2020). Harnessing the collective potential of GDPR access rights: towards an ecology of transparency. *Internet Policy Review*, *6*; Christl, W. (2021). Digitale Überwachung und Kontrolle am Arbeitsplatz. *Von der Ausweitung betrieblicher Datenerfassung zum algorithmischen Management*.

[651] See the section on Selected national legislation and policies.

[652] Adams, Z., & Wenckebach, J. (2023). Collective regulation of algorithmic management. *European Labour Law Journal*, *14*(2), 211-229. Abraha, H. (2023, 8 October). Automated Monitoring in the Workplace and the Search for a New Legal Framework: Lessons from Germany and Beyond. Available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4595760

[653] Abraha, H.H. (2023), p. 281

[654] Simitis, S. (1999). Reconsidering the premises of labour law: Prolegomena to an EU regulation on the protection of employees' personal data. *European Law Journal*, *5*(1), 45-62; Freedland, M. (1999). Data protection and employment in the European Union: An analytical study of the law and practice of data protection and the employment relationship in the EU and its Member States. European Commission.

practices can lead to unconventional and abstract classifications, which are not easily covered, considered, protected or associated with existing grounds.

- An expansive interpretation of the grounds protected from discrimination, as well as the notions of discrimination by proxy and association, are set to strengthen the suitability of the current model.
- Mobilising anti-discrimination legislation in courts in cases of algorithmic discrimination is challenging. However, the evidentiary regime eases the burden on victims, with anti-discrimination directives allowing for a partial shifting of the burden of proof from claimant to employer. Workers initially only need to establish plausible facts of discrimination, shifting the burden of proof on to the employer to demonstrate non-discrimination through evidence or valid justifications.
- The datasets on which AM systems are designed are crucially important, too. Data from a structurally imperfect world could be accurate, relevant, representative, free from errors and complete, and yet still result in discrimination.
- The current gaps result both from the distinctive and novel nature characteristic of algorithmic discrimination, and from some structural limitations of the equality framework. These structural limitations include *inter alia* a closed set of protected grounds, the regime of exceptions and justifications in cases of indirect discrimination, the relevance of causal links, and the implicit obstacles to evidence-gathering to bring a case before courts.

3.4.1. EU legal framework

The EU legal framework pursues the goal of combating social exclusion and discrimination, promoting social justice and protection, and equality between women and men (Article 3(3) Treaty on European Union (TEU)). The principle of equality is also a key element of the Charter of Fundamental Rights of the EU (CFREU). Moreover, the EU directives on equality and non-discrimination are among the first instruments that have been adopted to **promote fairness between genders and to avoid discrimination against people who hold protected characteristics**.[655] There are several AM-related opportunities and challenges pertaining to this dimension that can be addressed by relying on instruments in the EU *acquis.*[656]

EU non-discrimination directives create a **composite and comprehensive model**,[657] with important differences related to personal scope and the protection reserved to certain grounds and certain contexts. As noted by Kelly-Lyth, 'the employment sector has long been a key forum for the development of equality law'.[658] The large corpus of EU anti-discrimination instruments is **very relevant and apt to promote the creation of bias-free workplaces, including when decisions are made using the assistance of AM systems**.[659] Xenidis and Senden find that 'algorithmic discrimination collides with and is

[655] Ellis, E., & Watson, P. (2012). *EU anti-discrimination law*. OUP Oxford.

[656] Bartoletti, I., & Xenidis, R., (2023), Study on the impact of artificial intelligence systems, their potential for promoting equality, including gender equality, and the risks they may cause in relation to non-discrimination, Strasbourg, Council of Europe.

[657] Bell, M. (2002). *Anti-Discrimination Law, and the EU*. Oxford University Press. Strikingly, gender equality and sex discrimination are not mentioned in Recital 71 of the GDPR ('racial or ethnic origin, political opinion, religion or beliefs, trade union membership, genetic or health status or sexual orientation'), which misaligns the two groups of protected factors. Similarly, the broad notion of sensitive data set out in Art. 9(1) of the GDPR ('racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, […] genetic data, biometric data, data concerning health, or data concerning a natural person's sex life or sexual orientation') does not overlap with the list included in Art. 19 TFEU, failing to explicitly mention sex, disability and age. See also: Article 22(4) of the GDPR, according to which decisions based solely on automated processing, including profiling, must not be based on special categories of personal data.

[658] Kelly-Lyth, A. (2023). Algorithmic discrimination at work. *European Labour Law Journal*, 14(2) at 153

[659] Allen, R.Q.C., & Masters, D. (2020). Regulating for an equal AI: a new role for equality bodies - Meeting the New Challenges to Equality and Non-discrimination from Increased Digitalisation and the Use of Artificial Intelligence, Equinet, p. 30.

captured by EU non-discrimination law'.[660] Discriminatory impacts that occur during those phases of the working relationship in which AM systems may be used, such as in recruitment, promotions, working conditions and professional training, are addressed by existing EU tools, as detailed below.

A. Protected grounds

To thoroughly evaluate the relevance and effectiveness of the EU legal framework, it is essential to introduce certain foundational concepts. Only discrimination that is based on a closed list of legally protected grounds is addressed by EU law. **Sex, racial or ethnic origin, religion or belief, disability, age, and sexual orientation** are protected under Art. 19 Treaty on the Functioning of the European Union (TFEU).[661] When it comes to **algorithmic discrimination** (see 2.3.3. Fairness and discrimination), the risk is that such a limited selection of protected characteristics may not be ideally suited to addressing the new forms of data-driven classifications and categorisations made by AM systems. Art. 21 of the CFREU prohibits discrimination based on any ground such as sex, race, colour, ethnic or social origin, genetic features, language, religion, or belief, political or other opinion, membership of a national minority, property, birth, disability, age, or sexual orientation.[662] Given the use of the phrase 'such as', this catalogue can be understood as an open-ended and non-exhaustive list of protected factors. Commentators suggest that such a broad reading would **widen the range of grounds upon which the fight against algorithmic discrimination can be anchored, thus strengthening the effectiveness of the current framework**.[663] In addition, Art. 21 CFREU could serve as an 'umbrella provision contributing to improved coherence between the separate directives'.[664]

The personal scope of EU antidiscrimination legislation includes **workers** (Directive 2006/54/EC for the grounds of sex, Directive 2000/43/EC for the grounds of race or ethnic origin, and Directive 2000/78/EC for the grounds of disabilities, religion or belief, age, and sexual orientation). With regard to **self-employed workers**, some scholars have warned that the directives only protect to right of access to a profession or activity, while they do not guarantee equal treatment during the exercise of such activity.[665] However, in its ruling in the *J.K. v TP S.A*. case, the CJEU clarified that interpreters must opt for a broad, teleological reading of EU Framework Equality Directive 2000/78/EC to protect 'the professional relationship concerned in its entirety'.[666] In the case of sex (Directive 2004/113/EC) and race or ethnic origin (Directive 2000/43/EC), **service providers** are also covered.

Gender discrimination. The principle of equal pay for equal work or work of equal value between men and women (the 'principle of equal pay') is enshrined in Article 157 TFEU.

[660] Xenidis, R., & Senden, L. (2019). EU non-discrimination law in the era of artificial intelligence: Mapping the challenges of algorithmic discrimination, in: U. Bernitz et al (eds.), *General Principles of EU law and the EU Digital Order,* Kluwer Law International, 2020, pp. 151-182.

[661] Nationality is also covered in relation to the freedom of movement of persons, goods, services, and capital – see Art. 18 TFEU.

[662] See also Arts. 2 and 3 of the TEU; Arts. 8, 10, 19 153 and 157 of the TFEU. See also Arts. 21 and 23 of the EU Charter of Fundamental Rights.

[663] Nevertheless, the Court of Justice of the EU (CJEU) seems reluctant to depart from a closed catalogue, thus curtailing the potential of Art. 21 as a flexible catalogue. C-354/13 *Fag og Arbejde (FOA) v Kommunernes Landsforening (KL)* 18 December 2014 EU:C:2014:2463; C-13/05 *Sonia Chacon Navas v Eurest Colectividades* SA 11 July 2006 EU:C:2006:456.

[664] De Stefano, V., & Wouters, M. (2022). AI and digital tools in workplace management and evaluation: An assessment of the EU's legal framework, p. 46, citing Ward, A. (2018). The Impact of the EU Charter of Fundamental Rights on Anti-Discrimination Law: More a Whimper than a Bang? *Cambridge Yearbook of European Legal Studies*.

[665] Barnard, C., & Blackham, A. (2018). 'Discrimination and the Self-Employed: The Scope of Protection in an Interconnected Age', in: H. Collins (ed.), *European Contract Law and the Charter of Fundamental Right,* Cambridge University Press, 2018, p. 206.

[666] Paragraph 56, Case C-356/21 *J.K. v TP S.A.* EU:C:2023:9. See Aloisi A. (2023), '*J.K. v TP S.A.* and the 'universal' scope of EU anti-discrimination law at work: A paradigm shift?', *Industrial Law Journal*, 52(4), 977-996

The Gender Recast Directive 2006/54/EC[667] is the cornerstone of the EU legal framework prohibiting gender discrimination. It contains provisions to implement the principle of equal treatment in relation to: '(a) access to employment, including promotion, and to vocational training; (b) working conditions, including pay; (c) occupational social security schemes' (Art. 1). This framework is complemented by Directive 2010/41/EU regarding equal treatment between men and women engaged in an activity in a self-employed capacity.[668] Directive 2004/113/EC concerns equal treatment between men and women in terms of access to and supply of goods and services.

In addition, the recently adopted Directive 2023/970[669] strengthens the application of the principle of equal pay for equal work or work of equal value between men and women, in particular through pay transparency and reinforced enforcement mechanisms (Art. 1).[670] It requires Member States to implement measures ensuring that employers establish payment systems that guarantee **equal pay for equal work or work of the same value** (Art. 4 (1)). A central chapter of the Directive is devoted to transparency (Arts. 5-13). Job applicants must be informed by potential employers as regards (a) the starting salary or its range, based on objective and gender-neutral criteria, designated for the relevant position, and (b) when applicable, the pertinent sections of the collective agreement enforced by the employer related to the position. Criteria applied to determine workers' pay, pay levels, and pay progression, which must be objective and gender-neutral, need to be made easily accessible to workers (Art. 6). Workers must have the right to request and receive written information on their individual pay level and the average pay levels, broken down by sex, for categories of workers performing the same work as them, or for work of equal value to theirs (Art. 7). Moreover, the Directive requires employers to report on the pay gap between male and female workers (Art. 9).

The comprehensive nature of the equality norms in the area of gender discrimination allows for the avoidance of pay discrimination, including in the case of AM tools used for administering some essential aspects of the employment contract – for instance, when it comes to computing the variable parts of wages.

Racial or ethnic origin discrimination. Directive 2000/43/EC implements the principle of equal treatment irrespective of racial or ethnic origin in employment matters and even beyond.[671] It lays down a framework for combating discrimination on the grounds of racial or ethnic origin, with a view to putting into effect in the MS the principle of equal treatment. It applies to '(a) conditions for access to employment, to self-employment, and to occupation, including selection criteria and recruitment conditions, whatever the branch of activity and at all levels of the professional hierarchy, including promotion; (b) access to all types and to all levels of vocational guidance, vocational training, advanced vocational training, and retraining, including practical work experience; (c) *employment and working conditions, including dismissals and pay*; (d) membership of and involvement in an organisation of workers or employers, or any organisation whose members carry on a particular profession, including the benefits provided for by such organisations; (e) social protection, including social security and healthcare; (f) social advantages; (g) education; (h) access to and supply of goods and services, which are available to the public, including housing' (Art. 3, emphasis added).

[667] Directive 2006/54/EC of the European Parliament and of the Council of 5 July 2006 on the implementation of the principle of equal opportunities and equal treatment of men and women in matters of employment and occupation (recast).

[668] Art. 8, Directive 2010/41/EU of the European Parliament and of the Council of 7 July 2010 on the application of the principle of equal treatment between men and women engaged in an activity in a self-employed capacity and repealing Council Directive 86/613/EEC.

[669] Directive (EU) 2023/970 of the European Parliament and of the Council of 10 May 2023 to strengthen the application of the principle of equal pay for equal work or work of equal value between men and women through pay transparency and enforcement mechanisms (Text with EEA relevance).

[670] Hooton, V.E., & Pearce, H. (2023). As clear as mud: Assessing the relationship between proposed pay transparency mechanisms and data protection obligations in EU law, *European Labour Law Journal*, 20319525231178985.

[671] Council Directive 2000/43/EC of 29 June 2000 implementing the principle of equal treatment between persons irrespective of racial or ethnic origin.

Rules aimed at combatting discrimination based on racial or ethnic origin are very relevant to AM systems, especially those used in hiring processes.[672]

Religion or belief, disability, age and sexual orientation are protected under the employment-specific Directive 2000/78/EC, also known as the Framework Directive.[673] This Directive applies to all persons, in both the public and private sectors, including public bodies, in relation to: '(a) conditions for access to employment, to self-employment or to occupation, including selection criteria and recruitment conditions, whatever the branch of activity, and at all levels of the professional hierarchy, including promotion; (b) access to all types and to all levels of vocational guidance, vocational training, advanced vocational training, and retraining, including practical work experience; (c) employment and working conditions, including dismissals and pay; (d) membership of, and involvement in, an organisation of workers or employers, or any organisation whose members carry on a particular profession, including the benefits provided for by such organisations' (Art. 3).

In most cases, **AM tools operate by standardising individual variations and disadvantaging profiles that diverge from the norm**. Thus, the existing laws are not entirely effective in guaranteeing equality in this context.

People with disabilities, in particular, could be seriously impacted. Art. 5 of Directive 2000/78/EC mandates that employers must provide '**reasonable accommodations**' to workers with disabilities, understood as limitations, which 'in interaction with various barriers may hinder the full and effective participation of the person concerned in professional life on an equal basis with other workers',[674] to facilitate their equal participation in the workforce.[675] Employers must take appropriate measures to enable persons with disabilities to access, participate and advance in employment. It is not clear whether AM tools make it easier or more difficult to comply with the law in this regard. On the one hand, AM tools could be designed to provide individual-level responses.[676] In these circumstances, such accommodations must be reasonable, meaning that their deployment cannot 'impose a disproportionate burden' on the employer. As argued by Kelly-Lyth,[677] within the EU legal framework, employers are required to address the distinctiveness, fluidity, and context-specific nature of disabilities through the obligation of reasonable accommodations. On the other hand, reasonable accommodations depend on disclosure, while evaluating the appropriateness of an accommodation can be challenging and opting out of an unsuitable assessment may not always be a perfect solution if the alternative process lacks fairness in practice. While it is crucial to acknowledge that these hurdles, although substantial, are not exclusive to AM systems, one cannot overlook a topic that often lacks attention in studies: **the versatility and adaptability of certain AM software**. This feature could effectively simplify the implementation of reasonable accommodation measures with regard to administrative tasks in both blue- and white-collar industries, to cite just one instance.

B. Types of discrimination

Direct discrimination occurs when 'one person is treated *less favourably* than another is, has been, or would be treated in a comparable situation on any of the protected

[672] Ajunwa, I., & Greene, D. (2019). Platforms at work: Automated hiring platforms and other new intermediaries in the organisation of work. In: *Work and labor in the digital age* (Vol. 33, pp. 61-91). Emerald Publishing Limited.

[673] Council Directive 2000/78/EC of 27 November 2000 establishing a general framework for equal treatment in employment and occupation.

[674] Joined Cases C-335/11 and C-337/11 HK Danmark (*Ring* and *Skouboe Werge*), ECLI:EU:C:2013:222, para 38. See also: Waddington, L., & Broderick, A. (2018). Combatting Disability Discrimination and Realising Equality: A Comparison of the UN CRPD and EU Equality and Non-Discrimination Law. Luxembourg: Publications Office of the European Union.

[675] Case C-485/20. *XXXX v HR Rail SA*, 10 February 2022.

[676] Kelly-Lyth, A. (2023). Algorithmic discrimination at work. *European Labour Law Journal,* 14(2), 165.

[677] Ibid., p. 166.

grounds'.[678] The notion of differential treatment is central to direct discrimination. Notably, direct discrimination cannot be justified and is entirely prohibited in most employment cases.[679] However, the concept is geared around the identification of a **single protected ground** (or a proxy thereof, as discussed below), on which the less favourable treatment is based. **It is unlawful to input one or more protected criteria as key variables in the AM system's internal model**. Based on initial assessments, it appears highly improbable that AM programmers, providers, or adopters would choose to 'encode' discriminatory treatment predicated on the presence of a protected characteristic.[680] There is minimal practicality in pursuing such a course. Should a single category become the principal decision-making factor, AM tools might underperform, resulting in **diminished accuracy**. Moreover, software normally undergoes a phase of validation that allows engineers to assess the model's operation in practice, predict the likelihood of errors, and determine their relevance to the model's performance.[681] Thus, **non-discrimination rules apply even where a variable emerges as explaining a disparate impact. despite the source code being 'blind' to protected grounds**.[682] To prevent the occurrence of direct discrimination, any single element of AM systems' underlying code must be made neutral to protected grounds. Simultaneously, algorithms function by interweaving multiple factors and attributes in a dynamic and complex manner. This layering could make it more challenging to pinpoint a clear-cut case of direct discrimination, thereby suggesting that the inflexible model of unjustifiability might not be applicable. As a result, the effectiveness of this type of discrimination could be impacted. As Xenidis and Senden remark, 'if the concept of direct discrimination is fit to capture certain situations of algorithmic discrimination, **its relevance is likely to be less important than that of indirect discrimination**'.[683]

Indirect discrimination is defined as 'an apparently neutral provision, criterion or practice [that] would put [persons with a membership of a protected category] *at a particular disadvantage* compared with other persons unless that provision, criterion or practice is justified by a legitimate aim and the means of achieving that aim are appropriate and necessary' (emphasis added).[684] The focus of this definition is on the disadvantageous effects generated by a standardised provision, criterion or practice (PCP), rather than on the possession of a protected ground.

Indirect discrimination is generally relevant and effective in challenging algorithmic discrimination. It is also well suited to dealing with decisions made by algorithms which affect groups and communities. The model can capture a broad spectrum of scenarios in which algorithms are not directly designed to make choices based on protected groups, and even instances in which algorithms have been intentionally designed to be blind to these groups, ensuring they are not selected as significant variables. The **substantive and procedural rules associated with indirect discrimination can be effective in addressing some of the challenges associated with AM**, such as the opacity of the internal code, the complexity of the inner workings of AM systems, and the

[678] Art. 2(1)(a) Directive 2006/54/EC. See also Art. 2(2)(a) of Directive 2000/43/EC, Art. 2(2)(a) of Directive 2000/78 and Art. 3(a) of Directive 2010/41.

[679] Direct discrimination can be justified by a genuine occupational requirement. Gender Equality Directive (recast) Art. 14(2); Racial Equality Directive Art. 4; Employment Equality Directive Art. 4(1). See European Union Agency for Fundamental Rights and Council of Europe, Handbook on European Non-Discrimination Law (Publications Office of the European Union 2018) 91–109.

[680] Hacker, P. (2018). Teaching fairness to artificial intelligence: Existing and novel strategies against algorithmic discrimination under EU law. *Common Market Law Review*, *55*(4).

[681] Williams, B.A., Brooks, C.F., & Shmargad, Y. (2018). How algorithms discriminate based on data they lack: Challenges, solutions, and policy implications. *Journal of Information Policy*, 8, 78-115.

[682] Barbera, M. (2021). Discriminazioni algoritmiche e forme di discriminazione. *Labour & Law Issues*, *7*(1), I-1.

[683] Xenidis, R., & Senden, L. (2019). EU non-discrimination law in the era of artificial intelligence: Mapping the challenges of algorithmic discrimination, in: U. Bernitz et al. (eds.), *General Principles of EU law and the EU Digital Order,* Kluwer Law International, 2020, pp. 151-182.

[684] Art. 2(1)(b) of Directive 2006/54, Art. 2(2)(b) of Directive 200/43, Art. 2(2)(b) of Directive 2000/78 and Art. 3(b) of Directive 2010/41.

complex allocation of liability along the chain of command.[685] Specifically, the notion of indirect discrimination enables a pivotal shift in emphasis from the rules, parameters and content of AM, to their **effects**. Simultaneously, it redirects the analysis from the 'discriminators' to the individuals affected by the discrimination. To give an example, AM systems that are 'sartorially neutral', meaning that they provide for standardised treatment being reserved to all workers, without consideration for their protected characteristics, may disproportionately affect members of vulnerable groups,[686] and consequently be captured by the concept of indirect discrimination.

A broad set of justifications apply to this form of discrimination. An important element of indirect discrimination involves the potential for it to be objectively justified through a proportionality test. Reference must be made to the existence of a **legitimate aim being achieved through appropriate and necessary means**. First, the output of the given AM system must pursue a legitimate objective. Employers can explain that AM models intrinsically serve valid business purposes, for instance, **predictive accuracy** in making algorithm-based decisions at the workplace level. Second, if the objective is deemed legitimate, the measure in question must also be considered 'appropriate and necessary, that is effective and proportionate'.[687] Such a **regime of justifications, however, could affect the overall effectiveness of the indirect discrimination framework**, especially when the accuracy and performance of AM are presented as a means of limiting the biases embedded in human decision-making. However, to pass the proportionality test, there must be 'no other means of achieving [the same] aim that imposes less of an interference with the right to non-discrimination'.[688]

In this respect, it must be noted that **algorithmic discrimination, subtle as it might be, could nevertheless be easier to expose and identify by looking at the rules that have been codified by programmers**. By and large, algorithmic discrimination often leaves a more discernible trail. Conversely, understanding the reasons behind human decisions can be even more challenging. It has been argued that 'in the absence of a smoking gun, it will be difficult to prove discrimination on the part of any manager, even if a particular ethnic group consistently performs more poorly in that manager's assessment'.[689]

C. Flexibility and applicability of existing general provisions

One prominent aspect of EU equality legislation is that the notion that **intent is irrelevant** to both direct and indirect forms of discrimination. This means that the **emphasis is placed on the effects**, rather than on the intention of a presumed perpetrator, corroborating the suitability of the current legal framework in cases of personal biases and prejudices involuntarily creeping into the code behind AM tools. Notably, the effects-oriented way in which the legal framework operates could provide an opportunity to 'bypass' the need to reconstruct the internal operation of algorithmic 'black boxes'.[690] This implies that demonstrating the discriminatory outcomes of a decision is sufficient to establish the accountability of designers, developers and adopters. This facet might heighten the legal exposure of employers who implement AI tools from third-party providers, as they could

[685] Gerards, J., & Xenidis, R. (2021). *Algorithmic discrimination in Europe: Challenges and opportunities for gender equality and non-discrimination law*. European Commission. P. 71. However, see Adams-Prassl, J., Binns, R., & Kelly-Lyth, A. (2023). Directly discriminatory algorithms. *The Modern Law Review*, *86*(1), 144-175.

[686] De Vos, M. (2020). The European Court of Justice and the march towards substantive equality in European Union anti-discrimination law. *International Journal of Discrimination and the Law*, *20*(1), 62-87.

[687] Xenidis, R., & Senden, L. (2019). EU non-discrimination law in the era of artificial intelligence: Mapping the challenges of algorithmic discrimination, in U. Bernitz et al (eds.), *General Principles of EU law and the EU Digital Order*, Kluwer Law International, 2020, p. 22.

[688] European Union Agency for Fundamental Rights and Council of Europe (2018). *Handbook on European nondiscrimination law,* European Union and Council of Europe, p. 45.

[689] Kelly-Lyth, A. (2023). Algorithmic discrimination at work. *European Labour Law Journal*,14(2), p. 161

[690] Grozdanovski, L. (2021). In search of effectiveness and fairness in proving algorithmic discrimination in EU law, *CML Rev* 58: 99-136.

bear the repercussions of an inaccurate and biased AI design. However, concurrently, it could lead to greater efforts at interventions aimed at adopting bias-free AI tools.

Correlations and proxies. AM software can be built on metrics that are designed to be evolvable and be instructed to manage an infinite number of factors. Even in a case in which AM tools have been designed to avoid screening on the basis of a protected ground, they could end up selecting a correlation or proxy that is a substitute for the protected characteristic. **Novel and counterintuitive substitutes for traditionally protected characteristics could arise** as a result of AM, but these may not always be detected. The potential of AM systems is unprecedented in this respect, as they can statistically assess and factor in a large array of parameters. This challenge becomes even more complex when the presumed victims struggle to identify the protected ground involved. Moreover, the complexity and opacity of AM can complicate the process of reconstructing its inner workings in order to detect the protected characteristic affected, making it easier to conceal any reliance on a protected ground. In addition to this, the uncertainties surrounding algorithmic discrimination based on proxies create difficulties in categorising it as direct discrimination within the scope of EU law. This is because the definition of **direct discrimination requires a causal link between a specific treatment and a protected ground**, while inferential analytics mostly depend on correlations.[691]

Equality law has offered means to address these issues **prior to the advent of AM systems**, using case law as a driving force to strengthen the general framework.[692] The CJEU has employed a broad interpretation of what constitutes a protected characteristic, to include individuals who may not innately possess such a trait in a permanent manner. Several judgments have expanded the concept of direct discrimination to include situations in which individuals are treated unfavourably due to their **association** with a protected characteristic, even if they do not personally possess that characteristic. In simple terms, the concept of discrimination has been extended to cases in which decisions are made on the basis of factors related to, but different from, protected grounds. In *Coleman*, discrimination by association was recognised in favour of a mother based on her child's disability.[693] In *CHEZ*, residency was used as a '**proxy**'[694] for ethnicity.[695] These cases demonstrate that, far from being static, EU non-discrimination law can also apply to individuals who do not themselves hold protected characteristics.

As reported by Gerards and Xenidis, **existing notions 'can potentially extend to some cases of algorithmic proxy discrimination and miscategorisation'**.[696] In other words, they can address situations in which algorithms, designed or trained to use some specific categories of data, select and utilise characteristics that are closely linked to protected categories, for the purposes of classification and decision-making.[697] Thus, they can also

[691] Gerards, J., & Xenidis, R. (2021). Algorithmic discrimination in Europe: Challenges and opportunities for gender equality and non-discrimination law, European Commission, p. 74.

[692] Xenidis, R., & Senden, S. (2019). EU non-discrimination law in the era of artificial intelligence: Mapping the challenges of algorithmic discrimination, in: U. Bernitz et al. (eds), *General Principles of EU law and the EU Digital Order.* Kluwer Law International, 2020.

[693] Case C-303/06 *S. Coleman v Attridge Law and Steve Law 17* July 2008 ECLI:EU:C:2008:415. A prominent example is discriminatory treatment based on pregnancy, which is classified as sex discrimination, because pregnancy is considered a proxy for 'being a woman'. C-177/88 *Elisabeth Johanna Pacifica Dekker v Stichting Vormingscentrum voor Jong Volwassenen (VJV-Centrum) Plus* 8 November 1990 EU:C:1990:383. See De Vos, M., & Giegerich, T. (2020). Substantive formal equality in EU non-discrimination law, in: *The European Union as Protector and Promoter of Equality*, p. 250.

[694] Case C-457/17 *Heiko Jonny Maniero v Studienstiftung des deutschen Volkes eV* 15 November 2018 EU:C:2018:912. See also Prince, A.E.R., & Schwarcz, D. (2019). Proxy discrimination in the age of artificial intelligence and big data. *Iowa L. Rev.*, *105*, 1257.

[695] More specifically, in a claim brought by a non-member of the Roma group, residents of an urban district where many Roma people lived were found to have been discriminated against through technical impediments due to their electricity consumption meters being located at an unusual height. Case C-83/14 *'CHEZ Razpredelenie Bulgaria' AD v Komisia za zashtita ot diskriminatsia* 16 July 2015 ECLI:EU:C:2015:480.

[696] Gerards, J. & Xenidis, R. (2021). Algorithmic discrimination in Europe: Challenges and opportunities for gender equality and non-discrimination law, European Commission, p. 68.

[697] Kim, P.T., & Bodie, M.T. (2021). Artificial intelligence and the challenges of workplace discrimination and privacy. *Journal of Labor and Employment Law*, *35*(2), 289-315.

cover a situation in which individuals are algorithmically discriminated against on the basis of the inference of a characteristic that they do not actually possess. However, proxies are based on the rule of thumb. It is not entirely clear what threshold must be achieved in order to 'approximate' a neutral factor employed by AM systems used for decision-making in the context of employment, with a protected characteristic. In some cases, this is a rather simple exercise. For example, pregnancy is clearly related to sex. In other cases, statistical overlapping with protected grounds can be more difficult to prove. However, such challenges already existed prior to the advent of AM systems.

D. Procedural rules and applicability of presumptions

Mobilising the anti-discrimination legislation in the courts seems 'feasible yet far from easy', because workers face an imbalance of contractual and information power.[698] This challenge is exacerbated in the context of AM, given that **algorithmic discrimination is by nature more abstract, sophisticated and elusive than traditional forms of discrimination**.[699] Workers might never realise they have been put at a disadvantage, which leaves them without a basis on which to make a claim. To pursue legal action, workers first need to recognise the unfairness of the decision impacting them. Shedding light on the inner workings of AM systems is essential to increasing the ability of workers who consider themselves victims of algorithmic discrimination to pursue a legal remedy.

From a procedural perspective, the evidentiary regime gives the presumed victim a set a simplifications that eases their position. The **employer** appears to be the clear party held **accountable** when discrimination influences an employment decision. A method that places responsibility on employers when algorithmic discrimination impacts decisions is suitable, in order to reduce doubts regarding their relationship with the AM tool (whether it is developed in-house, purchased or licensed). Given that the definition of discrimination is based more on its effects than on its sources, liability may persist even if the employer has taken appropriate measures to prevent discrimination.

The anti-discrimination directives incorporate **unique rules of evidence**,[700] wherein the responsibility for presenting proof is partially shifted or shared between the claimant and the respondent who, in this scenario, is the employer using the AM system.[701] This rule represents a partial exception to the general principle by which 'each party bears the burden of proving the facts it alleges, and from which it derives favourable legal consequences'.[702] Workers who consider themselves discriminated against need only to establish in court **plausible yet not conclusive facts from which it can be presumed that direct or indirect discrimination has occurred**. They must present sufficient evidence (a *prima facie* case of discrimination) before the burden of proof shifts, at which point the employer is required to demonstrate that his decisions are based on objective

[698] De Stefano, V., & Wouters, M. (2022). AI and digital tools in workplace management and evaluation: An assessment of the EU's legal framework, p. 47.

[699] Wachter, S., Mittelstadt, B., & Russell, C. (2021). Why fairness cannot be automated: Bridging the gap between EU non-discrimination law and AI. *Computer Law & Security Review*, *41*, 105567, 5.

[700] Grozdanovski, L. (2021). In search of effectiveness and fairness in proving algorithmic discrimination in EU law. *Common Market Law Review*, *58*(1).

[701] Ringelheim, J. (2019). The burden of proof in antidiscrimination proceedings. A focus on Belgium, France, and Ireland. A Focus on Belgium, France, and Ireland (September 4, 2019), *European Equality Law Review*, (2).

[702] Farkas, L., & O'Farrell, O. (2015). Reversing the burden of proof: Practical dilemmas at the European and national level, Luxembourg: Publications Office of the European Union; Directive 2006/54/EC of the European Parliament and of the Council of 5 July 2006 on the implementation of the principle of equal opportunities and equal treatment of men and women in matters of employment and occupation (recast): 'Member States shall take such measures as are necessary, in accordance with their national judicial systems, to ensure that, when persons who consider themselves wronged because the principle of equal treatment has not been applied to them establish, before a court or other competent authority, facts from which it may be presumed that there has been direct or indirect discrimination, it shall be for the respondent to prove that there has been no breach of the principle of equal treatment' (Art. 19). Similar provisions can be found in the Racial Equality and Employment Equality Directives adopted in 2000, as well as in the 2004 Gender Equality in Access to Goods and Services Directive.

factors not related to any kind of discrimination.[703] This can be achieved by using **statistical or testimonial evidence, or even common-sense assessment and inferences drawn from circumstantial evidence**. For instance, the numerical over-representation of women among those taking parental leave can be used to trigger a presumption of discrimination when less favourable treatment is adopted in relation to someone who requests time off.[704] It falls on the defendants to prove that they did not violate the principle of equal treatment. They need to disprove the causal link between the harm, the conduct, and the protected characteristic or, in cases of indirect discrimination allegations, present a valid justification that passes the tests of appropriateness and necessity.[705]

The success of individual litigation in this domain significantly depends on the ease with which the burden of proof is shifted.[706] However, it must be stressed that this procedural system does not establish a presumption of discrimination. Rather, it still requires **plausible proof of the occurrence of discriminatory acts**.[707] As emerged in *Danfoss*,[708] the purpose of shifting the burden of proof is to ensure that workers who seem to be victims of discrimination are not deprived of effective ways to uphold the principle of equal pay.

Reversing the burden of proof only partially mitigates the obstacles faced by plaintiffs. Individuals who have experienced discrimination through AM systems may not have access to the necessary data to support their claims in court. Indeed, gathering evidence can prove arduous for victims and, in some cases, monitoring entities, due to the lack of transparency. The lack of the **accessibility of data is a long-standing issue in EU non-discrimination**. In general, the EU non-discrimination system is prevalently 'adjudicative, […] based on individual litigation'.[709] The burden is placed on the presumed victim, who is required to first uncover the discrimination and then to bring a case to court. Undeniably, the asymmetries that characterise professional contexts make these fact-finding and litigation tasks particularly burdensome. The emerging fields of **algorithmic auditing** and explainable AI (XAI) could provide more effective instruments to address these procedural limitations. Thanks to these techniques, it may become possible to alleviate some of the challenges faced by plaintiffs in terms of access to information, ultimately enhancing the pursuit of justice in discrimination cases involving AM systems. However, these fields are still in their early stages, and require further testing and refinement to be effective.

As providing *prima facie* evidence of discrimination could be enough from a procedural standpoint, **there is no need to open the 'black box' to prove algorithmic discrimination in court**. Moreover, a claimant can rely on the GDPR provisions on access and transparency, such as Art. 13(2)(f), Art. 14(2)(g) and Art. 15, to be informed and gain access to information in the case of AM systems that are considered to have discriminatory effects. Grozdanovski explains that, far from overburdening respondents, access rights would offer them the opportunity to effectively participate in a debate regarding evidence of

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[703] Fredman, S. (2016). 'Pasts and futures: EU equality law', in: Bogg, A., Costello, C. & Davies, A.C.L. (eds.), *Research Handbook of European Labour Law,* Edward Elgar Publishing, 2016, pp. 391-421.

[704] Xenidis, R., & Senden, L. (2020), n. 47 (indicating 'average working hours' as a potentially discriminatory proxy used against female applicants in case of promotion also when past successful candidates' data has been made blind to applicants' gender). See also C-170/84 *Bilka-Kaufhaus GmbH v Weber von Hartz,* 13 May 1986 ECLI:EU:C:1986:204.

[705] Allen, R. (2020). Artificial Intelligence, Machine Learning, Algorithms and Discrimination Law: The New Frontier, *Cloisters*.

[706] De Stefano, V., & Wouters, M. (2022). AI and digital tools in workplace management and evaluation: An assessment of the EU's legal framework, p. 61.

[707] Grozdanovski, L. (2021). In search of effectiveness and fairness in proving algorithmic discrimination in EU law. *Common Market Law Review*, *58*(1).

[708] Case No. 109/88 *Handels-og Kontorfunktionærernes Forbund I Danmark v. Dansk Arbejdsgiverforening, acting on behalf of Danfoss* 17 October 1989 ECLI:EU:C:1989:383 ('where an undertaking applies a system of pay, which is totally lacking in transparency, it is for the employer to prove that his practice in the matter of wages is not discriminatory, if a female worker establishes, in relation to a relatively large number of employees, that the average pay for women is less than that for men' [para. 16]).

[709] Gerards, J., & Xenidis, R. (2021). Algorithmic discrimination in Europe: Challenges and opportunities for gender equality and non-discrimination law, European Commission, p. 75.

algorithmic discrimination.[710] In *Meister*[711] – a case dealing with the rejection during the recruitment process of a candidate who met the criteria for the post – the CJEU confirmed that **there is no specific requirement to share data with the presumed victim.** Defendants are not obliged to reveal the factors applied by AM, as it is sufficient to show that a certain practice serves a legitimate aim and is proportionate and necessary. The CJEU stated that 'refusal to grant any access to information [to the applicant meeting the requirements of the job advertisement] may be *one of the factors* to take into account in the context of establishing facts from which it may be presumed that there has been direct or indirect discrimination' (emphasis added).[712] Consequently, **hesitancy or reluctance to provide information may be used in court as circumstantial evidence supporting a *prima facie* case of discrimination**.[713] However, a significant complicating factor is that the data feeding the algorithm may be classified as a **trade secret**, protected under Directive (EU) 2016/943.[714] To date, striking balance between the need for transparency in discrimination cases and the protection of trade secrets when it comes to the inner workings of AM systems remains a nuanced challenge for the courts. This poses a legal conundrum that requires careful consideration and potential future legislation or jurisprudential guidance.[715]

In summary, **redress mechanisms may be scarcely accessible, and thus ineffective**. For example, research finds that the enforcement of gender equality law in the MS is fraught with problems of various natures.[716] These include institutional issues, such as the length of proceedings; lack of expertise and assistance, lack of trust in the judiciary, lack of sufficient compensation, financial issues such as costs of litigation; lack of legal aid, due to uncertainty about the outcome; and fear of victimisation by the employer, family, and peers. These problems are not specific to AM systems, but they may be aggravated by the way in which AM systems operate.

D. Underlying dataset and training data

The draft **AI Act** has also been proposed to complement 'existing Union law on non-discrimination, with specific requirements that aim to minimise the risk of algorithmic discrimination, in particular in relation to the design and the quality of data sets used for the

[710] Grozdanovski, L. (2021). In search of effectiveness and fairness in proving algorithmic discrimination in EU law. *Common Market Law Review*, *58*(1): Recital 71 GDPR lays down a requirement to 'implement technical and organisational measures' that 'prevent, inter alia, discriminatory effects on natural persons on the basis of racial or ethnic origin, political opinion, religion or beliefs, trade union membership, genetic or health status or sexual orientation, or that result in measures having such an effect'.

[711] Case C-415/10 *Galina Meister v Speech Design Carrier Systems GmbH.*

[712] Case C-415/10 (para. 47). The employer's refusal to disclose information can 'make his decisions virtually unchallengeable. […] In the context of a recruitment procedure, […] the position of the applicant—inevitably external to the undertaking in question—makes obtaining evidence or facts from which it may be presumed that there has been discrimination even more difficult than if the applicant sought to prove that the employer applies discriminatory measures in respect of conditions of employees' pay, for example' (para 32), Opinion of Advocate General Mengozzi, delivered on 12 January 2012 in Case C-415/10 *Galina Meister v Speech Design Carrier Systems GmbH*.

[713] See also Case C-104/10 *Patrick Kelly v National University of Ireland (University College, Dublin)*, ECLI:EU:C:2011:506, in which the CJEU held that 'a refusal of disclosure by the defendant, in the context of establishing such facts, could risk compromising the achievement of the objective pursued by [the burden of proof] directive and thus depriving, in particular, Art. 4(1) thereof of its effectiveness (para 39).

[714] Directive (EU) 2016/943 of the European Parliament and of the Council of 8 June 2016 on the protection of undisclosed know-how and business information (trade secrets) against their unlawful acquisition, use and disclosure (Text with EEA relevance)

[715] For a broader discussion on the balancing exercise between algorithmic transparency and trade secrets, see de Torres Bóveda, N. (2022). Business Secrecy as a Limit to the Right to Information on Algorithms, *Hungarian Labour Law E-Journal* 2022/2.

[716] Chopin, I., Conte, C., & Chambrier, E. (2019). A comparative analysis of non-discrimination law in Europe 2018: The 28 EU Member States, the former Yugoslav Republic of Macedonia, Iceland, Liechtenstein, Montenegro, Norway, Serbia, and Turkey compared, *Luxembourg: European Commission,* and Timmer, A., & Senden, L. (2019). A comparative analysis of gender equality law in Europe 2018: A comparative analysis of the implementation of EU gender equality law in the EU Member States, the former Yugoslav Republic of Macedonia, Iceland, Liechtenstein, Montenegro, Norway, Serbia, and Turkey. Luxembourg: European Commission.

development of AI systems complemented with obligations for testing, risk management, documentation and human oversight throughout the AI systems' lifecycle'.[717]

Choices at the workplace level may be influenced by many factors, including prejudices, biases, and even discriminatory elements. Conventional managerial prerogatives are not immune from these risks. In this respect, AM could be used to reduce arbitrariness and favouritism. For this to happen, the data on which decisions are based must be free from bias and discrimination. This is a technical challenge which, until recently, has received little attention in the legal framework. As mandated by the draft AI Act, the providers of AI systems must comply with a 'set of horizontal mandatory requirements for trustworthy AI' listed in Chapter 2 of the Draft.[718] One precondition, seen as an uphill task by some commentators,[719] is the use of 'sufficiently' high-quality datasets for which the training, validation and testing are relevant, representative, free from error, and complete. Nevertheless, **factors such as accuracy, relevance, representativeness, freedom from error and completeness do not guarantee that the training data are free from bias and discrimination**. The intricate and interconnected disparities that form the foundations of the social fabric invariably influence the data that fuel predictive analytics and machine learning.[720] The risk lies in the possibility that discrimination, reflected in statistical correlations, could be further solidified by algorithms, thereby deepening the roots of exclusion and inequality.[721] **Data from a historically and structurally imperfect and unfair world** could be accurate, relevant, representative, free from errors and complete, and yet result in discrimination. This would result in an entrenched downward spiral of the aggravation of existing discrimination. Programmers are expected to clean and de-bias the dataset on which AM software is trained and run. Appropriate datasets can be produced with intensive worker involvement.[722] In this regard, the process of reviewing data by experts can only be iterative and continuous.[723] Data protection law provides the right to rectify inaccurate or incomplete personal data under Art. 16 GDPR. Meanwhile, Art. 29 WP Guidelines on automated individual decision-making and profiling suggests that **companies should assess datasets for biases, regularly review the accuracy and relevance of decisions, deploy systems that audit algorithms, and use 'appropriate procedures and measures to prevent errors, inaccuracies, or discrimination'** based on sensitive data.[724] In this respect, the DPIA should ensure algorithmic legibility and accountability, by allocating a set of duties to the data controller, who must explain 'the measures envisaged to address the risks'.

In the realm of AM, other solutions lean more towards socio-technical approaches than strictly legal ones. For instance, by **training** both workers and managers in mitigating the influence of their implicit biases, the potential adverse effects of algorithms that reflect such biases can be reduced.

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[717] AI Act, Exploratory Memorandum, p. 4.

[718] See European Commission High-Level Expert Group on Artificial Intelligence (2019)., Sectoral Considerations on Policy and Investment Recommendations for Trustworthy AI.

[719] Smuha, N., & Morandini, A. (2023). Trustworthy AI through Regulation? Sketching the European Approach, *The Digital Constitutionalist*, available at: https://digi-con.org/4-trustworthy-ai-through-regulation-sketching-the-european-approach/. The AI Act states that '[t]he use of AI with its specific characteristics (e.g., opacity, complexity, dependency on data, autonomous behaviour) can adversely affect a number of fundamental rights enshrined in the EU Charter of Fundamental Rights'. AI Act, p. 11.

[720] Xenidis, R. (2020)., Tuning EU equality law to algorithmic discrimination: Three pathways to resilience, *Maast J Eur & Comp L*, 27: 740. See also Gebru, T. et al. (2021). Datasheets for Datasets, ArXiv 1803.09010.

[721] Xenidis, R., & Senden, L. (2020). EU non-discrimination law in the era of artificial intelligence: Mapping the challenges of algorithmic, in: U. Bernitz et al (eds.), *General Principles of EU law and the EU Digital Order*, Kluwer Law International, 2020.

[722] Marsh W, Nur K, Yet B et al. (2016) Using operational data for decision making: A feasibility study in rail maintenance. *Safety and Reliability* 36(1): 35–47.

[723] Asatiani A, Malo P, Nagbøl PR et al. (2020) Challenges of explaining the behavior of black-box AI systems. *MIS Quarterly Executive* 19(4): 259–278; van den Broek E, Sergeeva A and Huysman Vrije M (2021) When the machine meets the expert: An ethnography of developing AI for hiring. *MIS Quarterly* 45(3): 1557–1580.

[724] Guidelines on Automated individual decision-making and Profiling for the purposes of Regulation 2016/679, 28.

Moreover, AM tools and their internal code may be protected as **trade secrets under Directive (EU) 2016/943,** which sets forth a series of criteria to be met in order to do so.[725] Since a request to reveal the underlying AM system might face such obstacles, this could also potentially complicate the already challenging task for the affected workers of proving algorithmic discrimination. In addition, several players are involved in the chain of AM, from engineers and coders to those who put the tool on the market, to the adopters (employers) and end users (managers, co-workers, or clients). Who should be held liable? The complex and yet fragmented sequence of decisions involved renders the allocation of liability and responsibility particularly problematic, with the employer bearing most of the brunt. In this respect, the current framework contains several gaps when it comes to assigning liability in situations where multiple agents are involved at different stages. In particular, the diffusion of responsibility for managerial decisions into the supply chain 'threatens the operation of regulatory systems, such as employment law, designed to condition managerial agency through a combination of ex-ante incentives, and ex-post liability attached to the exercise of employer functions'.[726]

3.4.2. Selected national legislation and policies

A. EU countries

There are currently no regulations in the EU MS that explicitly address the novel challenges relating to algorithmic discrimination at work. General national regulations on non-discrimination, although not AM-specific, apply to both 'conventional' and algorithmic settings. In some countries, labour law regulation has been successfully applied to litigate cases on indirect algorithmic discrimination (see the analysis of the Bologna Tribunal judgment in the *Deliveroo* case, below in subsection 3.4.5. Selected case law). A specific reference to AM in the context of algorithmic discrimination has been made in the **Portuguese** Law 13/2023, in force since May 2023, which explicitly clarifies that equality and non-discrimination provisions apply to decision-making based on algorithms or other artificial intelligence systems.[727] This is, however, merely a restatement of the already existing rules. Moreover, the **Italian** 'Transparency Decree' requires the employer or principal to inform the worker, prior to the start of the work activity, *inter alia* about 'the level of accuracy, robustness, and cybersecurity of the [AM] systems, and the metrics used to measure these metrics, as well as the potentially discriminatory impacts of the metrics themselves'.[728]

In some EU countries, the need for a reassessment of the existing anti-discrimination legislative framework is the subject of political debate. In **Germany**, the federal government committed itself to evaluating the AGG in the coalition agreement 2021-2025 between the SPD, BÜNDNIS 90/DIE GRÜNEN, and FDP.[729] This option has been advocated for by the German Data Ethics Commission (*Datenethikkommission*), which pointed out that the areas of application of the AGG in its current form are not comprehensively geared towards algorithmic discrimination, and that adjustments are necessary.[730] In **Finland**, the 2021 government plan included the Avoiding AI Biases project, which set out to identify risks related to fundamental rights and non-discrimination in existing and planned AI systems. To

[725] Directive (EU) 2016/943 of the European Parliament and of the Council of 8 June 2016 on the protection of undisclosed know-how and business information (trade secrets) against their unlawful acquisition, use and disclosure (Text with EEA relevance).

[726] Adams-Prassl, J., Abraha, H., Kelly-Lyth, A., Silberman, M. 'Six,' & Rakshita, S. (2023). Regulating algorithmic management: A blueprint. *European Labour Law Journal*, *14*(2), 140.

[727] See: https://www.macedovitorino.com/New-Legal-Framework-For-Portuguese-Labour-Code-ndash-Decent-Work-Agenda/6371/.

[728] Legislative Decree 27 June 2022, N. 104 Implementing Directive (EU) 2019/1152 of the European Parliament and of the Council of 20 June 2019 on transparent and predictable working conditions in the European Union.

[729] https://algorithmwatch.org/en/algorithmic-discrimination-law/.

[730] The full text of the Opinion is available in English at https://www.bmi.bund.de/SharedDocs/downloads/EN/themen/it-digital-policy/datenethikkommission-abschlussgutachten-lang.pdf?__blob=publicationFile&v=5. See p. 128.

ensure that AI applications are non-discriminatory, an assessment framework was developed to identify and manage discrimination risks and promote equality in AI use.[731]

B. Non-EU countries

Innovative regulatory approaches towards algorithmic discrimination in employment can be found in the **US**. A prime example is the local regulation introduced by **New York City Council** on automated employment decision tools ('AEDT')[732], effective as of January 2023. This requires employers and employment agencies to carry out mandatory bias audits of automated decision-making tools, no more than one year prior to the use of such tools. This audit must be conducted by an independent auditor and should assess the tool's potential disparate impact on sex, race and ethnicity. Before using automated decision-making tools, employers and employment agencies must publicly disclose the date of the technology's most recent bias audit and a summary of the results. The law defines AEDTs as 'any computational process, derived from machine learning, statistical modelling, data analytics, or [AI], that issues simplified output, including a score, classification, or recommendation' that is used to 'substantially assist or replace discretionary decision making for making employment decisions that impact natural persons'. Violation of the law could result in an employer being fined between USD 500 and USD 1,500 per violation, per day. Employers are navigating the nuances of the law, seeking clarity on its specific coverage.

Another relevant US example is the Stop Discrimination by Algorithms Act of 2021, proposed in **Washington**. This Act has not yet been passed. If adopted, it will prohibit companies and organisations from using algorithms that produce biased results. Annual audits will be made mandatory to ensure the algorithms used in companies are not discriminatory. Moreover, companies using AM systems will have to document and share with the Office of the Attorney General how their algorithms are built, how they make decisions, all of the decisions made, and the audit results. Lastly, transparency will increase, as companies will be obliged to disclose the personal information that is being collected, and how their algorithms reach decisions. They will also have to provide detailed explanations to consumers if an algorithm makes an unfavourable decision and allow consumers to submit corrections.[733] Failure to comply with this Act will result in fines of USD 10,000 per individual violation.

In **Brazil**, data protection legislation in force since 2018[734] permits audits to be conducted by the data protection authority in cases where there is a potential risk that automated decisions in the workplace will lead to unfair outcomes that accentuate existing biases, particularly against protected groups and union activists. Such audits can be initiated if an employer fails to disclose the criteria employed in their automated processes.

3.4.3. Social partner agreements and initiatives

Most examples of collective bargaining agreements explicitly addressing the issue of (algorithmic) discrimination concern **platform work**. A report issued by the European Trade Union Institute (ETUI) identified nine collective bargaining agreements that included anti-discrimination clauses or provisions, as of September 2022.[735] General anti-discrimination clauses have been included, e.g. as is the case of two Spanish agreements for TRADEs, and the Italian UGL-AssoDelivery agreement, all of which concern food delivery. For example, discrimination is expressly forbidden in the agreement signed in March 2021 in Tuscany between the food delivery platform Just Eat, platform workers, and trade union

[731] See: https://practiceguides.chambers.com/practice-guides/artificial-intelligence-2023/finland/trends-and-developments

[732] See: https://rules.cityofnewyork.us/wp-content/uploads/2023/04/DCWP-NOA-for-Use-of-Automated-Employment-Decisionmaking-Tools-2.pdf

[733] See: https://www.holisticai.com/blog/washington-dc-stop-discrimination-algorithms-law. The full text of this draft Act is available at https://lims.dccouncil.gov/downloads/LIMS/52282/Introduction/B25-0114-Introduction.pdf?Id=155626.

[734] Lei Geral de Proteção de Dados Pessoais, or LGPD; Lei 13709/2018.

[735] For a full list, see Lamannis, M. (2023). Collective Bargaining in the Platform Economy: A Mapping Exercise of Existing Initiatives. *ETUI Research Paper-Report*, *2* p. 45, Figure 9, p. 27.

CGIL (the Italian General Confederation of Labour), the CISL (the Italian Confederation of Workers' Trade Unions), and UIL Trasporti (the Italian Union of Transport Workers).[736] An agreement signed in December 2021 between Takeaway Express Spain and the Federation of Services CCOO, FSC-CCOO, and FeSMC-UGT, refers specifically to gender equality.[737] A collective agreement between Voocali and the Danish Union HK Privat from September 2018 requires objective reasons for being excluded from the platform.

Beyond the context of the platform economy, an example of a sectoral collective agreement addressing algorithmic discrimination can be found in **Spain**. The XXIV Collective Banking Agreement 2019/2023 (*Convenio Colectivo de Banca* 2019/2023) provides that employees have the right to non-discrimination in relation to decisions and processes when these are based solely on algorithms. In such cases, workers can request the assistance of persons designated for this purpose by the company, in the event of a discrepancy.[738]

3.4.4. Other relevant initiatives

A. EU countries

The European Trade Union Confederation (ETUC), which comprises 93 national trade union confederations in 41 countries, plus 10 European trade union federations, adopted a **Resolution on the European strategies on artificial intelligence and data**.[739] In this official position, the ETUC advocates for 'an independent fundamental rights and equality impact assessment'.

In March 2023, the **German** Ethics Council published a detailed opinion on 'Humans and Machines – Challenges of Artificial Intelligence'.[740] The Council noted that AI systems should be appropriately supervised to guard against discrimination. It argued for a gradual approach: the greater the depth of the AI intervention and the more indispensable the AI systems, the higher the requirements for minimising discrimination ought to be.[741] This principle can be applied in AM settings, too. The Council also suggests countering pervasive surveillance through the use of the principles of privacy-by-design,[742] which is also of considerable relevance in the context of AM.

In **Poland**, the 'Statement of Purpose for the AI Strategy in Poland. Action Plan of the Ministry of Digital Affairs'[743] has drawn attention to the ability of AI systems to take discriminatory decisions that may result especially from erroneously formed sets of training data. This document identified the need to develop anti-discriminatory legal regulations in this regard. No specific legislative proposals have followed, however.

B. Non-EU countries

Several innovative initiatives have been put in place overseas. In the **US**, the Equal Employment Opportunity Commission (EEOC) in January 2021 launched the Artificial Intelligence and Algorithmic Fairness Initiative. Within the framework of this initiative, the EEOC issues technical assistance to provide guidance on algorithmic fairness and the use

[736] See: https://apps.eurofound.europa.eu/platformeconomydb/collective-agreement-between-just-eat-trade-union-cgil-cisl-and-uil-trasporti-and-workers-in-italy-105068.

[737] The full text of this agreement is available in Spanish at https://www.ccoo-servicios.es/archivos/Acuerdo%20Sindicatos%20JUST%20EAT(1).pdf

[738] Available at https://bit.ly/3DOWm5l.

[739] ETUC (2020). Resolution on the European strategies on artificial intelligence and data, https://bit.ly/ETUCres.

[740] Deutscher Ethikrat [German Ethics Council], Mensch und Maschine – Herausforderungen durch Künstliche Intelligenz, 2023, https://www.ethikrat.org/fileadmin/Publikationen/Stellungnahmen/deutsch/stellungnahme-mensch-und-maschine.pdf; English translation available at: https://www.ethikrat.org/en/publications/.

[741] Deutscher Ethikrat (German Ethics Council) (2023). Mensch und Maschine – Herausforderungen durch Künstliche Intelligenz, p. 283.

[742] Ibid, p. 48.

[743] Statement of purpose for the AI strategy in Poland. The action plan of the Ministry of Digital Affairs Recommendations, prepared *pro bono*, at the invitation and under the direction of the Ministry of Digital Affairs by stakeholders interested in the development of AI in Poland, available at: https://bit.ly/44TQC6l.

of AI in the area of employment; identifies good practices; holds sessions with key relevant stakeholders; and collects information about the adoption, design, and impact of employment-related technologies.[744] Another US soft-law initiative targeting algorithmic discrimination is the **Blueprint for an AI Bill of Rights**. One of the five principles established by this non-binding instrument is concerned specifically with protections against algorithmic discrimination.[745] It provides that the designers, developers and deployers of automated systems should take proactive and continuous measures to ensure protection against algorithmic discrimination, and to use and design systems equitably. They should also carry out proactive assessments as part of the system design. Representative data and protection against proxies for demographic features should be used. Moreover, independent evaluation and plain language reporting in the form of an impact assessment, including disparity testing results and mitigation, should be performed and made public whenever possible.[746]

In **Japan**, the AI Utilisation Guidelines issued by the Ministry of Internal Affairs and Communications in 2019 have drawn specific attention to the issue of possible algorithmic discrimination. One of the principles of AI utilisation is the principle of fairness, which provides that 'AI service providers, business users, and data providers should pay attention to the possibility of bias inherent in the judgements of AI systems or AI services and take into consideration that individuals and groups will not be unfairly discriminated against by their judgements'. The guidelines explain that 'AI service providers, business users, and data providers are expected to pay attention to the representativeness of data used for AI learning or other methods and the social bias inherent in the data according to the social context in utilising AI, considering how the results of AI judgements may be determined by learning data. AI service providers and business users shall pay attention to the possibility of bias inherent in AI judgements, due to the algorithm used in it. In machine learning, in particular, the majority tends to be adopted, and the minority is less likely to be done. AI service providers and business users are expected to intervene with human judgment on whether to adopt or how to use, the judgement of AI, as well as with consideration for the social context and the reasonable expectations of people when utilising the AI, to ensure the fairness of the judgement result from it. They are expected to consider the necessity of human intervention based on examples of criteria from a viewpoint of fairness.'[747]

3.4.5. Selected case law

Bologna Tribunal – 31 December 2020 – Deliveroo

This case concerned the 'self-service booking system' that served Deliveroo food-delivery couriers to book time slots during which they would be available to accept delivery requests. Access to this scheduling system was not equal for all riders, however, but depended on a score (the so-called 'rating') given to them by the platform. Riders with the highest scores could avail themselves of a slot of their preference (11 a.m., 3 p.m., or 5 p.m.), while those with a lower score were left with no choice but to accept a shift that had not been reserved by the others. Riders' scores were calculated according to so-called 'reliability' and 'participation' factors. These were decisive for their actual access to work opportunities and, as a consequence, key to determining the relevant remuneration to they were entailed. The score was automatically reduced when riders failed to participate in a previously booked shift without having cancelled it 24 hours in advance. This was the case regardless of the specific reason for their absence from a given shift. Riders had to suffer the consequences of the lower score, even if their late cancellation or refusal to perform

[744] See: https://www.eeoc.gov/ai. For example, it provided guidance on the Americans with Disabilities Act in the context of the use of software, algorithms, and AI to assess job applicants and employees. See https://www.eeoc.gov/laws/guidance/americans-disabilities-act-and-use-software-algorithms-and-artificial-intelligence.

[745] Other principles include safe and effective systems through adherence to standards and ongoing monitoring; and ensuring data privacy.

[746] See: https://www.whitehouse.gov/ostp/ai-bill-of-rights/algorithmic-discrimination-protections-2/.

[747] See: https://www.soumu.go.jp/main_content/000658284.pdf.

tasks during a reserved slot was due to a salient and legitimate reason, such as illness or care duties or even participation in a lawful strike. **The fundamental issue with this 'impartial' algorithm**, which was claimed to act without bias in rating those riders who failed to show up for work, **was that it disadvantaged those who were engaging in activities that are legally protected, such as participating in a strike**.

The local confederation of Italy's most representative union, the *Confederazione Generale Italiana del Lavoro* (CGIL), brought a complaint against Deliveroo's booking system on behalf of the riders. The claimants sought a judicial declaration of the **discriminatory nature of the algorithm**, which did not differentiate between the legally protected and non-protected reasons behind the worker's absence from a reserved slot, thus putting those riders exercising collective rights at a particular disadvantage compared with other workers.

The Court in Bologna was confronted with three key legal questions. The first concerned the applicability of anti-discrimination provisions to self-employed riders. The second related to whether the interpretation of the notion of 'personal belief' as a protected ground against discrimination also encompassed trade union affiliation. The third contested issue related to the legitimacy of a trade union to act on behalf of workers. For this to be the case, the victims of alleged discrimination cannot be directly identifiable. The Court answered all three legal questions in the affirmative. It also found that Deliveroo's algorithm had an indirectly discriminatory effect on riders who were exercising their constitutional right to engage in strike action. The employment status of riders, i.e. the fact that they were engaged as self-employed, did not hinder the application of the anti-discrimination provisions.[748] The Court found that the supposedly neutral practice of using the 'self-booking system' met the criteria of the definition of **indirect discrimination** set out in Article 2(1) of the Legislative Decree No. 216/2003, since it put persons holding a particular belief at a disadvantage compared with other persons. The uniform application of the score thus had a detrimental effect on workers who were exercising their constitutional rights.

This is the first judicial decision declaring an algorithm in a workplace to be discriminatory. The judgment demonstrates that indirect discrimination is a highly suitable basis on which to challenge algorithmic biases, as it is inherently concerned with the disadvantaging impact of a *de facto* 'standardised' decision, measure, or policy, rather than with the mere membership of a community identified by the presence of a shared protected ground. It also confirms the applicability of the anti-discrimination regulations regardless of employment status and endorses the active legitimacy of trade unions to act on behalf of alleged victims. The legal basis for this judgment was Legislative Decree No. 216/2003, an Italian regulation implementing Directive 2000/78/EC into the domestic legal system. **By and large, this case shows the potential of the EU legal framework for the protection of workers against algorithmic discrimination.**

3.4.6. Conclusions

The way in which AM tools operate does not neatly fit the substantive and procedural provisions of the equality and non-discrimination legal framework. First, some AM tools are often almost **undetectable**, thereby negatively affecting workers' awareness of potentially discriminatory behaviours put in place by employers or co-workers.[749] Second, individuals and groups can be classified on the basis of information and factors that have **no clear and direct protection** in equality law. Relatedly, algorithmic discrimination can be dynamic and multidimensional, and can be based on insights that are not accessible based on ordinary human observations.[750] At the same time, algorithms can base their decisions on

[748] The Court relied on two domestic legal instruments: Legislative Decree No. 81/2015, and Legislative Decree No. 216/2003. The first of these is a specific piece of legislation with a narrow personal scope, covering only self-employed platform workers providing services through platforms in the delivery sector. According to Article 47 bis para. 1 of this Decree, anti-discrimination law provided for employees, including access to the platform, also applies to also to self-employed riders.

[749] Allen, R., & Masters, D. (2021). Technology Managing People: the legal implications, a report for the Trades Union Congress by the AI Law Consultancy.

[750] Kim, P.T., & Bodie, M.T. (2021). Artificial intelligence and the challenges of workplace discrimination and privacy. *Journal of Labor and Employment Law*, *35*(2), 289-315.

correlations and proxies. Third, AM systems often **lack transparency**, an aspect that experts consistently consider one of the primary obstacles to pursuing legal action and enforcement measures.[751] Fourth, the challenges of **securing evidence and overcoming obstacles** in cases of algorithmic discrimination often stem from the opacity of algorithms, limitations regarding data access, and the need for specialised expertise to uncover instances of bias. Without access to evidence and data, **proving algorithmic discrimination becomes very challenging**. Another barrier might stem from employers' reluctance to disclose details about the underlying technology and the variables that inform a certain decision, compounded by the issue that the algorithm could be safeguarded as a trade secret.

At the same time, the adoption of AM systems can make the **enforcement of fairness principles more accurate and efficient**.[752] Indeed, automating decision-making processes and their related instructions in the form of code can, in principle, make discriminatory criteria easier to identify and outcomes more measurable. Hence, the current shift towards AM systems holds considerable promise in **addressing long-standing inequalities in professional contexts**, not least those resulting from implicit and structural biases.

The EU equality framework is **flexible when it comes to addressing discrimination stemming from AM systems**. Challenges that arise are typically attributable either to **unresolved matters or interpretative gaps** regarding discrimination as a human phenomenon.[753] It follows that '[m]any of the issues relating to the discriminatory potential of algorithms are not unique to algorithms, but are problems that have been a staple of antidiscrimination law'.[754] In addition, several aspects of the EU corpus of anti-discrimination law, in principle, are not perfectly suited to the way in which AM systems operate. The EU legal model 'has a number of gaps and grey zones that are problematic in light of the phenomenon of algorithmic discrimination',[755] which, in turn, exacerbates the problems related to the applicability and enforcement of existing legislation.

Importantly, the constitutive features of algorithmic discrimination are barely compatible with the 'traditional' forms of discriminatory practices, thus reducing the effectiveness of the existing rules and redress mechanisms. AM systems design individualised instructions, treatments, and requirements for workers by aggregating legitimately collected data, which are not necessarily aligned with the set of characteristics protected under EU law. Moreover, AM systems can profile workers and make distinctions that may be completely 'artificial and nonrepresentational' from a legal perspective.[756] Unintuitive and abstract classifications can also be the product of AM practices. Yet, it is not easy to associate proxies and other characteristics with protected grounds. Hence, they are barely captured by the current EU legal framework.[757] Some of the recent examples provided in the literature include commuting distance from one's home to a job; browsing behaviour; interactions on social media; clicks on webpages; sports, nutritional and sleep habits; lifestyle and other hygiene-related data; facial expressions; tone of voice; the use of

[751] Kelly-Lyth, A. (2023). Algorithmic discrimination at work. *European Labour Law Journal*, 14(2), p. 167.

[752] In this regard, Art. 12 of the proposed AI Act foresees record keeping for high-risk systems, with the aim of ensuring traceability. See also Lobel, O. (2022). *The equality machine: harnessing digital technology for a brighter, more inclusive future*. Hachette UK.

[753] Ibid 166.

[754] Selmi, M. (2021). Algorithms, discrimination, and the law. *Ohio St. LJ*, *82*, 611. See also: Gerards, J., & Xenidis, R. (2021). Algorithmic discrimination in Europe: Challenges and opportunities for gender equality and non-discrimination law, European Commission, p. 9: novel questions 'shine a new light on […] 'traditional' problems […] of EU gender equality and non-discrimination law'.

[755] Gerards, J. & Xenidis, R. (2021). Algorithmic discrimination in Europe: Challenges and opportunities for gender equality and non-discrimination law, European Commission, p. 53.

[756] Leese, M. (2014). The new profiling: Algorithms, black boxes, and the failure of anti-discriminatory safeguards in the European Union. *Security Dialogue*, *45*(5), pp. 502, 504.

[757] Xenidis, R. (2020). Tuning EU equality law to algorithmic discrimination: Three pathways to resilience. *Maastricht Journal of European and Comparative Law*, *27*(6), 736-758.

specific words; sentence length; and talking speed during recruitment procedures.[758] Socioeconomic or health status, education and income, to give but a few salient examples of factors that could be assessed by AM systems, are not included in the list of protected grounds.

In the presence of a **closed catalogue of protected grounds**, the new forms of discrimination enacted by AM are at risk of not being captured. For instance, a recruiter could assume information about a candidate for a job position: if the applicant is using an outdated browser, for example, they might be connecting from a public library in a given neighbourhood, while the latter factor could be associated with a certain educational background.[759] Other aspects such as average working hours, career consistency, contractual length and retention prospects can provide a rather effective substitute for protected characteristics, thus making the application of anti-discrimination safeguards more complex. While patterns and correlations may be as effective as direct identifiers, factors such as tone of voice, IP address, location and education are not directly covered by EU equality law.[760]

EU law does not provide explicit protection against **discrimination by perception, ascription, or assumption**, despite this being the most common way in which AM systems work.[761] Algorithms are designed to identify patterns and recurrences, meaning that they **depend on statistical inferences that could potentially mirror discriminatory correlations**. Another incongruity relates to the **causal nature of the relationship between discrimination and specific protected categories**, when contrasted with the dependence of machine learning algorithms on statistical inferences and correlations. These uncertainties create hurdles in categorising algorithmic proxy discrimination as direct discrimination, as the definition of direct discrimination in EU law requires a causal link between a specific treatment and a protected ground, whereas inferential analytics depend on correlations.[762]

The above limitations are coupled with, and exacerbated by, a rather **static consideration of protected grounds**, which fails to address the very granular way in which AM might operate, posing several difficulties in 'identifying differential treatment on the basis of protected grounds, especially when they are abstracted, or intersectional, or emergent'.[763] The recognition of either direct or indirect forms of discrimination are at risk of being scarcely effective in the case of intersectional discrimination.[764] Not even the recognition of 'multiple discrimination' in Directives 2000/78/EC and 2000/43/EC would be enough to tackle intersectional discrimination, which can occur even in the absence of discrimination based solely on one of the protected grounds.[765] In addition, drawing a boundary between direct and indirect forms of discrimination is becoming increasingly complex.

At the level of enforcement, **an important gap pertains to remedies**. In the case of algorithmically induced discrimination, complaints and lawsuits can be more complex, both for those exposed to such systems and for judiciaries. The procedural model is designed on

[758] Ajunwa, I. (2023). *The Quantified Worker: Law and Technology in the Modern Workplace,* Cambridge University Press.

[759] Kelly-Lyth, A. (2021). Challenging biased hiring algorithms. *Oxford Journal of Legal Studies*, *41*(4), 899-928.

[760] As in the example of the Amazon hiring algorithm that taught itself to discriminate against female candidates, machine learning applications may elaborate multiple variables or detect patterns in large datasets, thereby 'elevating' a discernible protected trait in decision making. See Kullmann, M. (2018). Platform work, algorithmic decision-making, and EU gender equality law. *International Journal of Comparative Labour Law and Industrial Relations*, *34*(1).

[761] Gerards, J. & Xenidis, R. (2021). Algorithmic discrimination in Europe: Challenges and opportunities for gender equality and non-discrimination law, European Commission.

[762] Xenidis, R. (2020). Tuning EU equality law to algorithmic discrimination: Three pathways to resilience. *Maastricht Journal of European and Comparative Law*, *27*(6).

[763] Mann, M., & Matzner, T. (2019). Challenging algorithmic profiling: The limits of data protection and anti-discrimination in responding to emergent discrimination. *Big Data & Society*, *6*(2), 2053951719895805.

[764] Atrey, S. (2019). *Intersectional discrimination*. Oxford University Press, USA.

[765] Xenidis, R. (2020). Tuning EU equality law to algorithmic discrimination: Three pathways to resilience. *Maastricht Journal of European and Comparative Law*, *27*(6), 742.

the basis of individual rights-based approaches that may be scarcely effective in the field of AM. Workers are seldom in a position to feel or realise that they have been included in a 'risky' group at the systemic level, because data-driven matching processes are fed with innumerable variables.[766] Another important limitation pertains to the **individual rights-based approach**. It is crucial to increase the role of workers' representatives in monitoring the potentially discriminatory impact of AM tools on workers. This is all the more important, given the limited awareness of workers to understand whether or not they have been discriminated against. Trade unions are believed to have a fuller view of how algorithms operate in a given organisation, and how such decisions might affect them.[767] Even so, an information asymmetry still persists between unions and companies. Strengthening information and consultation rights is key to overcoming this imbalance of power and to strengthening the enforcement of the anti-discrimination framework.[768]

EU law's current approach encounters problems in guaranteeing fairness and legal certainty when it comes to AM systems. Beyond achieving cohesive legislation, as with other areas of the EU *acquis*, there is a pressing need for **improved and effective enforcement mechanisms at both EU and national levels**. This calls for the establishment of more robust public enforcement, monitoring, and compliance systems.

3.5. Workers' collective rights

Key findings:

- Information and consultation rights, social dialogue, collective bargaining and other participatory methods play a pivotal role in the realm of AM. Actively informing and consulting with workers' representatives fosters transparency, making algorithm-based systems more understandable and trustworthy. Social dialogue and collective bargaining offer a channel for negotiations on the adoption and specific uses of AM tools, depending on the 'bottom-up' capacity to exercise these rights.
- The EU framework on information and consultation faces challenges regarding its enforcement, which is highly dependent on the strength and quality of industrial relations at national level. The novel nature of AM further complicates this, widening interpretive gaps concerning certain fundamental elements of the legislation, such as the concept of 'substantial changes in work organisation or contractual relations'.
- The role of social partners and, more specifically, workers' representatives is acknowledged and promoted in several pieces of EU legislation that are also relevant to AM systems at work, such as the GDPR and the PWD.
- Collective agreements are gradually starting to include terms relating to information and consultation concerning AM systems at work. Rarely do they exclusively focus on AM systems, but instead deal with the impacts of such systems in connection with occupational health and safety, privacy, the evaluation of work performance, etc.
- Several initiatives are being developed at EU and national levels. One of the most prominent examples is the European Social Partners' Framework Agreement on Digitalisation, signed in 2020.

[766] Leese, M. (2014). The new profiling: Algorithms, black boxes, and the failure of anti-discriminatory safeguards in the European Union. *Security Dialogue*, *45*(5), 494-511.

[767] Gramano, E., & Kullmann, M. (2022). Algorithmic discrimination, the role of GPS, and the limited scope of EU non-discrimination. *A Research Agenda for the Gig Economy and Society*, p. 67.

[768] Ibid, p. 68.

3.5.1. EU legal framework

3.5.1.1. Collective rights

The Charter of Fundamental Rights of the EU (CFREU) states that 'workers or their representatives must, at the appropriate levels, be guaranteed information and consultation in good time in the cases and under the conditions provided for by Union law and national laws and practices' (Article 27). Article 28 CFREU protects broadly defined collective rights, providing for 'the right of workers and employers, or their organisations, to negotiate and enter into collective agreements, as well as to take collective action, including strike'. Principle 8 of the Pillar of Social Rights affirms that the right of workers or their representatives to be informed and consulted 'in good time on matters relevant to them'.[769] However, under art. 153(5) TFEU, the right of association and the right to strike are excluded from the list of fields in which the EU will support and complement the activities of the MS. This partially explains the scarce development at EU level of legislation regarding collective rights.

The formula 'collective rights' is very broad. It includes a wide range of prerogatives, from **information and consultation prerogatives** to **co-determination**[770] and **collective bargaining**. As explained by Rainone, information and consultation rights are key to pursuing two interconnected goals.[771] On one level, they serve as substantive rights that empower workers by giving them a voice within the company, thereby helping them to overcome their weaker bargaining position. In this context, these rights align with the ideals of industrial and workplace democracy. On a second level, information and consultation rights can be seen as **instrumental in enforcing other labour rights**. From this perspective, they can be considered 'enabling rights', as workers' representatives and trade union delegates not only hold information and consultation prerogatives, but also act to promote the effective enforcement of a broad variety of workers' rights in the areas of working conditions, occupational health and safety, data protection, and non-discrimination. Participation by workers, their representatives and social partners is also recommended from a human rights and ethical point of view.[772]

Directive 2002/14/EC[773] **requires information and consultation (I&C)** concerning 'decisions likely to lead to substantial changes in work organisation or in contractual relations' (Article 4(2)(c). This could be the case in the adoption of an AM system that significantly changes work organisation in terms of pace, time and the allocation of tasks or consequences for the contractual parties. At the same time, I&C covers 'the situation, structure, and probable development of employment within the undertaking or establishment and on any anticipatory measures envisaged, in particular, where there is a threat to employment' (Article 4(2)(b). AM-induced restructuring and reorganisation could thus fall within the scope of the Directive's provision.[774]

Depending on the functions of AM tools, **the introduction of such tools can fall under such I&C requirements**. However, such a horizontal rule does not necessarily apply to all decisions about the introduction and usage of data-gathering technologies and AM

[769] Interinstitutional Proclamation on the European Pillar of Social Rights (2017/C 428/09).

[770] Adams, Z., & Wenckebach, J. (2023). Collective regulation of algorithmic management. *European Labour Law Journal*, *14*(2), 211-229. See also Hans Boeckler Foundation (2019). Why Codetermination?, available at: https://www.boeckler.de/pdf/mbf_praes_arguments_co_determination

[771] Rainone, S. (2022). Enforcing EU information and consultation rights, in: Z. Rasnača, A. Koukiadaki, N. Bruun, & K. Lorcher (eds.), *Effective Enforcement of EU Labour Law*, Hart, p. 243.

[772] High-Level Expert Group on Artificial Intelligence, Policy and Investment (2019). Recommendations for Trustworthy AI, European Commission, p. 13; ILO Global Commission on the Future of Work (2019). Work for a brighter future, International Labour Organisation, pp. 41-42.

[773] Directive 2002/14/EC of the European Parliament and of the Council of 11 March 2002 establishing a general framework for informing and consulting employees in the European Community - Joint declaration of the European Parliament, the Council, and the Commission on employee representation.

[774] De Stefano, V. (2019). Negotiating the Algorithm: Automation, Artificial Intelligence, and Labor Protection. *Comp. Lab. L. & Pol'y J.* 41, 43.

systems. The interpretation of the meaning and scope of the phrase 'decisions likely to lead to substantial changes' presents a challenge, both in conventional contexts and in workplaces where AM systems operate. Even if the AM tool does not lead to substantial changes, I&C rights are needed to ensure transparency and maintain trust within the organization. Often, these general principles are reinforced at the domestic level through more detailed provisions. In summary, **the I&C framework is also relevant to AM and, despite the attendant interpretive complexities, the exercise of such rights remains unimpeded in the digital workplace**. It is up to social partners to leverage this framework to its fullest potential.

Rainone highlights the scarce enforcement of the EU I&C framework 'in relation to the introduction of technology into the work environment'.[775] Based on the findings of a survey conducted by ETUC between 2017 and 2018, it emerges that the EU legal framework on I&C rights is often disregarded when it comes to the introduction of technology into the workplace.[776] Two-thirds of the companies examined had implemented I&C procedures to address the effects of digitalisation on company structure and work conditions. While more recent data are missing, it is reasonable to speculate that the awareness of and inclusion of AM topics within I&C processes may have grown.

In addition, the Directive on information and consultation applies, according to the choices made by individual MS, to '(a) **undertakings employing at least 50 employees** in any one Member State, or (b) **establishments employing at least 20 employees in any one Member State**' (Art. 3). The method for calculating the threshold is also left to the MS. Crucially, it must be also noted that '[t]he effectiveness of representatives' right to information and consultation will partially depend on how domestic legislation and courts have implemented the Directive'.[777]

Directive 2009/38/EC on the European Works Council (EWC)[778] pursues the goal of improving the right to information and consultation of employees only in EU-scale undertakings and EU-scale groups of undertakings. Moreover, it leaves it to the social partners at company level to define the subject matter of I&C in European Works Councils. Directive 2009/38 does not lay down I&C regarding data-gathering technologies in particular. However, **questions related to AM systems can be integrated into this process**. Only if the social partners do not reach an agreement on the EWC, the subsidiary requirements in the Annex of Directive 2009/38/EC apply, according to which informing and consulting EWCs must, *inter alia*, relate to 'substantial changes concerning the organisation, introduction of new working methods or production processes'. Such substantial changes can be related to the adoption of AM tools, but not all AM systems necessarily bring about a substantial change.

One challenge posed to the I&C model could be the introduction of 'mixed-use' or multifunctional AM tools. There is a risk that the I&C phase might be neglected in cases where AM functions are embedded into ordinary tools or software, since is the law does not mandate informing and consulting with workers when managerial decisions are not likely to lead to substantial changes in work organisation (e.g. an email client in a collaborative or cloud-based suite, an indispensable working instrument that could be *also* used to track time spent online, thus facilitating the reorganisation of workloads with potential labour-saving effects). The bypassing of such procedural steps could negatively affect workers' awareness and undermine the logic of workers' involvement.

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[775] Rainone, S. (2022). Enforcing EU information and consultation rights, in: Z. Rasnača, A. Koukiadaki, N. Bruun, K. Lorcher (eds.), *Effective Enforcement of EU Labour Law,* Hart, p. 244.

[776] Voss, E., & Riede, H. (2018). Digitalization and Workers' participation: What Trade Unions, Company- Level Workers, and Online Platform Workers in Europe Think*,* Report to the ETUC.

[777] De Stefano, V. (2019). Negotiating the Algorithm: Automation, Artificial Intelligence, and Labor Protection. *Comp. Lab. L. & Pol'y J.* 41, 43.

[778] Directive 2009/38/EC of the European Parliament and of the Council of 6 May 2009 on the establishment of a European Works Council or a procedure in Community-scale undertakings and Community-scale groups of undertakings for the purposes of informing and consulting employees (Recast).

Information and consultation rights are also provided for in the proposed **Platform Work Directive**, which operates **without prejudice to Directive 2002/14/EC**. According to Art. 9(1), 'MS shall ensure information and consultation of platform workers' representatives or, where there are no such representatives, of the platform workers on decisions likely to lead to the introduction of or substantial changes in the use of automated monitoring and decision-making systems referred to in Art. 6(1)'.[779] Given the intricacies of data-driven tools, the PWD also provides for the possibility of **assistance by an expert chosen by persons performing platform work or the platform workers concerned to examine the matter that is subject to information and consultation, and formulate an informed opinion**. Where a digital labour platform has more than 500 platform workers in an MS, the expenses involved in engaging the expert must be covered by the digital labour platform, as long as they are proportionate (Art. 9(3)). Such information and consultation provisions do not, however, apply to self-employed persons performing platform work.[780] Furthermore, Art. 15 demands that MS ensure that platforms enable **persons performing platform work to contact and communicate with each other, and be contacted by their representatives, through the platforms' digital infrastructure** 'or similarly effective means'. The creation of such a communication channel could be very useful for overcoming the existing barriers to union organising in the platform economy.[781] This provision has a true 'enabling potential', as it provides a means for workers to organise around a 'collective interest'.[782] However, the proposed PWD makes no explicit mention of the potential role of trade unions.[783] In addition, Art. 20 PWD empowers Member States to apply or introduce laws, regulations, administrative provisions or collective agreements that are more favourable to platform workers, opening the possibility for dedicated collective agreements on this matter.

AM systems have the potential to be leveraged to predict various other aspects of a worker's actions and decisions, including whether they might **exercise their (individual and collective) rights**. For instance, these systems could analyse historical data and patterns to predict if a particular worker is likely to bring a claim against the company, submit complaints, or engage in other forms of legal action related to workplace issues. Furthermore, AM systems can go even further by attempting to predict the likelihood of a worker applying for another job or contemplating quitting their current position. Such predictions might involve the use of diverse data points, such as performance metrics, work habits, job satisfaction levels, and personal characteristics. By analysing various data points and patterns, AM systems could also be utilised to predict the likelihood of unionisation or the occurrence of strikes or other forms of collective action among workers. However, the protective regime reserved for 'special categories of personal data' under Art. 9 GDPR would apply in the case of an employer attempting to infer current trade union membership from these data. Nevertheless, should a prediction concerning unionisation at the plant or establishment level be based on 'aggregated and anonymised data', the GDPR would not apply.[784] This example highlights the potential limitations of data protection provisions regarding collective rights.

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[779] For an analysis, see Otto, M. (2022). A step towards digital self-& co-determination in the context of algorithmic management systems. *Italian Labour Law e-Journal*, *15*(1), 51-64.

[780] For a critical discussion on this limitation see for example, De Stefano, V. (2022). The EU Commission's proposal for a Directive on Platform Work: an overview, *Italian Labour Law e-Journal,* Issue 1, Vol. 15, p. 113.

[781] Collins, P., & Atkinson, J. (2023). Worker voice and algorithmic management in post-Brexit Britain. *Transfer: European Review of Labour and Research*, *29*(1), 47.

[782] Purificato, I., & Senatori, I. (2023). The Position of Collective Rights in the 'Platform Work' Directive Proposal: Commission v Parliament, *Hungarian Labour Law e-Journal* 2023/1, p. 4.

[783] See, however, the Report on the proposal for a Directive of the European Parliament and of the Council on improving working conditions in platform work of 21 December 2022 - (COM(2021)0762 – C9-0454/2021 – 2021/0414(COD)), providing for such rights. For a detailed comparison between the Commission's and Parliament's versions of the PWD, see Purificato, I., & Senatori, I. (2023). The Position of Collective Rights in the 'Platform Work' Directive Proposal: Commission v Parliament, *Hungarian Labour Law e-Journal,* 2023/1.

[784] Adams-Prassl, J., Abraha, H., Kelly-Lyth, A., Silberman, M.S., & Rakshita, S. (2023). Regulating algorithmic management: A blueprint. *European Labour Law Journal*, *14*(2), 131.

Trade union representatives can also represent workers and act on their behalf when it comes to lodging complaints with DPAs or exercising their right to a proper legal remedy (Art. 80(1) GDPR). Not-for-profit bodies, organisations or associations also have the legal standing to bring claims before a court or exercise a data protection right before the employer or the DPA, 'independently of a data subject's mandate' (Art. 80(2) GDPR). Not many MSs have taken up this option to nominate unions, consumer organisations and other civil society organisations in their national data protection frameworks so that they can bring such complaints.[785]

Given its **legal basis** – i.e. Article 114 TFEU – the draft **AI Act** is not well aligned with information and consultation rights that might exist at national level. For instance, it neglects the role of social partners in negotiating the introduction of technological tools at work. Moreover, the liberalising legal basis of the AI Act risks superseding any domestic regulations, including work-related ones, that mandate higher standards of I&C procedures. As a result, there is concern that the AI Act would act as a 'ceiling' rather than a 'floor' for (labour) protection.

3.5.1.2. Social dialogue

In January 2023, the Commission presented an **initiative to 'further strengthen and promote social dialogue with concrete actions at national and EU level'**.[786] **Social dialogue is encouraged as a cornerstone of the EU social market economy and its competitiveness**. More concretely, the initiative boosts the ability of social dialogue to adjust to the evolving workplace and emerging trends in the job market. This is set against the context of shifts towards a digital and climate-neutral economy, along with the rise of new forms of work and atypical employment. The key soft law instrument presented in this respect is a Communication.[787] The Commission also advanced a proposal for a Council Recommendation.[788] This contains **no explicit references to AM systems, however**. The Commission recommends that Member States: (i) ensure the consultation of social partners regarding the design and implementation of economic, employment, and social policies according to national practices; (ii) encourage social partners to look at new forms of work and atypical employment and to communicate widely about the benefits of social dialogue and about any collective agreements put in place; and (iii) enable an increase in the capacity of workers' and employers' organisations, for instance ensuring they have access to relevant information and ensuring support from national governments. The MS are recommended to 'ensure an enabling environment for bipartite and tripartite social dialogue, including collective bargaining, in the public and private sectors, at all levels, including cross-industry, sectoral-, company-, or regional-level that: […] adapts to the digital age, promotes collective bargaining in the new world of work and a fair and just transition towards climate neutrality'.

The 2020 **European Social Partners' Framework Agreement on Digitalisation (FAD)**[789] is the first EU-wide compact dealing with AM practices and aimed at fostering the adoption of a **human-centred approach**. Concluded by the European cross-sectoral social partners,[790] the FAD applies to the whole of the EU/EEA and covers all workers and

[785] Nogarede, J. (2021). No Digitalisation Without Representation. An Analysis of Policies to Empower Labour in The Digital Workplace, FEPS Policy Study, p. 18.

[786] See: https://ec.europa.eu/commission/presscorner/detail/en/ip_23_290.

[787] Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, Strengthening social dialogue in the European Union: harnessing its full potential for managing fair transitions COM(2023) 40 final.

[788] Proposal for a Council Recommendation on strengthening social dialogue in the European Union, COM(2023) 38 final 2023/0012 (NLE).

[789] Available at https://ec.europa.eu/social/main.jsp?catId=521&langId=en&agreementId=5665. See also the European Social Dialogue Work Programme 2022-2024, available at https://bit.ly/3qMIAtp. For an analysis see, for example, Mangan, D. (2021, December). Agreement to Discuss: The Social Partners Address the Digitalisation of Work, *Industrial Law Journal*, 50, 4, pp. 689–705.

[790] Business Europe, SMEunited, the European Centre of Employers and Enterprises providing Public Services (CEEP), and the European Trade Union Confederation.

employers in the public and private sectors and all economic activities.[791] The agreement comprises four sections: (a) digital skills and securing employment; (b) modalities of connecting and disconnecting; (c) artificial intelligence and guaranteeing the human-in-control principle; and (d) the respect of human dignity and surveillance. Each of these sections includes broad principles and sets of voluntary measures for consideration but lacks specific details or prioritisation. Given this very broad framing, the FAD is designed to facilitate discussion, negotiation and bargaining at national level.

Social partners acknowledge the benefits and perils of AI. **They also encourage 'a partnership approach between employers, workers and their representatives'.** The FAD calls for the application of the **'human in control' principle**,[792] stating that fairness, and the safe adoption and usage of AI thanks to a risk-based method, as well as transparency and explicability with effective oversight, are identified as paramount principles (Art. 3). Transparency must be safeguarded through the provision of information when AI systems are used in HR procedures such as recruitment, evaluation, promotion and dismissal, and performance analysis. Workers should be able to 'make a request for human intervention and/or contest the decision along with testing of the AI outcomes' (Art. 3). A substantial number of these principles and requirements are already grounded in current EU legislation such as the GDPR. In particular, when it comes to 'the respect of human dignity and surveillance', the **FAD refers to Art. 88 GDPR**, stressing the importance of 'lay[ing] down by means of collective agreements, more specific rules to ensure the protection of the rights and freedom with regards to the processing of personal data of employees in the context of employment relationships' (Art. 4). To better attune traditional collective rights to the new reality of workplaces, the social partners consider the opportunity of adopting measures 'enabling workers' representatives to address issues related to data, consent, privacy protection, and surveillance; linking the collection of data to a concrete and transparent purpose […]; providing workers representatives with facilities and (digital) tools, e.g., digital notice boards, to fulfil their duties in a digital era'.

From a procedural perspective, the FDA adopts a **participatory model,** which is justified by the shifting landscape of the matters it aims to govern. Given the uncertainties raised by the emergence of new tech-based workplace tools and practices, the social partners opt for a reflexive procedure dubbed a 'digitalisation partnership process',[793] allowing them to constantly adjust their initiatives to the external context. To achieve the objectives of the FAD, this method implies a well-designed and consensual 'dynamic circular process' that draws on the theories on stakeholder engagement, encompassing phases from joint preliminary conversations to negotiation, the adoption of concrete actions and *ex-post* monitoring. Such a model is **adaptive** by default, and the social partners acknowledge that such a *modus operandi* must be tailored to the specificities of the sectors, companies, and actors where it is applied, by also considering what measures are already implemented and in force.[794]

This agreement aligns with the pattern seen in many cross-industry agreements. The force of the FAD resides in the ability of the signatory parties to implement the content of the agreement at a **decentralised level**, i.e., at the level of the MS or the workplace, by means of more specific initiatives and measures. While this approach grants parties the flexibility to choose their preferred implementation methods, the implementation of the FAD varies widely, based on the national legal framework, the representation level of the social

[791] In the context of Art. 155 of the Treaty, this autonomous European framework agreement commits the members of BusinessEurope, SMEunited, CEEP, and the ETUC (and the liaison committee EUROCADRES/CEC) to promote and to implement tools and measures, where necessary at national, sectoral and/or enterprise levels, in accordance with the procedures and practices specific to management and labour in the Member States and in the countries of the European Economic Area.

[792] See also: EESC (2017). Opinion of the European Economic and Social Committee on Artificial intelligence – The consequences of artificial intelligence on the (digital) single market, production, consumption, employment, and society, European Economic and Social Committee.

[793] FAD, p. 6.

[794] Aloisi, A. & Ribeiro, A.T. (2024), Framework Agreement on Digitalisation, in: C. Hiessl (ed.), *EU Labour Law,* Wolters Kluwer.

partners, and industrial relations traditions.[795] The deadline for implementing the FAD was June 2023. Up to this point, most national social partners have taken various actions to implement the DFA. These actions encompass translating the agreement; conducting meetings to discuss implementation strategies; arranging events such as seminars and webinars; publications to promote the agreement among employers and workers; forming working groups; and engaging in discussions to reach collective agreements on these issues.

3.5.2. Selected national legislation and policies

A. EU countries

Information and consultation rights, including on matters broadly related to technology, are enshrined in the law of many EU Member States.[796] In **Austria**, for example, the Labour Constitution Act (*Arbeitsverfassungsgesetz, ArbVG*) provides that works councils need to approve the introduction of 'control measures and technical systems for the control of workers', insofar as these measures touch upon human dignity (Art. 96 (1) (3) AT-ArbVG).

Under the **Italian Workers' Statute (WS)**,[797] employers can introduce equipment 'giving rise to the possibility of remote control of workers' activities' only for technical, organisational reasons, for reasons associated with safety at work, and for the protection of corporate assets. Before introducing such tools, employers need to reach an agreement with workers' representatives, such as trade unions or works councils (Art. 4 WS). If the parties are not able to reach an agreement, the employer can seek administrative authorisation from the local branch of a ministerial body. However, instruments used by workers to perform their work duties and record their attendance can be introduced even without getting the green light from worker representatives. Data gathered in this way can only be used for purposes related to the employment relationship if workers are adequately informed as to how such monitoring tools are used, as well as the aim and method of such usage. (Art. 4(3), WS).

Some countries have introduced collective rights strictly in connection to AM. **Germany** is an example of a country with a particularly strong 'tradition'[798] of negotiating over the use of digital technologies that can affect working conditions. On 18 June 2021, the Works Council Modernisation Act (*Betriebsrätemodernisierungsgesetz, BRMG*) entered into force, amending *inter alia* the German Works Constitution Act (*Betriebsverfassungsgesetz, BetrVG*).[799] Under this regulation, **the works council has a right of co-determination concerning the 'introduction and use of technical devices designed to monitor the behaviour or performance of the employees'**.[800] Employers are required to inform the works council in due time of any plans concerning working procedures and operations, including the use of AI.[801] Moreover, they must consult the works council in good time regarding the action envisaged and its effects on the employees, taking particular account of its impact on the nature of their work, the resultant demands on the employees, and established ergonomics findings relating to the tailoring of jobs to meet human requirements.[802] Guidelines for the selection of employees for recruitment, transfer,

[795] Nogarede, J. (2021). No Digitalisation Without Representation. An Analysis of Policies to Empower Labour in The Digital Workplace, FEPS Policy Study, p. 32.

[796] For an overview, see: Eurofound (2020). Employee monitoring and surveillance: The challenges of digitalisation, Luxembourg: Publications Office of the European Union, p. 16.

[797] Law 300, 20 May 1970.

[798] For a comprehensive overview, see Waas, B. (n.d.). Artificial Intelligence and Labour Law, *Hugo Sinzheimer Institut*. See also: Schwartz Jr, R.G. (1991). Privacy in German Employment Law. *Hastings Int'l & Comp. L. Rev.*, *15*, 135,152.

[799] Works Constitution Act (Betriebsverfassungsgesetz - BetrVG), 25 September 2001 (*Federal Law Gazette I*, p. 2518), as amended by Article 6d of the Act of 16 September 2022 (*Federal Law Gazette*, p. 1454). See also: Gesetz über die Mitbestimmung der Arbeitnehmer (Mitbestimmungsgesetz - MitbestG), German Co-Determination Act of 4 May 1976.

[800] Section 87(1) no. 6 BetrVG. Note that works councils do not have the right to strike.

[801] Section 90(1) no 3 BetrVG.

[802] Section 90(2) BetrVG.

regrading and dismissal require the approval of the works council.[803] In establishments with more than 500 employees, the works council may request the drawing up of guidelines regarding the technical, personal and social criteria to be applied in the case at stake. Notably, this co-determination model also applies 'if **artificial intelligence** is used when drafting the guidelines'.[804] Insofar as the works council has to assess the introduction or application of AI, in order to carry out such tasks, it is deemed necessary to call on the advice of an expert in this regard.[805] The employer and the works council may agree on asking experts to provide advice. The relevant expenses arising out of the activities of the works council are defrayed by the employer.[806] The **adoption of AI requires consent from the works council, and all decisions made concerning these systems must be verifiable by the works council**.[807] According to experts, 'when exercising these rights, works councils in Germany generally take a constructive approach. The reason for this is probably also to be found in their legal constitution and their operational focus: The company agreements concluded by works councils have a lower legal status than collective agreements reached by trade unions. […] Works councils, therefore, prove in practice not to be impeders, but often even drivers of digitalisation' at the company level.[808]

Spanish Law 12/2021 (the so-called **'Riders' Law'**) provided works councils with the right to be informed by the company about 'the parameters, rules, and instructions on which the algorithms or artificial intelligence systems are based, which are used for decision-making practices, including profiling, that may affect working conditions, access and maintenance of employment' (Art. 64 (4) of the Workers' Statute Law). This reform partly overcame the problem of the lack of a collective dimension to transparency rights under the GDPR.[809] There are, however, some limitations to this law. The revised Workers' Statute does not introduce the right of co-determination, and it does not include systems of objection and redress regarding such metrics.[810]

In **Poland**, a draft amendment to the Trade Unions Act was put forward in September 2022, proposing to add a provision obliging the employer to provide, at the request of the company trade union, information about the parameters, rules and instructions on which algorithms or artificial intelligence systems are based, if such systems or algorithms affect decision-making with regard to working conditions or pay, access to and job retention, including profiling. Access to information about algorithms is treated as the implementation of the general right of employees to information.[811]

Moreover, examples exist of national or regional regulations that address collective rights specifically in relation to the context of platform work. In **France**, a specific piece of regulation was introduced that was dedicated exclusively to the collective rights of self-employed platform workers.[812] This lays down terms and conditions for the representation of self-employed workers who use platforms for their activity and has helped to develop

[803] Section 95 BetrVG.

[804] Section 95 (2a) BetrVG.

[805] Sec. 80 (3) BetrVG.

[806] Section 40(1) BetrVG.

[807] Krzywdzinski, M., Gerst, D., & Butollo, F. (2023). Promoting human-centred AI in the workplace. Trade unions and their strategies for regulating the use of AI in Germany. *Transfer: European Review of Labour and Research*, *29*(1), 60.

[808] Klengel, E., & Wenckebach, J. (2021). Artificial intelligence, work, power imbalance and democracy–why co-determination is essential. *Italian Labour Law e-Journal*, *14*(2), 163.

[809] Todolí-Signes, A. (2019). Algorithms, artificial intelligence, and automated decisions concerning workers and the risks of discrimination: The necessary collective governance of data protection. *Transfer: European Review of Labour and Research*, *25*(4), 475; Todoli-Signes, A. (2021). Spanish riders law and the right to be informed about the algorithm. *European Labour Law Journal*, *12*(3), 401.

[810] Aloisi, A. (2022). Platform work in Europe: Lessons learned, legal developments and challenges ahead. *European Labour Law Journal*, 13(1), 4-29.

[811] See: https://www.sejm.gov.pl/.

[812] Ordinance No 2021-484 regarding the terms of representation of self-employed workers using platforms in the course of their activity, and the conditions for the exercise of this representation.

collective bargaining and establish representation for platform workers. It created the Employment Platforms Social Relations Authority (*Autorité des relations sociales des plateformes d'emploi*, ARPE), which was tasked with, *inter alia*, organising and supervising the election of platform workers' representatives on behalf of the state; financing the training of these representatives, and compensating for their training and delegation hours; and facilitating social dialogues between workers and platforms.[813] In **Greece**, Art. 7 of Law 4808/2021 (in force since 19 June 2021) guarantees self-employed platform workers the right to form organisations in order to promote their professional interests, as well as the right to strike, and the right to negotiate collectively and to conclude collective agreements. In **Lazio**, Italy, a Regional Digital Labour Council was established to monitor the conditions of platform work in the region and to facilitate social dialogue between stakeholders.

3.5.3. Social partner agreements and initiatives

Collective rights with regard to AM or new technologies more broadly are starting to be enshrined in various collective bargaining agreements. One example is the 'Collective agreement of Renault Spain, SA', which includes provisions on information and consultation as regards the application of technology. More specifically, prior to the application of new technologies, the firm's management undertakes to provide sufficiently broad information about them, as well as on their impact on employment and working conditions, to workers' representatives. To this end, a joint committee on new technologies has been set up.[814]

The **collective rights of platform workers** have also been addressed and reaffirmed in several collective agreements. To give an example, an SE Works Council at Delivery Hero (the owner of Foodora) was established by an agreement signed in April 2018 between the German Food, Beverages and Catering Union (*Gewerkschaft Nahrung-Genuss-Gaststätten*, NGG), the Italian Federation of Workers in Commerce, Hotels, Canteens and Services (*Federazione Italiana Lavoratori Commercio, Albergo, Mensa e Servizi, FILMCAMS – CGIL),* and the European EFFAT (European Federation of Food, Agriculture and Tourism). This agreement foresees that each European country in which Delivery Hero operates must have at least **one worker representative within the SE Works Council**. The SE Works Council should be informed about the company's strategies and all plans that may impact the work organisation and interests of the company's employees. Moreover, the agreement provides for the participation of employees in the supervisory board of the group, in equal numbers and with the same voting rights as the representatives of the shareholders. The supervisory board must be informed at least once every three months on progress and developments at the corporation. In case of any events likely to have a substantial effect on the corporation, the management board is to pass on information to the supervisory board. This initiative greatly facilitates direct exchanges between workers' representatives and management. However, it has limited powers to change working conditions for riders, since workers do not have the means to check whether any information has been withheld. It has been pointed out that the cooperation is carried out at a rather abstract' level and lacks actionable measures.[815]

3.5.4. Other relevant initiatives

The **European Declaration on Digital Rights and Principles for the Digital Decade**,[816] proclaimed in January 2023, features a section on 'Fair and just working conditions'. The Declaration asserts that 'everyone has the right to fair, just, healthy, and safe working conditions and appropriate protection in the digital environment as in the physical workplace, regardless of their employment status, modality, or duration'. Furthermore, it

[813] See: https://apps.eurofound.europa.eu/platformeconomydb/ordinance-no-2021484-of-21-april-2021-to-establish-representation-for-platform-workers-105062.

[814] Resolución de 19 de agosto de 2021, de la Dirección General de Trabajo, por la que se registra y publica el Convenio colectivo de Renault España, SA https://www.boe.es/eli/es/res/2021/08/19/(15).

[815] See: https://apps.eurofound.europa.eu/platformeconomydb/se-societas-europaea-european-company-works-council-delivery-hero-103127.

[816] See: https://bit.ly/43WTWfM.

highlights the important role of trade unions and employers' organisations in defining fair and just working conditions, including with regard to the use of digital tools at work (Arts. 5 and 6). The European Parliament, the Council and the Commission solemnly commit to 'ensuring respect for workers' fundamental rights in the digital environment, including their right to privacy and the right to associate, right of collective bargaining and action, as well as protection against unlawful and unjustified surveillance' (point c) and to 'ensuring that the use of artificial intelligence in the workplace is transparent and follows a risk-based approach and that corresponding prevention measures are taken to maintain a safe and healthy working environment' (point d).

A. EU countries

The **Spanish** General Workers' Union (*Unión General de Trabajadores, UGT*) has published reports that recommend various stages for negotiating the use of algorithms in the workplace: (i) negotiations on the need, purpose, and proportionality of the system; (ii) agreement on the tools and applications to be used; (iii) agreement on the implementation design; (iv) impact assessment with regard to data protection; (v) co-execution of the plan; and (vi) monitoring of the implementation and readjustments. In addition, a hands-on guide, 'Collective negotiation and digitisation', was published by the Spanish trade union Comisiones Obreras in 2020.[817]

In **Germany**, IG Metall has launched the '*IG Metall vom Betrieb aus denken*' project. The key purpose of this is to advise works councils on how to deal with new technologies, while at the same time supporting trade union organising activities in the workplace. Moreover, the 2021 collective agreement for the metal industry encourages the adoption of company-specific 'future-proof collective agreements' (*Zukunftstarifverträge*), building on existing 'supplementary collective agreements'. These can be drawn up if a crisis occurs at a plant – for example, caused by the introduction of labour-saving technologies. In such a case, the employer pledges investment and capacity utilisation, and the IG Metall union makes temporary compromises in order to maintain employment. Before such an agreement is signed, a discussion must take place about the strategy between the plant's management and works council.[818]

The European Trade Union Confederation (ETUC) has also adopted a **Resolution on the European strategies on artificial intelligence and data**.[819] This calls for the involvement of trade unions and workers' representatives at the stages of the design and development of AI systems. It proposes that trade unions and workers' representatives should have the right to review the AI system regularly, and to demand adjustments or restrictions. When it comes to the consultation rights of workers' representatives and trade unions, employers should be required to provide a 'complete and understandable overview [of] the algorithmic systems' that considers, among other aspects, the training data, the way in which data are processed, the purposes, the impacts, and the effects in terms of psychological or physiological risks and discrimination.

UNI Global, a global union federation for the skills and services sectors, with affiliated unions in 150 countries representing 20 million workers, has published **practical guides** to help workers and trade unions to better understand emerging technologies, which provides guidance on how to negotiate AM systems.[820]

B. Non-EU countries

In **Switzerland**, a Position Statement was issued by AlgorithmWatch Switzerland and Syndicom, in which the social partners called for new legislation to encourage employee participation in the adoption of these types of tools by companies. The position paper

[817] 'Negociación colectiva y digitalización', available at: https://www.ccoo.es/af6e35ab004a61334480e3b2bcae0e93000001.pdf.

[818] Krzywdzinski, M., Gerst, D., & Butollo, F. (2023). Promoting human-centred AI in the workplace. Trade unions and their strategies for regulating the use of AI in Germany. *Transfer: European Review of Labour and Research*, *29*(1), 62.

[819] ETUC (2020). Resolution on the European strategies on artificial intelligence and data, available at: https://bit.ly/ETUCres.

[820] See: https://uniglobalunion.org/wp-content/uploads/uni_pm_algorithmic_management_guide_en.pdf.

emphasised that the participation rights of trade unions and employee representatives need to be further developed.[821]

In the **UK**, unions have brokered agreements that establish subcommittees dedicated to scrutinising data practices and ensuring technology is not detrimental to workers. These agreements call for the centrality of managers as decision-makers, for mutually agreed definitions of productivity metrics, and for the adoption of measures to facilitate compliance with existing data policies and UK data protection laws.[822]

3.5.5. Conclusions

The EU legal framework establishes an environment for the exercise of collective rights (from information and consultation to social dialogue), yet **the actual realisation of such rights is heavily dependent on various factors including domestic regulations, the strength of social partners, and the nuances of the local systems of industrial relations**.

Information and consultation, the involvement of workers, social dialogue,[823] and other participatory methods are highly relevant in the case of AM systems.[824] In principle, industry-wide, territorial and company-level collective bargaining agreements that are comprehensive enough to include rules on the adoption and deployment of technologies, could prove more effective than retrospective and remedial approaches, while facilitating the improvement, accuracy and efficiency of AM systems.[825] Responsive methods can also succeed in **accelerating adaptations in response to the fast-moving transformation of the world of work** by shaping adaptive responses, rather than merely acting at the level of consequences.[826] Experts agree that collective bargaining agreements offer a flexible regulatory approach that can be tailored to suit the needs of individual firms or industries – unlike legislation, which imposes general standards across the entire labour market and struggles to keep pace with rapidly evolving practices.[827] At the same time, **AM systems may encroach upon collective rights** by predicting and discouraging the exercise of workers' rights' such as the right to association, the right strike, and the right to bargain collectively.

Works councils, trade union delegates and bodies representing workers can perform a number of functions in this context. First, they can take part in '**awareness-raising and training in the workplace**' emphasising the responsible use of technology in order to ensure unbiased and equitable practices.[828] Second, they can facilitate a **knowledge-sharing process**. Such a process of information transfer is all the more important in situations of information asymmetry, where staff turnover, short-term contracts, and decentralised teams impair workers' ability to understand whether unfair treatment has been or is likely to occur. Workers' representatives at company level can help to lower the barrier for individuals without specialised knowledge and can collect 'convenient evidence for a fact finder'[829] in relation to matters that are barely accessible, while also supporting judicial discovery thanks to their ample viewpoint. In addition, representatives and unions

[821] See: https://www.zh.ch/content/dam/zhweb/bilder-dokumente/themen/politik-staat/kanton/digitale-verwaltung.

[822] Ibid.

Krämer, C., & Cazes, S. (2022). Shaping the transition: Artificial intelligence and social dialogue. OECD Publishing.

[824] Janssen, R. (2020). A business case for social dialogue: How workplace representation and collective bargaining deliver better business performance, *Global Deal*; Belloc, F., Burdin, G., & Landini, F. (2020). Robots and worker voice: an empirical exploration.

[825] De Stefano, V., & Taes, S. (2023). Algorithmic management and collective bargaining. *Transfer: European Review of Labour and Research*, *29*(1), 21-36.

[826] OECD (2019). Negotiating Our Way Up: Collective Bargaining in a Changing World of Work. OECD Publishing.

[827] Collins, P., & Atkinson, J. (2023). Worker voice and algorithmic management in post-Brexit Britain. *Transfer: European Review of Labour and Research*, *29*(1), 40.

[828] Nogarede, J. (2021). No Digitalisation Without Representation. An Analysis of Policies to Empower Labour in The Digital Workplace, FEPS Policy Study, p. 18.

Waldman, A.E. (2019). Power, process, and automated decision-making. *Fordham L. Rev.*, *88*, 613.

can build relationships with the data protection officer (if one is present at the firm) and the competent DPA, 'to facilitate better enforcement of the law, not only after complaints have been raised, but also before'.[830] Third, workers' representatives are best placed to **codetermine** the introduction and deployment of AM systems. They can also **co-design** internal rules, due to their capacity-building and vast knowledge of operational practices and internal bottlenecks.

The active participation of workers and their representatives also confers several types of advantages on businesses. By involving workers in the setting of AM parameters, it becomes possible to exclude from such parameters noisy and misleading metrics that capture erroneous factors. For example, Art. 29 WP recommends that 'in all cases, a representative sample of employees is needed in assessing the necessity of the monitoring, as well as the logic and accessibility of the policy'.[831] Moreover, where meaningful involvement and co-design procedures are implemented, workers possess peer-knowledge of the AM systems, and are thus discouraged from attempting to manipulate the model because their colleagues would be likely to do the same. An overarching goal of participatory schemes is to **build an environment of trust**. High levels of mutual trust can, in turn, have a positive impact on perceptions of justice, fairness and privacy in the workplace.[832]

Social partners can be at the forefront of the negotiation of AM systems at work. The collective exercise of rights regarding data protection in the workplace, OSH provisions, working conditions, as well as equality and non-discrimination safeguards, is considered particularly effective in the context of AM. However, the effective exercise of collective rights in the context of AM is fraught with **difficulties linked to the complexity of the technologies involved, the novelty of the phenomenon, and the correspondingly low level of digital literacy among workers, employers, and representative bodies**.

The technical intricacy of AM systems, which necessitates specialised knowledge and training, represents a significant barrier in this regard. In addition, in some sectors in which AM systems are being adopted (e.g. warehouses and logistics), workers are often hired on non-standard contracts. This may represent a **hindrance to the full exercise of collective rights**, as workers may have scant opportunities to interact with each other and join forces. The temporary nature of their contracts might also conflict with any substantial interest they have in participation strategies. Thus, traditional obstacles to exercising collective rights by non-standard workers may become even more pronounced when dealing with AM systems.

Other challenges are connected to the **scope of the EU *acquis* when it comes to information, consultation, and negotiation rights**. Self-employed individuals often struggle to access formal channels of representation or existing remedies, which are typically limited to those in traditional employment relationships.

The main **interpretive challenge** resides with the notion of 'decisions likely to lead to substantial changes in work organisation or in contractual relations' (Article 4(2)(c) Directive 2002/14/EC). It might be beneficial to seek further guidance on whether AM tools align with the stipulations of the Directive, as such clarity could be pivotal in ensuring the full exercise of rights, and this assessment must be done on a case-by-case scenario.[833] The possibility of subjecting all AM tools to I&C requirements is rooted in the unique and evolving nature of AM. These tools can swiftly change how they operate and have a significant influence on organisational dynamics. The opacity of AM, in contrast to traditional methods, can raise

[830] Nogarede, J. (2021). No Digitalisation Without Representation. An Analysis of Policies to Empower Labour in The Digital Workplace, FEPS Policy Study, p. 18.

[831] Art. 29 WP, Opinion 2/2017 on data processing at work.

[832] Hovorka-Mead, A.D., Ross, W.H., Jr., Whipple, T., & Renchin, M.B. (2002). Watching the detectives: Seasonal student employee reactions to electronic monitoring with and without advance notification. *Personnel Psychology*, 55, 329-362. As discussed in Chapter 2, the low unionisation rate and scarce coverage of collective agreements represent a key practical obstacle.

[833] Klengel, E., & Wenckebach, J. (2021). Artificial intelligence, work, power imbalance and democracy–why co-determination is essential. *Italian Labour Law e-Journal*, *14*(2): 168.

concerns among employees. However, there is a potential risk in such a strategy: continuous consultations might lead to 'consultation fatigue'. And if only AM tools were singled out for such scrutiny, it might seem inconsistent when compared to other managerial decisions.

A review of MS national laws is essential in order to confirm that the Directive is properly put into practice. **Preliminary research has found that MS often protect and promote information and consultation rights that derive from EU law 'somewhat unsatisfactorily'**.[834] In a 2009 report, Cottigny concurred with the Commission that the enactment of Directive 2002/14/EC 'has taken place in a minimal or deliberately vague manner or not at all'.[835] This insufficient implementation is also evident in discussions about AM systems in workplaces.[836]

Besides information and consultation, codetermination and collective bargaining, worker representatives can **file collective legal complaints** on behalf of worker groups before courts, administrative bodies, and data protection authorities. A crucial gap here lies in the complete application of these provisions, as their effectiveness largely hinges on the commitment and actions of the parties involved.

With regard to other initiatives such as the FAD, one significant constraint has been **minimal implementation at the decentralised level**, despite more than three years having passed since it was signed. This challenge is further compounded by the open-textured and general nature of some of its clauses, which require elaboration within collective agreements.

Because AM represents a shared challenge that transcends specific sectors and types of contracts, it opens up an opportunity for social partners to influence this evolving field. Collective rights can be facilitated by legal measures, creating a framework for their protection. However, to be fully exercised, **these rights require bottom-up energy**. At national level, this is increasingly hindered by the declining rate of unionisation and the reduced scope of collective bargaining. The law can lay the groundwork, but **without the engagement of workers' representatives and businesses themselves, the realisation of these rights remains stymied**.

3.6. Effects on broader society

Key findings:

- There is notable activism both at EU and national levels in the domain of AI and digitalisation, underscoring the growing momentum surrounding AM and other emerging technologies. Not only does this surge in interest attest to the increasing prominence of these phenomena, but it also signifies a collective recognition of their implications for the future of work.
- Amid these developments, the emphasis on skills and upskilling has emerged as a top priority, highlighting the imperative for citizens to be equipped and adaptable in the face of rapid advances in technology. In a similar vein, the regulation of AI – a technology that can support AM systems – is taking centre stage.
- The increasing significance of AI is universally recognised as a factor in society and in the labour market. The fact that it is being regulated at multiple levels corroborates its

[834] De Stefano, V., & Wouters, M. (2022). AI and digital tools in workplace management and evaluation. An assessment of the EU's legal framework, European Parliamentary Research Service Scientific Foresight, p. V.

[835] Cottigny, J.L. (2009). The explanatory memorandum to the report on the implementation of Directive 2002/14/EC establishing a general framework for informing and consulting employees in the European Community (2008/2246(INI)), *European Parliament*. Available at https://www.europarl.europa.eu/doceo/document/A-6-2009-0023_EN.html.

[836] De Stefano, V., & Wouters, M. (2022). AI and digital tools in workplace management and evaluation. An assessment of the EU's legal framework, European Parliamentary Research Service Scientific Foresight, p. 60.

importance. This multi-tiered regulatory approach ensures that AI is not only being developed responsibly, but also implemented in ways that uphold legal standards.

- Some initiatives lack a binding force. Moreover, the lack of coordination between them can result in gaps and overlaps when addressing the societal impacts of AM systems.

3.6.1. Initiatives at EU level

The legal and soft law initiatives discussed below are not directly connected to AM; they tangentially touch upon it. While they could influence or be influenced by the field of AM, their core focus is not on these systems. Their connection to AM is more indirect, at the **intersection of broader legal concerns, governance, and advances in technology**.

AM systems can cause **income inequality**. In March 2021, the European Commission adopted an **Action Plan to implement the European Pillar of Social Rights (EPSR)**, which had been presented in 2017 in two legal forms with identical content: a Commission Recommendation,[837] adopted on the basis of Article 292 TFEU, and a Proposal for an interinstitutional proclamation by the Parliament, the Council and the Commission.[838] The Action Plan is informed by the 20 principles of the EPSR, and reads as a pragmatic 'to-do list', primarily addressing the creation of a sustainable and inclusive growth model and a fair digital and ecological transition. The initiatives outlined range from labour market inclusivity and upskilling to health and long-term care, measures to combat homelessness, working conditions and education, and other aspects of social protection and social assistance. The EPSR and its roll-out via the Action Plan were publicly endorsed at the Social Summit in Porto on 7-8 May 2021, organised by the Portuguese Presidency of the European Council. Importantly, the EPRS Action Plan includes **various initiatives aimed at reducing inequality, such as the Adequate Minimum Wage Directive**,[839] **the Recommendation on Minimum Income**, and guidance on increasing the use of *ex-ante* distributional impact assessments in budgetary processes and the planning of reforms.

AM systems could reshape **skills development**, potentially leading to implications for citizens' adaptability. In 2021, the Commission proposed a **Digital Compass** to translate the EU's digital ambitions for 2030 into concrete terms. The four cardinal points identified therein may be relevant to the underlying conditions supporting the development of AM systems. First, by 2030, at least 80 % of all adults should be **digitally skilled citizens**, and there should be 20 million employed ICT specialists within the EU – while more women should take up such jobs. Second, among other things, all EU households should have secure, performant, and sustainable gigabit connectivity, and all populated areas should be covered by 5G. Third, by 2030, **three out of four companies should use cloud computing services, big data, and AI**; more than 90 % of SMEs should reach at least basic levels of digital intensity. Fourth, all key public services should be available online in a digital format.

The **Digital Skills and Jobs Coalition** is an initiative of the European Union that unites Member States, companies, social partners, non-profit organisations, and educational institutions in tackling the shortage of digital skills in Europe.[840] By participating in this coalition, organisations that strive to enhance digital skills across Europe have the chance to share insights, learn from fellow members, and highlight their contributions and effects. The work performed within the scope of the Coalition is crucial to achieving the objectives set for the Digital Decade. Moreover, the **Digital Education Action Plan 2021-27**,[841] including the forthcoming European Digital Education Hub, aims to support the

[837] Commission Recommendation of 26 April 2017 on the European Pillar of Social Rights, C(2017)2600final.

[838] Interinstitutional Proclamation on the European Pillar of Social Rights (2017/C 428/09). To this must be added the Communication of 26 April 2017, Establishing a European Pillar of Social Rights, COM/2017/0250final.

[839] Directive (EU) 2022/2041 of the European Parliament and of the Council of 19 October 2022 on adequate minimum wages in the European Union.

[840] See: https://digital-skills-jobs.europa.eu/en/about/digital-skills-and-jobs-coalition.

[841] Digital Education Action Plan 2021-2027.

development of a high-performing digital education ecosystem in Europe, as well as enhancing digital skills and competences, to address the digital transformation for all.

In general, **liability rules establishing how damage can be compensated could prove particularly complicated to enforce when it comes to AM systems**. In consistency with the AI Act, the European Commission published a proposal for a directive on adapting non-contractual civil liability rules to artificial intelligence (the proposed **AI Liability Directive**).[842] The goal of this proposed Directive is to ensure greater legal certainty, thereby enhancing consumer trust in AI and fostering successful innovations across the EU – while, at the same time, seeking to avoid the emergence of fragmented AI-specific reforms to national civil liability rules. The proposed Directive concerns **non-contractual civil liability rules**, i.e. rules providing for a compensation claim irrespective of a contractual link between the victim and the liable person. The proposed Directive aims to guarantee that individuals within the EU who experience harm due to AI systems receive the same level of protection as those affected by other technologies. Victims must be compensated if they are harmed by the fault or omission of a provider, developer, or user of AI, resulting in damage covered by national law. The AI Liability Directive is set to introduce a rebuttable 'presumption of causality', thereby simplifying the burden of proof for victims seeking to establish harm caused by an AI system. In addition, the Directive will grant national courts the authority to compel the disclosure of evidence concerning high-risk AI systems that are suspected of causing harm. According to the proposal, national courts should, in the course of civil proceedings, be able to order the disclosure or preservation of relevant evidence relating to the damage caused by high-risk AI systems from persons who are already under an obligation to document or record information under the AI Act, be they providers, persons under the same obligations as providers, or users of an AI system – either as defendants or third parties to the claim.

The widespread adoption of AM systems could also affect **public health spending** due, for example, to increased accidents at work. **EU4Health, a new initiative established by Regulation (EU) 2021/522**,[843] was launched in reaction to the COVID-19 crisis. EU4Health aspires to address long-term health hurdles by fostering more robust, adaptable, and inclusive healthcare systems. It targets four overarching goals: (i) to improve and foster health; (ii) to protect people; (iii) to ensure access to medicinal products, medical devices, and crisis-relevant products; and (iv) to strengthen health systems. The action plan envisages reinforcing health data, digital tools and services; the digital transformation of healthcare; enhancing access to healthcare; developing and applying EU health legislation and evidence-based decision-making; and conducting integrated work among national health systems.

3.6.2. Initiatives at national level

Many EU countries have developed **national AI strategies** that can be also relevant to AM systems. These strategies are broad policy documents setting out high-level strategic goals and highlighting opportunities and risks in various fields in relation to AI. They outline ways to increase national economic growth and competitiveness in the field of AI, investing in the digitalisation of enterprises, and creating a safe, trusted AI ecosystem. However, they do not focus in detail on issues relating to AI/AM in the workplace. Rather, they provide an overall direction for the development, adoption, and implementation of AI in their respective countries. They also emphasise the role of the **protection of fundamental rights**, ensuring transparency and a human-centred approach. For example, the strategy developed by the Austrian federal government recognises that the use of AI should be 'oriented towards the common good is aimed for, which is carried out in a responsible manner on the basis of

[842] Proposal for a Directive of the European Parliament and of the Council on adapting non-contractual civil liability rules to artificial intelligence.

[843] Regulation (EU) 2021/522 of the European Parliament and of the Council of 24 March 2021 establishing a Programme for the Union's action in the field of health ('EU4Health Programme') for the period 2021-2027, and repealing Regulation (EU) No 282/2014 (Text with EEA relevance).

fundamental and human rights, European fundamental values and the upcoming European legal framework'.

Much attention is given to the development of the **skills** needed in the era of AI, and to the risks of technological unemployment and/or deskilling. Practically every country is placing emphasis on the need to invest in research on AI and increasing digital literacy. To provide some examples, in the **Czech National AI strategy**,[844] released in May 2019, the government committed itself to monitoring the labour market by commissioning analyses to predict job creation and losses due to AI. These predictions for **future labour requirements** will be registered in the National Register of Professions and the Central Competence Database, for a systematic publishing of updates.[845] In **Malta**, the national AI strategy outlines the societal aspects of AI, and promotes it as an enabler, rather than something to fear. It encourages collaboration being facilitated between industry, educational and research institutes on AI-related initiatives, and promotes the reskilling of the workforce to prepare for AI. The Maltese strategy also fosters the adoption of Artificial Intelligence in Education (AIEd) as one of its priority areas. In **Finland**, a Competence and Innovations Committee has been established under the national Artificial Intelligence Programme, to support education reforms. Policy recommendations in Finland range from extending vocational training and lifelong learning opportunities, to introducing specialised Master's programmes on data science, digital humanities and AI.[846] Similar initiatives fostering AI-oriented education are taking place in **France**. The French AI strategy recommends the creation of a public laboratory on the transformation of work to better understand the impact of automation on occupations, and to provide support for professional transitions.[847]

Some strategies highlight the broader impact of AI on the labour market. The Polish 'Policy for the Development of Artificial Intelligence in Poland', issued by the Council of Ministers in December 2020,[848] is an example of a document emphasising the threats resulting from the progress of work automation, including unemployment. It declared the establishment of an AI Observatory for the Labour Market, which would be tasked with studying the impact of AI systems in the labour market.[849]

Lastly, challenges relating to the **use of AI in labour and social administration feature in some AI strategies**. In Germany, the Network for AI in Labour and Social Administration at the Federal Ministry of Labour and Social Affairs issued self-commitment guidelines in October 2022 regarding the use of AI in the official practice of labour and social administration. The guidelines focus on human-centred processes, particularly during the initial deployment phase, impact assessments, and risk evaluation.

3.6.3. Other relevant initiatives

In **Spain**, **the Agency for the Oversight of Artificial Intelligence** (*Agencia Española de Supervisión de la Inteligencia Artificial*, AESIA), launched in late 2023, will be responsible for promoting trustworthy AI and supervising AI systems that might pose significant risks to health, security and fundamental rights. It is the first AI oversight agency in Europe.

The Malta College of Arts, Science & Technology has released an AI strategy and action plan for 2020-2025. This outlines policy actions towards the introduction of AI into education systems. It is connected with eight relevant initiatives, such as Enhanced e-Learning

[844] See: https://ai-watch.ec.europa.eu/countries/czech-republic/czech-republic-ai-strategy-report_en#ai-to-address-societal-challenges.

[845] Ibid.

[846] See: https://ai-watch.ec.europa.eu/countries/finland/finland-ai-strategy-report_en.

[847] See: https://ai-watch.ec.europa.eu/countries/france/france-ai-strategy-report_en.

[848] Polityka dla rozwoju sztucznej inteligencji w Polsce od roku 2020. Załącznik do uchwały nr 196 Rady Ministrów z dnia 28 grudnia 2020 r. (poz. 23); (from 2020. Appendix to the Resolution No. 196 of the Council of Ministers of 28 December 2020 (item 23)).

[849] See: https://www.gov.pl/web/ai/polityka-dla-rozwoju-sztucznej-inteligencji-w-polsce-od-roku-2020.

Environments; Personalised Learning through Smart Content; CPD in AI for Academics; Intelligent Tutoring Systems; Simplification in Administration; SMART Campus Buildings; Alignment of LOs to Job Profiles; Curriculum & AI Specific Programmes & Projects; and AI Collaboration with Industry.

In some countries, such as Slovakia, **manuals for companies** on the introduction of AI have been developed, providing suggestions for and examples of AI implementation, as well as highlighting the ethical approach. Meanwhile, the German Trade Union Confederation (*Deutscher Gewerkschaftsbund, DGB*) outlined a six-step process for the deployment of trustworthy AI in workplaces, with each step containing a set of crucial questions to be asked before proceeding.[850]

Some transnational cooperation initiatives are also becoming increasingly relevant. One example is the **Declaration on AI in the Nordic-Baltic Region**, signed in May 2018 by the ministers responsible for digital development from Denmark, Estonia, Finland, the Faroe Islands, Iceland, Latvia, Lithuania, Norway, Sweden and the Åland Islands. The countries agreed to collaborate, to 'develop and promote the use of artificial intelligence to serve humans on the improvement of opportunities for skills development, enhancing access to data, developing ethical and transspainparent guidelines, standards, principles, and values, developing standards for hardware and software that enable privacy, security, and trust, ensuring AI gets a prominent role in European discussions of the Digital Single Market, and avoiding excessive regulations'.[851]

3.6.4. Conclusions

The legal measures examined in this subsection do not squarely align with AM systems at work, but rather allude to it in a more peripheral manner. Importantly, most of these instruments **lack a legally binding character**. Their primary concern is not to holistically address AM; instead, they gravitate more towards tackling specific elements of AM that are tied to AI or to the broader landscape of digitalisation. Furthermore, coordination between these instruments is limited.

3.7. Final remarks

The impact of data-driven technologies and AM systems on the existing legal framework is **ambivalent**. Hence, the concrete ways in which AM tools are designed, introduced, and used correlates with distinct effects. **Undoubtedly, regulation can play a pivotal role as regards the impacts of AM at work, on both businesses and workers.**

The in-depth analysis of the relevance and effectiveness of the EU *acquis* with regard to AM can be summarised as follows. **Many legal tools are still relevant in this context**, primarily because AM systems reproduce classical managerial functions in the workplace that influence many areas, from working conditions in a broad sense to data protection, and from equality to collective rights. **Nonetheless, the effectiveness of these legal instruments varies widely**. This is mostly due to the unprecedented nature of the way in which such technologies operate. The sheer volume, velocity, and variety of information they process is unparalleled. Their capability to execute tasks either serially or discretely is also a novel feature. **Moreover, there are intrinsic, 'architectural' limitations in the existing legal framework relating to personal and material scope, and these limitations are accentuated when confronted by the functionalities of AM systems. While these gaps do not solely result from the emergence of AM, their inadequacies become apparent when assessed within the context of AM.**

Elaborating on specific strategies to bridge the regulatory gaps goes beyond the remit of this exploratory study, which adopts a descriptive rather than a normative stance. However, it might be prudent for policymakers to identify distinct legal domains – data protection

[850] Artificial Intelligence (AI) for Good Work, Deutscher Gewerkschaftsbund, 2020.

[851] See: https://www.regeringen.se/49a602/globalassets/naringsdepartementet/20180514_nmr_deklaration-slutlig-webb.pdf

being a prime example – in which **targeted regulatory efforts might be particularly fruitful in addressing current deficiencies**.

Numerous pieces of legislation incorporate provisions that empower MS or social partners to enhance and reinforce the EU legal framework. Recent interventions including the proposed PWD, along with national initiatives in countries such as Germany, Italy, Portugal, and Spain, **highlight the room for manoeuvre that is enabled by EU legal tools**. This demonstrates that the general EU framework can be both complemented and enriched at the national level. However, any 'race' to regulate AM might spawn unintended outcomes, notably legal fragmentation. This also poses potential pitfalls: it can create avenues for arbitrage by tech providers and establish an uneven playing field for adopters such as employers. Therefore, **stronger coordination at EU level** could ensure more positive outcomes by minimising overlaps and potential conflicts between the parallel initiatives that either directly or indirectly aim to regulate AM systems within work environments.

4. Future Trends

The overall objectives of this chapter of the report are to:

- Identify key drivers and barriers to the future evolution of AM usage.
- Provide a quantitative forecast of the prevalence of AM in the short term (two years), medium term (5 years) and long term (10 years).
- Highlight the areas of managerial functions in which the use of AM will grow the most.

On the basis of these objectives, this section is structured as follows:

- 4.1. Methodology – a short overview of the methodology highlighting the most important aspects of it, while a more extensive discussion is provided in *Annex 1 – Methodological annex.*
- 4.2. The possible future evolution of AM – discussion on the drivers and barriers that might affect the prevalence of AM in the future, possible future scenarios of AM evolution, and insights on which of the scenarios is the most likely.

The main findings of the chapter are summarised in the box below.

Key findings:

- In the *short to medium term*, the promise of AM cutting costs and improving efficiency and productivity will be the key driver of wider AM adoption. However, cost could also be a barrier for some, as AM might require large-scale initial investments in terms of finance, time and learning. In addition, a lack of knowledge and skills concerning how such tools work could further hinder AM adoption.
- In the *medium to long term*, new regulations appear to be among the most vital factors influencing AM adoption. On the one hand, they could dampen AM usage by increasing the cost of compliance, limiting AM functions through laws governing the use of personal data, and restricting the use of AM altogether for certain managerial functions (e.g. dismissing workers). On the other hand, if regulation clearly defines how AM can be used and for what purpose, it could boost the adoption of AM by alleviating uncertainty.
- In the *long term*, technological megatrends serve as primary drivers for more widespread AM adoption. These include growing hyperconnectivity, as well as rapid research into and development of AI and machine learning, allowing AM to constantly improve. The potential exists for possible human enticement to use these technologies, whereby many workers are already starting to rely on a symbiotic relationship with such technologies.
- Based on the Delphi survey, it is most likely that in the *next 10 years*, AM will grow between 3 % and 6 % annually. This means that around half of companies in the EU will be using AM in the next five years (i.e. by 2028), while 10 years from now, around three-quarters of companies in the EU-27 might be using some form of AM. However, to highlight that this prediction is based predominantly on insights from the Delphi survey, which does not provide completely robust results.

Respondents to the Delphi survey believe that among the various types of AM, the use of algorithms to allocate worker shifts will increase the most, with 64.9 % of the 130 respondents stating that it will increase significantly in the next 5-10 years. In addition, between 55 % and 59 % of respondents also believe that the use of algorithms to monitor workers, collect workers' personal data, provide recommendation/instructions to workers, and recruit workers will also increase significantly over the next 5-10 years. Meanwhile, 38.8 % of respondents believe that the use of algorithms for talent management will

increase significantly in the future. Lastly, 29.2 % of respondents believe the use of AM will increase in connection with rewarding workers – while just 20.2 % believe it will be increasingly used for firing/dismissing workers.

4.1. Methodology

Forecasting trends and scenarios for future evolution in AM and their likely impacts is **very challenging due to several limitations**, including but not limited to:

- The prevalence of AM not being uniform. For example, some types of algorithms are used more frequently by specific types of employers and workers than others, and *vice versa*.
- It is challenging to estimate the future of algorithms by focusing on specific managerial functions, given that most AM implementation, as identified in the AM case studies, often comes in the form of 'packages of tools' that cover several algorithms.
- Different data sources provide conflicting information on the usage of algorithms for similar managerial functions, as discussed in Section 1.2. 1.2. Prevalence of AM in the EU.
- The scarcity of data on trends in AM usage, and a lack of credible proxies, hinder our ability to build accurate and robust econometric forecasting models. However, the possibility of running an econometric forecasting model using one of the existing proxies was explored. The results derived were, unfortunately, not robust enough (for a summary of these results, see *Annex 17: Trends in AI, robot, and digital technology usage*).

Due to the limitations listed above, a multifaceted approach was therefore employed. Namely, the study team began by developing qualitative scenarios for the future evolution of AM, focusing on three scenarios – baseline, growth, and slowdown. Each of these scenarios depicts a probable future with regard to critical drivers of change, potential outcomes in terms of the prevalence of AM, and likely impacts. The scenarios were developed and analysed using the following five-step methodology:

- **Step 1: Discussion of the drivers of and barriers to the future evolution of AM** –based on a literature review, consultations with experts, and workshops, the main drivers and barriers that could shape its future were identified and discussed. Insights into the various drivers are provided in subsection 4.2.1. Drivers and barriers affecting the future evolution of AM.
- **Step 2: Data collection, aggregation and initial analysis** – based on the quantitative and qualitative outputs discussed previously, initial estimates regarding the current usage of AM have been established, collecting some insights on the evolution of AM and technologies associated with it, to develop possible future trends. These are discussed in Section 1.2. 1.2. Prevalence of AM in the EU.
- **Step 3: Scenario building** – to control for a lack of robust data, it was decided to analyse three scenarios in the study: baseline, growth and slowdown. What exactly each scenario entails is discussed in subsection 4.2.2. Possible future scenarios for AM's . Meanwhile, more insights on the methodology employed in this step are provided in section *1.4. WP1 Task 4: Future trends* of *Annex 1: Methodological annex*.
- **Step 4: Delphi survey programming and launching** – to gather insights on the scenarios, a Delphi survey approach was adopted. A total of 130 respondents fully completed the survey, while around 80 partial answers could be analysed. Respondents included academics, employer and employee representatives, AM tool developers, employers, and regulators. The majority of answers (60.5 %) came from

academics, while other groups are represented to a much smaller extent. Hence, the discussions below predominantly reflect the views of academics. A more in-depth discussion as to why the Delphi survey approach was selected, and how it was employed, is provided in *1.4. WP1 Task 4: Future trends* in *Annex 1: Methodological annex*.

- **Step 5: Final analysis** – based on the results of the Delphi survey, with corroboration from other sources, Section 4.2. The possible future evolution of AM. This includes a discussion of the future use of AM based on managerial functions (see subsection 4.2.3. The future of AM usage, by specific managerial function), and what opportunities and challenges the future of evolution of AM could foster (see subsection 4.2.4. Likely effects of further growth in AM).

4.2. The possible future evolution of AM

4.2.1. Drivers and barriers affecting the future evolution of AM

Short- to medium-term drivers and barriers

One of the primary drivers behind the adoption of AM in the workplace, as evidenced by academic literature, expert interviews and the Delphi survey carried out for the present report, is the potential it offers for increased efficiency/productivity and reduced costs. More specifically:

- Economic stagnation and the possibility of a significant recession in the coming years (even though October 2023 data provides no clear signs of a recession)[852] due to the disruptions caused by the wars in Ukraine and in other regions, high inflation, and the energy crisis, could put pressure on some organisations to integrate AM solutions as a cost-saving measure.
- Growth in wages and the costs of inputs results in higher prices for producers, which companies in competitive industries have difficulties in passing on to consumers. This provides a strong impetus towards cost savings, including those that may be achieved via AM.
- 'Re-shoring', and the questioning of the resilience of global supply lines due to geopolitical uncertainties, has resulted in previously offshored manufacturing and services increasingly being moved back to the EU, creating cost pressures for the companies involved, which AM could help to mitigate.[853]

However, although the cost-cutting benefits of AM are clear in theory, based on the Delphi survey and other sources, costs can also be a barrier to the more widespread adoption of AM. More specifically, 45.9 % of respondents to the Delphi survey stated that cost-related considerations would mostly speed up adoption, while the same number of individuals – 45.9 % – said such considerations were likely to have a mixed effect (while 8.2 % stated that cost-related aspects would slow down adoption). Hence, **while AM has the potential to reduce employers' costs, its introduction also requires high levels of upfront investment,** which might prevent some from using such tools. This is also supported by the literature and the VA survey of employers, which found that 60 % of the 159 companies surveyed strongly or partially agreed that high costs prevent them from adopting AM. This is despite the increased accessibility of various AM tools, which implies that the cost of implementation remains a significant barrier to their adoption.[854] In addition as Burton et al. (2021) note, integrating algorithmic decisions into existing workflows requires substantial

[852] See: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Eurostatistics_-_data_for_short-term_economic_analysis

[853] European Commission (forthcoming). Study exploring the social, economic and legal context and trends of telework and the right to disconnect, in the context of digitalisation and the future of work, during and beyond the COVID-19 pandemic.

[854] Christin, A. (2017). Algorithms in practice: Comparing web journalism and criminal justice. *Big Data & Society* 4(2): 1–14.

time and energy, including building transparency and trust between how employers and workers view the algorithms as being used.[855] Lastly, the current economic environment of growing interest rates might reduce borrowing potential, thus dampening the ability of some companies to integrate AM solutions.

In addition to efficiency and costs, **current developments in the market and significant shifts towards telework and gig work might also be drivers for wider AM adoption**. Telework became significantly more prevalent during the COVID-19 pandemic when, at its peak, around 48 % of workers worked from home.[856] This also increased the usage of AM tools, especially those that monitor workers' performance at home or help manage work.[857] Based on the upcoming European Commission study on telework,[858] the prevalence of telework is expected to fall compared with the levels seen at the peak of the pandemic, but is still expected to remain above pre-COVID-19 levels, and will continue to grow. The persistence of telework can be expected to lead to the broader usage and development of AM tools for remote working, as was highlighted in the literature and by some of the experts interviewed for the present study. The Delphi survey supports this to some extent, with 58.4 % of respondents agreeing that the shift to telework will speed up the adoption of AM. Similarly, 90 % of respondents believe technological developments will increase AM adoption.

AM is also frequently utilised in **gig/platform work**. However, the future of the gig economy, and hence the usage of AM in this context, is relatively unclear. For example, there is currently much debate in the EU over how to regulate gig work,[859] on which the future deployment of AM will depend. Nonetheless, the use of AM for monitoring and rating workers, as well as scheduling work, can be expected to persist within the gig economy – especially given that gig work makes heavy use of such technologies already [860] and because, according to one expert interviewed, such work is often used as a testbed for AM. Hence, its evolution will continue to have broader implications for work and management in other parts of the economy.

In addition, despite the world's rapid digitalisation, a **lack of know-how and a deficit of skills regarding how AM works and how it may be used could also hinder company decisions to implement new AM technologies.** Managers may be ill-equipped to use AM due to their existing experience of traditional management functions. They may require new skills in order to use AM effectively. Even when trained to use AM, managers may fail to grasp the full impact of its use. They may not realise, for example, its effects on workers' autonomy and privacy, which may risk violating workers' rights and impairing their well-being. Compared with previous management technologies and strategies, these tools are considerably more intricate and impactful. As such, they require upskilling to leverage their potential advantages responsibly. Closing skills deficits could be seen to include not just more technical training, but also more legal and ethical training focusing on the responsible use of AM.

Evidence based on interviews with experts and the results of the surveys conducted supports the above claim. First, many of the experts in the field who were interviewed agreed that a lack of skills would be a significant hindrance. In connection with this, 61.5 %

[855] Jarrahi, M.H., Newlands, G., Lee, M.K., Wolf, C.T., Kinder, E., & Sutherland, W. (2021). Algorithmic management in a work context. *Big Data & Society*, 8(2), 20539517211020332.

[856] Eurofound (2020b). Living, working and COVID-19, COVID-19 series. Luxembourg: Publications Office of the European Union.

[857] Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek, E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work

[858] European Commission (forthcoming). Study exploring the social, economic and legal context and trends of telework and the right to disconnect, in the context of digitalisation and the future of work, during and beyond the COVID-19 pandemic.

[859] Webster, C., & Ivanov, S. (2020). Robotics, artificial intelligence, and the evolving nature of work. In Digital transformation in business and society. Cham: Palgrave Macmillan, pp. 127-143).

[860] For details, see: https://joint-research-centre.ec.europa.eu/jrc-news-and-updates/platformisation-work-accelerated-pandemic-phenomenon-likely-grow-2023-04-17_en

of 182 employers surveyed in the VA survey strongly or partially agree that they lack knowledge, and 41 % out of 178 respondents strongly or partially agree that they lack trust to integrate AM. However, only 19.8 % of respondents in the Delphi survey stated that know-how as a factor would slow down AM adoption, as the current growing discourse related to the use of algorithms, AI, digital technologies, and so on at work already showed many of the benefits such tools might bring. Hence, similarly to costs, it seems there is no one answer to whether or not current levels of know-how will slow down the adoption of AM.

Lastly, in the short term, at least – but also probably in the longer term – **strong pushback from employees and ethical considerations might hinder AM adoption**. For example, around 47 % of employers in the VA survey stated that they strongly or partially agree that negative employee perceptions and attitudes towards algorithms are a barrier to implementing an AM tool. Similarly, 40 % of respondents to the Delphi survey believe employee preferences will slow down AM adoption, while 52.4 % think they will have a mixed effect. While these results support the findings of the VA employer survey, they also highlight the fact that some employees might welcome AM, due to some of the benefits such tools might bring them, as discussed in Chapter 2. Opportunities and challenges. Similarly, while 37.8 % of the respondents in the Delphi survey believe ethical consideration will slow down the adoption of AM, 40.5 % think it will have a mixed effect. This again implies that most drivers can also serve as barriers to AM adoption in some cases, which complicates estimations regarding its future.

Medium- to long-term drivers and barriers

In addition to the short-term factors above, several trends are likely to shape AM adoption in the medium and long term. First, **sustainable development goals such as greening the economy might create new markets and drive (or course-correct) the development of AM.** For example, the megatrend of climate change and environmental degradation has increased the need for green skills. These include, but are not limited to, skills in retrofitting, the cyclical economy, energy efficiency, sustainable construction, waste management and conservation, sustainable agriculture, and the creation and distribution of new sources of energy. These, in turn, can create opportunities for the usage of AM tools. For instance, tools that enable talent management and skills development could guide workers' career development, highlighting needs and opportunities for the development of green skills. They could also help to reduce work time, an acknowledged mechanism to reduce carbon emissions.[861] Work time reduction would need to be formalised as an objective of AM and used to drive a broader shift towards a low-carbon economy.

However, based on the Delphi survey, experts and relevant stakeholders disagree on the effects of green goals on AM adoption. First, around 23 % of respondents believe green goals will accelerate AM adoption in the next five years, while 16.7 % think they will slow it down. In addition, 60.4 % think green goals will have a mixed effect (see Figure 4-1 below for more details). With regard to its adverse impact on AM, greening could slow down AM adoption if it requires an expansion of data centres, as this might negatively affect the climate due to their large energy requirements and heat emissions.[862] Hence, depending on how AM will be used, it could be boosted by greening goals or slowed down heavily.

Second, **the changing security paradigm, such as the increasing cost of protecting AM technologies from potential cyberattacks, data security issues and similar**, might also significantly influence how AM tools are developed and used. For example, to safeguard AM tools from cyberattacks, they might become more complex, creating a barrier to some employers wishing to implement them in the workplace. In addition, constant data

[861] Knight, K.W., Rosa, E.A., & Schor, J.B. (2013). Could working less reduce pressures on the environment? A cross-national panel analysis of OECD countries, 1970–2007. *Global Environmental Change*, *23*(4), 691-700.

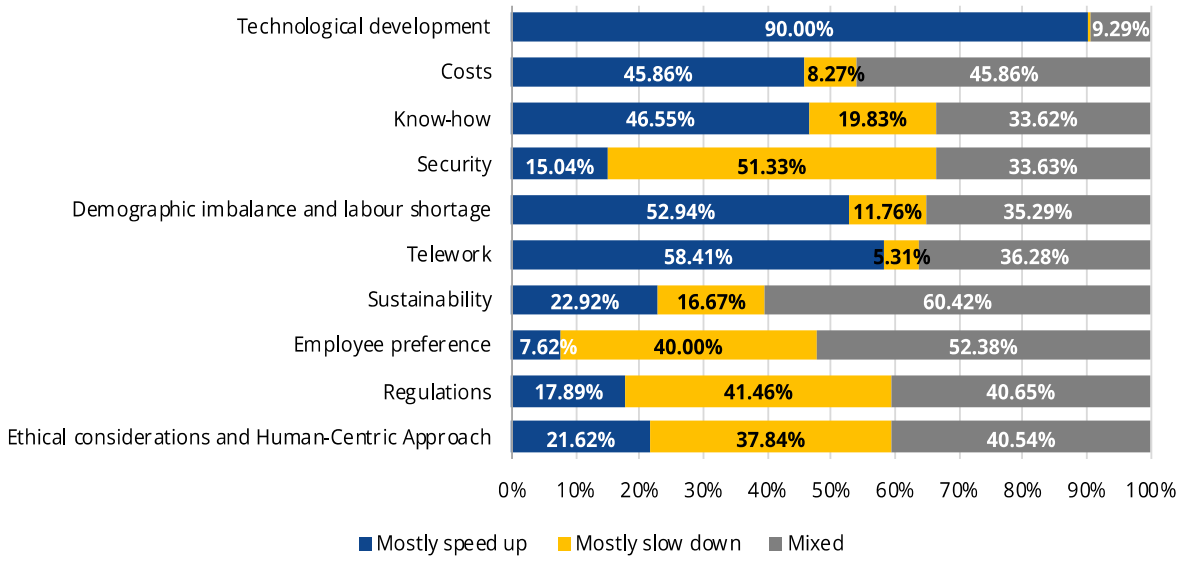
[862] Orikpete, O. & Ewim, D. (2023). Impact of Data Centers on Climate Change: A Review of Energy Efficient Strategies. *Journal of Engineering and Exact Sciences*. 9, 1-15.

privacy and access issues could continue to have several effects on AM's development and usage. Lastly, as highlighted by many of the academics interviewed, a lack of trust in such tools and possible transparency issues might further reduce the uptake of such technologies. The results of the Delphi survey also support this, with 51.3 % of respondents believing that security concerns will negatively affect AM tool adoption, while 33.6 % think that such concerns might have a mixed effect.

Third, introducing **new EU/national regulations and increasing the cost of compliance might significantly impact the adoption of AM across the EU**. Legal issues, including those linked to data access and ownership, are mentioned in the literature as having significant impacts on whether AM systems are implemented at the workplace level.[863] In addition to the traditional costs of technology implementation, interviewees highlighted the expenses relating to regulatory compliance issues that may arise when adopting AM tools. This is a particular issue with some AM tools developed outside the EU (e.g. in China, the USA, etc.), as these may not adhere to the high standards required by EU social rights frameworks and regulations. In addition, there are ongoing debates regarding the regulation of digital tools and business models (see the discussion on the gig economy above). New regulations could potentially be introduced, and some technologies even banned (e.g. Chat-GPT was recently and temporarily banned in Italy[864]); this could slow down the potential evolution of AI-based systems and AM.

The results of the Delphi survey largely reflect these concerns, highlighting that likely regulations will slow down adoption, though many also saw the benefits of more regulations. For example, around 41.5 % of respondents believe regulations will slow down AM adoption, around 40.7 % believe that regulations will have a mixed effect while 17.9 % have a positive effect. Respondents predominantly highlighted that if regulations provide clear guidelines for AM usage, they can promote its adoption, as companies will face less uncertainty. Hence, it appears the effect of regulation on AM will depend heavily on *how* such technologies are regulated.

Figure 4-1: Factors that will speed up or slow down AM tool adoption in the next five years



[863] Alsheibani, S.A., Cheung, D., & Messom, D. (2019). Factors inhibiting the adoption of artificial intelligence at organizational level: A preliminary investigation.

[864] See: https://www.bbc.com/news/technology-65431914.

N = 140, which includes insights from 130 respondents who answered all questions and 10 who did not answer all but did answer this one.
Source: Authors' elaboration based on Delphi survey.

In addition to these, other factors could serve both as drivers and barriers to more widespread adoption. These include technological development, the growing prevalence of telework, and the growing demographic imbalance, as shown in Figure 4-1 above. An extensive discussion of technological developments, including telework, is provided below under the heading 'Long-term drivers and barriers'. In response to demographic imbalance and labour shortages – currently a severe issue for all EU Member States[865] – some might see AM as part of a potential solution. First, it provides tools for automating certain tasks; second, it can foster efficiency and effectiveness in task performance, as well as work scheduling; and lastly, as discussed, AM can be used to keep track of employee's well-being. This could allow employers to avoid their workers suffering from fatigue and burnout, and they operate at the highest level. Respondents to the Delphi survey also anticipate barriers to the wider adoption of AM in the next 5 years that have not yet been mentioned. These include employee preferences, which were selected by 40 % of respondents as a factor slowing down AM adoption. This might imply that many workers will not want AM tools being used to monitor, evaluate them, provide recommendations, etc., in their workplace. This also relates to ethical considerations, which 37.8 % of respondents believe will heavily slow down AM adoption.

However, it is also important to highlight that most respondents to the Delphi survey believe the various factors will have a mixed effect on AM adoption, accelerating it in some cases while dampening growth in others. Hence, different factors may have different effects on the prevalence of AM, depending on how they materialise themselves.

Long-term drivers and barriers

The main long-term drivers of AM are the rapid development of new digital technologies and business models, leading to widespread application and acceptance of digital technologies in society (strengthened by generational change), and the wide availability of 'ready-to-use' AM technologies in the market. Several of these technological trends emerge from the European Commission's[866] discussions on global megatrends, with three of them being especially relevant to AM:

- **Growing hyperconnectivity at work:** the increasing prevalence of the Internet of Things (IoT), predicted to grow at around 19 % annually until 2027,[867] allows 'smart products' to remain constantly connected and exchange information. As a result, IoT has the potential to enable everyday work tools, including laptops, office chairs, desks, etc., to detect and respond to their usage[868], including through different AM solutions. For example, if a smart office chair detects that a worker has bad posture, it might send a message urging the worker to adopt a more ergonomically correct position. Similarly, hyperconnectivity could also be fostered by developments in 5G, allowing faster communication and the possibility of collecting more accurate indicators from workers– for example, through emotion recognition, which is sometimes used in worker monitoring.[869] For instance, experts anticipate that the

[865] EURES (2022). Report on labour shortage and surplus. Available at: https://www.ela.europa.eu/sites/default/files/2023-03/eures-labour-shortages-report-2022.pdf

[866] See: https://knowledge4policy.ec.europa.eu/foresight/tool/megatrends-hub_en#explore

[867] See: https://iot-analytics.com/iot-market-size/#:~:text=Overall%20enterprise%20Internet%20of%20Things,plays%20in%20reaching%20sustainability%20goals.

[868] De Stefano, V., & Wouters, M. (2022). AI and digital tools in workplace management and evaluation: An assessment of the EU's legal framework. Osgoode Legal Studies Research Paper, forthcoming.

[869] Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek,E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work, available at: https://op.europa.eu/en/publication-detail/-/publication/c326ad8b-12d1-11ed-8fa0-01aa75ed71a1/language-en

emotion recognition industry will reach USD 37 billion by 2026, tripling its value since 2018.[870] The danger with hyperconnectivity, however, is that it will lead to more intrusive forms of worker monitoring. If employers can not only see what workers are doing but also, to an extent, detect what they are feeling, it will prove more difficult for workers to determine their own space, time, and thoughts at work.

- **Rapid research and developments in AI and machine learning:** based on expert feedback, and with some estimates projecting that the global AI market will increase by around 29.6 % a year between 2022 and 2026,[871] it is expected that AI will be used in almost all facets of life and work. This increase will not just be limited to the private sector; many governments and public companies will also use AI.[872] A large market will mean more tools, driving their costs down and making algorithmic management that uses AI more accessible to employers. In addition, the range and application of AM tools is also likely to change as AI develops, which might also significantly change the AM landscape. For example, at the end of November 2022, ChatGPT – a natural language processing tool able to mimic human conversation, write high-quality texts, answer complex questions and more – was released to the public, highlighting the powers of AI.[873] Recently, Google released its own chatbot, Bard, which boasts similar capabilities.[874] In turn, such tools might be used by companies to, for example, speed up recruitment, where they might be used to 'chat' with potential hires to assess their capabilities and how well they fit the company.
- **Human enhancement through technologies:** given the growing interest in AI and similar technologies, significant research is being dedicated to brain-computer interfaces – technology allowing individuals to control and communicate with computers directly.[875] In relation to AM, this might enable workers to transform thoughts into outputs, such as writing text, creating presentations and programming,[876] allowing them to complete their tasks more quickly. Such technologies might also be used by managers to quickly convey 'tips and tricks' on how to solve a problem or complete a task more efficiently.

Of course, these new technologies and business models will take years or even decades to reach peak adoption. It is not expected that such trends will have an immediate and significant impact in the short term (i.e. in the next few years). However, the development of these megatrends is likely to be fuelled by research and innovation investments. As discussed in a recent EU-OSHA report on AI-based worker management, nearly all EU Member States have planned their AI development strategies.[877]

In the longer term, the views of Delphi survey respondents regarding the factors that will foster or hinder AM adoption in the next 10 years (see Figure 4-2) largely mirror those discussed previously regarding the next five (see Figure 4-1). This implies that the same factors will continue to foster and dampen down AM adoption in both the medium-term (five years) and long-term (10 years) horizons.

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[870] Wright, J. (2021, 24 May). 'AI' is being used to profile people from their head vibrations – But is there enough evidence to support it? The Conversation, available at: https://theconversation.com/ai-is-being-used-to-profile-people-from-their-head-vibrations-but-is-there-enough-evidence-to-support-it-160566

[871] See: https://www.idc.com/getdoc.jsp?containerId=prEUR250505223#:~:text=AI%20spending%20in%20Europe%20will,than%20%2470%20billion%20in%202026.

[872] See, for example: https://ai-watch.ec.europa.eu/index_en

[873] For more information, see: https://openai.com/

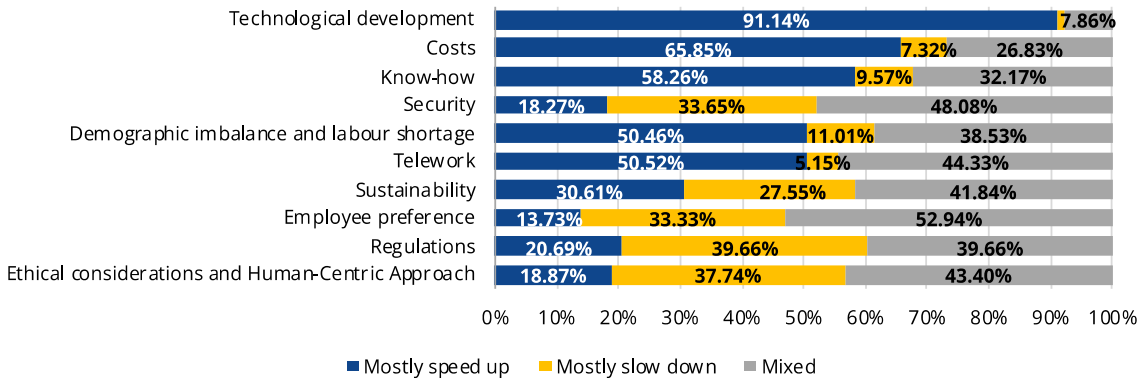
[874] See: https://bard.google.com/

[875] Saha, S., Mamun, K.A., Ahmed, K., Mostafa, R., Naik, G.R., Darvishi, S., Khandoker, A.H., & Baumert, M. (2021). Progress in brain computer interface: Challenges and opportunities. *Frontiers in Systems Neuroscience*, 15.

[876] See: https://hbr.org/2020/10/what-brain-computer-interfaces-could-mean-for-the-future-of-work.

[877] Christenko, A., Jankauskaitė, V., Paliokaitė, A., Leon van den Broek, E., Reinhold, K., & Järvis, M. (2022). Artificial intelligence for worker management: An overview. European Agency for Safety and Health at Work.

Figure 4-2: Factors that will speed up or slow down AM tool adoption in the next 10 years



N = 140 , which includes insights from 130 respondents who answered all questions and 10 who did not answer all, but did answer this one.
Source: Authors' elaboration based on Delphi survey.

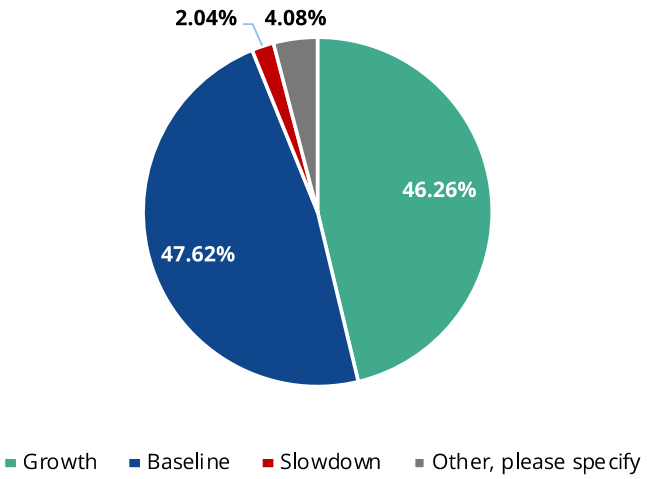
4.2.2. Possible future scenarios for AM's evolution

Given the lack of robust data, this section explores three possible scenarios for the evolution of AM, comprising a **baseline scenario**, an optimistic **growth scenario**, and a **slowdown scenario**. These three scenarios allow us to acknowledge different possible futures that might become realities due to some of the drivers and barriers previously discussed becoming more or less prevalent. Each initial scenario was derived from the literature review, interviews with experts and stakeholders, and quantitative data covered in previous parts of the report. Following this, each scenario was validated through the Delphi survey.

The discussion of these scenarios predominantly centres on the possible evolution of AM in general, although some insights into likely trends relating to specific managerial functions are also provided later in the report. The decision to focus on AM in general rather than on specific functions is motivated by the fact that, according to case studies with AM tool creators, many AM tools cover several managerial functions. Hence, it is not easy to decouple the trajectories of different algorithms. In addition, given that AM is still evolving and changing, this focus on the future of AM in general ensures some coverage is given in this report to certain managerial functions that are not yet automated, but will become automated in the future, (e.g. due to companies starting to integrate Large Language Models and similar technologies).

The remainder of this section thus explores the three scenarios, highlighting which of them is the most probable, and why. The meaning concerning the prevalence of AM over the next five- and 10-year periods is also discussed. For a summary of the results, see the figure below.

Figure 4-3: Percentage of respondents who selected which scenario is most likely to become reality



N = 147, which includes insights from 130 respondents who answered all questions and 17 who did not answer all but did answer this one.
Source: Authors' elaboration based on Delphi survey.

Baseline scenario

This scenario assumes that the prevalence of AM will grow in the short term, but will be slowed down by relatively high costs, skills requirements, a desire for human-centric management, and potential new regulations. In other words, it assumes a continuation of the current conditions, and that no significant disruptive events ('wild cards') will occur in the near future.

Based on current trends, historical data, insights from the literature, and opinions from experts collected during interviews, **the baseline scenario assumes that AM will grow at a pace of 3-4 % annually**. **In the survey, respondents are also asked to assume that 25-35 % of organisations currently use AM**. Based on this assumption, **the respondents were provided with an estimate that in around five years, 35-55 % of companies will be using AM in the EU-27**. The relatively wide range reflects the high volatility in predictions concerning drivers and barriers that affect it. **In turn, it is estimated that around 45-60 % of companies will use AM 10 years from now (i.e. in 2033).** The large percentage here reflects the assumption that, according to the baseline scenario, most EU-27 (2020) companies will use algorithmic for at least one managerial function. However, the prediction for 10 years diverges from the linear trend of a 3-4 % increase annually. This divergence is because it is likely that some small, and especially micro-enterprises, which comprise 98.9 % of companies in the EU-27,[878] will still not use AM due to cost constraints, a lack of know-how, lack of trust, or the belief that it is more effective for humans to carry out all managerial tasks in their field of work.

In the Delphi survey, there is much support for this scenario. **Out of the 147 respondents who answered this question, 47.6 % selected the baseline scenario as the most likely scenario regarding change in the usage of AM**. In terms of the types of respondents, 42.3 % of academics selected this scenario (51 respondents), compared with around 61.5 % of workers' representatives (16 respondents). The main arguments that respondents provided for selecting the baseline scenario include:

- Many large and medium-sized companies are already using AM, and more are likely to follow, especially given that financial and learning costs are declining.
- However, some companies, especially micro- and small enterprises, will be unable to use such tools even at lowered costs.

[878] See: https://ec.europa.eu/eurostat/web/products-eurostat-news/-/edn-20220627-1

- Currently, there is ‘hype’ and excitement surrounding AI, which might prompt some companies to integrate AM tools that use AI, but some might find it challenging to find good uses for them.
- Some companies will be scared away from adopting AM by existing regulations such as the GDPR, as well as the prospect of new ones.
- Ethical and social issues and strong pushback from employees may dampen rapid adoption.

Among those respondents who selected the baseline scenario, 91.4 % agree that in five years (by 2028), AM will be used in around 35-55 % of companies in the EU-27. Furthermore, **82.9 % of respondents who selected the baseline scenario agreed that in 10 years (by 2033), 45-60 % of companies in the EU will use AM**. It is essential to highlight that not all academics who selected the baseline scenario agreed with these rather broad estimates. This is predominantly because some believe it is impossible to predict how extensively AM will be used in the next 10, or even five years. This is due to the high level of uncertainty regarding its current usage, as well as which factors will serve as drivers, and which barriers to adoption are the most widespread. There is also uncertainty regarding regulations, including whether new ones will shape the usage of AM. Hence, the estimates provided only represent forecasts in a case where AM usage continues at a similar trajectory, and there are no large-scale shocks that that might affect this trajectory (e.g., a new very restricting regulation is introduced).

Growth scenario

The growth scenario assumes significantly faster growth in the prevalence of AM driven by four factors: cost-saving pressures; the potential for increased efficiency and productivity; the rapid growth of new digital technologies and business models; and non-restricting regulations. Regarding the first two points, as discussed in Chapter 2. Opportunities and challenges, AM allows an increase in efficiency and productivity and reduces costs by automating some managerial functions. The development of AM tools might also lead to growing competition in this sector which could, in turn, lower cost-related barriers to entry for employers. Regarding the third point, AM might be prompted by increasing levels of telework, automation, and novel AI technologies fostered by the AI funding strategies adopted in most EU countries and globally. In addition, the rapid development of Large Language Models, such as ChatGPT, might also open new avenues for using AM tools (e.g. they might provide better tools for analysing workers' emails, as well as tools to create high-quality chat-bots that can explain tasks to workers in a more human manner). Lastly, under this scenario, EU and national/international regulations will not create barriers to growth, and AM will globally transform business management. This does not mean that new rules will not be implemented. Rather, it means that they will foster AM or at least not hinder the spread and adoption of AM.

Based on this scenario, in the Delphi survey, the assumption that **AM will grow by roughly 4-6 % annually** was presented. Taking the same baseline of 25-35 % of companies currently using AM, it was estimated that **in five years (i.e. by 2028), around 55-65 % of companies will use AM.** This further implies that **10 years from now (i.e. in 2023), around 70-85 % of companies will use AM**. As can be seen in these predictions, there is a divergence from the linear growth observed previously, with the growth of AM usage likely to decline in the future, as some small and micro-enterprises will not use such technologies because they may not see the need for them.

Similarly to the baseline scenario, the growth scenario attracted a lot of support among respondents. **Around 46.3 % of respondents to the Delphi survey selected this scenario**. Notably, 52.6 % of academics selected this scenario, while 30.8 % of workers did the same. In addition, 85.7 % of respondents agree that in the next five years, AM usage will grow to 55-65 % of companies using such tools, while 83 % also agree that in 10 years, 70-85 % of companies will use AM. Here, similarly to the baseline scenario, not everyone

agreed with these predictions of the future, mainly because they disagreed with the notion of predicting the future using relatively sparse data.

Respondents who selected this scenario predominantly explained their decision as follows:

- More companies will realise the potential of such tools to cut costs and improve productivity.
- Rapid developments in digital technologies, including Large Language Models, are likely to continue for the foreseeable future, creating many avenues to introduce AM.
- Only new EU legislation can hinder the rapid growth of AM, and such legislation would be difficult to adopt as it would require many social partners to agree on a complex topic.
- AM and general automation tools could be used to tackle the issue of labour shortages, which appear to be on the rise.

Given that the vast majority of respondents (a total of 93.9 %) selected either the baseline or the growth scenario, this implies that a rapid expansion of AM usage is most likely in the coming decade. Hence, **it is likely that in the next five years around half of organisations in the EU will use at least some form of AM technology, while in 10 years, the share is likely to grow to three-quarters of organisations using AM.** This, in turn, implies that AM will become a big part of the workplace in the upcoming years, with most employers having to integrate such tools in order not to be left behind. Similarly, most workers will also have to learn how to use such technologies, at least at a rudimentary level, to adapt to the new realities of the world of work.

Slowdown and other scenarios

Under the slowdown scenario, AM costs will continue to be too substantial for many organisations, especially smaller ones. AM tool developers might exacerbate this issue by focusing more on giant corporations, creating tools that fit their needs but ignoring the needs of smaller enterprises. Lastly, under this scenario, newly adopted regulations in the EU will become a significant barrier preventing some algorithms from being used, such as those that rely heavily on personal data or lead to entirely automated decisions affecting workers. However, this does not mean that AM usage will not grow. Even under the slowdown scenario, more AM tools will be developed, and more companies will use them, but growth will be slow and linear.

Under the slowdown scenario, in the survey, it was assumed that AM will grow by around 1-2 % annually, at least in the upcoming two years. Given current usage, respondents were asked if they believe that in the next five years, around 30-40 % of companies will use AM to some extent, while in 10 years this percentage could increase to 35-45 %. In addition, if the slowdown scenario becomes a reality, the uptake of AM is likely to hover at around 50 % after the 10-year horizon, as regulations will limit what kind of AM can be used. At the same time, such tools might be too expensive for small and micro-enterprises.

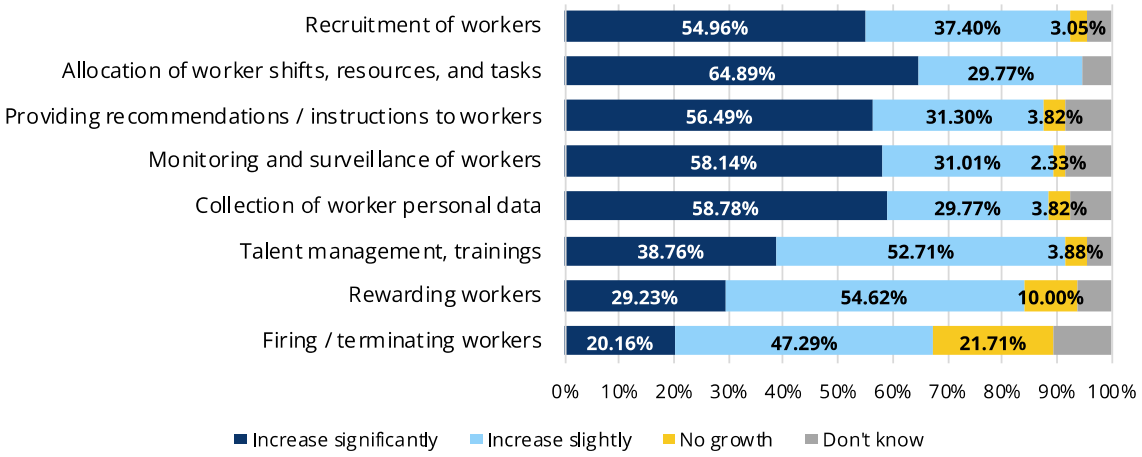
However, **although the logic regarding how the slowdown scenario could come to fruition is plausible, respondents to the Delphi survey felt that it was not very likely, with only 2 % of respondents selecting this option**. A further 4 % of respondents chose the 'Other' option, but most stated that it is difficult to predict the future. Hence, a slowdown or some other scenario seems unlikely to come to fruition without a big 'shake-up' of the current status quo.

4.2.3. The future of AM usage, by specific managerial function

Beyond general trends, some insights can also be derived from the Delphi survey regarding the growth potential of algorithms that automate particular managerial functions. Based on this, 64.89 % of respondents say they expect that the usage of AM for the allocation of worker shifts, resources and tasks will grow significantly in the next 5-10 years (see Figure

4-4 for more details). This is the most significant prediction, implying that this category of tools will see the biggest growth in the coming years.

Figure 4-4: How the usage of algorithms for the relevant managerial functions will change in the next 5-10 years



N = 130.
Source: Authors' elaboration based on Delphi survey.

Next, respondents predicted that the usage of AM tools for collecting workers' private data, monitoring workers, providing recommendations/instructions, and recruitment would grow substantially, and to a similar extent. More specifically, around 55-59 % of respondents believe that usage of AM for such managerial functions will increase significantly in the next 5-10 years. Furthermore, 38.76 % of respondents believe that the use of AM for talent management will grow substantially in the next 5-10 years; 29.23 % share a similar opinion regarding tools to reward workers, while a comparatively smaller number of respondents – 20.16 % – believe that the use of AM for dismissing will increase significantly.

The results above highlight that the growth potential for AM tools is not uniform and, to some extent, depends on the types of managerial functions such tools are used to automate. On the one hand, recruitment, as well as directing workers and controlling their activities will be the areas most often automated. On the other hand, activities not directly related to how workers do their jobs, apart from recruiting workers, seem less likely to be automated by AM in the future. These results are, to a large extent, consistent with the findings in Section 1.2 1.2. Prevalence of AM in the EU. Based on the results discussed in that section, it seems that AM is used least often for rewarding workers, or for terminating their employment – a trend that is likely to continue, based on the Delphi survey.

4.2.4. Likely effects of further growth in AM

The widespread and growing adoption of AM is likely to accelerate and exacerbate certain opportunities and challenges brought by such technologies. In the Delphi survey, the respondents were asked what they think the future holds regarding opportunities and challenges for each scenario, the results of which are summarised in Table 4-1 below. Specifically, the differences between the two most prominent scenarios are presented in bold. These show that individuals who selected the baseline scenario were less optimistic about the effects of AM than those who selected the growth scenario. In addition, respondents most frequently highlighted that AM will have a positive economic effect, some specifying that it will lead to large cost savings and improvements in efficiency and productivity. However, respondents predominantly believe that AM will have a mixed or negative effect on social factors. Specifically, they believe it may have mixed effects on aspects relating to decision-making and accountability, while it will most likely harm aspects related to workers, such as workers' privacy, pressure to perform, mental health, and collective rights. Some respondents also highlighted that AM will negatively affect work-life balance, but that it might be used to combat the issue of labour shortage, as discussed

previously. Moreover, respondents believe AM is likely to have a mixed effect on the carbon footprint. These results are largely consistent with the findings discussed in Chapter 2. Opportunities and challenges, implying that the benefits and issues identified there will persist into the future.

Table 4-1– Impact of growing usage of AM under the alternative scenarios*

Impact on	Baseline	Growth
Economic impacts		
Cost savings for employers	Mostly positive effect	Strong positive effect
Efficiency, productivity	Mostly positive effect	Strong positive effect
Overall competitiveness of companies	Mostly positive effect	Mostly positive effect
Workplace innovation and digital transition	Mostly positive effect	Mostly positive effect
Automation of managerial activities	Mostly positive effect	Mostly positive effect
Social impact		
Transparency in decision making	Mixed effect	Mixed effect
Unbiased and accurate decisions	Mixed effect	Mixed effect
Accountability for biased/inaccurate decisions	Mixed effect	Mixed effect
Routinisation and standardisation of work	Mixed effect	Mixed effect
Deskilling of workers	Mostly negative effect	Mixed effect
Autonomy of workers	Mostly negative effect	Mostly negative effect
Workload and work intensity	Mixed effect	Mostly negative effect
Work scheduling	Mostly positive effect	Mostly positive effect
Physical health and safety of employees	Mixed effect	Mixed effect
Mental health	Mostly negative effect	Mostly negative effect
Performance pressure, burnout, anxiety	Strongly negative effect	Strongly negative effect
Social isolation	Mostly negative effect	Mostly negative effect
Worker privacy	Strongly negative effect	Strongly negative effect
Workplace relationships	Mostly negative effect	Mostly negative effect
Discrimination	Mixed effect	Mixed effect
Collective rights	Strongly negative effect	Strongly negative effect
Other	Negative effect on work-life	Positive effect on labour shortage

Impact on	Baseline	Growth
	balance	
Environmental impacts		
Carbon footprint	Mixed effect	Mixed effect

(*) Results from the slowdown scenario are excluded, given that very few respondents selected this.
Source: Authors' elaborations, based on the Delphi survey.

Conclusions

Conclusion 1: Based on estimates, around a quarter of organisations in the EU-27 are arguably using AM tools, and usage is likely to grow rapidly in the next 10 years.

Based on preliminary quantitative data, the literature review, interviews with experts, surveys of employers and workers, and a Delphi survey involving various stakeholders including experts in the field, around a quarter of EU companies use AM for at least some sort of activity. Organisations most frequently use AM to monitor and evaluate workers and use it least often to terminate/fire or reward workers. Based on several estimates, it is likely that AM usage will see an annual growth rate between 3 % and 6 % over the coming 10 years. However, these estimates are only tentative, given that there is no one database that robustly covers AM; existing datasets covering different aspects of AM often provide conflicting results, and businesses/employers are often reluctant to share their experiences of using AM. This also explains the relatively vague nature of the estimates.

Conclusion 2: AM systems can improve the speed, efficiency, and productivity of workers, in turn reducing costs and increasing profits, but they can also reduce worker autonomy, result in deskilling, and lead to psychosocial risks.

AM can directly benefit companies by automating the performance of some managerial tasks. It can also be used to create support mechanisms that help workers to perform their tasks by providing information or instructions when needed. In addition, AM can foster Occupational Safety and Health (OSH) by alerting workers to dangers, hazards, or ergonomic risks such as providing warnings about bad posture, or if they are not wearing the right protective equipment in a dangerous environment.

However, heavy oversight and control through algorithms can reduce workers' autonomy and might force some to behave like machines to 'please the algorithm'. This might lead to work intensification and numerous psychosocial risks such as fatigue and burnout, as well as deskilling (e.g. if algorithms prescribe the exact way in which work needs to be performed), social isolation, and more. Similarly, the dehumanisation that AM can bring might lead to a decline in meaningful communication between workers and managers, resulting in an erosion of trust, fractured relationships, and miscommunication issues.

Conclusion 3: If used appropriately, AM can foster fairness and good decision-making, but it can also create challenges in relation to data protection and the possible exacerbation of existing biases.

AM provides a means for companies and organisations to collect, analyse and use workers' personal data on an unprecedented scale. If such data are used appropriately, this can enrich a company's decision-making process by, for example, allowing it to make fast decisions on complex questions; identifying hidden talents within the workforce; highlighting if a worker is experiencing burnout, and more. AM can also foster fairness if such tools are created using unbiased data, and hence do not perpetuate any prejudice against a particular individual or group holding protected characteristics.

However, even if such tools are used appropriately, many concerns remain regarding workers' privacy. These concerns extend beyond the mere processing of extensive data, as some AM solutions can be used to predict workers' emotional state, including aspects employers might consider a 'weakness' and hence find a reason to disadvantage such workers. In addition, if algorithms are built using biased data, they could perpetuate discriminatory and biased practices. This issue is also particularly difficult to address, as many AM tools that use AI constantly learn and adapt, allowing a once unbiased system to begin exhibiting biases. However, the opposite may also occur.

Conclusion 4: AM can directly empower workers' collective rights, but it can also provide employers with tools to undermine collective bargaining.

Given that AM tools collect extensive data about workers, if such data are made available to worker representatives and works councils, they could enable them to determine if a

worker was demoted or fired without just cause, or if issues occur concerning unpaid work, lack of breaks, procedural unfairness or similar. Even where workers' representatives do not have access to data, the very existence of such information might, in some cases, prevent some employers from abusing their power, as access to such evidence could be requested by a court if a worker decides to file a claim in the event of a labour dispute. However, there is a risk that employers could also use AM to undermine collective bargaining. For example, AM tools could be used to predict which workers are likely to instigate unionisation and exercise collective rights. In addition, even in cases where workers' representatives have access to data, the complex nature of AM tools might hinder their ability to make sense of and use them.

Conclusion 5: Although current EU legislation does not specifically address the usage of AM, existing provisions can safeguard workers from many of its negative effects. However, some gaps remain.

AM systems function in ways that diverge from the way in which managerial functions are exercised in conventional work relationships, in terms of volume, velocity and variety. This means that the effectiveness of the various relevant legal instruments can vary considerably. The existing EU legal framework is suited to addressing the foundational aspects of a range of issues related to the effects of AM on workers' working conditions. These encompass aspects such as work intensity, transparency, predictability, and questions related to OSH. For example, although the current OSH Framework Directive does not specifically address AM, it requires employers to take preventive and protective measures, provide information and training programmes, and adapt proactively to changing circumstances regarding safety and health at work – all of which apply to AM usage. In the area of working time legislation, some structural challenges that may be relevant to AM systems concern the binary distinction between working time and rest time, the lack of indications as to how to compensate for productive and unproductive working time, and insufficient protection against underemployment. Lastly, the existing legal *acquis* often fails to cover self-employed persons, who may be subject to AM. Admittedly, these concerns are not unique to AM but represent broader challenges.

Conclusion 6: The General Data Protection Regulation (GDPR) provides a solid backbone to govern the processing of workers' personal data, but it also has its limitations.

The direct applicability of the GDPR compels AM providers and adopters (i.e. employers) to meet the general principles contained therein. The key principles and rules, such as grounds for lawful processing, principles of purpose limitation and data minimisation, the prohibition of solely automated processing and the required safeguards when the ban does not apply, protect also against abusive use of AM systems at the workplace. This makes data protection and privacy rules a primary, yet not exclusive, building block for the effective regulation of AM at work. However, existing EU data protection rules do not cover all functions of AM comprehensively. For example, under the GDPR, unless a decision is based on *entirely* automated processing, employees do not possess the right to receive information about the existence of the decision-making system, or the reasoning, relevance, and implications of the decision-making process. This is a significant gap, given that many AM systems that impact individuals combine both automated and human-driven components. Moreover, the general nature of many GDPR provisions represents a shortcoming in relation to the use of AM at work, as it leads to an increasing number of interpretive difficulties as regards some provisions and their defining features (e.g. 'solely automated' and 'significant effects' in Art. 22 GDPR; 'provided by the data subject' in Art. 20 GDPR, etc.). Other limitations pertain to the scarcely developed collective character of the GDPR, making it somewhat inadequate for effective governance of the complexities inherent in AM.

Conclusion 7: The ways in which AM tools operate do not fit within the provisions of the EU legal framework for equality and non-discrimination.

The EU equality framework is in principle suitable to address discrimination stemming from AM systems. However, the constitutive features of algorithmic discrimination are barely compatible with the 'traditional' notions of discriminatory practices, thus reducing the effectiveness of the existing rules and redress mechanisms. AM systems design individualised instructions, treatments, and requirements for workers by aggregating legitimately collected data, which are not necessarily aligned with the set of grounds protected under EU law. Moreover, some AM tools are often almost undetectable, making it challenging to identify and rectify discriminatory behaviour. This is exacerbated by the fact that AM systems often lack transparency. In addition, it is often a challenge to secure evidence in cases of algorithmic discrimination due to the opacity of algorithms, data access limitations, and the need for specialised expertise. Finally, mobilising anti-discrimination legislation in courts in cases of algorithmic discrimination can be arduous. However, the evidentiary regime eases the onus on victims, with anti-discrimination directives allowing for a partial shifting of the burden of proof from claimant to the alleged perpetrator.

Conclusion 8: Collective bargaining is crucial to ensure the appropriate usage of AM

Given the ever-evolving nature of AM and the constantly transforming world of work, collective rights such as information and consultation and collective bargaining provides a flexible framework to mitigate the negative effects of AM, even in the absence of regulations directly governing AM. This requires awareness-raising and training in the workplace, emphasising the responsible and law-abiding use of technology to ensure unbiased and fair practices. Worker representatives are also in the best position to codetermine the introduction and deployment of AM systems via social dialogue. They can also co-design internal rules due to their capacity-building and vast knowledge of operational practices and internal bottlenecks. In addition to information and consultation, codetermination and collective bargaining, workers' representatives can file collective legal complaints on behalf of groups of workers before courts, administrative bodies, and data protection authorities. Collective agreements are gradually starting to include terms relating to information and consultation concerning AM systems at work. They generally deal with the impacts of such systems in connection with occupational health and safety, privacy, and the evaluation of work performance.

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